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Mathematical Foundations for Computer Science

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Introduction

The interconnectedness of mathematics and computer science is generally accepted as relevant and necessary, especially in the undergraduate curricula on technical schools. However, this linking often remains in the form of general statements and is not quite understood by the computer science students themselves. They sometimes feel that the mathematics they are required to study is not relevant for computer science and is a nuisance in their studies. Oftentimes in such cases, the problem is not the mathematics itself, but rather the content selected and the way it is presented.

Still, the well known experts, such as Donald Knuth in his textbook [4], consider teaching mathematics to future computer scientists and programmers as extremely useful. Also in practice, it has been shown that it is not sufficient to add a mathematician to a team of programmers. To find a good solution to an informatics problem, we need more than a rudimentary appreciation of mathematical recipes. Understanding the underlying mathematical concepts can significantly improve the work and effectiveness of the created solutions and tools and in many cases can help spot errors and inaccuracies.

The inspiration for our teaching of mathematics for computer science was an MIT course by Professor Albert R. Meyer [6]. In our own course, we chose a lower difficulty level to enable the first-year bachelor' studies in computer science also for students coming from high schools that do not primarily prepare students for higher education. The problems in this textbook are typically at the level of the senior year at high school, interspersed with more difficult tasks and challenges for those who are familiar with the basics.

Our goal is not just to teach the students how to solve the problems provided in this textbook. In addition to this, it is important that the students learn how to approach a problem, how to discuss it with others, how to verify that a solution (either one's own or somebody else's) is correct.

The topics and problems were selected in such a way that they show many

common concepts and analogies between mathematics and computer science. This is the reason why every topic includes a set of problems meant to be worked out using a computer and link both mathematical and algorithmic skills of the students.

The book contains 12 chapters which represent the material covered in 12 weeks of semester – one chapter per week. Each chapter has four sections. The first one is **Lecture Overview**. It is not an exhaustive theoretical treatment of the week's topic. We tried to avoid the traditional high formalism, which sometimes makes the content incomprehensible for students, but rather the text is provided in the form of notes from an introductory lecture. New terms and ideas are explained as we encounter them and are denoted in *italics*.

The second section in each chapter is a list of **Questions**. The answers to these questions should be clear after going through the lecture overview and serve as a confirmation of understanding. We do not provide the answers explicitly next to the questions, but recommend going back to the lecture text for clarification, if needed. After answering these questions, a reader should be ready to tackle the problems.

The **Problem Set** is meant for practice, to help the reader gain confidence in mastering the material and clear up ambiguities if there are any. It is not meant to be read all at once, therefore we start with stating the problems and then in the next sections we give **Hints**. These should only serve as *inspiration* or *kick-off* points if the reader does not know how to start or how to continue with a problem. The reader should follow the hints only to such extent as minimally needed and then continue on their own.

We also provide more detailed **Solutions** for selected problems but reading all the steps should be the last resort. We consider them almost unnecessary, because reading the solutions cannot replace working out the problems but we agree they can be useful for people who solved the problem and want to compare the results. We also recommend working on these problems in groups with fellow students. In our experience, this is very beneficial for mastering the material.

The final section in each chapter is called **Lab problems**. This is a second problem set, structured similarly to the previous one. The difference is that these problems should be worked out on computer, mostly using Excel. The choice of Excel over the option to work in some programming language, e.g. Java or Matlab, was made for clarity. It allows us to focus less on the technical solution and more on the mathematical concepts we want to demonstrate.

While we provide some introduction to Excel functions as we go, this is not meant to teach Excel. We view Excel as a tool, which lets the reader experiment. In these problems, it is often not sufficient to create a particular

Excel worksheet that does something, but it is necessary to use it to answer (mathematical) questions. Again, the reader should attempt to create their own solution before reading the hints and steps provided in the text, which should only be read to the extent needed to for continuing.

Challenges

Throughout the text we have included boxes like this one. They contain more advanced questions or more challenging problems. As a warmup, try to estimate which chapters contain material that you know and in which are you going to encounter new ideas.

With few exceptions noted below, the individual topics are self-contained and in principle, could be re-arranged, but we recommend using them in the order presented in the textbook. This order resulted from our experience teaching the course in a few different ways. We have decided to use some concepts even before they are formally introduced and only make them more precise later. This can be seen for example with proofs, which are theoretically discussed at the very end of the book. The reason behind this is that we wanted the students to have some experience with proofs before we discuss them theoretically.

We want to thank our students who already took this course and iteratively lead us in development of this material. We thank our colleagues who taught the course with us and shared their experience. Our sincere thanks also belong to our reviewers who offered detailed constructive feedback that helped us significantly improve the manuscript and to Professor Martin Klimo, who has always suggested and supported the idea that mathematics and computer science are interconnected and stood behind the creation of this course. Finally, we acknowledge the support of Ministry of Education, Science, Research and Sport of the Slovak Republic as part of the KEGA project (contract No. KEGA 052ŽU-4/2018) and the project University of Žilina Enhancing Quality of Education based on Internationalization (contract No. 04ŽU-2/2016).

We hope this textbook will be a helpful, to this day not very common, crosslink between the worlds of mathematics and computer science.

Abbreviations

AD = Anno Domini

FSA = Finite State Automaton

GCD = Greatest Common Divisor

IBAN = International Bank Account Number

ID = IDentification

LCM = Least Common Multiple

PC = Personal Computer

RSA = encryption algorithm developed by Rivest, Shamir, Adleman

TM = Turing Machine

XOR = eXclusive OR operator

ZFC = Zermelo–Fraenkel set theory with the axiom of Choice

Chapter 1

Combinatorics

1.1 Lecture overview

1.1.1 Rules for counting

Problem 1.1 Castles from blocks:

Suppose we want to build all possible castles using seven cube blocks while respecting the following rules. 1. Castle has several towers in a row. 2. The towers can have different height. 3. When looking at two towers that are next to each other, a tower on the left cannot be smaller than the tower on the right, see Figure 1.1. So going from the left to right, the towers stay the same or decrease in height. How many possible castles are there?



Figure 1.1: Castles from blocks. a) Correct castle. b) Incorrect castle, because the tower on the right is taller than the tower immediately to the left of it.

Solution: To solve this problem, we split it into smaller subproblems. We calculate how many castles with seven floors are there, how many with six, all the way to how many with just one floor. We add the results to find out that there are 15 castles. Alternatively, we could have split the problem by the number of towers in a castle.

Good advice for solving problems

- If the problem is too difficult, split it into several smaller simpler problems.
- Often it is helpful to solve the problem for smaller numbers first. Even if it means writing out all the possibilities.
- When solving combinatorial problems by writing out all the possibilities, it is useful to create some sort of system using which it will be clear whether a given possibility has already been counted or not.

Addition rule

Let $A_1, A_2, \dots, A_n$ be disjoint sets. Then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n|.$$

Here we see some notation that will be explained later in more detail. For now, we simply say that *set*, e.g. A_1 , is a collection of items, $|A_1|$ says how many items there are and $\cup$ denotes *set union* – two sets joined into one. *Disjoint* means that every item can be only in one set. If two or more sets share an item, this rule can no longer be used and we return to such situations in Section 6.1.4. For our Problem 1.1, $n = 7$, A_1 may denote the set of castles with just one floor, $|A_1| = 1$ because there is just one such castle and $|A_1 \cup A_2 \cup \dots \cup A_7| = 15$.

Problem 1.2 Licence plates:

Licence plates have 6 to 8 characters. They have to start with a letter, but the remaining characters can be either letters or digits. How many different licence plates are there?

$$L = \{A, B, \dots, Z\}, \quad C = \{A, B, \dots, Z, 0, 1, \dots, 9\}$$

Solution: Here we again have some new notation that will be revisited later when we talk about sets. For now only briefly: $A \times B$ denotes a *Cartesian product of sets*. The result is a new set of *ordered pairs*. The first item in the

pair comes from the set A , the second from the set B . All possibilities are included. A generalisation of this concept is A^n , where we repeat the product operation several times, e.g. $A^4 = A \times A \times A \times A$. We write the solution to the licence plates problem in this notation:

$$\begin{aligned} |(L \times C^5) \cup (L \times C^6) \cup (L \times C^7)| &= |(L \times C^5)| + |(L \times C^6)| + |(L \times C^7)| \\ &= |L| \cdot |C^5| + |L| \cdot |C^6| + |L| \cdot |C^7| \\ &\doteq 1.3 \cdot 10^{12}. \end{aligned}$$

Multiplication rule

Let $A_1, A_2, \dots, A_n$ be distinct sets. Then

$$|A_1 \times A_2 \times \dots \times A_n| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_n|,$$

where $\times$ denotes the Cartesian product.

Problem 1.3 Passengers on a bus:

- a) 28 passengers have arrived at a bus station and formed a queue to buy tickets. How many different queues are possible?
- b) 28 passengers boarded a bus which has 40 seats. How many possibilities are there for them to sit down?
- c) 28 passengers exited the bus and nobody else has entered. In the 40-seat bus, there are now 28 used and 12 unused seats. How many possibilities are there to have 28 used seats out of 40?
- d) How many possibilities are there to have 12 unused seats out of 40?
- e) 28 passengers exited the bus at stops K, L, M, N, O. Leaving the stop O, the bus was empty. How many possibilities are there for this to happen? Note that this question can be understood in two ways. In one we care which person got off at which stop, in the other we only care how many people got off at each stop. Find both answers.

Hint: We solve the 28 passenger problem for 3 passengers first. Let us call them A, B, C. Using the second advice mentioned before, we write out all the possible queues: ABC, ACB, BAC, BCA, CAB, CBA. Looking at this, we can formulate

a strategy. We start with one passenger, from the remaining ones we select one for the second place and then put the remaining person at the third place. It means we have 3 options for the first place, 2 for the second and 1 for the third. Altogether $3 \cdot 2 \cdot 1 = 6$ possibilities. Can you now extend this to 28 passengers and use similar strategy for questions b)-e)?

Problem 1.4 Passwords:

- a) How many possible four-letter passwords can be created using four letters of the word POKE? (The password must contain each letter exactly once.)
- b) How many possible four-letter passwords can be created using four letters of the word BOOK? Words BOOK and BOOK (two letters O have switched places) are the same.
- c) How many possible six-letter passwords can be created using six letters of the word KEEPER?
- d) How many possible ten-letter passwords can be created using ten letters of the word BOOKKEEPER?

Problem 1.5 Codes with zeros and ones:

- a) How many possibilities are there to place two 1s in a 4-bit word?

Solution: For one digit 1, we have 4 possibilities for placement. For the other digit 1, we only have 3 (one is occupied), so at the first glance it might seem that there are $4 \cdot 3 = 12$ possibilities. However, the 1s are identical (similar to letters O in the previous problem, part b)), so the order does not matter and each possibility was counted twice. Therefore the actual number of possibilities has to be halved, bringing us down to 6. We do not have to consider the placement of 0s now, because for each placement of 1s, there is just one way how to fill in the 0s. We could have however started by placing the two 0s first and then filling the remaining bits by 1s, the outcome would have been the same.

- b) How many possibilities are there to place four 1s in an 8-bit word?
Hint: Carefully consider how many placements of those four 1s result in the same possibility.

Problem 1.6 Encoding sets by sequences:

For counting something, it might be easier to count something else, e.g. instead of directly counting the number of subsets of a given set with n elements, we can count the number of different sequences of 0s and 1s. The problems then are how do we know that we have the same number of items (subsets/sequences) in each case? And how do we transform a set into a sequence? Let us look at these step-by-step.

- a) Let us have a set $\{x_1, x_2, x_3\}$. What are its subsets?

Solution:

$$\emptyset = \{\}, \{x_1\}, \{x_2\}, \{x_3\}, \{x_1, x_2\}, \{x_1, x_3\}, \{x_2, x_3\}, \{x_1, x_2, x_3\}$$

- b) Write these subsets as bit sequences (sequences of 0s and 1s).

Solution: We use 1 for those bits that represent elements present in the subset and 0 for those bits that are not included in the subset:

$$(0, 0, 0), \quad (1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1)$$

$$(1, 1, 0), \quad (1, 0, 1), \quad (0, 1, 1), \quad (1, 1, 1)$$

- c) Can we find exactly one 8-bit sequence for each subset of $\{x_1, x_2, \dots, x_8\}$?

Solution: We can use the same encoding as in the previous case, e.g.:

$$\{x_2, x_3, x_6, x_7, x_8\} \rightarrow (0, 1, 1, 0, 0, 1, 1, 1)$$

- d) How many subsets does an n -element set have?

Solution:

$$\{0, 1\} \times \{0, 1\} \times \dots \times \{0, 1\} = \{0, 1\}^n \Rightarrow |\{0, 1\}^n| = |\{0, 1\}|^n = 2^n$$

Problem 1.7 Hockey match outcomes:

The CAN:SVK hockey match score was 6:3. Write all possible courses of play as sequences of letters C and S, representing goals scored by individual teams. Re-write these sequences using 1s and 0s. How many are there?

Problem 1.8 Streets and avenues:

Suppose that going from home to school in Manhattan, where there is a rectangular grid of *streets* and *avenues*, means going 3 blocks north and 5 blocks east. How many different possibilities are there to get to school if we always take the shortest route (only 3 blocks north and 5 blocks east)? Write these routes as sequences of letters N and E and then as sequences of 0s and 1s. How many are there?

Problem 1.9 Donuts and their flavors:

Donuts can have five different flavors and there is always a dozen of them in a box. We do not have to include every flavor in every box and the position of the donut in the box does not matter. How many different boxes can we have?

Solution: Suppose the flavors are P – plain, L – lemon, C – chocolate, N – nougat, B – berry. One possibility is: PPPLLLLCCNBB. Another possibility is: PPLCCCCCBBB. We map the possibilities to 16-bit words using 1s and 0s, while keeping the order of P L C N B. The 1s separate the individual flavors:

$$\text{PPPLLLLCCNBB} \rightarrow 0001000010010100$$

$$\text{PPLCCCCCBBB} \rightarrow 0010100000011000$$

For example, if we order the 0000000000001111 box, we are going to get 12 plain donuts. We have created a bijection between boxes of dozen donuts and 16-bit words with four 1s.

Bijection rule

Bijection is a one-to-one correspondence between two sets. If $f : A \rightarrow B$ is a bijection, then $|A| = |B|$. By $f : A \rightarrow B$ we denote a *mapping* f which gives an element of B for an element of A.

Injection rule

Injection is such a mapping that cannot assign the same elements of B to different elements of A. If $f : A \rightarrow B$ is an injection, then $|A| \leq |B|$.

Surjection rule

Surjection is such a mapping that each element of B is assigned to some element of A, i.e. the whole B is covered. If $f : A \rightarrow B$ is a surjection, then $|A| \geq |B|$.

Problem 1.10 What do these questions have in common?

- a) There were 5 teams in a round-robin tournament, i.e. every team played with every other team. How many matches were there?
- b) Five people at a birthday party clinked their wine glasses with each other. How many clinks were there?
- c) How many edges are there in a complete graph with five vertices?
- d) How many airline routes can there be to connect five cities?
- e) How many sides and diagonals does a regular pentagon have?

Problem 1.11 Coding contest (reducing possibilities):

There were 12 participants in the latest hackathon. The top three received a new laptop. How many possibilities are there for the best trio out of 12 coders?

Solution:

$$\frac{12 \cdot 11 \cdot 10}{6}$$

Problem 1.12 Chessboard (reducing possibilities):

How many possibilities are there to place two rooks (towers) of the same color on a chessboard so that they do not protect the same row or column?

Solution: The position (r_1, c_1, r_2, c_2) is the same as (r_2, c_2, r_1, c_1) .

$$\frac{8 \cdot 8 \cdot 7 \cdot 7}{2}$$

Division rule

If f is a mapping $A \rightarrow B$ which maps exactly k elements of A to each element of B , (*mapping k -to-1*), then it is true that

$$|A| = k|B|, \quad |B| = \frac{|A|}{k}.$$

Problem 1.13 Zeros and ones (revisited):

How many possibilities are there to place four 1s in an 8-bit word?

Solution: We have already looked at this problem once. Now we know that we need a reduction for all those placements of 1s that result in the same possibility. What kind of reduction? For any given 4 bits we have $4 \cdot 3 \cdot 2 \cdot 1$ possibilities how to place four distinct 1s. Since they are not distinct for us in this setting, the reduction is 24-to-1. The number of possibilities is then

$$\frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1}.$$

Factorial

There is a useful notation for shortening expressions such as $4 \cdot 3 \cdot 2 \cdot 1$ and it is $4!$. It is called *factorial* and it denotes the product of all natural numbers not larger than the given number, i.e. $n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1$. Note the special case defined separately: $0! = 1$.

Incidentally, the statement $0! = 1$ is also true in many programming languages, since 0 is clearly not equal to 1.

Combinatorial number

Many of the previous problems came down to computing how many ways we can choose a set of k items out of n options. The notation for the number of such selections is $\binom{n}{k}$ and it is called a *combinatorial number* or a *binomial coefficient*. It is calculated as

$$\binom{n}{k} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1}{(k \cdot (k-1) \cdot \dots \cdot 2 \cdot 1) \cdot ((n-k) \cdot (n-k-1) \cdot \dots \cdot 2 \cdot 1)} = \frac{n!}{k!(n-k)!}.$$

1.1.2 Pigeonhole (Dirichlet) principle

Problem 1.14 Drawers, pigeons, socks and Prague:

- a) Suppose we have books and notebooks for five different subjects. Our desk has five drawers and so we put all items for each subject into a separate drawer. Then we find out that a mouse has moved into one of the drawers and so we have only four drawers left. As a consequence, we have to move the items from that drawer to another one and thus there will certainly be a drawer that contains items for two different subjects.
- b) Suppose we have 15 pigeons and a dovecot with 10 holes. There is certainly be at least one hole through which at least two pigeons enter the dovecot.

- c) We are taking socks out of a drawer. There are black and white socks but it is so dark in the bedroom that we cannot tell them apart. Since we do not see them, we have to take at least three socks to be sure that we have a pair of the same color.
- d) There are at least six people in Prague, who have the same number of hairs on their heads.

Solution: We have to make some assumptions about the number of hairs and the number of people in Prague. Suppose that a person has at most 200 000 hairs and that there are 1 391 000 people in Prague. If we group the people in Prague by their hair number and try to do it in such a way that there are no more than five people in each group, we end up with $200\,000 \cdot 5 = 1\,000\,000$ people in groups. There is a group of five people with one hair each, a group of five people with two hairs each, with three hairs etc. There are still 391 000 people left. We could make a one more group of five bald people, but everybody else must join one of the existing groups and so we know that there will be at least one group of six people with the same number of hairs.

- e) Can we say that there are at least seven people in Prague who have the same number of hairs?

Pigeonhole principle

- If the number of elements of set A is larger than the number of elements of set B , i.e.

$$|A| > |B|,$$

then for each mapping f from A to B it is true that we can find two distinct elements a_1 and a_2 from A for which $f(a_1) = f(a_2)$.

- If the number of elements of set A is larger than k -times the number of elements of set B , i.e.

$$|A| > k|B|,$$

then for each mapping f from A to B it is true that we can find $k + 1$ distinct elements of A which get mapped to the same element of B by the mapping f .

1.1.3 Brief review of related concepts

Binomial theorem

This theorem tells us about expansion of powers of a binomial:

$$(a + b)^n = \binom{n}{0}a^n b^0 + \binom{n}{1}a^{n-1}b^1 + \dots + \binom{n}{n}a^0 b^n.$$

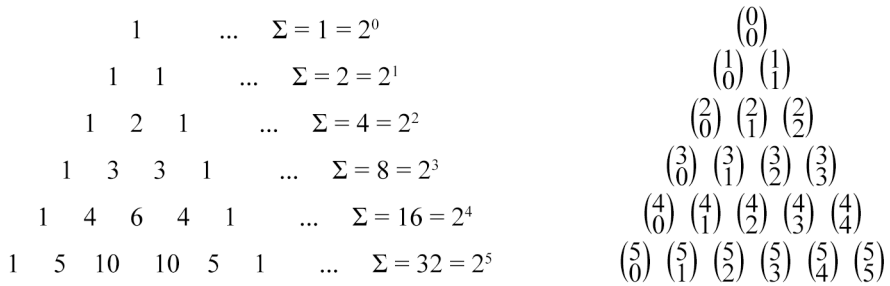


Figure 1.2: Pascal's triangle.

Pascal's triangle

Binomial coefficients are also present in the *Pascal's triangle*. It is a triangular array of combinatorial numbers in which the numbers in n^{th} row are the same as the coefficients in front of the terms of $(a + b)^n$. The sum of numbers in the n^{th} row of Pascal's triangle is 2^n , Figure 1.2. Every number in the Pascal's triangle except the 1s that are on both sides can be calculated as the sum of the two numbers directly above it.

1.2 Questions

1. A castle consists of several towers built next to each other. A tower cannot be taller than the tower immediately to its left. How many different castles can be built using 7 cubes?
2. There are 28 people in a queue for tickets. How many different queues can there be?
3. A bus has 35 seats. How many ways are there for 28 passengers to sit down?
4. 28 passengers got on an empty bus in Žilina. They all got off at one of stops Teplička, Strečno, Vrútky, Martin. How many possible ways were there for them to get off the bus?
5. How many 4-letter passwords can be created using the letters SONG?
6. How many 4-letter passwords can be created using the letters LOOK?
7. How many 10-bit words can be created using exactly four 1s (e.g. 1111000000)?
8. How many subsets does the set $\{2, \text{sun}, @\}$ have?
9. What is the connection between the number of subsets of a 4-element set and the number of 4-bit words?
10. What is the connection between the number of different matches that ended 3:2 and the number of 5-bit words with three 1s?
11. Going home from school, I have to go 3 blocks east and 6 blocks south. What is the connection between these routes and the set of all sequences of three 0s and six 1s? Re-write the route ESSSEESSS using 0s and 1s. Re-write the sequence 101111001 using directions E and S.
12. How many ways are there to fill a box with 10 donuts when there are 3 flavors available?
13. How many games are there in a round-robin tournament with 6 players (everybody plays all other players)?
14. How many different triplets of winners can there be in the top coder contest, when all 800 students at the faculty participate?

15. How many ways are there to place 3 white rooks (towers) on a chessboard 8×8 so that they do not attack each other?
16. What does the binomial theorem say?
17. What is the sum of coefficients in the binomial theorem (i.e. in one row of Pascal's triangle)?
18. What does the pigeonhole principle say?
19. There are white, black and pink socks in a drawer. When in the dark, how many of them do we have to take out to be sure that we have a pair of the same color?

1.3 Problem set

1. Horse-race

- There were three horses in a horse-race: Alejandra, Beatrix and Claire. Write all possible orders how they can arrive at the finish line. (We assume that two horses never arrive at exactly the same time.)
- What are the possible orders if a horse called Pedro joins the race as well? You can use the orders from the previous question.
- There were eight horses in a race. How many different orders are there at the first three places (we do not care about the remaining places)?
- There were eight horses in a race. How many possible triplets of winners are there? Winners here are the first three horses regardless their specific medal position.

2. Shortest path in a grid

How many different shortest paths are there from the upper left corner to the lower right corner in a grid 5×3 , see Figure 1.3?

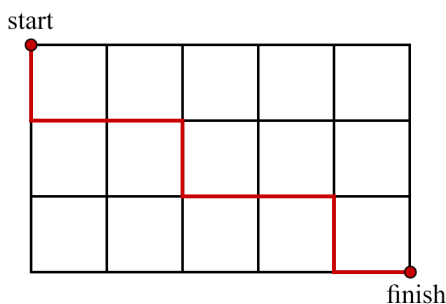


Figure 1.3: One of the shortest paths in the grid is shown in red.

3. Hockey match

The result of a hockey match between Slovakia and Canada was 4:3.

- How many different matches can there be? (Here different matches are those that have a different order of Slovak and Canadian goals.)

- b) In the next match, there were also 7 goals. In how many ways could these goals be scored in the 1st, 2nd and 3rd period of the game? (Here we do not care which team scored which goal.)
- c) Overall, there were 22 matches at the tournament and none of them had more than six goals total. At most, how many matches could there (certainly) be which had the same number of goals?

4. Licence plates

A licence plate in our town can have these forms:

- Letters ZA, then three digits, then two letters
- Letters ZA, then five letters

Let us denote by L the set of all licence plates in our town.

- a) Express the set L using the sets $C = \{A, B, C, \dots, Z\}$ and $D = \{0, 1, 2, \dots, 9\}$ and set operations.
- b) Compute $|L|$, i.e. the number of all possible licence plates.

5. Queues

There are six boys (B) and four girls (G) who are asked to form a queue BBGGBBGGBB. They all have different names.

- a) In how many ways can they arrange themselves into such a queue?
- b) In how many ways can they arrange themselves into such a queue if we do not care about the order in the BB, GG pairs?

6. Oranges

Parents want to give seven oranges to their four children.

- a) In how many ways can they do that?
- b) In how many ways can they do that if each child should get at least one orange?
- c) In how many ways can they do that if the youngest child should get at least two oranges? (It may happen that some of the older children get zero oranges.)

7. Grandpa's bunnies

Grandpa has 21 bunnies and five stalls. Determine whether the following statements are always true, sometimes true or always false. In case the statement is sometimes true, give an example with placement of bunnies in stalls that makes it true and another example that makes it false:

- a) There is a stall that has two bunnies.
- b) In every stall there is at least one bunny.
- c) There is a stall that has at least two bunnies.
- d) There is a stall that has exactly four bunnies.
- e) In at least two stalls there are at least four bunnies.
- f) There is a stall that has at least four bunnies.
- g) In at most three stalls there are at most three bunnies.
- h) There is a stall that has 22 bunnies.
- i) In at most two stalls there are at least six bunnies.
- j) There is a stall that has exactly five bunnies.
- k) There is a stall that has at least five bunnies.
- l) There is a stall that has zero bunnies.

8. Grandpa's bunnies one more time

Grandpa has 21 bunnies and

- a) four stalls
- b) five stalls
- c) six stalls

Based on this information, what can be said about the number of bunnies in the individual stalls?

9. Hiking in a forest

- a) Five students went hiking in a forest in the shape of square with 4 km side. Each student has a transmitter with range $2\sqrt{2}$ km. How do you show that at least two of them can contact each other?
- b) Solve the same problem if the forest is an equilateral triangle and the range of the transmitter is 2 km.

1.3.1 Hints and solutions

1. Horse-race

- a) There were three horses in a horse-race: Alejandra, Beatrix and Claire. Write all possible orders how they can arrive at the finish line. (We assume that two horses never arrive at exactly the same time.)
- b) What are the possible orders if a horse called Pedro joins the race as well? You can use the orders from the previous question.
- c) There were eight horses in a race. How many different orders are there at the first three places (we do not care about the remaining places)?
- d) There were eight horses in a race. How many possible triplets of winners are there? Winners here are the first three horses regardless of their specific medal position.

Hints:

- When writing out all possibilities it is useful to stick to a reasonable system and not write the possibilities in random order. Using a system ensures that we do not forget any possibility and do not write any possibility twice.
- We can verify the number of possibilities by a calculation. The following questions can help us with that: How many possibilities are there for the first place? How many possibilities remain for the second place and subsequently for the third place? The multiplication rule might help.
- When looking for the triplets of medalists, we are not interested in specific orders in each triplet, therefore what we need to do is to suitably reduce the number of orders. How many different possibilities are there for ordering the horses in each triplet?

Solution:

- a) We can start with the possibility that they arrive in the alphabetical order A, B, C and then switch the second and third place. We repeat this for the cases when the other horses finish first.

$ABC \quad ACB \quad BAC \quad BCA \quad CAB \quad CBA$

An alternative method for writing out all possibilities is to use a *tree structure* as depicted in Figure 1.4. We have all the possibilities for the first place written on top. Then we have all options for second places and finally all third places. The lines connecting the names of the horses represent the individual possibilities.

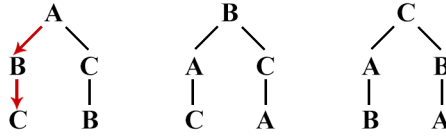


Figure 1.4: All possibilities of the orders of three horses at the finish line.

We can do the following calculation to verify that we have all the possibilities. There were three options for the first place. Since a horse cannot take two places simultaneously, there are only two options left for each second place and consequently just one option for the third place. The multiplication rule helps us calculate that there are six possible orders.

1 st place	2 nd place	3 rd place	$\Rightarrow 3 \cdot 2 \cdot 1 = 6$
3 possibilities	2 possibilities	1 possibility	

- b) The new horse Pedro (P) can finish at any place. We obtain the orders for four horses from the orders for three horses by including Pedro consecutively at all possible places:

PABC PACB PBAC PBCA PCAB PCBA
APBC APCB BPAC BPCA CPAB CPBA
ABPC ACPB BAPC BCPA CAPB CBPA
ABCP ACBP BACP BCAP CABP CBAP

The visualisation using the tree method is shown in Figure 1.5.

We calculate the number of possible orders the same way as before. This time, there are four possibilities for the first place (one of the horses A, B, C and P). Regardless of who won, there are three possibilities for the second place. The same reasoning leads to the fact that only one of the remaining two horses could be at the third

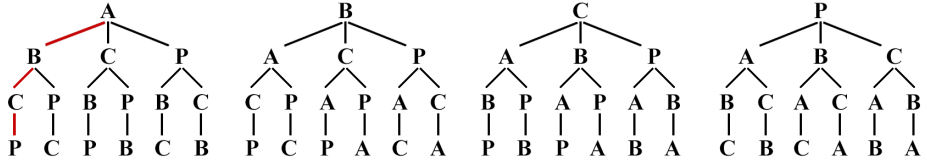


Figure 1.5: All possible orders for four horses.

place in each case and the last horse finished fourth. Altogether, this gives 24 possible orders.

1 st place	2 nd place	3 rd place	4 th place
4 poss.	3 poss.	2 poss.	1 poss.

↓

$$4 \cdot 3 \cdot 2 \cdot 1 = 24$$

- c) There were eight possibilities for the winner, seven possibilities for the second place and one of the remaining six horses finished third, therefore there are 156 different possible orders.

1 st place	2 nd place	3 rd place	
8 poss.	7 poss.	6 poss.	$\Rightarrow 8 \cdot 7 \cdot 6 = 156$

- d) We know from the previous problem that there are $8 \cdot 7 \cdot 6 = 156$ possibilities for first three places in races of eight horses. Since now we are only interested in medal placements, not a specific medal, we need to suitably reduce this number. For example, if Alejandra, Beatrix and Pedro were the first three, then we consider all orders in which they could have finished (ABP , APB , BAP , BPA , PAB , PBA) as one triplet. Therefore we calculate the number of all winner triplets as

$$\frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{156}{6} = 56.$$

2. Shortest path in a grid

How many different shortest paths are there from the upper left corner to the lower right corner in a grid 5×3 ?

Hints: In order to understand the problem better, we can draw a grid with three rows and five columns. Since we are looking for the shortest path, there are only two directions allowed: south and east. We need to go south three times and east five times. If we denote the directions by 0s (for south) and 1s (for east), the individual possible trips can be written as sequences of 0s and 1s. We can thus map the grid problem into binary words and continue using the approach shown in the lecture.

Solution: We start by drawing the grid, Figure 1.6.

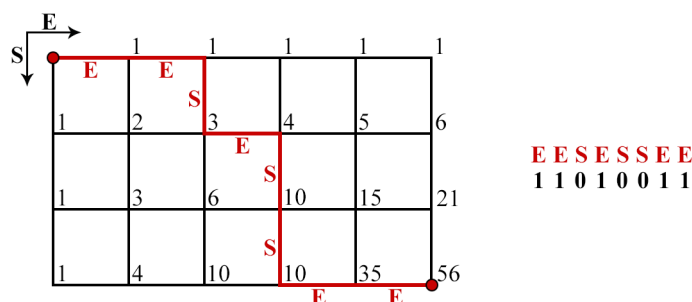


Figure 1.6: One possible shortest path in the grid. The numbers at the crossings denote how many ways there are to get to each crossing.

We are going to calculate how many possible ways are there to get to the individual crossings when we can travel only east (E) or south (S). There is only one way how to get to the crossing immediately to the right of the starting point and we denote it E. Similarly, there is only one way how to get to the first crossing to the south and we denote it S. Now there are two possible paths to the crossing that lies south-east: SE and ES. The crossing that can be reached via SSE can also be reached through SES and ESS, which means three different paths. This can also be determined by adding the numbers of paths that lead to the immediate predecessors of these crossings, i.e. one way to SS and two ways to SE/ES. We continue this way until we get to the lower right corner.

Another solution: Instead of using S and E to denote the directions, we can use 0s and 1s. If we denote the south direction $\downarrow$ as 0 and the east direction $\rightarrow$ as 1, we can express each path as a sequence of three 0s and five 1s. 00011111 is an example of one such path, which first takes us three blocks south and then five blocks east. We can now transform the problem

into a different question. How many different 8-bit words are there that contain exactly three 0s? How many different 8-bit words are there that contain exactly five 1s?

$$\frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{156}{6} = 56 \text{ or } \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{6720}{120} = 56$$

This approach is revisited in more detail in Problem 5 in Section 1.4.

3. Hockey match

The result of a hockey match between Slovakia and Canada was 4:3.

- a) How many different matches can there be? (Here different matches are those that have a different order of Slovak and Canadian goals.)
- b) In the next match, there were also 7 goals. In how many ways could these goals be scored in the 1st, 2nd and 3rd period of the game? (Here we do not care which team scored which goal.)
- c) Overall, there were 22 matches at the tournament and none of them had more than six goals total. At most, how many matches could there (certainly) be which had the same number of goals?

Hints:

- There are many possible ways how a match can go. If the number of goals is small, we can try to write them all out. When doing so, it is good to think about how to go through them in a logical fashion so that we always know which is the next one to write and so that we do not forget any of them.
- If we want to calculate the number of possibilities, both questions a) and b) can be transformed into problems using bit sequences, but the 1s and 0s do not denote the same things in these two cases.
- Question c) is of completely different nature. It does not ask about possibilities, but rather about an upper bound (*at most*) on something that has to happen (*certainly*). This should remind us of the pigeonhole principle from the lecture. We just need to realise what are the pigeons and what are the holes here.

Solution:

- a) One possible score order in a match could have been SSSSCC. This means that Slovakia scored the first four goals, the score was 4:0 and they thought the game was theirs and stopped paying attention. Canada woke up, seized the opportunity and managed to score three times. A completely different game would be CSCSCSS. This one would keep the fans on their toes, because the score was 0:1, then 1:1, then 1:2, then 2:2 and so on. Always close.

So how many different orders could there be? To answer this question, it is good to realise that we need four letters S and three Cs. With this realisation, the problem is transformed into finding the number of 7-bit sequences that contain four 1s and three 0s. Fortunately, we already know how to do that and thus the total number of possibilities is $\frac{7!}{4!3!}$.

- b) In the next match, we do not care what teams played, who won, or even how many goals they each scored. The only thing that matters is how the goals were divided into the periods. Again, we could systematically write them out, starting with e.g. 7+0+0 to denote that all goals happened in the first period of the game, then 6+1+0, 6+0+1, 5+2+0, 5+1+1, 5+0+2, etc. While certainly doable for seven goals, we should also try to calculate the number possibilities.

This time we can denote each goal with a 0 (a puck is round, is it not?) and separate the thirds using dividers – 1s. Thus 7+0+0 corresponds to 000000011, 6+1+0 to 000000101, 5+1+1 to 000001010, etc. For each division of goals we have one sequence and for each sequence, we have one division of goals, so that means we have one-to-one mapping and the number of possibilities is the same as the number of 9-bit sequences with seven 0s and two 1s, which is $\frac{9!}{7!2!}$.

- c) We start by thinking about how many different matches could we have. None of them had more than six goals total. This means, there are seven possibilities: matches without goals, matches with one goal, matches with two goals, ..., matches with six goals.

While unlikely, it is *possible* that all matches had six goals. Then we have 22 matches with the same number of goals. However, we cannot be sure, because it is also possible that only 21 had six goals and one had no goals at all. And there are many other possibilities how the tournament could go. We are looking for something we can

be *certain* of. We can surely say that there were two matches that had the same number of goals. If there were not, we could have at most seven matches – to get seven different outcomes. Since we have 22, there is surely an outcome that has to repeat.

And we can make an even stronger statement: In any such tournament there must be at least four matches that had the same number of goals. If there were only three (or fewer), we would have at most $3 \cdot 7$ matches total. Since we have 22, this means that one of the outcomes (we do not know which one) has to repeat at least four times. We need to remember that the pigeonhole principle that we used here tells us not what *can* happen, but rather what *has to* happen. In this case the matches were pigeons and the possible outcomes were pigeonholes.

4. Licence plates

A licence plate in our town can have these two forms:

- Letters ZA, then three digits, then two letters
- Letters ZA, then five letters

Let us denote by L the set of all licence plates in our town.

- a) Express the set L using the sets $C = \{A, B, C, \dots, Z\}$ and $D = \{0, 1, 2, \dots, 9\}$ and set operations.
- b) Compute $|L|$, i.e. the number of all possible licence plates.

Hints:

- Use the addition and multiplication rules from the lecture.
- Consider whether the letters and the numbers can repeat. Do we select each character from the whole set of letters and/or numbers?
- Since the licence plates can have two distinct formats, that result in two disjoint sets, we can use the addition rule to determine the total number of licence plates.

Solution: The letters ZA are the same on all the licence plates, therefore, we do not need to include them in the calculations and only consider those parts of the licence plates that can differ. The numbers and the letters can repeat, therefore we choose each character from the whole set that

is available. The individual types of licence plates can be written in the following way using the set operations:

$$\begin{aligned} L_1 &= D \times D \times D \times C \times C = D^3 \times C^2 \\ L_2 &= C \times C \times C \times C \times C = C^5 \end{aligned}$$

The set of all possible licence plates is then union of these two sets

$$L = L_1 \cup L_2 = (D^3 \times C^2) \cup C^5.$$

To determine the number of possible licence plates, we calculate

$$|L| = |(D^3 \times C^2) \cup C^5|.$$

Since the sets D and C are disjoint, we can use the addition rule and the multiplication rule. If we use a 26-letter alphabet for C , then the total number of available licence plates is

$$\begin{aligned} |L| &= |D^3 \times C^2| + |(C \cup D)^5| = |D|^3 \cdot |C|^2 + |C|^5 \\ &= 10^3 \cdot 26^2 + 26^5. \end{aligned}$$

5. Queues

There are six boys (B) and four girls (G) who are asked to form a queue BBGGBBGBB. They all have different names.

- In how many ways can they arrange themselves in such queue?
- In how many ways can they arrange themselves in such queue if we do not care about the order in the BB, GG pairs?

Hints:

- Clearly a person cannot be at two places at the same time, therefore we have fewer and fewer boys and girls to choose from once some of the places are already occupied.
- Since we know the order of the boys and girls, we only choose from one of the two groups when assigning people to specific places.

Solution: One of the six boys stands first in the line, one of the remaining five stands second. Then we have a pair of girls, so one of the four girls

stands third in the line and one of the remaining three stands fourth. Continuing like this, we get the overall number of possibilities:

order	B	B	G	G	B	B	G	G	B	B
possibilities	6	5	4	3	4	3	2	1	2	1

$$\Rightarrow 6 \cdot 5 \cdot 4 \cdot 3 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 6! \cdot 4!$$

To answer the second question, consider the first two places. If Tom is first, Jerry second and then they switch places, we consider this as one possibility. All other pairs of boys and girls can switch similarly, therefore we divide the number of possibilities for each pair by two:

$$\frac{6 \cdot 5}{2} \cdot \frac{4 \cdot 3}{2} \cdot \frac{4 \cdot 3}{2} \cdot \frac{2 \cdot 1}{2} \cdot \frac{2 \cdot 1}{2} = \frac{6! \cdot 4!}{2^5}.$$

6. Oranges

Parents want to give seven oranges to their four children.

- In how many ways can they do that?
- In how many ways can they do that if each child should get at least one orange?
- In how many ways can they do that if the youngest child should get at least two oranges? (It may happen that some of the older children get zero oranges.)

Hints:

- We can start by checking a few ways how parents can divide the oranges among their children, e.g. 7+0+0+0 (one child gets all seven oranges) or 2+2+2+1 (three children get two oranges each and one child gets one orange). Can these possibilities be coded into sequences of 0s and 1s? Could we use the *separators* here similarly as in the donut Problem 1.9?
- The additional requirement in question b) can be taken care of at the beginning by giving each child one orange first. The problem then transforms into a problem of dividing three oranges among four children.
- A similar approach can help in problem c).

Solution:

- a) Similarly to the donut problem in the lecture, we can use the encoding to binary words here. In this case, the 0s represent the oranges and the 1s represent the separators for splitting the oranges among the children. For example, the binary word (0101010000) means that three children got one orange each and the fourth child got four oranges. The problem of dividing seven oranges among four children is now transformed into the problem of how many binary words of length 10 with three 1s are there:

$$\frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = \frac{10!}{3!7!} = \binom{10}{3}.$$

- b) We can simplify the problem by giving each child one orange first and only then divide the remaining three oranges. Again we use 1s to represent the separators and we are looking for 6-bit words with three 1s:

$$\frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = \frac{6!}{3!3!} = \binom{6}{3}.$$

- c) We use the same idea as in the previous question. The youngest child gets two oranges and then we divide the remaining five oranges among the four children:

$$\frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = \frac{8!}{3!5!} = \binom{8}{3}.$$

7. Grandpa's bunnies

Grandpa has 21 bunnies and five large stalls. Determine whether the following statements are always true, sometimes true or always false. In case the statement is sometimes true, give an example with placement of bunnies in stalls that makes it true and another example that makes it false:

- a) There is a stall that has two bunnies.
 b) In every stall there is at least one bunny.

Hints:

- For each statement we should try to look whether we can place the bunnies into the stalls in a way that does not contradict the statement. If so, the statement cannot be always false. We should then also try to place the bunnies in such a way that it does contradict the statement. If we succeed, the statement cannot be always true.
- The words to look for and be careful about are *at least*, *at most* and *exactly*. Also there is a difference between *there is a stall ...* and *in every stall ...*

Solution:

- a) There is a stall that has two bunnies.

We can make the following placement A: stall 1 has 2 bunnies, stall 2 has 5 bunnies, stall 3 has 5 bunnies, stall 4 has 5 bunnies, stall 5 has 4 bunnies (denoted by $2+5+5+5+4$). Thus the statement is true.

It is also true, if there are two such stalls, i.e. placement B: $2+2+5+5+7$, since this does not contradict the statement.

Is it also true for placement C: $3+3+3+3+10$? There are at least two bunnies in each stall, but none of the stalls has *exactly* two bunnies. If not stated otherwise, we interpret this as requirement to have exactly two bunnies and thus placement C contradicts the statement.

Therefore we conclude that the statement is sometimes true.

- b) In every stall there is at least one bunny.

Placement D: $1+2+3+4+11$ makes the statement true, placement E: $0+5+5+5+6$ makes it false. So this statement is sometimes true.

8. Grandpa's bunnies one more time

Grandpa has 21 bunnies and

- four stalls
- five stalls
- six stalls

Based on this information, what can be said about the number of bunnies in the individual stalls?

Hints: We are interested in the strongest statements we can be sure about even when we do not know the precise partition of bunnies into the stalls. One statement which we can be sure about is that there will be at least one

bunny in at least one stall, regardless the overall placement of the bunnies. The pigeonhole principle guarantees that. Can we say something stronger? Do there have to be at least two bunnies together in at least one stall? Three bunnies? Try using the pigeonhole principle again to determine the largest number of bunnies that have to be together in one stall.

Solution:

- a) By placing the bunnies *uniformly* and thus trying to find the *least favorable placement* we have five of them in each of the four stalls. We can place the last one into any stall, Figure 1.7.

	Example 1	Example 2
Stall no. 1		
Stall no. 2		
Stall no. 3		
Stall no. 4		

Figure 1.7: Placing 21 bunnies into four stalls.

Therefore we can claim that that there is always going to be at least one stall that holds at least six bunnies. We do not know which one is it going to be. We certainly do not know the exact placement of all bunnies, but even if they were all together in one stall, it does not invalidate our statement. There might be a stall with seven bunnies or with eleven, but we cannot be sure of that. The only thing that is certain is that there is a stall with at least six of them.

- b) If there are five stalls available, in the least favorable case, we have four stalls with four bunnies and one with five, since:

$$21 = 5 \cdot 4 + 1.$$

Therefore we claim that there will be at least one stall with at least five bunnies.

- c) In this case we have:

$$21 = 6 \cdot 3 + 3.$$

We start by three bunnies in each stall and there are various possibilities for the remaining three bunnies. Each one of them can go to a

different stall or perhaps all can go together to one stall. We simply do not know. Therefore the strongest statement we can make is the *lower bound*: there is at least one stall that has at least four bunnies.

9. Hiking in a forest

- a) Five students went hiking in a forest in the shape of square with 4 km side. Each student has a transmitter with range $2\sqrt{2}$ km. How do you show that at least two of them can contact each other?
- b) Solve the same problem if the forest is an equilateral triangle and the range of the transmitter is 2 km.

Hints:

We can divide the forest into four equal parts. Thanks to the pigeonhole principle, there is certainly going to be a forest part in which there are two students. If the largest possible distance of this pair of students is shorter than the transmitter range, we can be certain that at least these two can communicate. The question thus is, how to divide the square and triangle forests into parts and what is the maximum distance in these parts relative to the transmitter range.

Solution:

- a) The square forest with 4 km side can be split into four equal squares with 2 km sides, Figure 1.8.

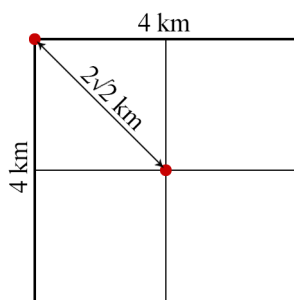


Figure 1.8: Partition of the square forest into four smaller squares. At least two students will be in the same smaller square.

The largest distance of two points in a smaller square is the length of its diagonal. It can be determined using the Pythagorean theorem as $\sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$. Since the pigeonhole principle implies that there are at least two students in the same smaller square, we know they be at most $2\sqrt{2}$ km apart. Therefore they will be able to communicate, since this distance corresponds to the transmitter range.

What would happen if we partitioned the square differently? For example into four triangles using the diagonals or into four rectangular stripes, Figure 1.9? Can we ensure connection also in these cases?

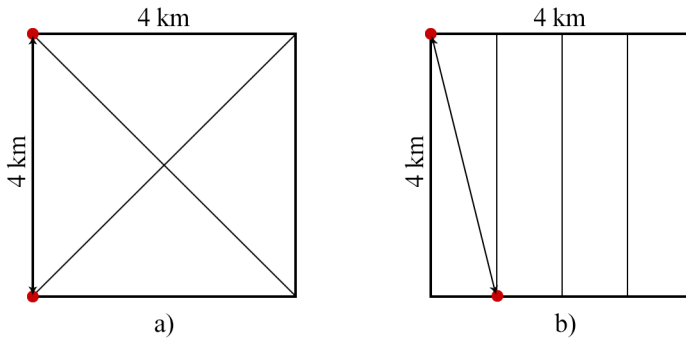


Figure 1.9: Two different partitions of a square forest.

If we tried to partition the square into triangles as shown in Figure 1.9, we see that the largest distance is 4 km, which is more than the transmitter range. Similarly, the diagonal of a rectangular stripe is longer than this range. Thus, with this reasoning we cannot guarantee connection. It does not mean that it is not there (it is as we saw when we partitioned the square into four smaller squares), it just cannot be proven this way. This hints at a strategy how to partition the forest: we try to partition it into parts with the smallest (minimal) maximal distance.

1.4 Lab problems

1. Snowmen

Draw all 3-level snowmen when we have 4 different colors of snow available.
Draw all 4-level snowmen if only 3 kinds of snow are available.

2. Pascal's triangle

Write the first 20 lines of Pascal's triangle using a suitable formula. Color the numbers in the triangle so that you use different colors for

- a) even and odd numbers,
- b) numbers that give different remainders when divided by 3.

3. Hockey match

A hockey match ended 4:3. Write (in a reasonable way) all possible orders how the teams scored and count how many there are.

4. Hockey match II

There were 7 goals in a match. Write (in a reasonable way) all possible ways how to divide the goals into thirds (how they could have been scored) and count how many there are.

5. Wall-e goes to Eve

Simulate in Excel the shortest path from robot Wall-e to Eve. The robot moves in a field 5×3 , it can go up and to the right. The graph will show a random path from $[0,0]$ to $[5,3]$, see Figure 1.10. Compute how many possible paths can the robot take.

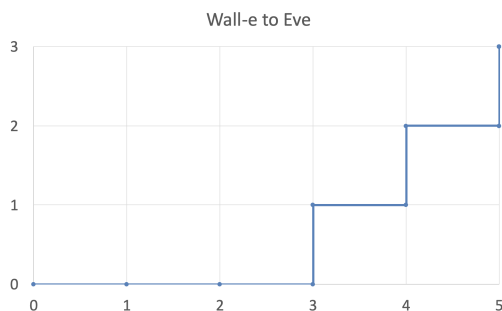


Figure 1.10: One possible path Wall-e can take on his journey to Eve.

1.4.1 Hints and solutions

1. Snowmen

Draw all 3-level snowmen when we have 4 different colors of snow available.
Draw all 4-level snowmen if only 3 kinds of snow are available.

Hints: What a weird problem! When we were growing up, the snow used to be just white. Well, sometimes yellow and that was it. But ok, let us suppose we have few more snow colors and that we want to draw some snowmen. 3-level snowman would be the regular kind as we know it and we can draw each level with different color, see Figure 1.11 A. Can we draw a different one? Of course, we just shuffle the colors somewhat or use the fourth color that we have available, as in Figure 1.11 B. Note that since we are working in Excel, we have to loosen the snowman-making rules just a tiny bit and allow snowman with rectangular levels. The color of the cells in Excel can be changed using the Home toolbar.

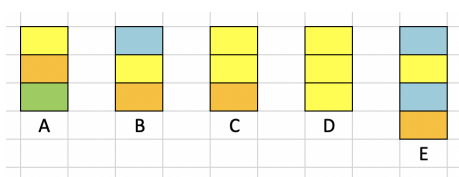


Figure 1.11: Very weird snowmen. From the left: A: 3-level snowman with 3 colors, B: 3-level snowman with 3 colors, C: 3-level snowman using just 2 colors, D: 3-level snowman using just one color and E: 4-level snowman when only 3 colors are available.

So how many different snowman can we build? It is good to settle on a system and not just jump arbitrarily between different colors. Using a system, we do not miss any combinations.

And what if we use the same color twice in the same snowman, e.g. Figure 1.11 C or even three times as in Figure 1.11 D? Of course, we can do that. How many snowmen are possible now?

Once we can answer these questions, it will be no problem to extend the same kind of thinking to 4-level snowmen, Figure 1.11 E, i.e. those that consist of four balls on paper or four rectangles in Excel. In this case, since we only have 3 colors available, it is certain that at least one of them will repeat.

2. Pascal's triangle

Write the first 20 lines of the Pascal's triangle using a suitable formula. Color the numbers in the triangle so that you use different colors for

- even and odd numbers,
- numbers that give different remainders when divided by 3.

Solution: To write the Pascal's triangle, we need to first think about how to make a(n isosceles) triangle in Excel. Some possibilities are shown in Figure 1.12. Note that we might run into problems for larger triangles if we try to keep the actual triangular shape, as in Figure 1.12 a).

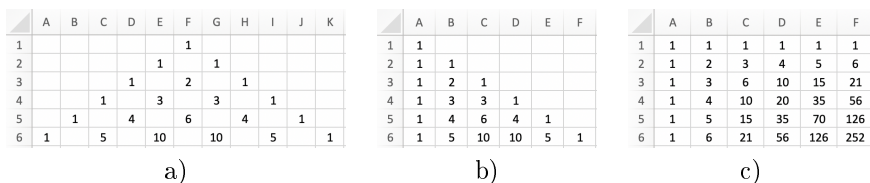


Figure 1.12: Different ways of creating Pascal's triangle in Excel.

The numbers in the Pascal triangle are not calculated using factorials (we would have to stop very quickly in such case) but rather by using addition – each number is a sum of the two numbers above it. To get the colors according to requirements, we can use *Conditional formatting* and setting a rule `=IF(MOD(A1;2)=0)`. This will ensure that even numbers will be colored as selected.

3. Hockey match

A hockey match ended 4:3. Write (in a reasonable way) all possible orders how the teams scored and count how many there are.

Hints: There are many possible ways (two of them are illustrated in Figure 1.13). When writing your own, it is good to think about the following:

- How to write one possibility? Into one cell? Into seven cells? Arrange them in rows? In columns?
- How to go through them in logical fashion so that we always know what is the next one to write?

- How can we be certain that we have them all and that they do not repeat?
- How to format the result (using color or something else) so that it is easy to check?

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Slovakia	Slovakia	Slovakia	Canada	Canada	Canada	Canada		0	0	0	0	0	0	0	0
2	Slovakia	Slovakia	Canada	Slovakia	Canada	Canada	Canada		0	0	0	0	0	0	1	1
3	Slovakia	Slovakia	Canada	Canada	Canada	Slovakia	Canada		0	0	0	0	0	1	0	1
4	Slovakia	Slovakia	Canada	Canada	Canada	Slovakia	Canada		0	0	0	0	0	1	1	2
5	Slovakia	Slovakia	Canada	Canada	Canada	Canada	Slovakia		0	0	0	0	1	0	0	1
6	Slovakia	Canada	Slovakia	Slovakia	Canada	Canada	Canada		0	0	0	0	1	0	1	2
7	Slovakia	Canada	Slovakia	Canada	Slovakia	Canada	Canada		0	0	0	0	1	1	0	2
8	Slovakia	Canada	Slovakia	Canada	Canada	Slovakia	Canada		0	0	0	0	1	1	1	3
9	Slovakia	Canada	Slovakia	Canada	Canada	Canada	Slovakia		0	0	0	1	0	0	0	1
10	Slovakia	Canada	Canada	Slovakia	Slovakia	Canada	Canada		0	0	0	1	0	0	1	2
11	Slovakia	Canada	Canada	Slovakia	Canada	Slovakia	Canada		0	0	0	1	0	1	0	2
12	Slovakia	Canada	Canada	Slovakia	Canada	Canada	Slovakia		0	0	0	1	0	1	1	3

Figure 1.13: Two possible ways how to write the match outcomes. The one on the left systematically goes through all correct possibilities. The one on the right shows all 7-bit sequences and highlights those that contain three 1s.

Solution: In Section 1.3.1 we already counted how many possible ways there are. Now we are asked to write them all out. One approach is to write out all 7-bit sequences of 0s and 1s in the first seven columns and then use the SUM function to determine which contain exactly three 1s. Those can be highlighted by conditional formatting.

Binary words using text processing functions

A more challenging variant of this solution is one that uses text processing and conversion from decimal to binary. The smallest binary number that satisfies our conditions is 0000111, which evaluates to 7 in decimal system and largest 1110000, which evaluates to 112. In this 7 to 122 range, we have all numbers we are looking for. The downside is that the range also includes some numbers, which do not fit the criteria, e.g. 0001111. We can use the Excel built-in functions to convert the number to string and count the number of 1s to reject the incorrect sequences.

An alternative way to solve this problem, would be to write a *tree*, as in Figure 1.14. It starts with the initial 0:0 score at the top of the Excel

from 0:0 to 4:3. This looks similar to the Pascal's triangle. So we can determine the number of possible matches by filling in the corresponding portion of the Pascal's triangle.

4. Hockey match II

There were 7 goals in a match. Write (in a reasonable way) all possible ways how to divide the goals into thirds (how they could have been scored) and count how many there are.

Solution: Continuing with the hockey topic from the previous problem, we try to answer a different question this time. We do not care if it was Slovakia or Canada who scored the goal this time. We only care in which period it was. One possibility is that all seven goals were scored in the first period and then it was a boring match when nothing happened till the end. Or maybe the goals were evenly spaced out – 2 in the first period, 3 in the second and 2 in the final period. Or there could have been 6 goals in the first period, nothing in the second and the final goal scored in the third period.

Again, we need some systematic way to write all of them without forgetting any, so the rules from the previous problem still apply. One way would be to break the task into several smaller subproblems. We could write out all possibilities with 7 goals in the first third (there will not be too many of those), then all possibilities with 6 goals in the first third, then 5, etc. Clearly, no possibility will be included twice, so to get to the answer, it is sufficient to just add up all the partial answers.

Note that as a check, we can also count how many possibilities do we expect. Looking at the lecture, we can treat the goals as 0s (the puck is round after all) and partition them into three periods using 2 dividers, e.g. 000100010 is a match with 3 goals in the first and second periods each and one in the final period. The problem is then reduced to counting how many 9-bit sequences with two 1s and seven 0s are there.

Goal orders separated into periods

What if we combined the previous two problems? How many ways are there to partition goals into periods, if we do care about who scored each goal? This time SS CCS SC is different from SS SSC CC, but both were counted as the same option 001000100 in the previous problem. The two options SS CCS SC and SS CCS CS represent the

same partition in this question.

5. Wall-e goes to Eve

Simulate in Excel the shortest path from robot Wall-e to Eve. The robot moves in a field 5×3 , it can go up and to the right. The graph will show a random path from $[0,0]$ to $[5,3]$. Compute how many possible paths can the robot take.

Solution: Two possible paths are shown in the Figure 1.15. The orange one is $[0,0]$ to $[0,1]$ to $[2,1]$ to $[2,2]$ to $[3,2]$ to $[3,3]$ to $[5,3]$. Perhaps it would be better to denote the individual steps: $[0,0] \uparrow [0,1] \rightarrow [1,1] \rightarrow [2,1] \uparrow [2,2] \rightarrow [3,2] \uparrow [3,3] \rightarrow [4,3] \rightarrow [5,3]$. Try writing out the coordinates of the green path.

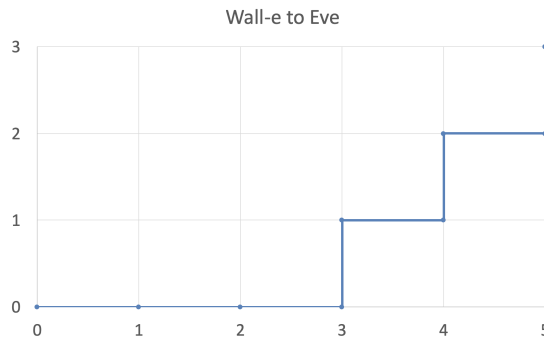


Figure 1.15: Wall-e starts at $[0,0]$ and is trying to get to Eve who waits at the position $[5,3]$. Only steps to the right and up are allowed. One possible path is shown.

How to draw this in Excel? Once we have the coordinates in two separate columns for x and y , we can insert a scatter plot with straight lines (x - y dependence chart) from the Insert toolbar.

For one specific path we would just write down the coordinates as in Figure 1.16.

Instead of re-writing the coordinates by hand to show a different path, let us do something different. We are going to generate Wall-e's steps at random, so that every time it will move either up or to the right. For that

x	y
0	0
1	0
2	0
3	0
3	1
4	1
4	2
5	2
5	3

Figure 1.16: Coordinates of points on Wall-e's path.

we can use the `=RANDBETWEEN(0,1)` function in Excel. It returns either 0 or 1 and each of these numbers is equally likely to appear.

We then use 0 to denote $\uparrow$ and 1 to denote $\rightarrow$. In each move up, the y -coordinate increases by 1 and x -coordinate remains constant. In each move right, the x -coordinate increases by 1 and y -coordinate remains constant. Note that we could have used `=RANDBETWEEN(7,8)` or any other two subsequent integers and everything would work the same way.

Each new generated number has to cause a change in coordinates, since it defines the direction of step. So we write the directions for the arrows: $\uparrow$ is $[0,1]$ and $\rightarrow$ is $[1,0]$. In each move, starting from $[0,0]$ we add the new generated step and build the path as in Figure 1.17.

D34					
	A	B	C	D	E
30		direction		coordinates	
31	random	x	y	x	y
32				0	0
33	1	1	0	1	0
34	0	0	1	1	1
35	1	1	0	2	1
36	1	1	0	3	1
37	1	1	0	4	1
38	0	0	1	4	2
39	1	1	0	5	2
40	0	0	1	5	3

Figure 1.17: One of the possible randomly generated paths for Wall-e.

We can now hit the reload key (F9) a few times. Each time, this will

generate a new set of 0s and 1s that will determine the steps and we will see a new path. A chart linked to the coordinates automatically updates and draws the new path. Have you noticed something? We also see paths that do not lead to Eve! This way it is possible to end up at coordinates [4,4] or (even though not very likely) [0,8].

So how do we fix this? There are a few options. In all of them, it is helpful to count how many times have we stepped up or right (or both). Once we reach the allowed number (e.g. three 0s), we can simply stop accepting that digit and the next time that digit is generated, we just force the use of the other one (1 in this case). This can be done by using the IF commands. Similarly, we can simply refuse to use the 1s that are generated once we have enough of them and only use the next 0s.

Probability of paths

Note that if we generate the paths using the steps described in this solution, not all of them are equally likely. Since we need more steps to the right, but both directions are generated with the same probability, we will see more solutions which end in (are padded with) *steps to the right*. Can you come up with a different way of generating the individual paths, so that they are all equally likely? (The term for this is generating them *uniformly*.)

So now that we have a functional simulation of Wall-e's walks, we can think about calculating how many possible paths there are.

We have seen that the sequences of 0s and 1s are a useful tool in various combinatorics problems. Even problems which originally do not have 0s and 1s, can often be transformed into equivalent problems that do. Here we do not even need to do that. We already have a sequence with five 1s and three 0s. So to count the number of possible paths it is sufficient to calculate how many 8-bit sequences with five 1s there are. Using the approach from the lecture, we get $\frac{8!}{5!3!}$.

General final coordinates

Can you solve this problem for general Eve's coordinates? Her position would be given as input in two cells in your Excel sheet and everything else would recalculate with respect to these coordinates.

Chapter 2

Divisibility and Algorithms in Number Theory

2.1 Lecture overview

2.1.1 Divisibility

Problem 2.1 Horse-racing:

A journalist is preparing an article about a horse-race with 10 horses. He has decided to prepare all possible articles for outcomes for the top four places in advance so that he does not have to worry about work after the race. How many articles does he have to prepare?

Solution: Using what we learned in the previous chapter, we know that the answer to this problem is

$$\frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = \binom{10}{4}.$$

Combinatorial fractions that simplify to integers

In the previous chapter we have solved many problems similar to Problem 2.1 and in all of them the answer was a natural number. Of course, it should be, since the number of possibilities is a whole number, but how do we know that is always the case, when we have a fraction of this sort?

Also, to calculate $\binom{n}{k}$ for large values of n and k we cannot use factorials, i.e. the formula

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}.$$

The computer will have trouble calculating the factorials for large n and k . Evaluating a combination number in such case means reducing the multiplication terms where possible and in order to do that, we need to understand divisibility.

In the following, we consider *natural numbers* with 0, denoted by $\mathbb{N}_0$ (see Section 5.1.1), but the properties also hold for *integers*, denoted by $\mathbb{Z}$ (Section 5.1.2).

Divisibility

Let us have $a, b \in \mathbb{N}$. We say that a *divides* b (and we write $a \mid b$) if and only if $b = a \cdot k$ for some suitable number $k \in \mathbb{N}$. We also say that a is a *divisor* of b and that b is a *multiple* of a . (The notation $a \in \mathbb{N}$ means that number a *belongs* to the set $\mathbb{N}$.)

Divisibility properties

- If $a \mid b$ and $b \mid c$, then $a \mid c$. We say that divisibility is *transitive*.
- If $a \mid b$, then for any number c , more precisely for any $c \in \mathbb{N}_0$, it is true that $a \mid b \cdot c$. So if a divides b , it also divides its multiples.
- If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- If $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.

Divisibility criteria

For some numbers we have simple criteria we can use to check whether a given number is divisible by them. For number two, the criterion says: *A number is divisible by two if and only if its last digit is divisible by two*. For three, the criterion says: *A number is divisible by three if and only if the sum of its digits is divisible by three*. The criteria simplify the situation by reducing the size of the problem. What other divisibility criteria do you know?

2.1.2 Primes

Prime numbers

We call a natural number p a *prime number* if and only if it has exactly two divisors: one and p . If it has more than two divisors, it is called a *composite number*. Number one does not fit into these categories and is neither prime nor composite.

Infinitely many prime numbers

There are infinitely many prime numbers. We can prove this statement by assuming the opposite: Suppose, there is a finite number of primes. Then we can label them $p_1, p_2, \dots, p_n$. And we can construct a new number $M = p_1 \cdot p_2 \cdot \dots \cdot p_n + 1$. The number M must be a composite number because it is larger than all known primes. Thus there must exist at least one prime number which divides M . But which one is it? The number M gives a remainder 1 after division by any prime p_i . We get a contradiction. Our assumption was false. Thus the opposite statement is true.

Goldbach's conjecture

This hypothesis states that every even number greater than 2 can be split into sum of two primes. Despite great efforts of many mathematicians and use of computers (at the time of writing this book, the conjecture has been tested up to 400 000 000 000 000), this statement remains a hypothesis. It has not been proven yet and no counterexample has been found yet.

Prime twins hypothesis

There exist infinitely many *prime twins*, i.e. prime numbers that differ by 2. Also this hypothesis has neither been proven nor rejected, yet.

Sieve of Eratosthenes – a method for searching for primes

This method comes from the ancient Greece. It involves writing out as many natural numbers as we want (in order) and then using the following steps to cross some of them out. Step 1: Cross out number 1. Step 2: Circle number 2 and cross out all its multiples. Repeat step 2 with every number has not been crossed yet (3, 5, ...). In the end, every number that is circled is a prime.

How to check if a number is prime?

For a large n , it is not efficient to try whether it is divisible by all numbers smaller than n . It is sufficient to try all numbers not greater than $\sqrt{n}$. Why?

Fundamental Theorem of Arithmetics

Every natural number a greater than 1, $a \in \mathbb{N} - \{1\}$, can be uniquely (up to the order of factors) written in the form

$$p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot p_3^{\alpha_3} \cdot \dots \cdot p_n^{\alpha_n},$$

where $p_1, p_2, p_3, \dots, p_n$ are distinct prime numbers and $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are natural numbers (not 0).

Problem 2.2 Factorisation into primes:

Find all divisors of the number 100? Write them as products of powers of primes.

Solution: First we factor 100 according to the Fundamental theorem of arithmetics: $100 = 2^2 \cdot 5^2$. The divisors are numbers $2^{\alpha_1} \cdot 5^{\alpha_2}$, where $\alpha_1 \in \{0, 1, 2\}$ and $\alpha_2 \in \{0, 1, 2\}$. There are nine possible pairs of α_1, α_2 , therefore 100 has 9 divisors.

Number of divisors

For any $p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \dots \cdot p_n^{\alpha_n}$ we can calculate the number of its divisors using the formula

$$(\alpha_1 + 1) \cdot (\alpha_2 + 1) \cdot \dots \cdot (\alpha_n + 1).$$

Problem 2.3 Numbers with given number of divisors:

Which number has 6 divisors? Which has 12? Which has 7?

Hint: Starting with the definition of a prime number, we know that any number with two divisors is a prime. To have three, we need 1, the number and something that multiplied by itself gives our desired number. That looks like a square, e.g. 9 with divisors 1, 3, 9. Now it is easy to continue and get a number with 4 divisors, e.g. 27 with 1, 3, 9, 27. However, the powers of primes are not the only way how to go about it. We could use two different primes and their product, e.g. 1, 2, 5, 10. Following this approach, it is possible to *build* numbers with the desired number of divisors.

Hasse diagram

The divisors of a number and relations among them can be displayed in a Hasse diagram, Figure 2.1. We start with 1 at the bottom and the given number at the top. We write and connect divisors from those that have the fewest prime factors (counting their powers) to those that have the most observing the rule that only direct multiples/divisors are connected, i.e. no lines going *over a level*. This diagram is related to *partial ordering* that is covered in Section 7.1.1.

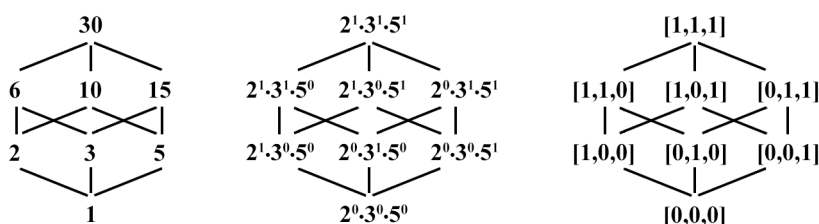


Figure 2.1: Hasse diagram. Note the correspondence between the divisors and sequences of 1s and 0s.

2.1.3 Common divisors

Problem 2.4 Dancers:

Suppose we have a group with 24 dancers. What kind of rectangles can they form? Now suppose we have a second group of 30 dancers. What kind of rectangles can this group form? In which rectangular formations can the groups join so that the resulting formation is also a rectangle?

Hint: The first group can stand in a line and face the audience, this corresponds to a rectangle 1×24 . They can also stand in a row so that the audience sees just the first person, this corresponds to a rectangle 24×1 . There are other rectangles, e.g. three rows of eight people each, 8×3 . Similarly, we can look at the second group. Ultimately, we are looking for such a formation that has the same number of rows (or lines) for both groups.

Greatest common divisor (GCD)

Take $a, b \in \mathbb{N}$. If there is a number $d \in \mathbb{N}$ for which $d \mid a$ and $d \mid b$, then we say that d is a *common divisor* of a and b . The largest such d is called the

greatest common divisor and is denoted $GCD(a, b)$. $GCD(a, b)$ can be found by factoring a and b into product of prime numbers. If $GCD(a, b) = 1$, we say that a, b are *relatively prime* or *coprime*.

Euclidean algorithm for finding GCD

Suppose we are looking for the greatest common divisor of numbers a and b . If $a > b$, then $GCD(a, b) = GCD(a - b, b)$. Similarly, if $a < b$, then $GCD(a, b) = GCD(a, b - a)$. We repeat these steps until we get the same numbers. These represent $GCD(a, b)$.

Here we illustrate how the algorithm works for two pairs of numbers:

$$\begin{aligned}
 GCD(247, 437) &= GCD(437 - 247, 247) = GCD(190, 247) \\
 &= GCD(247 - 190, 190) = GCD(57, 190) = GCD(133, 57) \\
 &= GCD(76, 57) = GCD(19, 57) = GCD(38, 19) \\
 &= GCD(19, 19) = 19 \\
 GCD(437, 110) &= GCD(327, 110) = GCD(217, 110) \\
 &= GCD(107, 110) = GCD(107, 3) = GCD(104, 3) \\
 &= GCD(101, 3) = GCD(98, 3) = \dots = GCD(2, 3) \\
 &= GCD(2, 1) = GCD(1, 1) = 1
 \end{aligned}$$

Note that in the second case, the number 107 is the remainder when dividing $437 : 110$ and 2 is the remainder when dividing $107 : 3$.

Improved Euclidean algorithm for finding GCD

We are again looking for the greatest common divisor of numbers a and b . Let $a > b$ and z be the remainder after division $a : b$. Then $GCD(a, b) = GCD(z, b)$. Similarly if $a < b$ and z is the remainder after division $b : a$, then $GCD(a, b) = GCD(a, z)$. We repeat these steps until we get $z = 0$. The non-zero number of the final pair represents $GCD(a, b)$. Returning to the example we already saw:

$$\begin{aligned}
 GCD(437, 110) &= GCD(107, 110) = GCD(107, 3) \\
 &= GCD(2, 3) = GCD(2, 1) = GCD(1, 0) = 1
 \end{aligned}$$

Note that we could use the same stopping criterion (one number equal to zero) for the slower, subtraction-based Euclidean algorithm if we performed one additional step. This would ensure that also for edge-cases, such as finding $GCD(x, 0)$, the algorithm finishes in finite number of steps.

Common multiples

Suppose $a, b \in \mathbb{N}$ and suppose there exists a number $n \in \mathbb{N}$ for which $a \mid n$ and $b \mid n$. Then we say that n is a *common multiple of numbers a and b* . We denote the smallest such number $LCM(a, b)$ and call it *least common multiple*. It can be found by factorisation of a and b into products of primes and is used, for example, when adding fractions. An interesting property is that

$$GCD(a, b) \cdot LCM(a, b) = a \cdot b.$$

Remainder after division

We have already used the remainder after division in the Euclidean algorithm. Here we specify it more precisely: Take $a, b \in \mathbb{N}$ and let $a = b \cdot k + z$, where $k \in \mathbb{N}$ is a suitable natural number and $z \in \{0, 1, 2, \dots, (b-1)\}$. Then we say that z is the *remainder after division* of number a by the number b . This is sometimes denoted by $z = \text{rem}(a, b)$.

Congruence

If the numbers $a, b \in \mathbb{N}$ have the same remainder after division by the number $m \in \mathbb{N}$, then we say that a and b are *congruent modulo m* and we write $a \equiv b \pmod{m}$. Some properties of congruence are:

- For all $a, m \in \mathbb{N}$ we have $a \equiv a \pmod{m}$.
- For all $a, b, m \in \mathbb{N}$, if $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.
- For all $a, b, c, m \in \mathbb{N}$, if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

The congruence $\pmod{m}$ is thus an *equivalence* relation in $\mathbb{N}$ (more on equivalence relation can be found in Section 7.1.1). The set of natural numbers can be split using the equivalence relation into groups of numbers that give the same remainder after division by m . We call these groups *congruence classes modulo m* .

Fermat's little theorem

Let p be a prime. Then for any natural number a is the number $a^p - a$ a multiple of p , i.e. $a^p \equiv a \pmod{p}$. A generalisation of this theorem is used in the proof of correctness of the RSA encryption.

Some uses of divisibility and prime numbers

- Personal ID numbers in Slovakia - The format of these is $yymmdd/aaab$, where the first six digits represent the date of birth and also encode gender in the mm digits (01-12 for males and 51-62 for females). The next three digits are related to the registrar office in the place of birth and the final digit is added so that the 10 digit number $yymmddaaab$ is divisible by 11. The divisibility by 11 can be verified as follows: the difference of sums of numbers on even and odd places has to be also divisible by 11 (problem reduction). The reason for using divisibility by 11 is that if just one digit of the ID number is compromised, e.g. a typo, the result is not another valid ID number.
- IBAN - The format of the international bank account numbers is such that a single typo or position exchange of two neighboring digits do not result in another correct IBAN. The check relies on division by 97.
- Cryptography - Large primes are used for encryption. Some encryption algorithms use the fact that factoring a product of two extremely large prime numbers is a very time-consuming task. Some simple encryption examples using primes inspired by Alan Turing, together with introduction into RSA (= Rivest, Shamir, Adleman - authors of the algorithm) encryption, can be found in [6].

Showing that the Slovak ID rule works

Prove that if we change a single digit in a correct Slovak personal ID number to a different digit, we do not get another valid ID number.

2.2 Questions

1. Decide whether the following statements are true:
 - a) 5 divides 10.
 - b) 10 divides 5.
 - c) 7 divides 490.
 - d) 10 is divisible by 5.
2. Decide whether the following statements are true:
 - a) If $(x \text{ divides } y \text{ and } z \text{ divides } y)$ then $x + z$ divides y .
 - b) If $(x \text{ divides } y \text{ and } x \text{ divides } z)$ then x divides $y + z$.
 - c) If $(x|y \text{ and } z|y)$ then $x - z|y$.
 - d) If $(x|y \text{ and } x|z)$ then $x|y - z$.
 - e) If $(x \text{ divides } y \text{ and } z \text{ divides } y)$ then $x \cdot z$ divides y .
 - f) If $(x|y \text{ and } x|z)$ then $x|y \cdot z$.
3. Write the criterion for divisibility by 11.
4. Explain how is IBAN protected against common typos.
5. How many different rectangular formations can 24 dancers form?
6. Every even natural number can be written as a sum of two prime numbers. Check this statement for numbers from 10 to 30. For each of them, find all possible partitions.
7. How does the Sieve of Eratosthenes work?
8. Is the number 119 prime? Is 117 prime? Is 2017 prime?
9. How many divisors does the number 72 have? Draw the Hasse diagram for them.
10. What does the Fundamental Theorem of Arithmetics say?
11. How many divisors does the number $p_1^3 \cdot p_2^5 \cdot p_3^2$ have? (p_1, p_2 and p_3 are distinct primes.)
12. Find the greatest common divisor of numbers 54 and 72.

13. Find the smallest common multiple of numbers 54 and 72.
14. Find two natural numbers that are not primes but do not have a common divisor other than 1.
15. How does Euclidean algorithm work and what is it for?
16. What is the largest remainder after division by 13?
17. What is the largest remainder after division by 12?
18. What is the smallest remainder after division by 6?
19. Two natural numbers m and n both give remainder 4 when divided by 5.
 - a) What is the remainder after division of $m - n$ by 5?
 - b) What is the remainder after division of $m + n$ by 5?
 - c) What is the remainder after division of $m \cdot n$ by 5?
20. For each natural number, we are only interested in its remainder after division by 4. What groups will the numbers form, if we group them by their remainder after division by 4?

2.3 Problem set

1. Euclidean algorithm

Find all common divisors for the following pairs and triplets of numbers using

- simple, subtraction-based, Euclidean algorithm
 - faster, remainder-based, Euclidean algorithm
- a) (350, 460, 820)
 - b) (85, 340)
 - c) (360, 450, 840)
 - d) (65, 420)
 - e) (1260, 300)
 - f) (84, 210)
 - g) (3850, 198, 1764)

2. Number of divisors

For each of these numbers determine how many divisors it has:

- a) 120
- b) 760
- c) 504
- d) 1650
- e) 840

3. Factorisation

Using the factorisation into prime numbers find the greatest common divisor and the least common multiple for the pairs and triplets of numbers:

- a) (84, 210)
- b) (70, 165, 220)
- c) (64, 180)
- d) (50, 145, 310)

4. Reverse engineering

Write at least 3 numbers that have 2, 4, 5, 8, 10, 12, 15, ... divisors.

5. Modulo equations

For which n are the following statements true?

- a) $16 \equiv n \pmod{7}$
- b) $n \equiv 13 \pmod{13}$
- c) $4 \equiv 13 \pmod{n}$
- d) $171 \equiv n \pmod{71}$
- e) $7 \equiv 18 \pmod{n}$
- f) $10 \equiv 34 \pmod{n}$
- g) $6 \equiv 61 \pmod{n}$

6. Systems of modulo equations

For which n are both statements true?

- a) $12 \equiv 4 \pmod{n}$ and $15 \equiv 30 \pmod{n}$
- b) $10 \equiv 2 \pmod{n}$ and $10 \equiv 34 \pmod{n}$
- c) $4 \equiv 13 \pmod{n}$ and $7 \equiv 18 \pmod{n}$

7. Hasse diagram

Find the divisors of numbers 6, 12, 15, 18, 24, 28, 36, ... and display them using a Hasse diagram.

2.3.1 Hints and solutions

1. Euclidean algorithm

Find all common divisors for (85, 340) and (350, 460, 820) using

- simple, subtraction-based, Euclidean algorithm,
- faster, remainder-based, Euclidean algorithm.

Hints:

- To find all common divisors we need to first find the *GCD* and then all its divisors.
- In order to find the *GCD*, the pairs of numbers can be treated directly as shown in the Section 2.1.3.

Solution: Using the subtraction-based Euclidean algorithm we calculate:

$$\begin{aligned}
 GCD(85, 340) &= GCD(85, 340 - 85) = GCD(85, 255) \\
 &= GCD(85, 255) = GCD(85, 255 - 85) \\
 &= GCD(85, 170) = GCD(85, 170 - 85) \\
 &= GCD(85, 85) = 85
 \end{aligned}$$

Then we find all divisors of 85 by trying out numbers smaller than $\sqrt{85}$. Of those, only 1 and 5 divide 85. The remaining divisors are then $85:1 = 85$ and $85:5 = 17$.

These four numbers (1, 5, 17, 85) then also divide the number 340 and there are no other common divisors.

If we wanted to use the remainder-based Euclidean algorithm, we would be able to combine several steps of subtracting 85 into one step:

$$\begin{aligned}
 GCD(85, 340) &= GCD(85, \text{rem}(340, 85)) = GCD(85, 0) \\
 &= 85
 \end{aligned}$$

After that we proceed the same way as before by searching for all divisors of 85.

Note that in this case, the remainder was 0, which means that we found the answer in one step. If the remainder was non-zero, we would divide again. In this case, we stop when one of the numbers is 0 (we cannot divide any further). The other number is then the *GCD*. This stopping criterion is universal and will work also for more general cases.

What should we do when we have three numbers to consider? We would like to modify the Euclidean algorithm to work with three numbers too. Certainly, we can reduce the problem, by looking at a pair of numbers and performing the subtraction:

$$GCD(350, 460, 820) = GCD(350, 460 - 350, 820) = GCD(350, 110, 820)$$

Whatever the GCD for the initial triplet of numbers was, it is the same for the new triplet. As with the pair of numbers, we subtract a smaller number from a larger number, but here we have more options to choose a pair. We could consider 350 and 460. We could also start with 460 and 820. And of course, we could also start with 350 and 820. To save work, it makes sense to perform such subtractions that lead to answers in small number of steps, i.e. use the largest numbers, however, keep in mind, that choosing any pair should lead to the same final result. (Give it a try.)

$$\begin{aligned}
 GCD(350, 460, 820) &= GCD(350, 460, 820 - 460) \\
 &= GCD(350, 460, 360) = GCD(350, 460 - 360, 360) \\
 &= GCD(350, 100, 360) = GCD(350, 100, 360 - 350) \\
 &= GCD(350, 100, 10) = GCD(350 - 100, 100, 10) \\
 &= GCD(250, 100, 10) = GCD(250 - 100, 100, 10) \\
 &= GCD(150, 100, 10) = GCD(150 - 100, 100, 10) \\
 &= GCD(50, 100, 10) = GCD(50, 100 - 50, 10) \\
 &= GCD(50, 50, 10) = GCD(50, 50 - 10, 10) \\
 &= GCD(50, 40, 10) = GCD(50 - 40, 40, 10) \\
 &= GCD(10, 40, 10) = GCD(10, 40 - 10, 10) \\
 &= GCD(10, 30, 10) = GCD(10, 30 - 10, 10) \\
 &= GCD(10, 20, 10) = GCD(10, 20 - 10, 10) \\
 &= GCD(10, 10, 10) = 10
 \end{aligned}$$

All common divisors of numbers (350, 460, 820) are then divisors of number 10, i.e. numbers 1, 2, 5, 10.

How do we adapt the remainder-based Euclidean algorithm in a similar

manner?

$$\begin{aligned}
 GCD(350, 460, 820) &= GCD(350, 460, \text{rem}(820, 460)) \\
 &= GCD(350, 460, 360) \\
 &= GCD(350, \text{rem}(460, 360), 360) \\
 &= GCD(350, 100, 360) \\
 &= GCD(350, 100, \text{rem}(360, 350)) \\
 &= GCD(350, 100, 10) \\
 &= GCD(\text{rem}(350, 100), 100, 10) \\
 &= GCD(50, 100, 10) \\
 &= GCD(50, \text{rem}(100, 50), 10) \\
 &= GCD(50, 0, 10) \\
 &= GCD(\text{rem}(50, 10), 0, 10) \\
 &= GCD(0, 0, 10) = 10.
 \end{aligned}$$

And again we proceed to find all divisors of number 10.

The algorithms for three numbers presented here are suitable for computer implementation. However, a human doing the calculations, would probably work slightly differently. Once we encounter $GCD(50, 50, 10)$ in the subtraction-based version, we can eliminate one 50 and continue with $GCD(50, 10)$. Similarly, once we encounter $GCD(350, 100, 10)$ in the remainder-based version, where we see that 100 is a multiple of 10, we can eliminate the multiple and continue with $GCD(350, 10)$.

Generalisations of Euclidean algorithm

Can you now state the generalised versions of both Euclidean algorithms for three numbers?

How about a generalised version of a generalised version: Euclidean algorithm for any finite set of numbers?

2. Number of divisors

Determine how many divisors does the number 120 have.

Hint: to determine the number of divisors, it would certainly be helpful to know the factorisation of the number into prime numbers.

Solution: So let us first factor the number 120 into prime numbers. We start by the smallest prime numbers, factor them out if possible and continue towards the larger ones until we have factored the whole number.

$120 = 2 \cdot 60 = 2 \cdot 2 \cdot 30 = 2 \cdot 2 \cdot 2 \cdot 15 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 5$. This can alternatively be written as $120 = 2^3 \cdot 3 \cdot 5$. In all the divisors, the factor 5 may or may not be present – these are two options. Similarly, the factor 3 may or may not be present and this gives us two more options. As for the 2s: we can include none of them, one of them, two of them or all three of them – four different possibilities. Altogether, to build a divisor, we have $2 \cdot 2 \cdot 4$ options, which means that the number 120 has 16 divisors. (To make sure, here they are: 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. If we pick any of them, e.g. 24, we can determine which prime numbers are included – here 2, 2, 3.)

3. Factorisation

Using the factorisation into prime numbers find $GCD(84, 210)$ and also $LCM(84, 210)$.

Hint: The LCM has to include all prime factors of both numbers in question, but not more than necessary.

Solution: We factor the numbers $84 = 2^2 \cdot 3 \cdot 7$ and $210 = 2 \cdot 3 \cdot 5 \cdot 7$. The LCM definitely has to include 5, otherwise it would not be a multiple of 210. It has to include two 2s, since one would be sufficient for the result to be a multiple of 210, but we need another one for it to also be a multiple of 84. Finally, the LCM also has to include 3 and 7, but it is sufficient to include them only once each, because they appear only once in each factorisation.

To solve problems with more than two numbers, we need to collect all the necessary primes across all numbers in the problem.

To find the GCD using the factorisations, we simply collect the primes (including their powers) that are in both factorisation (Unlike the *union* we performed for LCM , now we are performing an *intersection*.) and we calculate their product. In this case, $2 \cdot 3 \cdot 7 = 42$.

4. Reverse engineering

Write at least 3 numbers that have 2, 4, 5, 8, 10, 12, 15, ... divisors.

Hint: First try this by manually looking for such numbers, then try to *build* them.

Solution: To find numbers that have 2 divisors, all we need to do is recall that every number is divisible by 1 and by itself. Therefore to have only 2 divisors, it cannot be divisible by anything else. We even have a name for

this kind of numbers – they are prime. So for example 2, 3, 83, ... have 2 divisors.

To find numbers that have 3 divisors, we build on that. We already know that two divisors will be *for free*, so we are looking for a number that has one more divisor. Clearly if we have a divisor a , there is a co-divisor b , such that $a \cdot b = \text{number}$. And we must have $b = a$, because otherwise, we would have another different divisor and their total would be 4. So we can have 4 (divisors are 1, 2, 4) or 9 (divisors are 1, 3, 9) or any other number of the form p^2 (divisors are 1, p , p^2).

To find numbers that have 4 divisors, we can follow the same thought process and arrive at p^3 with divisors $(1, p, p^2, p^3)$, such as 8. But notice that there are other numbers that do not fit this recipe, e.g. divisors of 10 are 1, 2, 5, 10. This is due to the fact that 10 is not a power of a single prime. Since its factorisation is $2^1 \cdot 5^1$, its number of divisors is $(1 + 1) \cdot (1 + 1)$.

If we want to find numbers that have 12 divisors, we first factor $12 = 1 \cdot 12$. This will be one type of solutions – powers of a single prime. Then we can factor $12 = 2 \cdot 6$. This will be second type of solutions $p_1 \cdot p_2^5$. We can factor in yet another way $12 = 3 \cdot 4$ and this will be the third type of solutions $p_1^2 \cdot p_2^3$.

Building numbers that have exactly 12 divisors

Can you find still another type of numbers (not mentioned above) that also have 12 divisors?

5. Modulo equations

For which n are the following statements true?

- a) $16 \equiv n \pmod{7}$
- b) $n \equiv 13 \pmod{13}$
- c) $4 \equiv 13 \pmod{n}$

Hints:

- Try plugging in a few numbers for n and see whether the statement holds. Find a few for which it holds and few for which it does not.
- For these problems it is very helpful to read them properly. For example in the first one, we are looking for such a number n which gives

the same remainder as 16, when divided by 7. The *same remainder* part is denoted by the sign $\equiv$.

- What is the difference between $16 \equiv n \pmod{7}$ and $n \equiv 16 \pmod{7}$?
- Can the number n be larger than 16?
- What is special about problem b)? If we read it carefully, we see that we are looking for numbers that give the same remainder after division by 13 as 13 does. In other words, we are looking for multiples of 13.
- We would read problem c) as *Which number do we need to divide 4 and 13 by in order to get the same remainder?*

Solution: Looking at the first problem, the $\text{rem}(16, 7) = 2$. Which other number gives the same remainder? Number 9, since $7 \cdot 1 + 2 = 9$. Clearly, this is not the only solution, we also have $7 \cdot 3 + 2 = 23$ and many more. Number 2 is among them, since $7 \cdot 0 + 2 = 2$. Can you write these numbers using one common description? The set of solutions is thus $\{7 \cdot k + 2, \text{ for all } k \in \mathbb{Z}\}$.

Looking at the third problem, we see that we are now looking for such a number that 4 and 13 give the same remainder when divided by it. This means that $4 = a \cdot n + \text{rem}$ and $13 = b \cdot n + \text{rem}$. To get rid of the unknown rem , which is the same for both numbers, we can subtract these two equations and get $13 - 4 = 9 = (b - a) \cdot n$. Therefore n has to be a divisor of 9. We can check them all to see that they indeed make the statement $4 \equiv 13 \pmod{n}$ true. Thus this problem has three solutions: 1, 3 and 9.

Unknown n at two places

Combine the insights from the two types of problems just discussed and find all values of n which make $15 \equiv n \pmod{n}$ true.

6. Systems of modulo equations

For which n are both statements true?

$$12 \equiv 4 \pmod{n} \text{ and } 15 \equiv 30 \pmod{n}$$

Hint: This problem consists of two equations of the type discussed in the previous problem. Try solving them individually. Once you have the two sets of solutions, the correct values of n have to appear in both of them, since both of the statements have to be true at the same time.

7. Hasse diagram

Find the divisors of 28 and display them using a Hasse diagram.

Solution: The divisors of 28 are 1, 28, 2, 14, 4, 7. The corresponding Hasse diagram, Figure 2.2 is constructed by placing 1 at the bottom of the diagram and 28 at the top. We place the prime divisors above 1. The next level are the numbers composed of two primes. We would proceed in this fashion, if we had more divisors. In our case, the second level is connected to 28.

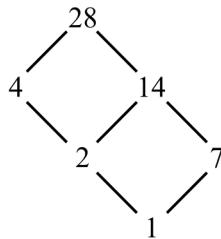


Figure 2.2: Hasse diagram for divisors of 28.

2.4 Lab problems

1. Subtraction-based Euclidean algorithm

- Create an Excel sheet that will compute the GCD for any pair (triplet) of positive integers. Optimise the algorithm speed for the triplet of numbers.
- Indicate when the algorithm found the GCD and how many steps were necessary.

2. Remainder-based Euclidean algorithm

Solve the same problems for finding GCD with the Euclidean algorithm that uses remainders after division.

3. Spirolaterals

Create an Excel sheet that for a given n draws its spirolateral. Spirolateral is a polyline which changes direction with a constant angle at each step, see e.g. [2]. For this problem we consider 90 degrees angle, i.e. four directions.

4. Prime numbers

- Write a program that tests whether a number n is prime. Give a statement about computational complexity of your algorithm.
- In a given time limit, write down the largest prime number that you can and a method for verification (using a PC) that it is indeed a prime.
- In a given time limit, write down the largest composite number and its factorisation and a method for verification (using a PC) that the factorisation is correct.

2.4.1 Hints and solutions

1. Subtraction-based Euclidean algorithm

- Create an Excel sheet that will compute the GCD for any pair (triplet) of positive integers. Optimise the algorithm speed for the triplet of numbers.
- Indicate when the algorithm found the GCD and how many steps were necessary.

Hints:

- When looking for $GCD(a, b)$ using the Euclidean algorithm, we intuitively subtract the smaller number from the larger number. (Here we are using the property that $GCD(a, b) = GCD(b, a)$.) For this to work in Excel, we need to specify which number is subtracted from which. It is also possible (but not necessary) to write the formulas in such a way that larger numbers appear in the same column and smaller numbers in the next column.
- The whole algorithm should proceed until $a = b$, therefore, we can check the previous line for this and if this happens, stop and report the result.
- The Excel function COUNT can be used to determine how many cells in a given column are non-empty and thus indicate the number of steps needed.
- When working with three numbers, for the final result, it does not matter which two of them we choose for subtraction. This choice manifests itself in the number of steps needed. If we choose numbers that are close to each other, the result after subtraction is small.

Solution:

- Start with numbers a, b , see Figure 2.3.
- In the next line, we can for example use the formulas $=MAX(ABS(a-b);MIN(a,b))$ and $=MIN(ABS(a-b);MIN(a,b))$, where a, b are the numbers in the previous line. This ensures that the difference is always positive and always written on the left. (Other formulas are possible.)
- Copy this formula down.

- In the next column, check if the work should stop $=\text{IF}(a = b; a; "$ "). This will write the greatest common divisor. This condition can also be combined with the previous formulas and the process will stop at the *GCD*. If we do it this way, we can use COUNT to find out number of steps necessary for calculation.

B3 $\times$ $\checkmark$ f_x $=\text{IF}(A2=B2;"";\text{MIN}(\text{ABS}(A2-B2);\text{MIN}(A2;B2)))$						
	A	B	C	D	E	F
1	a	b				
2	99	45				
3	54	45				
4	45	9				
5	36	9				
6	27	9		GCD		
7	18	9		9		
8	9	9				
9				no of steps		
10				6		

Figure 2.3: Calculation of the greatest common divisor and the number of steps needed. Input is denoted in yellow (row 2), output in green (column D).

2. Remainder-based Euclidean algorithm

Solve the same problems for finding *GCD* with the Euclidean algorithm that uses remainders after division.

Hint: Modify the sheet from previous problem and use division instead of subtraction. The function $\text{MOD}(a, b)$ gives $\text{rem}(a, b)$. Note that it is also important to change the stopping criterion. This time we finish, when one of a, b is 0, as in Figure 2.4. The other number is then the *GCD*.

B5 $\times$ $\checkmark$ f_x $=\text{IF}(B4="";""; \text{IF}(A4*B4=0;"", \text{MIN}(\text{MOD}(\text{MAX}(A4;B4); \text{MIN}(A4;B4)); \text{MIN}(A4;B4))))$									
	A	B	C	D	E	F	G	H	I
1	a	b							
2	99	13							
3	13	8							
4	8	5							
5	5	3							
6	3	2		GCD					
7	2	1		1					
8	1	0							
9				no of steps					
10				6					

Figure 2.4: Calculation of the greatest common divisor and the number of steps needed using the remainder-based Euclidean algorithm. Input is denoted in yellow (row 2), output in green (column D).

Most steps needed

Using the created sheets for the two algorithms, find the pair (triplet) of numbers from 1 to 100 (1 to 500, ...) for which the algorithms needed the most steps.

3. Spirolaterals

Create an Excel sheet that for a given n draws its spirolateral. Spirolateral is a polyline which changes direction with a constant angle at each step, see e.g. [2]. For this problem we consider 90 degrees angle, i.e. four directions.

Hints:

- Write multiples of n and the sums of their digits.
- Cyclically add the sum of their digits to the current position of the *pen* as multiples of the directional vector N/E/S/W. The origin is at $[0,0]$. Draw the spirolateral as an x - y chart.
- Examine the shapes and symmetries of the chart for different n . For which n do we get the same charts? For which n do we get symmetric graphs?

Sum of digits vs. divisibility by 9

Prove that for all integers n that are not multiples of 9, the *repeated* sum of their digits is equal to the remainder after division $n:9$. By repeated sum of digits we mean that we repeat the sum of digits until we get a one-digit result, e.g. $64 \rightarrow 10 \rightarrow 1$.

Solution:

- While it is easy to write a formula for multiples of n (column B in Figure 2.5), the sum of digits (column C) is a trickier problem. There is no simple function for this in Excel. We could write a complicated expression that relies on several built-in Excel functions, but instead, we are going to use the property described in the previous grey box. The *repeated* sum of digits of any integer n that is not a multiple of 9 is equal to the remainder after division $n:9$. For multiples of 9, we can use a logical test $=\text{IF}(\text{MOD}(\text{B4};9)=0;9;\text{MOD}(\text{B4};9))$. This number now represents the length of the step.
- To determine the direction of the step, we first assign the first four directions: $[0,1]$ for North, $[1,0]$ for East, $[0,-1]$ for South and $[-1,0]$

for West. These will be cyclically repeated, so we can use formulas, e.g. in H7 use =H3 and copy down. The same can be done for the I column. The directions now repeat every four steps.

- The origin is at [0,0]. The new x and y coordinates are then calculated by taking the previous coordinates and adding to them a step of a given length in a given direction. Note that due to the direction vectors, each of which has one 0 coordinate, only one of the x , y changes in each step. This means that for example in line 5, we are at the position [8,1] and move 7 steps south by adding $7 \cdot [-1,0]$. After this step, we are at coordinates [1,1], line 6.
- Once the trajectory returns to [0,0], we draw a scatter plot with lines. The chart should look similar to the one in Figure 2.5.

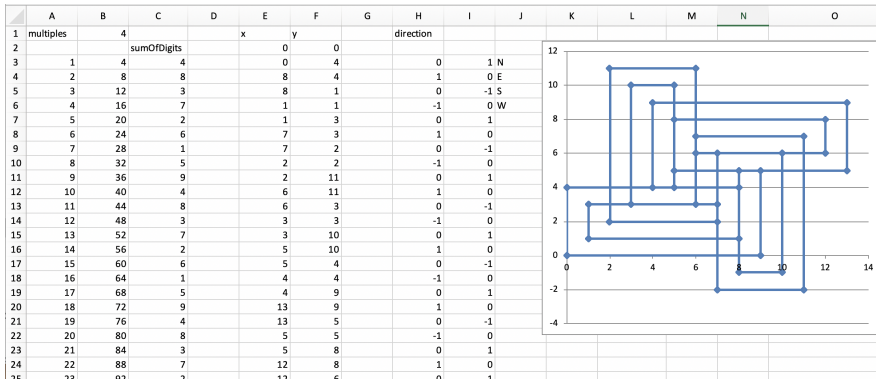


Figure 2.5: Calculation and chart of the spirolateral for $n = 4$.

Other interesting things to do with spirolaterals

What happens (and why), when we use a decimal number instead of an integer n ?

Also try changing the directions from N/E/S/W to NE/SE/SW/NW to see what happens with the resulting charts.

4. Prime numbers

Hints:

- This task can be done better by programming, e.g. in python, than in Excel. The numbers to check are all the prime numbers smaller than $\sqrt{n}$. If we checked all numbers smaller than $\sqrt{n}$, the computational complexity (in the worst case) would be $\mathcal{O}(\sqrt{n})$. For more information about the computational complexity and the $\mathcal{O}$ -notation, see Section 12.1.1. Since we are checking only primes, we expect the algorithm to be better than $\mathcal{O}(\sqrt{n})$.

Note that in practice, the primes are tested by even more sophisticated and efficient methods.

- Good place to start looking for primes is <https://primes.utm.edu/largest.html>.

Chapter 3

Finite state automata and Turing machines

3.1 Lecture overview

Problem 3.1 River crossing problem:

A farmer (F) wants to get across the river in a boat. He or she also needs to take a wolf (W), a goat (G) and a cabbage (C). Only one of these can be with the farmer in the boat at any given time, since the boat is small and the farmer is the only one who can row. Without the supervision of the farmer, the wolf eats the goat and the goat eats the cabbage. How should the farmer transport W, G and C so that nothing gets eaten in the process?

Solution: First we write out all admissible situations/states on the riverbanks:

1. (FWGC, $\emptyset$)
2. ($\emptyset$, FWGC)
3. (FWG, C)
4. (FWC, G)
5. (FGC, W)
6. (C, FWG)

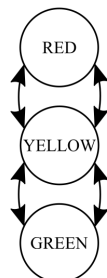


Figure 3.2: A traffic light with three colors that represent the states. The arrows denote the transitions between the states.

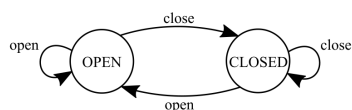


Figure 3.3: An open/close valve.

Solution: If the valve is open, it remains open until somebody/something closes it. It then remains closed until somebody/something opens it, see Figure 3.3.

Problem 3.4 Parity problem:

Alice is reading a very long sequence of 0s and 1s. Bob is listening to her, but he has a very bad memory. There is a light with a switch in the room. How can Bob use it so that when Alice is done reading, he will be able to whether there was an even or an odd number of 1s?

Solution: This light has two states: on (E) for even and off (O) for odd number of 1s. At the beginning we are in state E (Bob switches the light on), because so far there are zero digits and that means even number of 1s. As Alice reads the individual digits, Bob switches the light on and off, transitioning between the states E and O depending on whether Alice read 0 or 1. The digit 1 means a transition to the other state (using the switch), the digit 0 means that we remain in the current state (doing nothing). If the light is on (state E) at the end of the reading, the sequence had an even number of 1s. If the light is off

(state 0), the sequence had an odd number of 1s.

3.1.1 Finite state automata

Finite state automaton

A *finite state automaton* (FSA), sometimes also called a *finite state machine*, is a simple theoretical model of a computer that solves a task using a given algorithm. The task is to read an input *word* and correctly determine whether it belongs to the *acceptance set*. FSA has

- an *alphabet* of *letters*, this is usually $\{0, 1\}$, which are used in *words*.
- a finite set of *states* $S = \{s_0, s_1, s_2, \dots, s_n\}$. The automaton starts in the *initial state* s_0 , reads the whole word, changes states in the process and then stops. Some of the states are *acceptance states* – if the automaton stops in these, it means that the input word is accepted.
- a set of *transition rules* that let it get from one state to another. A transition depends on the combination of the current state and the current letter of the word that is being read. The transition rules are either written directly or in a table (matrix) or they can be drawn in the form of a diagram. The diagram is more readable for humans, the table for computers.

Problem 3.5 Automaton which accepts words ending with 0:

Find the transition rules for an automaton which accepts the words ending with 0. The alphabet is $\{0, 1\}$.

Solution: We use states s_0 and s_1 . s_0 corresponds to *last read digit was 0* and s_1 corresponds to *last read digit was 1*. The state s_0 is an acceptance state. The transition rules are:

$$\begin{aligned} [s_0, 0] &\rightarrow s_0 \\ [s_0, 1] &\rightarrow s_1 \\ [s_1, 0] &\rightarrow s_0 \\ [s_1, 1] &\rightarrow s_1 \end{aligned}$$

The same rules can also be written in a table:

	0	1
s_0	s_0	s_1
s_1	s_0	s_1

And the same rules can also be depicted in a diagram, Figure 3.4. Note that there have to be both arrows with 0 and with 1 coming out of each state. It is worth checking this property for the diagrams you draw to prevent mistakes.

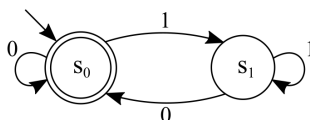


Figure 3.4: FSA that accepts words ending in 0. Note how similar this Figure is to Figure 3.3.

Problem 3.6 Automaton which accepts words starting with 110:

Find the transition rules for an automaton which accepts the words starting with 110. The alphabet is $\{0, 1\}$.

Solution: The states are s_0 =I do not know anything, s_1 =word starts with 1, s_2 =word starts with 11, s_3 =input is ok and s_4 =input is not ok. The state s_3 is an acceptance state. The transition rules are:

$$\begin{aligned}
 [s_0, 0] &\rightarrow s_4 \\
 [s_0, 1] &\rightarrow s_1 \\
 [s_1, 0] &\rightarrow s_4 \\
 [s_1, 1] &\rightarrow s_2 \\
 [s_2, 0] &\rightarrow s_3 \\
 [s_2, 1] &\rightarrow s_4 \\
 [s_3, 0] &\rightarrow s_3 \\
 [s_3, 1] &\rightarrow s_3 \\
 [s_4, 0] &\rightarrow s_4 \\
 [s_4, 1] &\rightarrow s_4
 \end{aligned}$$

The diagram of this automaton is depicted in Figure 3.5.

Problem 3.7 Automaton which accepts words ending with 110:

Find the transition rules for an automaton which accepts the words ending with 110. The alphabet is $\{0, 1\}$.

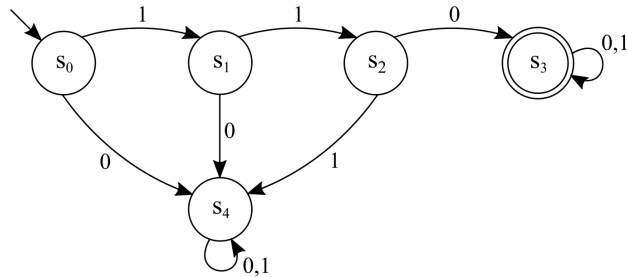


Figure 3.5: FSA that accepts words starting with 110.

Solution: The states are s_0 =last digit was 0 or nothing, s_1 =last digit was 1, s_2 =last two digits were 11 and s_3 =last three digits were 110. The state s_3 is an acceptance state. The transition rules are:

$$\begin{aligned}
 [s_0, 0] &\rightarrow s_0 \\
 [s_0, 1] &\rightarrow s_1 \\
 [s_1, 0] &\rightarrow s_0 \\
 [s_1, 1] &\rightarrow s_2 \\
 [s_2, 0] &\rightarrow s_3 \\
 [s_2, 1] &\rightarrow s_2 \\
 [s_3, 0] &\rightarrow s_0 \\
 [s_3, 1] &\rightarrow s_1
 \end{aligned}$$

The diagram of this automaton is depicted in Figure 3.6.

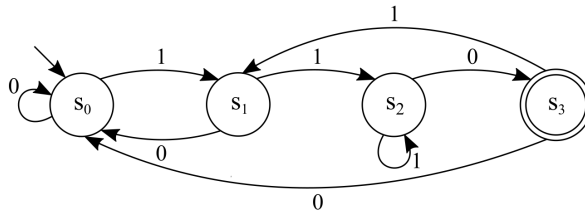


Figure 3.6: A diagram of a finite state automaton that accepts words ending with 110.

Problem 3.8 Automaton which represents volleyball matches with two sets:

Consider the alphabet $\{H, G\}$ (home team, guests). Then consider the words with letters H, G that represent the sequences of sets won by home team and by guests. Find the transition rules for an automaton which accepts all correct outcomes of a best-of-three match.

Solution: The states are 0:0, 0:1, 1:0, 1:1, 2:0, 0:2, 1:2 and 2:1. The states with 2 are acceptance states. The transition rules are:

$$\begin{aligned}
 [0:0, H] &\rightarrow 1:0 \\
 [0:0, G] &\rightarrow 0:1 \\
 [0:1, H] &\rightarrow 1:1 \\
 [0:1, G] &\rightarrow 0:2 \\
 [1:0, H] &\rightarrow 2:0 \\
 [1:0, G] &\rightarrow 1:1 \\
 [1:1, H] &\rightarrow 2:1 \\
 [1:1, G] &\rightarrow 1:2 \\
 \end{aligned}$$

The diagram of this automaton is depicted in Figure 3.7.

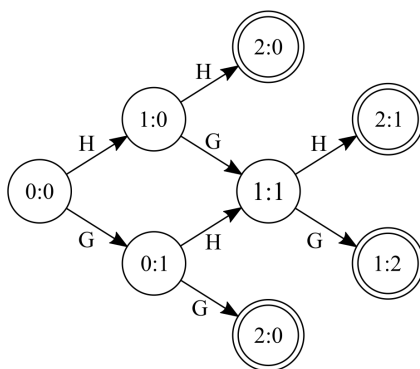


Figure 3.7: FSA which represents best-of-three matches.

There are problems a FSA cannot solve

An example of problem which a FSA cannot solve, is to recognise the set of words:

$$0^n 1^n = \overbrace{00 \dots 0}^n \overbrace{11 \dots 1}^n$$

for arbitrary n . In order to do that, we would have to use the states (we do not have anything else at our disposal) to count the number of 0s at the beginning of the word. The thing is that n can be arbitrarily large, which is in conflict with the finite number of states.

Can you design a FSA that accepts words of the type $0^n 1^n$, where n is at most 100?

3.1.2 Turing machines

What is a Turing machine?

The idea of *Turing machine* (TM), Figure 3.8, as a theoretical computer model dates back to the year 1937. Similarly to the FSA, a Turing machine can accept words that have given properties (even some properties that were too complicated for FSA) but it can do more than that. It can perform a calculation or operation and write their results on an input/output tape. The Church-Turing hypothesis states that anything we can calculate by any algorithm, can be done using a Turing machine.

- The *alphabet* of a TM is usually $\{0, 1, \sqcup\}$, where $\sqcup$ represents an empty space.
- The *input/output tape* is as long as needed (sometimes it said that it is infinite). At the beginning of the calculation, there is a finite *word* written on the tape. The rest of the tape is empty.
- The TM has a *head* that can read and write the symbol at its current location. At the beginning of the calculation, the head is at the first symbol of the input word. The head may stay in place, but it can also move one place to the left and one place to the right (N, L, R).
- The TM has a finite number of *states* among which is the *initial state* s_0 and some *final states* $s_{k_1}, s_{k_2}, s_{k_3}, \dots$. Some final states may be *acceptance states*.

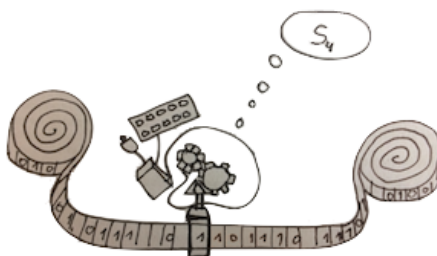


Figure 3.8: A fictitious illustration of Turing machine with an infinite tape, a reading head and an engine that performs instructions and remembers a state in which it is.

- The TM has unambiguous *instructions* (rules) for each pair (state, symbol) in the format:
(state, symbol on the tape) $\rightarrow$ (new state, write symbol, move)
- There are several ways how a TM can finish a task. If there is no instruction for a given combination of a state and a symbol, TM stops. In some cases it might be important in which state it stopped – whether it was an acceptance state or not. In other cases, the final state is not important but the output written on the tape is.

Problem 3.9 Turing machine for negation:

Create a Turing machine that can negate each bit of a binary number, i.e. it replaces 1s with 0s and 0s with 1s.

Solution: It is sufficient to have two states. The state s_0 is an initial state and we remain in it while we traverse and overwrite the input word. When we read either 0 or 1, we replace them with the other symbol and move the head onto the next symbol. If we read an empty symbol, we know we are at the end of the word. At this point, the head does not do anything more, except it changes its state into the final state s_1 for which we do not have any more instructions. The result is written on the tape.

state	symbol on the tape	new state	write symbol	move
s_0	0	s_0	1	R
s_0	1	s_0	0	R
s_0	$\sqcup$	s_1	$\sqcup$	N

Problem 3.10 Turing machine for multiplication by two:

Create a TM that multiplies a binary number by two.

Solution: Binary multiplication essentially translates into adding a 0 at the end of the number. The alphabet is $\{0, 1, \sqcup\}$, states are s_0, s_1 , where s_0 is initial. The instructions are:

state	symbol on the tape	new state	write symbol	move
s_0	0	s_0	0	R
s_0	1	s_0	1	R
s_0	$\sqcup$	s_1	0	N

These instructions can also be written as:

(state, symbol on the tape)	$\rightarrow$	(new state, write symbol, move)
$(s_0, 0)$	$\rightarrow$	$(s_0, 0, R)$
$(s_0, 1)$	$\rightarrow$	$(s_0, 1, R)$
$(s_0, \sqcup)$	$\rightarrow$	$(s_1, 0, N)$

Problem 3.11 Turing machine for adding one:

Create a TM which adds 1 to a binary number.

Solution: This machine moves to the end of the word first, then adds 1 and then carries over if necessary. The alphabet is $\{0, 1, \sqcup\}$, states are s_0, s_1, s_2 , where s_0 is initial and s_2 is acceptance state. Once TM reaches s_2 , the result is written on the tape. The instructions are:

state	symbol on the tape	new state	write symbol	move
s_0	0	s_0	0	R
s_0	1	s_0	1	R
s_0	$\sqcup$	s_1	$\sqcup$	L
s_1	1	s_1	0	L
s_1	0	s_2	1	L
s_1	$\sqcup$	s_2	1	N

Problem 3.12 Turing machine that can do more than a FSA:

Create a Turing machine that accepts numbers in the format:

$$\overbrace{00\dots 0}^n \overbrace{11\dots 1}^n.$$

Solution: The alphabet is $\{0, 1, \sqcup\}$, states are $s_1, s_2, s_3, s_4, s_5, s_6$, where s_1 is an initial state and s_6 is an acceptance state. If the automaton stops in any other state, the word is not accepted. This machine erases pairs of numbers – always one 0 and one 1. It gets to the acceptance state if there is nothing left when it cannot erase another pair. The instructions are:

state	symbol on the tape	new state	write symbol	move
s_1	0	s_2	$\sqcup$	R
s_2	0	s_2	0	R
s_2	1	s_2	1	R
s_2	$\sqcup$	s_3	$\sqcup$	L
s_3	1	s_4	$\sqcup$	L
s_4	0	s_4	0	L
s_4	1	s_4	1	L
s_4	$\sqcup$	s_5	$\sqcup$	R
s_5	0	s_2	$\sqcup$	R
s_5	$\sqcup$	s_6	$\sqcup$	N

Infinite cycles

It is possible to create TMs that fail to halt, e.g. a TM that moves to the right without writing anything or changing its state. Moreover, in general it is not possible to determine whether an arbitrary TM halts for an arbitrary input. In computability theory, this is called the *halting problem*.

A brief reminder about the binary system

When performing conversion from decimal to binary system we have: 1_{10} is 1_2 , 2_{10} is 10_2 , 3_{10} is 11_2 , 4_{10} is 100_2 , 5_{10} is 101_2 , etc. Note that powers of two are all of the form $100\dots 000_2$. Why?

State

Here we return to one concept which we encountered both when working with the FSA and with TMs. It is difficult to precisely define what a state can

represent. As we saw in the problems, the states and transition rules comprise the inner logic of the automaton, which depends on the task for which it was designed. And for the same task, it is possible to design different machines. The *state* is the inner situation or setting of the FSA or TM. It can represent a typical activity being currently performed (e.g. a TM head is moving to the end of the binary word), some quantity (e.g. the parity of the number of 1s in the word being currently read), current situation of the object represented by the automaton (e.g. a closed valve) or something completely else depending on how we designed the machine.

3.2 Questions

1. Write all admissible states for the problem of transferring a man, a goat and a wolf to the other side of the river using a boat for two.
2. Write at least four admissible states for the problem of transferring a man, a goat, a wolf and a cabbage to the other side of the river using a boat for two.
3. Write at least four admissible states for the problem of transferring a man, a goat, a sheep, a cabbage and a wolf to the other side of the river using a boat for three. (Wolf eats both goats and sheep and both goats and sheep eat cabbage.)
4. Write at least four admissible states for the problem of transferring a man, a woman, a cabbage, a goat, a sheep and a wolf to the other side of the river using a boat for three. (Wolf eats both goats and sheep and both goats and sheep eat cabbage. Both people can row and take care that nothing gets eaten on their watch.)
5. Draw all possible transitions between the *closed* and *open* states of a valve.
6. Propose a FSA for determining the parity of number of 1s in any binary word.
7. Propose a FSA for determining the parity of number of 0s in any binary word.
8. What is *finite* in a FSA? What is its *alphabet*?
9. What do *state* and *initial state* mean in the definition of a FSA?
10. What does *acceptance state* mean in the definition of a FSA?
11. What does a Turing machine consist of?
12. How long is the input/output tape of a Turing machine?
13. What is the most commonly used alphabet of a Turing machine?
14. What is the difference between a Turing machine and a FSA?
15. What does Church-Turing hypothesis say about a Turing machine?

3.3 Problem set

River-crossing problems

1. A mother M with two children – a son s and a daughter d – stand at a river bank. They have a small boat, which can carry at most one adult person or two children. Children can row. How can they all get to the other side of the river?
2. Two people, a big ape and a small monkey stand on the riverbank. Only people and the ape can row. In any moment there cannot be more animals than humans standing on a riverbank. Only two creatures fit into the boat. Animals can jump in and out of the boat by themselves. How can they all get to the other side of the river?
3. Two parents with two children – younger and older son – stand on a riverbank. The boat is small and can carry at most one adult person or two children. Children can row. How can they all get to the other side of the river?
4. Three people, one big ape and two small monkeys stand on a riverbank. Only people and big ape can row. In any moment there cannot be more animals than humans standing on a riverbank. Only two creatures fit into the boat. Animals can jump in and out of the boat by themselves. How can they all get to the other side of the river?
5. Three missionaries and three cannibals stand on a riverbank. The boat can carry two people. We cannot allow more cannibals than missionaries at the riverbank, because the cannibals would eat the missionaries. How can they all get safely to the other side of the river?

Finite state automata

6. Write and draw a FSA which accepts words starting with 1010.
7. Write and draw a FSA which accepts words ending with 101.
8. Write a FSA which accepts words containing 111.
9. Write a FSA which accepts words containing 010.
10. Write a FSA which accepts words containing 010 and not containing 011 (both conditions must hold).

11. Write a FSA which accepts words starting with 1010 and ending with 111.
12. In the previous problems, change the text *FSA which accepts words* to *FSA which does not accept words*.
13. Write a FSA which accepts words with even number of 0s and odd number of 1s.
14. Write a FSA which recognises correct game outcomes, when playing volleyball for three winning sets.

Turing machines

15. Write instructions for a Turing machine which performs binary XOR operation of the word on the tape with a (fictitious) word 111...111 of the same length.
16. Write instructions for a Turing machine which performs binary XOR operation of the word on the tape with a (fictitious) word 000...000 of the same length.
17. Write instructions for a Turing machine which multiplies the binary number written on the tape by 4.
18. Write instructions for a Turing machine which adds 2 to the binary number written on the tape.
19. Write instructions for a Turing machine which divides the binary number written on the tape by 2 (and rounds down).

3.3.1 Hints and solutions

River-crossing problems

First we need to write admissible states, the transitions between them and find a path in the graph from the initial to the final state. If it is not clear how to start or continue with the problem, we can try the following steps. Read them only until it is clear how to continue!

Hints:

- a) We simplify the problem by translating it into the language of states and transitions between them. The states correspond to different arrangements of people (and animals/things) on the riverbanks. The transitions are done using admissible boat transfers.
- b) Which state represents the initial situation and which state corresponds to a successful solution?
- c) We need to think about the representation of all possible states which can arise in the problem, including all theoretical options. Some of them might not be actually used but we will take care of them later. What does each state have to clearly determine? We should also decide if we need to include only people or also animals/things.
- d) Then we make a list of all the states, thinking carefully about its organisation. We should be able to explain and confirm that the list is complete and that there is nothing missing from it. It should be transparent enough for easy examination of transitions between states. There should be approximately 10 to 20 states in this problem.
- e) Once our list of states is complete, we draw them as vertices of a simple graph into which we can also draw the transitions. If we do not know the structure of the graph in advance, we can draw the vertices in a circle.
- f) Using our list, we start examining from which state can we get to which other state(s). We join the respective vertices by an edge. We should do this examination in such a way that we do not forget any transition and also do not examine the same transition too many times. Are the transitions one-way or two-way? Does this distinction help with examining the transitions?

- g) Once we have a graph, we can find a path from the initial to the final state. Graphs in these problems are not too complicated and it should be possible to find the path without going into graph theory algorithms.
- h) Check carefully whether it is truly possible to follow the path through the states that we have found. When solving problems such as these, it is a common mistake to underestimate the state description part. If necessary, we should return to points b) and c).
1. A mother M with two children – a son s and a daughter d – stand at a river bank. They have a small boat, which can carry at most one adult person or two children. Children can row. How can they all get to the other side of the river?

Solution:

In this solution we translate the problem into the language of states, transitions between them and finding the path from the initial to the final state. We show our steps that lead to a surprise, but this is not the only solution.

- a) The individual states correspond to the placement of people on the riverbanks. It is helpful to look for a reasonably sorted and clear way to write the states. We have created the following list:

s_0 :	M, d, s		...
s_1 :	M, d		s
s_2 :	M, s		d
s_3 :	d, s		M
s_4 :	M		d, s
s_5 :	d		M, s
s_6 :	s		M, d
s_7 :	...		M, d, s

- What kind of system have we used to write out these states?
- Are we certain that we did not miss any state (that we have them all)? Why?
- What is the initial state? What is the acceptance state? Do we know if there is a final state?

It is possible to create a different ordered list of these same states and solve the problem with it.

- b) The transitions between pairs of states correspond to the allowed boat transfers. We can draw the states as vertices of a simple graph and represent the permissible transitions as edges. Let us first examine the transitions from state s_0 . In this situation M , d , s or $d + s$ can row to the opposite riverbank. So we can get to states s_3 , s_2 , s_1 or s_4 . The second approach is to go through the state list s_1 to s_7 and for each state check whether we can get into it from s_0 . The result should be the same. Then we continue with s_1 and the rest. Before we do that though, try to answer the following questions:

- We have noted that it is possible to transition from s_0 to s_1 . Does it mean we can also go from s_1 to s_0 ?
- If we join s_0 and s_1 in the graph by an edge, do we need to denote the *orientation* of this edge?
- When we analyze the state s_1 (and then s_2 , s_3 , ...), is it necessary to check also the transitions to the previous states?

Finish the analysis of states and draw them into a graph. Repeat the steps for your own list of states.

- c) For our list of states, we got a graph shown in Figure 3.9. Our task now is to find a path from s_0 to s_7 . There are multiple such paths, e.g. $s_0 \rightarrow s_3 \rightarrow s_7$ or $s_0 \rightarrow s_4 \rightarrow s_7$ or a longer one $s_0 \rightarrow s_2 \rightarrow s_6 \rightarrow s_7$. We see that the problem has more than one solution. Can you estimate the number of solutions?
- d) Now we check the solutions that we found. First, let us look at $s_0 \rightarrow s_3 \rightarrow s_7$. In this case, the M transfers from the left riverbank to the right, and in the second step $d + s$ row from the left riverbank to the right. However, they cannot do that, since they do not have a boat! The boat is on the other side of the river, with their mother. Our solution has a huge flaw and for the same reason also the other solutions will not work. Why is our solution incorrect? What did we forget when we were describing the states?
- e) Create a list of states which also consider the boat position. If we leave out unreasonable states (a boat alone on the riverbank), we should get 14 states. Solve the problem using the steps a) to d) with these correct states. There will be more transitions but they should result in the correct solution.
- f) Consider if it is important to distinguish between the son and the daughter. Will the solution be simpler, if we just have two children

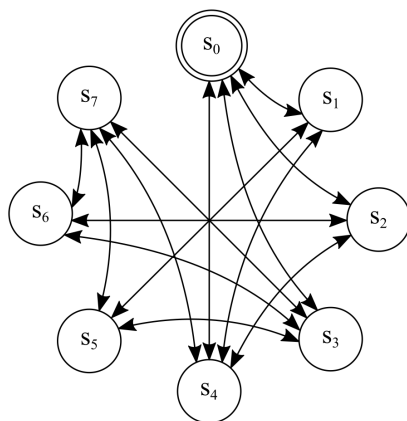


Figure 3.9: Graph that represents the states and transitions in the boat problem.

and do not know anything about their gender? If we only consider the number of children on the riverbanks, 10 states will suffice. Solve the problem for this situation.

Finite state automata

Hints:

When drawing graphs of state automata, there are no simple and universal hints. To create the automaton is similarly difficult to (and sometimes even more difficult than) writing a corresponding computer program. We offer the following hints and recommendations:

- The initial state s_0 corresponds to the beginning of the reading. We can stay in this state until *something interesting starts happening*.
- The state should *start paying attention* when it reads the characters of the desired sequence and use subsequent states to monitor how far into the sequence has the automaton gotten into.
- Once the FSA is done, check if it is complete, i.e. whether for each state and each admissible letter of the alphabet there is a transition rule.

- d) If we are interested in a sequence at the beginning of a word, once we found it, we are done and we can remain in the acceptance state. On the contrary, if the search was *unsuccessful*, we can go to the *reject* state and remain there.
 - e) If we are looking for a sequence at the end of the word, even if we find such a sequence, the next letter can throw away this success. Still it does not land us in a reject situation, we just need to return to one of the previous states, not necessarily the initial one. Consider a search for an end sequence 0011 and a *hopeful* state 001. If we now get the letter 0, we do not return to the initial state because we already have the first letter of the desired sequence.
 - f) There are other types of problems besides those mentioned in d) and e) and there is no universal solution. The approach should be a modification of the reasoning shown above.
6. Write a FSA that accepts words starting with 1010.

Solution: Follow these steps only to the point from which you can continue on your own:

- a) The initial state s_0 corresponds to the beginning of the reading. Since we are interested in the sequence 1010 at the beginning of the word, we go to the state s_1 when we read the first letter 1. How to proceed from here?
- b) From the state s_1 we go to state s_2 , when we read the next 0, then to state s_3 , when we read 1 and then to state s_4 , if we read letter 0. What have we read so far?
- c) If we are in s_4 , this means we have read the sequence 1010 at the beginning of the word. This is the *final accepting* state. However, the description of states and transitions between them is not complete yet.
- d) If the beginning of the word is 1010, the outcome will not change no matter what the subsequent letters are. So we remain in the state s_4 even if we read another 0 or 1. Is the description of the FSA complete now?
- e) For each state s_0 , s_1 , s_2 and s_3 we only have one transition that corresponds to reading the correct sequence 1010. If at any point we read anything else, the word cannot be accepted. In those cases, we

transition into a common *garbage* state s_5 . This is also a *final* but *not accepting* state. Therefore we cannot leave it whether we get a 0 or 1 afterwards. Can you write a list of transition rules for this automaton?

- f) The transition rules are: $[s_0, 1] \rightarrow s_1, [s_0, 0] \rightarrow s_5, [s_1, 0] \rightarrow s_2, [s_1, 1] \rightarrow s_5, [s_2, 1] \rightarrow s_3, [s_2, 0] \rightarrow s_5, [s_3, 0] \rightarrow s_4, [s_3, 1] \rightarrow s_5, [s_4, 0] \rightarrow s_4, [s_4, 1] \rightarrow s_5, [s_5, 0] \rightarrow s_5, [s_5, 1] \rightarrow s_5$.
- g) The diagram of this FSA is depicted in Figure 3.10.

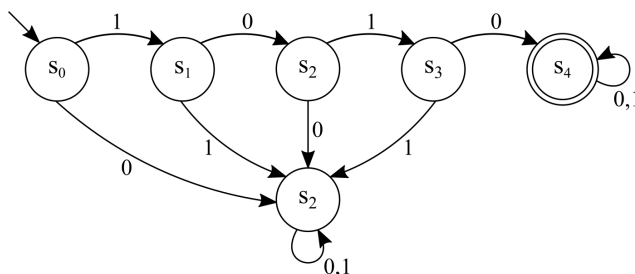


Figure 3.10: FSA that accepts words starting with with 1010.

7. Write a FSA which accepts words ending with 101.

Solution: Follow these steps only to the point from which you can continue on your own:

- a) The initial state s_0 corresponds to the beginning of the reading. Since we are interested in the sequence 101 at the end of the word, we only start paying attention when we read the letter 1. In such case we get to state s_1 . In case we are in state s_0 and read a 0 (at the beginning or later on), we remain in the state s_0 .
- b) If we are in s_1 and read a 0, we get to s_2 which corresponds to having read 10. If we are in s_1 and read a 1, we remain in s_1 , since we still only have the first letter of the desired sequence.
- c) If we are in state s_2 and read a 1, we go to s_3 which means we have read the desired sequence 101. A lot of work is lost though, if the word continues. If we are in state s_2 and read a 0, the *hopeful*

sequence 10 has changed to 100, which is unusable. We return to state s_0 and wait for a 1 again. What can you say about s_3 ?

- d) The state s_3 is accepting state and corresponds to the fact we were looking for a 101 sequence. If the word continues, we leave the state s_3 . In case the next letter is 1, we currently have 1011. Therefore we go to state s_1 , which *waits* for a 0. If we are in s_3 and the next letter is 0, we currently have 1010. Therefore we go to state s_2 , which *waits* for a 1.
- e) We have finished the description of this FSA, see Figure 3.11. Since we are interested in the sequence 101 at the end of the word, *there is always hope* until we read the last letter, so we keep moving through the graph. We do not have any *hopeless garbage* state. However, even the arrival to the acceptance state does not guarantee that we have found our word yet. We can only be certain, if we are in the acceptance state after reading the last letter.
- f) Write the transition rules.

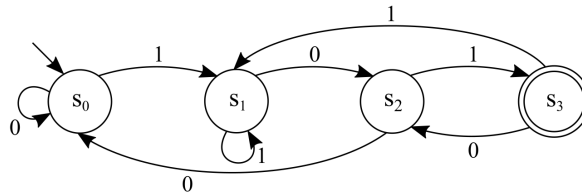


Figure 3.11: FSA that accepts words ending with 101.

10. Write a FSA which accepts words containing 010 and not containing 011 (both conditions must hold).

Solution:

- a) This problem is more difficult than the previous one because it has two requirements that have to hold at the same time. Note that both sequences that determine whether the word is accepted or not start with the same digits 01. This will help significantly when designing the FSA.

- b) We start and remain in the initial state s_0 while we read *uninteresting* digits 1 at the beginning of a word. Only once we get a 0, which is the start of both important sequences 010 and 011, we switch to state s_1 .
- c) We remain in s_1 while reading more 0s – always a fresh potential start of the important sequences. Once we get a 1, we can switch to state s_2 . This means we have already encountered a sequence 01. The next digit will determine the fate of the word.
- d) If the next digit is 1, the word contains the forbidden sequence 011 and we do not accept it no matter what comes next. Therefore we switch to state s_4 and remain in it. This state is not an acceptance state.
- e) If we are in state s_2 and the next digit is 0, we know we read the *good* sequence 010 and we would like to accept the word. We can switch to state s_5 and remain in it, see Figure 3.12.

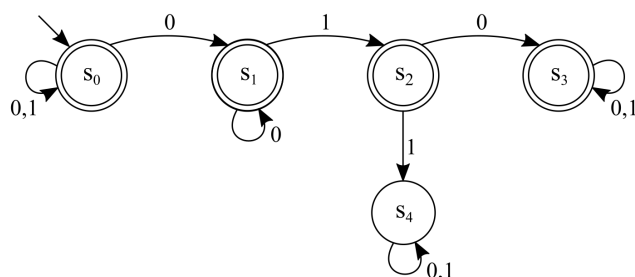


Figure 3.12: First attempt at FSA that accepts words containing 010 and not containing 011.

- f) Are we happy with this result? We should carefully consider the properties of this FSA. What would it do with the word 01011? This word starts with the *good* sequence 010 and ends with the *bad* sequence 011. We should reject it, but after reading the first three digits, our FSA switches to the acceptance state s_5 and stays in it. Where do we have a mistake?
- g) Even though the FSA correctly identifies the sequence 010 and switches to the state s_5 , it is premature to remain in it. We need to remain

on the lookout for the forbidden sequence 011 and (definitely) leave the acceptance state if it arrives. We need to extend our FSA design.

- h) While reading digits 0, we remain in s_5 , but if we encounter the digit 1, we are again the 01 situation (and new state s_6) and wait for the final outcome.
- i) Subsequent 0 indicates another *good* sequence 010 and brings us (again temporarily) back to the acceptance state s_5 . If we read the digit 1, the sequence 011 is complete and this leads us to the final state s_4 .
- j) We should verify that the modified FSA shown in Figure 3.13 works as it is supposed to.

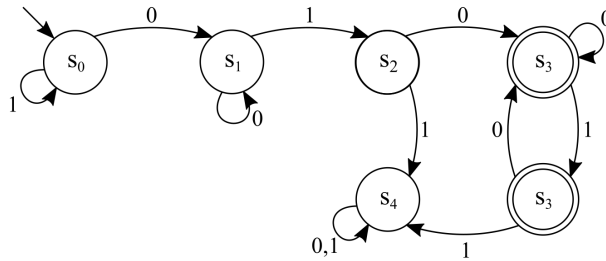


Figure 3.13: FSA that accepts words containing 010 and not containing 011.

Turing machines

Hints:

According to the Church-Turing hypothesis (which we assume holds), the Turing machines can perform each computational task that can be described algorithmically. The scope is so broad that we cannot expect a general set of steps for problems in which we are supposed to write instructions for a Turing machine. We offer some basic advice instead:

- a) Consider the type of the problem. Is the machine supposed to perform a calculation and write the result on the tape? Or is the machine supposed to accept a word of given properties? In the first case, the transition rules are primarily steps how to read the input and write output. The output

can be written on a completely different part of the tape or can be a modification of or addition to the original input. In this case, the final acceptance state has to be defined more precisely to distinguish among the cases when the calculation did not go correctly, result does not exist or is incorrect. In the second case, the resulting text on the tape is not important (often the input gets modified), but the important distinction is whether the machine ended up in an acceptance state.

- b) Remember that a standard means for stopping a Turing machine is a situation in which for a given current state and letter on the tape there is no instruction for the next move. This current state is then the final state of the machine and if necessary, we need to be able to decide, whether it is also an acceptance state.
 - c) When designing a Turing machine, think about the mathematical foundation of the algorithm it is supposed to represent. A good start of the analysis is to try the required calculation by hand, ideally for all types of input, result or method of calculation.
 - d) Based on this experience, think about the basic stages and steps of the algorithm.
 - e) Formalise these steps into the machine rules. Often the machine work regimes get represented by the states. It is common to encounter new problems and ambiguities in this phase.
15. Write instructions for a Turing machine which performs binary XOR operation of the word on the tape with a (fictitious) word 111...111 of the same length.

Solution: Follow this solution only to the point from which you can continue on your own:

The result in this problem should be a word written on the tape. The calculation can always be performed and thus the machine will stop once the operation is done correctly. Consequently, we do not need to separately specify acceptance states. The result will be written in place of the original input word.

The idea of the algorithm lies in the way we modify the letter we have read. Since $0 \text{ XOR } 1 = 1$ and $1 \text{ XOR } 1 = 0$, the input letter 0 is changed to 1 and the input letter 1 is changed to 0. It is thus sufficient to change

the 0s to 1s and vice versa, as we read the word until we get to its end. Two states will suffice for this. The state s_0 is the initial state and we remain in this state while reading and modifying the input word. If we are in state s_0 and read an empty space, we know that we are at the end of the word. The head will not do anything else but change the state to the final state s_1 for which we do not need any more instructions.

Note that in the lecture overview, there is a description of a Turing machine which performs binary negation. Have you noticed any similarities with this problem?

16. Write instructions for a Turing machine which performs binary XOR operation of the word on the tape with a (fictitious) word 000...000 of the same length.

Solution: Follow this solution only to the point from which you can continue on your own:

The problem is very similar to the previous one. The only difference is in the way we do the *change* of the letter. Since $0 \text{ XOR } 0 = 0$ and $1 \text{ XOR } 0 = 1$, the input 0 should be changed to 0 and the input 1 should be changed to 1. We can easily modify the Turing machine from the previous problem to do this.

However, when we look at this more closely, we see that the machine does not actually have to do anything. For every binary word it is true that binary XOR with the word 000...000 does not change the input. How to write a Turing machine which does not do anything is more of a theoretical question. The machine should have a set of instructions. We could leave the set empty, but that is unusual and confusing.

What we can do instead, is for s_0 and letters 0, 1 and empty space move to final state s_1 , which does not have any further instructions.

17. Write instructions for a Turing machine which multiplies the binary number written on the tape by 4.

Solution: The result of this machine's work is again a word written on the tape and the calculation can always be performed.

Algorithmically, multiplication by 4 is easy: we add two zeros at the end of the word. This is because multiplication by 4 corresponds to two multiplications by 2, which have been described in Section 3.1.2.

First, while in state s_0 , we move to the end of the word without any changes to the current 0s and 1s. At the end of the word (when we read

an empty space), we change it to 0 and switch to state s_1 . We move to the right once and write the second 0. We switch to state s_2 and finish.

3.4 Lab problems

1. **Finite state automata with alphabet $\{0, 1\}$.**

If a word is accepted, the output is *accepted word*, if not, the output is *rejected word*.

- a) Design a FSA that accepts words ending with 1.
- b) Design a FSA that accepts words ending with 10.
- c) Design a FSA that accepts words ending with 11.
- d) Design a FSA that accepts words containing (anywhere) a sequence of at least five 1s.
- e) Design a FSA that accepts words containing (anywhere) a sequence 101.

2. **Bunny invasion**

The Australian prairie with dimensions 5 miles $\times$ 5 miles is invaded by a growing number of bunny colonies. The colonies gradually cover 25 square regions 1 mile $\times$ 1 mile. In the first year, two colonies settled in two of the regions. When colonising regions, we consider neighboring regions to be those which share either a side or a corner. Every year, new colonies appear in every region, which has at least two neighboring regions colonised. Existing colonies survive indefinitely.

Create an Excel sheet with 5×5 input. Two numbers 1 denote the two regions initially colonised by bunnies and 0s in all other regions represent free prairie. The output is a sequence of sufficiently many 5×5 squares corresponding to the next years of colonies' spreading.

Using the created sheet, answer the following questions:

- a) In which initial setups, there will be no new colonies?
- b) Are there any initial setups that result in colonisation of the whole prairie?
- c) Find such initial setups which colonise the whole prairie in the shortest and in the longest time.
- d) Are there any initial setups that stop spreading after some time?
- e) Use the conditional formatting to denote the colonised and free regions.

- f) Find geometrically and graphically interesting examples of colonisation.
- g) How do the solutions (and corresponding formulas) change, if the colonisation condition changes from *at least two* to *exactly two*?
- h) Answer the same questions for different colonisation rules.

3. Game of life

There are several Game of Life applets available online. We are going to create our own in Excel. The game environment is 5×5 . At the beginning, some cells are occupied, some are not. In the next generation, some cells die, some stay alive and some come alive depending on the number of neighbors. Neighbors share either a side of the square or a corner. If a populated cell has 2 or 3 neighbors, it will survive. If it has 0 or 1 neighbor, it dies from loneliness, if 4 or more, it dies from overpopulation. If the cell is free and has 3 neighbors, it will come alive because of suitable surroundings.

Create an Excel sheet with 5×5 input. Denote the occupied cells by some character, e.g. x or o. The output are 5×5 squares representing the next generations that arose from the first one using the above-stated rules. Using the Excel sheet, solve the following problems:

- a) Find the largest population that will die out fast (in a given number of steps).
- b) Find an *oscillator* with the smallest initial number of cells. Oscillator is a setup which repeats after some time.
- c) Find oscillators with a period of 2, 3, 4, ...
- d) Find an oscillator with a known period but a different initial regime.
- e) Find the smallest/largest *stationary state*. Here we are looking for an initial setup that does not change in the next generation.
- f) What happens in a given configuration if you symmetrically add/delete certain cells? Does it help or harm the population?

3.4.1 Hints and solutions

1. **Finite state automata with alphabet $\{0, 1\}$.**

If a word is accepted, the output is *accepted word*, if not, the output is *rejected word*.

- a) Design a FSA that accepts words ending with 1.
- b) Design a FSA that accepts words ending with 10.
- c) Design a FSA that accepts words ending with 11.
- d) Design a FSA that accepts words containing (anywhere) a sequence of at least five 1s.
- e) Design a FSA that accepts words containing (anywhere) a sequence 101.

Hints:

- The automata a) to c) are relatively simple and of the same type. They test the end of the word.
- We first decide whether we are using the vertical or horizontal representation of the input word. In either case, its end is indicated by an empty space after a sequence of 0s and 1s. This is a suitable place for evaluation of the word and making the acceptance decision.
- To test the *short ends* of words, it is possible to make the decision with a single condition (=IF) at the end of the word.
- For more complex conditions, e.g. problem d), we can use auxiliary calculations next to the input word. In problem d) this would be a calculation of continuous sequences of 1s.
- The auxiliary calculation can be further improved for more complicated automata, such as the one in problem e). The auxiliary line or column can contain the individual (numerically coded) states and in the condition that calculates them, we write the transition rules using logical operators.
- If we are using the auxiliary line for calculation, in certain sense, we are solving the problem using a TM, since we write and evaluate necessary data. This can replace the need for additional states.
- The most difficult and also the most universal approach is to state the transition rules in a table and then generate the states in the

auxiliary line or column by searching this table using the functions VLOOKUP and MATCH.

a) Design a FSA that accepts words ending with 1.

Solution:

- We designate a part of a row for input consisting of 0s, 1s and empty cells. The length of the input region is fixed, e.g. 20 cells. The input is entered manually. Be careful when using an empty cell ("") and a space (" "), which appear to be the same but behave differently.
- We evaluate the input word in the next line. The value *z* corresponds to the part under the input letters. Under the first empty cell, we evaluate whether the previous letter was 1 and accept the word in the state *a* or otherwise transition to a non-acceptance final state *k*. This can be achieved using a command =IF(I4<>"";"z";IF(H4=1;"a";"k")).
- The final result can be displayed by checking whether there is the state *a* in the state row, e.g. =IF(COUNTIF(C3:W3;"a")>0;"accepted word";"rejected word"), as in Figure 3.14.

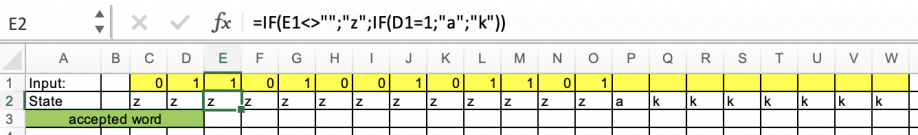


Figure 3.14: FSA that accepts words ending with 1.

Clever input

If we want to enter the input in a more clever way, we can use a random number generator. In such case, we also need to generate the random ending of the word in addition to generating a random sequence of 0s and 1s. To do that, we can use additional row with commands =RANDBETWEEN(0;2), where 2 corresponds to a space, as in Figure 3.15. Using the command =IF(OR(C2=2;B4="");"",C2) we achieve that the sequence of 0s and 1s ends with the first occurrence of 2 after which there are only empty cells.

E12																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O			
11	Input:		0	1	1	0	1	0	1	1	1	1	1	0	1			
12	State		0	1	2	0	1	0	1	2	3	4	5	0	1			
13	accepted word																	

Figure 3.16: FSA that accepts words containing a sequence of at least five 1s.

=IF(C1<0,6;1;IF(C1<0,9;0;2)), as shown in Figure 3.17. Now in approximately 60% of cases we generate a 1, in 30% of cases 0 and in 10% of cases 2.

A20																		
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
16	Random	0,68	0,68	0,52	0,19	0,19	0,26	0,47	0,47	0,98	0,98	0,74	0,87	0,07	0,08	0,43	0,75	0,17
17	Auxiliary	0	0	1	1	1	1	1	1	2	2	0	0	1	1	1	0	1
18	Input:	0	0	1	1	1	1	1	1									
19	State	0	0	1	2	3	4	5	6	0	0	0	0	0	0	0	0	0
20	accepted word																	

Figure 3.17: FSA that accepts words containing a sequence of at least five 1s. Input is randomly generated.

2. Bunny invasion

Hints:

- We create a sequence of tables 5×5 in which more and more new bunny colonies appear. The colonised regions are denoted by 1, the free regions are denoted by 0. To answer the questions, we should display at least 10 generations.
- The only place where we enter the colonies manually is the first table. In all others, a formula (the same one for all regions) determines whether there is a 0 or a 1 in the given region. The 5×5 tables should be displayed in a graphically clear manner.
- If we leave sufficient gaps between the tables, we can copy the formulas for the individual regions. In fact, it is sufficient to use the same formula for all regions, write it just once and then copy.

Solution:

- We create one 5×5 table for each generation in which 1 represents a colonised region and 0 represents a free region.
- We enter the 0s and 1s manually in the first table.
- To determine the population of the region in the next year, we need to check its current population (if it is 1, it is going to remain 1) and the number of neighbors. For this we add their values.
- If we have at least one row and one column *padding* around the table, we can use the same formula for all regions, even those that are at the sides. Their neighbors are empty and Excel interprets the empty cells as numerical value 0.
- We can use the formula

$$=IF(OR(B3=1,A2+B2+C2+C3+C4+B4+A4+A3>1),1,0)$$
- This formula is copied to all regions in the table and the whole table is copied below (or next to) the original one with one row (column) between them, as in Figure 3.18.

K5																																						
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI			
1		1								2						3							4							5								
2																																						
3																																						
4			0	0	0	0	0			0	0	0	0	0		0	0	1	1	0			0	1	1	1	1			1	1	1	1	1				
5			0	0	0	0	0			0	0	1	1	0		0	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
6			0	0	1	1	0			0	0	1	1	0		0	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
7			0	0	0	0	0			0	0	0	0	0		0	0	1	1	1			1	1	1	1	1			1	1	1	1	1				
8																																						
9		6								7						8							9							10								
10																																						
11		1	1	1	1	1				1	1	1	1	1		1	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
12		1	1	1	1	1				1	1	1	1	1		1	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
13		1	1	1	1	1				1	1	1	1	1		1	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
14		1	1	1	1	1				1	1	1	1	1		1	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
15		1	1	1	1	1				1	1	1	1	1		1	1	1	1	1			1	1	1	1	1			1	1	1	1	1				
16																																						

Figure 3.18: Bunny invasion.

3. Game of life

Hints:

- Our goal is to generate a sequence of 5×5 tables in which the next generations of life arise. Only the first table is entered manually, the rest of them are generated automatically. We are going to need around 10 generations.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1																												
2				X																								
3			x			x																						
4		x		X			x																					
5			x			x																						
6				X																								
7																												
8																												
9				X																								
10			X			X																						
11		X							X																			
12			X			X																						
13				X																								
14																												
15																												
16				X																								
17			X	X	X																							
18		X	X			X	X																					
19			X	X	X																							
20				X																								
21																												

Figure 3.19: Life

- It is useful to have a few auxiliary tables for determining the occupancy in the next generation, e.g. one for calculating the number of neighbors or one for applying the life rules.
- If there are more tables per generation, it is useful to find a clear graphical representation.
- It should be possible to copy the formulas when creating next generations and performing their calculations.

Solution:

- For each generation, we first create a table that displays the occupied cells. We use three auxiliary tables with numbers for calculations.
- In the first auxiliary table, we transform the occupancy into numerical values 1 (occupied) and 0 (empty).
- Based on this table we create a second auxiliary table into which we denote the number of occupied neighbors for each cell.

- In the third auxiliary table, we use the life rules and the second table to calculate the occupancy in the next generation and denote it by 0s and 1s.
- The following formulas may be used for calculating the values in the auxiliary tables:

$$=IF(D4="X",1,0)$$

$$=K3+L3+L4+L5+K5+J3+J4+J5$$

$$=IF(OR(AND(K4=1,OR(R4=2,R4=3)),AND(K4=0,R4=3)),1,0)$$
- The four tables together represent one generation of life starting with the display of initial state and ending with a *precalculation* of the next generation.
- The tables of the next generation again start with a display of the environment occupancy. Unlike in the first table, this one is obtained by redrawing the precalculation from the row above.
- We obtain the next generations by copying, see Figure 3.19.

Chapter 4

Mathematical logic

4.1 Lecture overview

4.1.1 Logical statements and operators

Logical statements and their truth value

A *logical statement* or a *proposition* is the basic building block of more complicated mathematical structures. It is any statement for which we can determine whether it is *true* (denoted by 1) or *false* (denoted by 0). Two examples are: "*The snow is cold.*" which is a true statement and "*My hair is green.*" which is a false statement.

Not a statement

Consider "*Let's go to Mars.*" We cannot determine whether this sentence is true or false, therefore it is not a logical statement.

Consider also the sentence "*Apples are the best fruit.*" Is it a statement? Is it true? We are using sentences about the world around us to illustrate the mathematical concepts, but while doing so, we have to work only with sentences for which it is clear whether they are true or false. Some people might agree that apples are the best fruit, some might not, so this is not an example of a logical statement.

The main use of mathematical logic is to examine the statements whose truth value can be determined unambiguously. Even if we stay in the mathematical realm, we have to be careful because the sentence "*For every x and y there*

exists a z such that $x+z=y$." is true if we are talking about integers, but not true if we are talking about natural numbers.

Negation

The *negation* replaces the original proposition by its opposite. It essentially means *"It is not true what the statement says."* We denote it by the symbol: $\neg$, or by an apostrophe or by a bar over the proposition we are negating. The proposition $\neg A$ is true if and only if the proposition A is not false and vice versa. For example a negation of a true statement *"The snow is cold."* is the false statement *"It is not true that the snow is cold."* which can also be stated as *"The snow is not cold."*

Note that the negation has to be the exact opposite of the original statement. For example, since there are things that are neither large nor small, the statement *"The mouse is large."* is not a correct negation of the statement *"The mouse is small."*

Two basic principles of classical logic

Since there are only two options for the truth value of a logical statement, in classical mathematical logic we have the following two principles. The first one is the *principle of excluded middle* also known as the *law of excluded third* which means that every statement which is not true has to be false and vice versa. There are no other options. The second principle is the *principle of double negation* which means that if we negate a proposition twice, we return to the original proposition.

Problem 4.1 Double negation:

Negate the following statement twice: *"A googol is a large number."*

Solution: The negation is *"A googol is not a large number."* The negation of the negation is *"It is not true that a googol is a not large number."* which is the same as the statement *"A googol is a large number."*

Problem 4.2 Positive auto-reference:

Consider *"This statement is true."*

Solution: If this sentence is true, everything is ok. If it is false, then its negation must be true. The negation is *"This statement is false."* Since we assumed it was, there is no contradiction. However, we cannot have it both ways. Since

we cannot determine whether the sentence is true or false, it is not a logical statement.

Problem 4.3 Negative auto-reference:

Consider "*This statement is false.*"

Solution: If this sentence is true, it has to also be false, since that is what it says. And obviously it cannot be both at the same time. On the other hand if it is false, then its negation "*This statement is true.*" must be true. So again, the sentence should be both true and false which cannot happen. So there is an inconsistency either way and the sentence is not a logical statement. We call a sentence like this one or the one in the previous problem a *paradox*.

Equivalence

The *equivalence* of two propositions means that either both of them are true or both of them are false. It is also often expressed using the words *if and only if*, sometimes shortened to *iff*. We denote it by $\iff$. So the proposition $A \iff B$ is true iff both A and B are true or both of them are false. It is false, when A and B have different truth values. The statement "*A number is divisible by two if and only if its last digit is divisible by two.*" is an example of an equivalence. We cannot have one property without the other.

Conjunction

The *conjunction* of two propositions is also called *and* or *logical product*. It is denoted by $\wedge$ or $\&$. The statement $A \wedge B$ is true if and only if both A and B are true. The negation of "*A and B are true.*" is thus "*A or B are not true.*", i.e. $\neg(A \wedge B) \iff \neg A \vee \neg B$. An example statement is: $x > 0 \wedge y > 0$.

Alternative

The *alternative* of two propositions is sometimes called *or*, *disjunction*, *logical sum* or *at least one of*. It is denoted by $\vee$. The statement $A \vee B$ is true iff at least one of A and B is true. Indeed, both of them can be true, this *or* is not excluding. The negation of "*A or B are true.*" is thus "*A is false and B is false.*", i.e. $\neg(A \vee B) \iff \neg A \wedge \neg B$. An example statement is "*I have a soup or I have a sandwich for lunch.*" I can certainly also have both.

Implication

This is the logical operator that often causes confusion. It is sometimes called *If... then...* If the premise is true, then the conclusion needs to be true. We denote it by $\Rightarrow$. The statement $A \Rightarrow B$ is false iff the premise A is true and the conclusion B is false. In all other cases, it is true. For the negation we have $\neg(A \Rightarrow B) \iff A \wedge \neg B$. The statement $B \Rightarrow A$ is called a *converse* and may or may not be true. The statement $\neg B \Rightarrow \neg A$ is called a *contrapositive* and is always true.

Let us look at an example statement: *"If it is snowing, I wear a scarf."* This is only false, if it is snowing and I do not have a scarf. If it is not snowing, I may or may not have a scarf, whatever I do, the original statement remains true. Since the negation is *"It is snowing and I do not have a scarf."* the original implication can also be written in the form of an alternative *"It is not snowing or I wear a scarf."*

Consider another example. Your Mom tells you *"If you get an A in this class, I will bake you a cheesecake."* There are four scenarios that can happen:

- You got an A, Mom baked a cheesecake. Life is great, implication is true.
- You did not get an A, Mom did not bake a cheesecake. Not that great but the implication is true.
- You did not get an A, Mom baked a cheesecake. Mom is great, wanted to cheer you up, the implication is still true.
- You got an A, Mom did not bake a cheesecake. This should not happen, implication is false.

It might seem that our intuition of how the world works is fairly aligned with the mathematical rules in this case, but we should not rely on it in general. Consider a similar situation, only this time, Mom says *"If you get FX in this class, I will bite your head off."* There are again four scenarios:

- You got FX, Mom bit your head off. Not that great but the implication is true.
- You did not get FX, Mom did not bite your head off. Life is great and the implication is true.
- You did not get FX, Mom bit your head off. Wait, what? This should not happen, but the implication is true!

- You got FX, Mom did not bite your head off. Mom is great and the implication is false.

4.1.2 Quantifiers and predicate logic

Predicates

Predicate logic examines the structure of *predicates* or atomistic logical formulas. These formulas contain variables denoting objects from an agreed-upon set, usually a number set, such as natural numbers, real functions, etc. and talk about (properly defined) properties of these objects. The properties can be related to individual objects, e.g. a number is even, a function is periodic or to groups of objects, e.g. a number a divides a number b , linear functions are continuous, etc. When we want these formulas to be exact, we have to determine for which objects are they true. To do this, we use two basic quantifiers.

Universal quantifier

A *universal quantifier* also called *for all* and denoted $\forall$ means that the property holds for all objects from the set under consideration. We read $\forall n \in \mathbb{N}$ as "*For all n in $\mathbb{N}$...*" or "*For every natural number n ...*" The negation of statement "*For every x in A , the following is true.*" is "*There exists an x in A for which the following is not true.*"

Existential quantifier

An *existential quantifier* denoted by $\exists$ means that there exists (at least one) object with the given property, e.g. $\exists n \in \mathbb{N} : 3|n$. The negation of statement "*There exists an x in A for which the following is true.*" is "*For all x in A , the following is not true.*"

4.1.3 Language of mathematics

The mathematics uses the language of formulas, predicates and quantifiers when it wants to be precise, especially in writing. Mathematical texts (books, articles, lectures) contain a formal, and for general public often unusual, language. To get oriented in it, here we mention some of its typical and important components.

Definition

Since the time of Aristotle it has been clear that for meaningful communication, it is necessary to establish common terms that are going to be used. To do this in mathematics, we use *definitions*. The definitions use the already known, previously defined terms to introduce the new ones. The good definitions have to be not only correct, but also understandable and should only be used for important key terms.

Axiom

Before we get to definitions of complex terms, we need to define simpler ones. Similarly, complex properties of mathematical objects are derived from simpler and simpler ones. However, this process has to start somewhere. The set of basic definitions of objects and their basic properties is usually captured in the *axioms of the given theory*. We consider the axioms to be true and we do not prove them. Conversely, the whole theory can be derived from the axioms. It is necessary to verify the correctness of the axiomatic system, e.g. whether the axioms contradict each other or whether there are theorems that cannot be derived from these axioms. The first axiomatic system is the planar theory described in Euclid's Foundations, which is the basis of the planar geometry.

Theorem

Important mathematical statements that describe properties of and relationships among mathematical objects are called *theorems*. Some of them are important enough that they earned names, e.g. Pythagorean Theorem, Fundamental Theorem of Arithmetics, Fundamental Theorem of Algebra, Fermat's Last Theorem, etc. The auxiliary theorems that are important but not focal in various mathematical fields are called *lemmas*. Some of these also have their own names, such as Zorn's lemma. Some other statements mathematicians use are *corollary*, *note*, etc.

Proof

With the exceptions of axioms and definitions, all statements in mathematics need to be proven. (And even for axioms and definitions it needs to be proven that they are not formally incorrect.) The proof of a theorem or a lemma has to start from the valid axioms or from previously proven theorems. Using the rules of mathematical logic, new theorems can be derived.

The proof can be more or less formal. A completely exact proof of the statement $1+1=2$ originating from the axioms of set theory supposedly contains approximately 20 000 steps [8]. While this is interesting, it is only interesting for mathematicians (and even for them for reasons other than to find out how much is $1+1$). For us, the proof is useful and necessary for the following reasons:

- It convinces us of correctness of the hypothesis, which holds for some cases but we cannot verify it for all cases one-by-one.
- It clarifies the meaning and background of the theorem. It shows how it is connected to and follows from other simpler theorems or concepts.
- A constructive proof includes steps how to create an object or calculate a quantity which is needed or useful for solving a problem.

4.2 Questions

1. Explain what inaccuracy (in language or in thinking) causes the paradox:
 - a) Girls are not boys. Girls are busy. Therefore boys are not busy.
 - b) Tea may be hot. Tea may be cold. Therefore hot is the same as cold.
 - c) Horse is a word. Horse can gallop. Therefore words can gallop.
 - d) 3 divides 18. 4 does not divide 18. 7 is 3 and 4. Therefore 7 divides and does not divide 18.
2. Which of these sentences are logical statements? Explain. If possible, determine whether they are true:
 - a) In the year 2020, there will be a person on Mars.
 - b) In the year 1970 there was a person on the Moon.
 - c) In the year 1670 there was a person on the Moon.
 - d) We will go to Mars!
 - e) Go to Mars!
 - f) This sentence is true.
 - g) This sentence is false.
3. Do the following equivalences hold? Prove or give a counterexample:
 - a) A number is divisible by 5 if and only if its last digit is 0.
 - b) A number is divisible by 5 if and only if its last digit is 0 or 5.
4. We have an implication *"If the last digit of a number is 0, then the number is divisible by 5."*
 - a) State its negation.
 - b) State this implication in terms of an alternative.
5. State the negations of these statements about all natural numbers:
 - a) If a number is divisible by 5, then its last digit is either 0 or 5.
 - b) If a number is divisible by 6, then it is divisible by 3 and divisible by 2.

- c) If a number is divisible by 2, then it is a multiple of 4. (Note that this statement is not true.)
- d) If a number is a multiple of 4, then it is divisible by 2.
6. State the negations of the statements:
- a) Everybody likes something.
- b) Everybody dislikes something.
- c) Sometimes everybody likes something.
- d) Sometimes everybody dislikes something.
7. Evaluate in binary:

$$\text{a) } \begin{array}{r} 110010110 \\ \text{AND } 001110100 \\ \hline \end{array}$$

$$\text{b) } \begin{array}{r} 100101100 \\ \text{OR } 001001110 \\ \hline \end{array}$$

$$\text{c) } \begin{array}{r} 111100100 \\ \text{XOR } 010011010 \\ \hline \end{array}$$

8. Solve equation in binary:

$$\text{a) } \begin{array}{r} 100111010 \\ \text{AND } \\ \hline 100100010 \end{array}$$

$$\text{b) } \begin{array}{r} 101101011 \\ \text{OR } \\ \hline 011010110 \end{array}$$

$$\text{c) } \begin{array}{r} 100110010 \\ \text{XOR} \\ \hline 100001101 \end{array}$$

9. Write a definition
 - a) of a geometric object or property
 - b) of an arithmetic object or property
10. Write
 - a) a theorem from geometry
 - b) a theorem from arithmetics
 - c) a theorem from algebra
 - d) Pythagorean Theorem in the form of two implications
11. Write a false statement (theorem)
 - a) from geometry
 - b) from arithmetics
 - c) from algebra
12. Write your best definition of
 - a) a right angle
 - b) an equilateral triangle
 - c) an isosceles triangle
 - d) a rectangle
 - e) a cube
 - f) a line
 - g) the number 0
 - h) the number 1
 - i) the number π

4.3 Problem set

1. Pseudo-scientific language

The following sentences use a pseudo-scientific language to describe well-known proverbs. What are the proverbs?

- a) Repeated execution of a task causes one not to make mistakes when performing the task later.
- b) The graphical representation of a certain object has three orders of magnitude larger value than a meaningful array of individual letters describing the said object.

2. Marbles in boxes

In each of two boxes there is exactly one marble, which may be black or white. Each box has something written on the lid. We know that the lids are either both true or both false. Determine whether they are true/false and what color marble is in each box.

- a) The writing on the first box says "*There is a white marble in the first box or there is a black marble in the second box.*" The writing on the second box says "*There is a black marble in the first box.*"
- b) The first box says "*In the first or in the second box there is a white marble.*" The second box says "*There are black marbles in both boxes.*"

3. Tools

$P(x)$ denotes *x is at the right place*. $O(x)$ denotes *x is in good order*. $T(x)$ denotes *x is a tool*. Let us assume that the domain of x are all things A . Write the following statements in *everyday language* and write their negations:

- a) $\exists x \in A : \neg P(x)$
- b) $\forall x \in A : T(x) \Rightarrow (P(x) \wedge O(x))$
- c) $\forall x \in A : P(x) \wedge O(x)$
- d) $\forall x \in A : \neg(P(x) \wedge O(x))$
- e) $\exists x \in A : (T(x) \wedge \neg P(x) \wedge O(x)) \wedge (\forall y \in A : (T(y) \wedge \neg P(y) \wedge O(y)) \Rightarrow (x = y))$
- f) $\forall x \in A : (P(x) \wedge T(x)) \Rightarrow O(x)$

4. Princesses and tigers

(These problems are inspired by the book *The lady or the tiger* [7].) We have princesses (P) and tigers (T). Behind each door, there is either a P or a T. There is also a statement (more or less helpful) on each door. A prince wants to get married and for that he needs to find a P (if he finds a T, he will be eaten). Which door should he choose in the following scenarios?

- a) There may be a T or P in either room (both rooms can even contain the same creature):
 Door 1: There is a P in this room and a T in the next room.
 Door 2: In one of the rooms there is a P and in the other a T.
 One statement is true, one is false.
- b) There may be a T or a P in either room (both rooms can even contain the same creature).
 If there is a P in the room 1, the statement on the door 1 is true, if there is a T, the statement is false. If there is a P in room 2, the statement on door 2 is false, if there is a T, the statement is true.
 Door 1: It makes a difference what you choose.
 Door 2: It is better to go next door.
- c) We have three rooms, in one of them is a P, in the remaining two are Ts:
 Door 1: There is a T in this room.
 Door 2: There is a P in this room.
 Door 3: T is in room 2.
 At most one statement is true.

5. Negating quantifiers

Read each statement and decide whether it is true. Then negate it and check if the negation has the opposite true/false value.

- a) $(\forall x \in \mathbb{N})(\forall y \in \mathbb{N}) \quad x > y$
- b) $(\forall x \in \mathbb{N})(\exists y \in \mathbb{N}) \quad x > y$
- c) $(\exists x \in \mathbb{N})(\forall y \in \mathbb{N}) \quad x > y$
- d) $(\exists x \in \mathbb{N})(\exists y \in \mathbb{N}) \quad x > y$
- e) $(\forall y \in \mathbb{N})(\forall x \in \mathbb{N}) \quad x > y$
- f) $(\forall y \in \mathbb{N})(\exists x \in \mathbb{N}) \quad x > y$

$$\text{g) } (\exists y \in \mathbb{N})(\forall x \in \mathbb{N}) \quad x > y$$

$$\text{h) } (\exists y \in \mathbb{N})(\exists x \in \mathbb{N}) \quad x > y$$

6. Autoreferencing tablets

There were a few stone tablets found during an archaeological research. Each of them contains some statements. Find out which statements are true and which are false. Find all solutions.

a) Tablet A:

There is exactly one true statement on this tablet.

There are exactly two true statements on this tablet.

There are exactly three true statements on this tablet.

...

There are exactly eight true statements on this tablet.

b) Tablet B:

There are exactly two true statements on this tablet.

There are exactly three true statements on this tablet.

...

There are exactly six true statements on this tablet.

c) Tablet C:

There are exactly zero false statements on this tablet.

There is exactly one false statement on this tablet.

There are exactly two false statements on this tablet.

There are exactly three false statements on this tablet.

...

There are exactly ten false statements on this tablet.

d) Tablet D:

There are exactly five false statements on this tablet.

There are exactly six false statements on this tablet.

There are exactly seven false statements on this tablet.

There are exactly eight false statements on this tablet.

e) Tablet E:

There are at least zero true statements on this tablet.

There is at least one true statement on this tablet.

There are at least two true statements on this tablet.

There are at least three true statements on this tablet.

...

There are at least ten true statements on this tablet.

f) Tablet F:

There are at least three true statements on this tablet.

There are at least four true statements on this tablet.

There are at least five true statements on this tablet.

...

There are at least ten true statements on this tablet.

4.3.1 Hints and solutions

1. Pseudo-scientific language

The following sentences use a pseudo-scientific language to describe well-known proverbs. What are the proverbs?

- a) Repeated execution of a task causes one not to make mistakes when performing the task later.
- b) The graphical representation of a certain object has three orders of magnitude larger value than a meaningful array of individual letters describing the said object.

Hint: The following sentence could serve as a hint how to solve this kind of problem: "*An asymmetric discussion that includes a barter transaction of personally obtained experience is of significant value.*" The language of mathematics is often perceived as pointlessly complicated. Sometimes such observation leads to the incorrect conclusion that the wisdom or importance of the mathematical statements lies in the incomprehensibility and difficulty of their language. This problem demonstrates how even a well-known simple proverbs can be obfuscated just by using a pseudo-scientific language.

To decode them, we can try to replace some of the expressions by simpler words and remember a proverb that contains them. For example *repeated execution* may be a *practice*. A *graphical representation* may be a *picture*.

Solution:

- a) Repeated execution of a task causes one not to make mistakes when performing the task later. (Practice makes perfect.)

2. Marbles in boxes

In each of two boxes there is exactly one marble, which may be black or white. Each box has something written on the lid. We know that the lids are either both true or both false. Determine whether they are true/false and what color marble is in each box.

Hint: There is more than one approach to this problem:

- a) Follow the logical line of the statements.
- b) Write out all possibilities for truth values of the statements. Choose from them those that meet the conditions stated in the problem.

These problems are used for development of reasoning and logical thinking. The logical correctness in thinking should come before the exactness of expression and formulation of mathematical statements. Therefore it is important to discuss these problems in pairs or in larger groups and to try to understand a different solution and compare it with one's own. That should be also a method for solving this kind of problems: be able to explain one's own reasoning, understand a different reasoning, confront the two and perhaps apply either of them in another problem.

Solution:

- Suppose it is true that *"There is a white marble in the first box or there is a black marble in the second box."* The marbles can then be placed either WW or WB or BB. The second statement has the same truth value as the first statement, so also the following has to be true: *"There is a black marble in the first box."* Of the available options, only BB fits. We now need to check what happens if the first statement is false. Then it is true that *"There is a black marble in the first box and a white marble in the second box."* Also *"There is not a black marble in the first box."* has to be true. However, these two statements cannot be true at the same time. Consequently, BB is the only solution.
- We now try to solve the problem in a slightly different way. We write all the options for truth values of the statements: 11, 10, 01 and 00. We know that both statements have the same truth value, therefore we can exclude the options 10 and 01. In the case 11, it is true that *"There is a white marble in the first box or a black marble in the second box."* and it is also true that *"There is a black marble in the first box."* This implies solution BB. In the case 00, it is not true that *"There is a white marble in the first box or a black marble in the second box."* and it is not true that *"There is a black marble in the first box."* This implies that the black marble both is and is not in the first box. Thus BB is the only solution.
- Yet another possible approach is to write out all possible marble placements: WW, WB, BW and BB. We evaluate the corresponding truth values of the statements on the boxes: 10, 10, 01 and 11. We know that both statements have the same truth value, so we have to exclude 10 and 01. Only BB with 11 remains, thus this is the only solution to the problem.

3. Tools

$P(x)$ denotes *x is at the right place*. $O(x)$ denotes *x is in good order*. $T(x)$ denotes *x is a tool*. Let us assume that the domain of x are all things A . Write the following statements in *everyday language* and write their negations:

- a) $\exists x \in A : \neg P(x)$
- b) $\forall x \in A : T(x) \Rightarrow (P(x) \wedge O(x))$
- c) $\forall x \in A : P(x) \wedge O(x)$
- d) $\forall x \in A : \neg(P(x) \wedge O(x))$
- e) $\exists x \in A : (T(x) \wedge \neg P(x) \wedge O(x)) \wedge (\forall y \in A : (T(y) \wedge \neg P(y) \wedge O(y)) \Rightarrow (x = y))$
- f) $\forall x \in A : (P(x) \wedge T(x)) \Rightarrow O(x)$

Hint: In mathematical statements, we use quantifiers to state the number of items with the given property. The basic rule when negating statements with quantifiers states that the negation of statement with universal quantifier is a statement with an existential quantifier. For example the negation of the statement "*All aliens are pink.*" is "*There exists an alien, which is not pink.*" Formally, this negation is OK, it is problematic only for those that do not think there are any aliens. In that we encounter the properties of an empty set, which we will examine in one of the following chapters. Here we are trying to make statements about objects we know exist. For example, the statement "*All dogs are dachshunds.*" is clearly not true. Its negation "*There exists a dog, which is not a dachshund.*" has the opposite truth value and thus it is true. We can use a similar approach in the problems with tools.

We need to be careful when negating statements *everything holds* as *nothing holds*. This is not a negation. We can see this clearly when we look at the statement "*All integers are even.*" This statement is false. However, also this attempt at negation is false "*No integers are even.*"

Solution:

- a) This statement says: *There exists a thing which is not at the right place.* To negate it, we would say *All things are in the right place* and write $\forall x \in A : P(x)$.

- b) This statement says that *All tools are in the right place and in order*. To negate it, we would say that *There exists a tool which either is not at the right place or is not in good order*. We would write it as $\exists x \in A : T(x) \wedge (\neg P(x) \vee \neg O(x))$
- e) This statement says *There exists a tool, which is in order and not at the right place and for all tools in order and not at the right place we have that they are the same as the first tool we mentioned*. In plain language we could simplify this statement to *There exists exactly one tool which is in order and not at the right place*. This can be written as $\exists! x \in A : (T(x) \wedge \neg P(x) \wedge O(x))$. For negation of this statement, we need to consider both cases: *There is no thing in A that ...* and *More than one thing in A ...*

More quantifiers

Use the quantifiers in the proverbs.

State a statement with three quantifiers about dogs and dachshunds.

4. Princesses and tigers

(These problems are inspired by the book *The lady or the tiger* [7].) We have princesses (P) and tigers (T). Behind each door, there is either a P or a T. There is also a statement (more or less helpful) on each door. A prince wants to get married and for that he needs to find a P (if he finds a T, he will be eaten). Which door should he choose in the following scenarios?

- a) There may be a T or P in either room (both rooms can even contain the same creature):

Door 1: There is a P in this room and a T in the next room.

Door 2: In one of the rooms there is a P and in the other a T.

One statement is true, one is false.

Hint: There is more than one approach to this problem. One is to consider the content of the statements written on the doors. Another is to write out all the four possibilities for the truth values of the statements on the doors, i.e. $(True, True)$, $(True, False)$, $(False, True)$ and $(False, False)$ and figure out what creatures can be behind each door. Yet another approach is to write out all the four possibilities of room occupancy behind the doors, i.e. $(Tiger, Tiger)$, $(Princess, Princess)$, $(Princess, Tiger)$ and $(Tiger, Princess)$ and evaluate the truth value of the statements on the doors based on these occupancies. We can try to solve the same problem

using all three of these approaches to verify the correctness of the solution and compare the complexity and clarity of the individual solutions and reasonings.

Solution: The solution of this problem follows from the fact that the first sentence implies the second. Therefore, if the first statement is true, the second must be true as well. However, having both statements true contradicts king's information. Therefore only the second sentence can be true and the first one is false. The only way how to put two different creatures into two rooms and contradict the first sentence is to have a tiger behind the first door and a princess behind the second. Our recommendation for the prince is to try the second door.

5. Negating quantifiers

Read the statement and decide whether it is true. Then negate it and check if the negation has the opposite true/false value.

$$(\forall x \in \mathbb{N})(\forall y \in \mathbb{N}) \quad x > y$$

Hint: In these problems, it is helpful to substitute specific numbers. A basic sanity check is that if the statement is true, its negation has to be false and vice versa. Remember that the order of quantifiers matters.

Solution: Since the statement must be true for all natural x and y , we can try numbers 5 and 3:

$$(5 \in \mathbb{N})(3 \in \mathbb{N}) \quad 5 > 3.$$

We see that the statement is true. However, it has to be true for all pairs of x and y . For the pair 7 and 9, the statement is not true. Therefore it is not true overall.

Its negation is

$$(\exists x \in \mathbb{N})(\exists y \in \mathbb{N}) \quad x \leq y.$$

This is true. There are such numbers, e.g. 2 and 22.

6. Autoreferencing tablets

There were a few stone tablets found during an archaeological research. Each of them contains some statements. Find out which statements are true and which are false. Find all solutions.

a) Tablet A:

There is exactly one true statement on this tablet.

There are exactly two true statements on this tablet.
There are exactly three true statements on this tablet.
...
There are exactly eight true statements on this tablet.

Hint: The purpose of these problems is to make clear how the truth value changes when a statement references itself and is bound also to other statements. Also remember that to solve a problem, we need to find all solutions.

Solution:

- We need to consider the connections between the statements. Since they talk about specific numbers of true statements, every two sentences are mutually exclusive. Once one of them is true, all the rest of them has to be false. This means, at most one statement on the tablet can be true. If one was true, obviously it has to be the first one (because it says that one statement is true) and the rest of them are false.
- There is also a second solution: all statements false.
- With any solution, we need to check that it is correct, i.e. that the truth values as we determined them correspond to the content of the statements.

4.4 Lab problems

1. Tool for verification of algorithms

Here are three simple algorithms. Would you give them a certificate of correctness? Check them on examples. If you find mistakes, try to correct them.

- a) We have a string that contains an algebraic expression. We want to check that it uses the brackets correctly. At the beginning, we set the variables L and R to 0. We read the string from left to right. If we encounter "(", we increment L , if we encounter ")", we increment R . If $R \neq L$ at the end, the use of brackets is not correct. If $R = L$, it is correct.
- b) Heron's formula calculates the area of a triangle using the lengths of its sides. We start with numbers x, y, z and calculate A using Heron's formula. The value A represents the area of triangle with sides x, y and z .

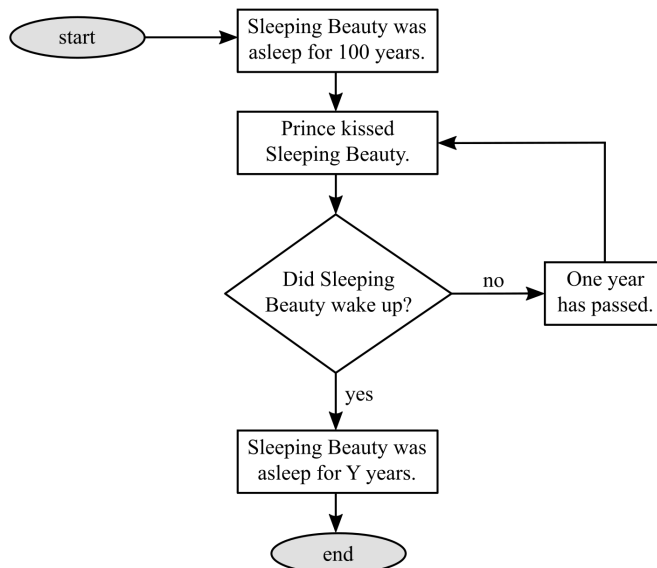


Figure 4.1: A diagram for Sleeping Beauty.

- c) Sleeping Beauty was asleep for 100 years. Then came a prince, kissed her and maybe she woke up. If she did not wake up, he returned again after one year and kissed her again. Create a table in Excel that counts the years of Sleeping Beauty's sleep as suggested in the Figure 4.1. Assume that there is a 0.5 probability that the Sleeping Beauty wakes up after a kiss. Generate randomness by repeated use of the F9 key and observe when the princess wakes up. What was the longest time your princess was asleep?

2. Tool for checking numeric games

Create an Excel checker for number games Latin square and Magic square which checks if the square is correct. For the Magic square it is sufficient to consider only rows and columns (not the diagonals).

- a) Latin squares 3×3 , 4×4

A Latin square of order n is a table $n \times n$ whose elements are n different symbols placed in such a way that every symbol appears in every row and every column exactly once, see Figure 4.2.

1	2	3
2	3	1
3	1	2

1	2	3	4
3	4	1	2
4	3	2	1
2	1	4	3

Figure 4.2: Examples of Latin squares 3×3 and 4×4 .

- b) Magic squares 3×3 , 4×4

A Magic square is a placement of natural numbers $1, 2, 3, \dots, n^2$ in a table $n \times n$ in such a way that every number is used only once and the sum of numbers in every row, every column and every diagonal is the same, see Figure 4.3.

3. Tool for solving logical problems

Princess Porcia has a golden, a silver and a bronze box. To select a husband, she places a portrait of herself in one of the boxes. Two boxes remain empty. She puts a sign on each box, but the signs do not have to be true. The future husband tries to find the box with the portrait.

4	9	2
3	5	7
8	1	6

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

Figure 4.3: Examples of Magic squares 3×3 and 4×4 .

In Excel, create a tool for solving specific logical problems about hiding things in boxes. This tool should conveniently display the question and test where the portrait is located. The Excel sheet should automatically evaluate whether the statements are true or false and determine whether the placement of the portrait fits the problem.

- a) At most one sign is true.
Golden: *The portrait is in this box.*
Silver: *The portrait is not in this box.*
Bronze: *The portrait is not in the golden box.*
- b) At least one sign is true and at least one is false.
Golden: *The portrait is not in the silver box.*
Silver: *The portrait is not in this box.*
Bronze: *The portrait is in this box.*
- c) There are two sentences on each box. On one box, both are true, on one box, both are false and on one box, one is true and one is false.
Golden: *Portrait is not in this box. Portrait is in the silver box.*
Silver: *Portrait is not in the golden box. Portrait is in the bronze box.*
Bronze: *Portrait is not in this box. Portrait is in the golden box.*
- d) There are two sentences on each box. At most one sentence on each box is true.
Golden: *Portrait is not in this box. Painter is from Venice.*
Silver: *Portrait is not in the golden box. Painter is from Florence.*
Bronze: *Portrait is not in this box. Portrait is in the silver box.*

4.4.1 Hints and solutions

1. Tool for verification of algorithms

Here are three simple algorithms. Would you give them a certificate of correctness? Check them on examples. If you find mistakes, try to correct them.

- a) We have a string that contains an algebraic expression. We want to check that it uses the brackets correctly. At the beginning, we set the variables L and R to 0. We read the string from left to right. If we encounter (, we increment L, if we encounter), we increment R. If $R \neq L$ at the end, the use of brackets is not correct. If $R = L$, it is correct.

Hint: We can indeed check the parentheses using this algorithm. In some situations, the algorithm gives a correct answer, in some it does not. The situations in which it does not give a correct answer are the situations when we have the same number of left and right parentheses but they are not in the correct places.

Solution: According to the problem, we should write an algebraic expression and create two variables L and R. We can write their values in the rows below the input expression. The sequence of values represents the changes in the variables and the last value in the row is the final value that gets evaluated.

C3																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	(	)															
2			5	+	4	-	(	3	+	6	)	+	(	2	-	3	)
3	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	R	0															

Figure 4.4: Variable L.

The characters + and − cannot be entered directly as text in Excel, therefore we input them as '+' and '-'. Initially, the variable L contains zero, see Figure 4.4, but then it changes according to the formula we entered in C3.

What to do if the formula does not seem to work properly? It is possible that it did not get copied correctly, see Figure 4.5. We check

another cell to see its formula. To display which cells enter the given formula, we click the formula and the cells are highlighted both in the spreadsheet and in the formula line. In this case, we find out that we need to fix the cell A1 (using the F4 key) and replace the zero with the cell's left neighbor in the condition, see Figure 4.6.

SUM		✖	✓	<i>fx</i>	=IF(F2=D1;E3+1;0)																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q				
1	(	)																			
2			5	+	4	-	(	3	+	6	)	+	(	2	-	3	)				
3	L	0	0	0		=IF(F2=D1;E3+1;0)	0	0	0	0	0	0	0	0	0	0	0				
4	R	0																			

Figure 4.5: Formulas can be checked in the formula line.

Then we create a similar line for the right parentheses and variable R.

F3		✖	✓	<i>fx</i>	=IF(F2=\$A\$1;E3+1;E3)																
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q				
1	(	)																			
2			5	+	4	-	(	3	+	6	)	+	(	2	-	3	)				
3	L	0	0	0	0	0	1	1	1	1	1	1	2	2	2	2	2				
4	R	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	2				

Figure 4.6: The formula is fixed by adding dollar signs.

Finally, we check whether the last values in L and R match, as in Figure 4.7.

We used the Excel IF command to evaluate the condition whether we have the same values in the green fields. If we want to check whether the algorithm works as desired, we break the parentheses rule and check what it says, see Figure 4.8.

What about some other expression?

The algorithm does not work properly on all inputs, see Figure 4.9. We need to modify it in such a way that there are never more right parentheses than left.

B7																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	(	)															
2			5	+	4	-	(	3	+	6	)	+	(	2	-	3	)
3	L	0	0	0	0	0	1	1	1	1	1	1	2	2	2	2	2
4	R	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	2
5																	
6																	
7		parentheses are correct															

Figure 4.7: Final check whether the final values of variables L and R match.

B7																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	(	)															
2			5	+	4	-	(	3	+	6	)	+		2	-	3	)
3	L		0	0	0	0	0	1	1	1	1	1	1	1	1	1	1
4	R		0	0	0	0	0	0	0	0	1	1	2	2	2	2	3
5																	
6																	
7			parentheses are incorrect														

Figure 4.8: Result of processing of an incorrect input.

B7																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	(	)															
2			5	+	4	-	)	3	+	6	(	+	(	2	-	3	)
3	L		0	0	0	0	0	0	0	0	1	1	2	2	2	2	2
4	R		0	0	0	0	0	1	1	1	1	1	1	1	1	1	2
5																	
6																	
7			parentheses are correct														

Figure 4.9: Apparently, we are not done yet, because in some cases an incorrect input is not discovered by our checks.

More complex input situations

How would we make sure that the first character inside parentheses is not a + and the last character is not a + or a -?
Another condition is that the whole expression cannot end with

a + or a -. How do we ensure that?
 And how would our algorithm change if the whole algebraic expression was given as a string in one cell?

- b) Heron's formula calculates the area of a triangle using the lengths of its sides. We start with numbers x, y, z and calculate A using Heron's formula. The value A represents the area of triangle with sides x, y and z .

Hint: Nobody remembers all mathematical formulas, therefore it is ok to look up the Heron's formula online. In the spreadsheet, we designate some cells as input for x, y, z that represent the lengths of triangle sides. We calculate the auxiliary value that represents half the triangle perimeter and use it to calculate the triangle area. We try various values of x, y, z to see whether this approach works correctly and fix any issues that arise.

Solution: The Heron's formula is $A = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = (a + b + c)/2$. In the Excel sheet, we designate cells for input of triangle sides x, y, z and for calculation of s and A , see Figures 4.10 and 4.11.

E2		fx		=(A2+B2+C2)/2			
	A	B	C	D	E	F	G
1	x	y	z		s		A
2	3	4	5		6		6

Figure 4.10: Calculation of s for Heron's formula.

G2		fx		=SQRT(E2*(E2-A2)*(E2-B2)*(E2-C2))			
	A	B	C	D	E	F	G
1	x	y	z		s		A
2	3	4	5		6		6

Figure 4.11: Calculation of A for Heron's formula.

Then we enter sides of various triangles for which we can easily compute the area, e.g. right triangles for which the area is half the product of the two short sides and the hypotenuse is calculated using the Pythagorean theorem, see Figure 4.12.

C3							
	A	B	C	D	E	F	G
1	x	y	z		s		A
2	3	4	5		6		6
3	2	3	3.605551		4.302776		3
4	7	4	8.062258		9.531129		14
5	8	6	10		12		24
6	4	5	6.403124		7.701562		10
7	4	4	5.656854		6.828427		8
8	7	3	7.615773		8.807887		10.5
9	6	1	6.082763		6.541381		3
10	2	9	9.219544		10.10977		9

Figure 4.12: Verifying that Heron’s formula works for the right triangles.

G9							
	A	B	C	D	E	F	G
1	x	y	z		s		A
2	3	4	6		6.5		5.3327
3	2	3	4		4.5		2.9047
4	7	4	7		9		13.416
5	8	6	9		11.5		23.525
6	4	5	8		8.5		8.1815
7	4	4	7		7.5		6.7777
8	7	3	9		9.5		8.7856
9	6	1	10		8.5		#NUM!
10	2	9	11		11		0

Figure 4.13: Verifying that Heron’s formula works for random triangles.

We can enter few more random triangles to see whether we get correct results, see Figure 4.13.

The results seem to be ok, except for the last two triangles. If we tried to construct those two, we would find out it is not possible. Therefore, we need to check the input values and display a message when they are inadmissible. We can write the message *Incorrect input of triangle sides* when the triangle inequality does not hold, see Figure 4.14.

Checking inputs for correctness

The sides of the triangle have to have positive lengths. Do we need to check for this separately or is the triangle inequality a sufficient check that takes care also of this issue?

- c) Sleeping Beauty was asleep for 100 years. Then came a prince, kissed her and maybe she woke up. If she did not wake up, he returned again after one year and kissed her again. Create a table in Excel

19												
	A	B	C	D	E	F	G	H	I	J	K	
1	x	y	z		s		A					
2	3	4	6		6.5		5.3327					
3	2	3	4		4.5		2.9047					
4	7	4	7		9		13.416					
5	8	6	9		11.5		23.525					
6	4	5	8		8.5		8.1815					
7	4	4	7		7.5		6.7777					
8	7	3	9		9.5		8.7856					
9	6	1	10		8.5		#NUM!		Incorrect input of triangle sides			
10	2	9	11		11		0		Incorrect input of triangle sides			

Figure 4.14: Incorrect input of triangle sides.

that counts the years of Sleeping Beauty’s sleep. Assume that there is a 0.5 probability that the Sleeping Beauty wakes up after a kiss.

Hints: We adjust the flowchart slightly so that we use a variable Y to denote the number of years asleep. We use the function =RANDBETWEEN(0;1) to generate randomness in such a way that both options are equally probable. Depending on this chance, we decide whether the Sleeping Beauty wakes up or not, i.e. whether we should add 1 to Y or end the algorithm. Next we verify if the algorithm works as expected for the inputs we enter. We can generate various random inputs by repeated use of the F9 key.

Solution: As usual, we should consider how to create the solution in such a way that it is comprehensible even if we return to it after some time. An example layout is shown in Figure 4.15. In addition to the command RANDBETWEEN(0;1), we used the commands IF(B2=0;"yes";"no"), IF(C2="no";"";"end") and IF(D2="end"; A2; ""). It is also possible to improve the solution in such a way that after the first ending, the algorithm does not re-start. This can be done in the next column of the table.

Unequal probabilities

How would we modify the formula for chance, if the Sleeping Beauty woke up after a kiss with probability 1/10 and kept on sleeping with probability 9/10?

B7					
		✕	✓	<i>f</i> x	=RANDBETWEEN(0,1)
	A	B	C	D	E
1	Y	chance	woke up	end	Years of sleep
2	100	1	no		
3	101	1	no		
4	102	1	no		
5	103	1	no		
6	104	0	yes	end	104
7	100	0	yes	end	100
8	100	0	yes	end	100
9	100	0	yes	end	100
10	100	0	yes	end	100
11	100	1	no		
12	101	0	yes	end	101
13	100	0	yes	end	100
14	100	0	yes	end	100
15	100	0	yes	end	100

Figure 4.15: Sleeping Beauty

Revisiting Wall-e

Draw a flowchart for the Wall-e problem from Section 1.4.

2. Tool for checking numeric games

Create an Excel checker for number games Latin square and Magic square which checks if the square is correct. For the Magic square it is sufficient to consider only rows and columns (not the diagonals).

Hints: We enter (or randomly generate) the numbers that represent the problem solution that needs to be checked. We use auxiliary rows and columns on the perimeter of the table to verify the sums according to the problem. We indicate the correct and the incorrect rows, columns and whole table. What can be checked? How should we highlight incorrect data?

a) Latin squares 3×3 , 4×4

Solution: See Figure 4.16 for an example of a simple check for a Latin square. The user enters values into the yellow cells. The OKs for rows and columns are determined using the formula =IF(AND (COUNTIF (B2:D2,1)=1, COUNTIF (B2:D2,2)=1, COUNTIF (B2:D2,3)=1), "OK", "NOT OK").

F6						
	A	B	C	D	E	F
1						
2		1	2	3	OK	
3		2	3	1	OK	
4		3	1	2	OK	
5		OK	OK	OK		
6						OK

Figure 4.16: Check for a Latin square.

b) Magic squares 3×3 , 4×4

Hint: See Figure 4.17 for an attempt at a check for a Magic square. The user enters values into the yellow cells. The numbers in the green cells are determined by the computer. Is this check sufficient? Will it work also for larger values of n ? How can it be done differently?

E5									
	A	B	C	D	E	F	G	H	I
1	4	9	2	OK					
2	3	5	7	OK					
3	8	1	6	OK					
4	OK	OK	OK		Result	Calculations:	n	sum of row	
5					OK		3	15	

Figure 4.17: Check for a Magic square.

3. Tool for solving logical problems

Princess Porcia has a golden, a silver and a bronze box. To select a husband, she places a portrait of herself in one of the boxes. Two boxes remain empty. She puts a sign on each box, but the signs do not have to be true. The future husband tries to find the box with the portrait.

In Excel, create a tool for solving specific logical problems about hiding things in boxes. This tool should conveniently display the question and test where the portrait is located. The Excel sheet should automatically evaluate whether the statements are true or false and determine whether the placement of the portrait fits the problem.

Hints: After writing the symbols into boxes, Excel translates the symbols into numbers and verifies whether the conditions are met. Using a suitable logical operation we can count the number of true statements and check that it is as it should be.

Generalisation for other rules

The algorithm can evaluate the problem according to the rule written in B3 and thus it can solve multiple problems at once. Can you write the (more complicated) condition for B7 for such case?

- a) At most one sign is true.
Golden: *The portrait is in this box.*
Silver: *The portrait is not in this box.*
Bronze: *The portrait is not in the golden box.*

Solution: It is useful to design the table header in a clear format so that it helps to perform the needed evaluations. We fill the second and third row according to the problem, see Figure 4.18. The user enters values into the white cells. The numbers in the green cells are generated by the computer. We use the fourth line to place the portrait into one of the boxes. Excel evaluates the statements in the fifth line and makes a control checksum. If the checksum is larger than one, it writes out *"This placement is incorrect."*, otherwise it writes out *"This placement is OK."*

B7							
	A	B	C	D	E	F	G
1		Golden	Silver	Bronze			
2	sentences	The portrait is in this box	The portrait is not in this box	The portrait is not in the golden box			
3	rule	at most one sentence true					
4	content	empty	empty	portrait	auxiliary calculation		
5	processing content	0	1	1	2		
6							
7	result	This placement is incorrect.					

Figure 4.18: Checker of statements.

Inverse problem

Is it possible to work in reverse, i.e. input the truth values of individual statements and the worksheet determines the contents of the boxes?

Chapter 5

Number sets

5.1 Lecture overview

The number sets did not arise all at once. They appeared historically as they were needed for practical, technical and scientific progress. Later, mathematicians defined them precisely and investigated their, often very abstract, properties. However, many of them are important for understanding practical problems appearing in computer science.

5.1.1 Natural numbers $\mathbb{N}$

The natural numbers are the simplest and basic number set. While this statement is true when we think about them from the position of the science of mathematics today, in the history of people, science and mathematics, their discovery was significant and not at all obvious.

Natural numbers defined using sets

In the set theory, we define the natural numbers in such a way that we model the first one as an empty set $\emptyset$. Every next number is created as a set containing all previous sets, e.g. $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$ etc. These sets are so special that using the set operations, it is possible to define addition of natural numbers. Using the addition it is then possible to define multiplication. This approach is only feasible when examining the inherent properties of natural numbers and theoretical foundations of mathematics.

Peano axioms

A more convenient approach for examining the natural numbers and their arithmetics are the *Peano axioms*. We call every set which has the following properties a set of *natural numbers*:

- Natural numbers $\mathbb{N}$ contain a special element, which we denote O . This essentially says that the set $\mathbb{N}$ is nonempty. The notation suggests that this special element is 0, which may be true but it is not necessary.
- With every element x , the set $\mathbb{N}$ also contains the element $S(x)$, its successor. As the following axioms show, the successor corresponds to the natural number following immediately after x , i.e. to the number $x + 1$.
- O is not a successor of any element in $\mathbb{N}$. Therefore, O is the first element of the set $\mathbb{N}$. Using this property, we can also show that $S(O) \neq O$, $S(S(O)) \neq S(O)$ etc. $\mathbb{N}$ is thus an ever growing set that continues to infinity.
- If $x \neq y$, then $S(x) \neq S(y)$. (We can also use the contrapositive $S(x) = S(y)$ implies $x = y$.) Thus it cannot happen that two different elements have the same successor. Together with the following axiom, this property lets us show that there is no other element between O and $S(O)$, or between x and $S(x)$ in general.
- Axiom of induction is a different beast. It says that if O has some property V and for each x in $\mathbb{N}$ we can prove (the implication) that if x has the property V , then also its successor $S(x)$ has this property, then all natural numbers have the property V . This axiom enables proofs by mathematical induction. The principle is simple and can be thought of as a *domino principle*. If we knock down one domino and we ensure that every domino knocks down the one that comes after it, then we know that all dominoes are going to fall, even if there are infinitely many of them. It also ensures that the natural numbers form one *line*, which can be traversed from the origin O by sequential addition of *one*.

Arithmetics and ordering

The arithmetics and ordering of natural numbers are described by the following properties, where O corresponds to the number 0:

- $x + 0 = x$

- $x + S(y) = S(x + y)$
- $x \cdot 0 = 0$
- $x \cdot S(y) = x \cdot y + x$
- $x \leq y$ iff there exists z such that $x + z = y$. (It can be shown that $\leq$ is a linear ordering, see Section 7.1.1.)

If we are content with a simple, not completely precise, definition we can say that natural numbers are those which can be used to denote the number of elements of sets. Even though the properties that we mentioned consider 0 to be the first natural number, other areas of mathematics typically say that the smallest natural number is 1. We also adopt this convention, therefore in the following, we will have $\mathbb{N} = \{1, 2, 3, 4, \dots\}$. In case we want to include also zero, we are going to use the notation $\mathbb{N}_0 = \{0, 1, 2, 3, 4, \dots\}$.

Base systems

If we want to use a number system, besides the definition, we also need some means to write the numbers down. Finding a suitable and generally accepted notation for natural numbers took thousands of years. From sticks, stones, knots or notches people transitioned to fingers and then to written forms and various counting tools. Even the Egyptian hieroglyphs, Sumerian cuneiforms or Roman numerals were not the definitive answers. After those, people adopted the Hindu, i.e. Arab numbers. (The Arabs adopted them from the Indians and the Europeans from the Arabs).

The various system brought with them also various bases. The base systems, such as decimal, binary or hexadecimal, all work on the same principle. They use n different characters, mostly $0, 1, 2, \dots, n - 1$, which also represent the amount $0, 1, 2, \dots, n - 1$. If there is not enough digits, e.g. in the hexadecimal system, some other symbols are included, e.g. a, b, c, d, e, f . The individual digits and their position in the written number denote the power to which the base n is raised. So for the decimal system, they represent ones, tens, hundreds, thousands, etc. For binary they represent ones, twos, fours, eights etc. If we also use the *decimal point*, we write the reciprocals after it, i.e. tenths, hundredths, etc. or halves, quarters, eighths, etc.

Problem 5.1 Binary to decimal:

Convert 100010 from binary to decimal.

Solution:

$$100010_2 = 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 32 + 2 = 34_{10}$$

Problem 5.2 Decimal to binary:

Convert 34 from decimal to binary.

Solution:

$$\begin{array}{rcl} 34 : 2 & = & 17 \text{ rem } 0 \\ 17 : 2 & = & 8 \text{ rem } 1 \\ 8 : 2 & = & 4 \text{ rem } 0 \\ 4 : 2 & = & 2 \text{ rem } 0 \\ 2 : 2 & = & 1 \text{ rem } 0 \\ 1 : 2 & = & 0 \text{ rem } 1 \end{array}$$

The numbers in the binary system are the remainders after division ordered from the end, i.e. $34_{10} = 100010_2$.

Problem 5.3 Decimal to hexadecimal:

Convert 43 from decimal to hexadecimal.

Solution:

$$\begin{array}{rcl} 43 : 16 & = & 2 \text{ rem } 11 \\ 2 : 16 & = & 0 \text{ rem } 2 \end{array}$$

The numbers in the hexadecimal system are the remainders after division ordered from the end, i.e. $43_{10} = 2b_{16} = 0x2b$, where the digit b represents 11 and $0x$ is used to denote a hexadecimal number.

Problem 5.4 Binary to hexadecimal:

Convert 100010_2 from binary to hexadecimal.

Solution: $100010_2 = 0010|0010_2 = 0x22$. Here we used the fact that $16 = 2^4$ and thus all we need is to split the binary number into groups of four and replace each of them with the corresponding hexa-digit.

5.1.2 Integers $\mathbb{Z}$

The natural numbers are a great invention but they are not perfect. As soon as we want to subtract a larger number from a smaller one, we encounter a problem. Many years ago, the merchants keeping track of their gains and losses and of course mathematicians in their calculations encountered this problem. Formally, it is not difficult to create integers from $\mathbb{N}_0$. To every number (besides 0) we add once a character $+$ and once a character $-$. We thus create a set $\mathbb{Z} = \{0, +1, -1, +2, -2, +3, -3, \dots\}$ and define the operations using the operations we had with natural numbers, e.g. $(-, 7) + (-, 4)$ is $(-, 7+4)$, or $(-, 3) \cdot (+, 4) = (-, 3 \cdot 4)$. We extend the ordering in a similar fashion: positive ($+$) numbers are ordered in the same way as natural numbers, negative ($-$) numbers are in reverse order and each negative number is smaller than each positive number.

The natural numbers $\mathbb{N}$ are part of integers. They correspond to the $+$ part. Compared to the natural numbers, integers are *infinite in both directions*. We use the notation $\mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Z}_0^+, \mathbb{Z}_0^-$.

5.1.3 Rational numbers $\mathbb{Q}$

The integers solved the problem with subtraction, but not with division of numbers. Various economic, geodesic and technical calculations required working with parts or fractions of integers. This work was hindered for a long time by the lack of a suitable notation. Egyptian root fractions and Babylonian fractions with base 60 were impractical and complicated the development of mathematics until the 16th century. That was the time when the decimal numbers and fractions finally arrived in Europe.

Fractions as ordered pairs

Formally, we can define the rational numbers as ordered pairs (z, n) that belong to the Cartesian product $\mathbb{Z} \times \mathbb{N}$, which can be thought of as *fractions* z/n . We define the operations as we are used to from working with fractions, e.g. $(z, n) + (u, m) = (z \cdot m + u \cdot n, n \cdot m)$. Note that if we define fractions in this way, we have too many of them. Many ordered pairs (fractions) $(2, 3)$, $(4, 6)$, $(-20, -30)$, etc. denote the same rational number. Therefore, we can say that if $n \cdot u = m \cdot z$ then we consider the numbers (z, n) and (u, m) equivalent and they merely represent a different notation for the same number.

Periodic decimal numbers as fractions

The integers are part of rational numbers and correspond to the pairs $(z, 1)$. If we write the rational numbers in the base 10 system, we use a *decimal point* and the digits to the right of it represent tenths, hundredths, etc. The decimal part of a rational number is either finite or infinitely periodic. The decimal numbers with infinitely periodic part can be written as fractions using the following method:

$$\begin{aligned} 0, \bar{7} &= x \\ 7, \bar{7} &= 10x \\ 7 + 0, \bar{7} = 7 + x &= 10x \\ 7 &= 9x \\ x &= 7/9 \end{aligned}$$

Ordering and being dense

The ordering of rational numbers remains linear but compared to integers they have a new property. They are *dense*: between every two rational numbers r, q , there is another rational number, e.g. their average (or middle point) $(r + q)/2$. Consequently, between every two rational numbers, there are infinitely many more rational numbers. Similarly to integers, we use the notation $\mathbb{Q}^+, \mathbb{Q}^-, \mathbb{Q}_0^+, \mathbb{Q}_0^-$.

Representation in computers

When representing natural numbers and integers in computers, we do not encounter many surprises. Of course, due to hardware limitations, it is necessary to limit their minimal and maximal values. Issues then arise when the results of some operations *overflow* the ranges of represented values, so different approaches have to be taken when computing with really large numbers with millions of digits, e.g. when searching for large primes and factorisations.

When working with rational numbers, we encounter also issues pertaining to *accuracy* – infinite decimal numbers do not fit into the computer memory. Moreover, since they are saved in binary, even seemingly simple numbers such as 0.1 have infinite binary expansions and calculation with such numbers leads to unpleasant and sometimes downright dangerous numerical inaccuracies. Sometimes the solution is to teach the algorithms to work with fractions.

It is not correct to say that for computers, engineering and science it is sufficient to only know the rational numbers to the extent to which they are restricted by hardware. If we restricted our knowledge in this way, the numerical errors could lead to collapse of buildings, traffic accidents, chaos in banks, etc. The knowledge of number sets leads us to solutions that eliminate these errors.

5.1.4 Real numbers $\mathbb{R}$

Irrational numbers

The path from rational numbers to real numbers was long and winding and ended only in the 19th century. The ancient Greeks knew that if the side of a square was a fraction, the diagonal was not a fraction. The reason is that the *square root* of 2 is *irrational*. However, if we only wanted to extend the rational numbers by all the numbers that can be created using a ruler and a compass, it would be sufficient to iteratively add just square roots of all already constructed numbers.

The ancient Greeks knew about other numbers, e.g. the golden ratio ϕ and more importantly π , whose significance was primarily geometric. Another area that needed extension of rational numbers was algebra. Solving polynomial equations with degree 2, 3, 4 and more required existence of square, cube, fourth etc. roots and expressions containing them. Today we call these numbers (roots of polynomials) *algebraic*. They represent a significant but still not sufficient extension of the rational numbers. Later, more and more irrational numbers appeared: a special singleton is e , large groups were values of the sine, cosine and logarithmic functions. What else are we going to need?

Dedekind cuts

The definitive answer to the previous question required quite advanced mathematics. One formulation was given by Richard Dedekind. If we restrict ourselves to the positive rational numbers q , we can consider the open intervals $(0, q)$ instead of them. These intervals have simple properties: 1. with every rational number, they also contain all smaller rational numbers and 2. we can define certain *arithmetics* on them. If $(0, q) + (0, r)$ is an interval containing all sums of any number from the first interval with any number from the second interval, it corresponds to the interval $(0, q + r)$.

However, there exist other sets that have these properties but do not correspond to any interval ending with a rational number. One of them is for

example the set of all positive rational numbers for which $q^2 < 2$ holds. Upper end-point of the corresponding interval suggests the irrational number $\sqrt{2}$. If we take all sets of positive rational numbers that have properties 1 and 2, their upper end-points correspond to all positive *real numbers*.

Limits of sequences

Another path to real numbers leads through sequences of rational numbers. If we write a sequence, such as $1/2, 1/3, 1/4, 1/5, \dots$ we often notice that it *tends* to some number, in this case to zero. *Tending to some number* has been precisely defined at the beginning of the 19th century and is called a *limit*. Considering special properties of sequences, see Section 8.1.3, it is possible to determine that a sequence must have a limit even if we do not know its value (yet). And here was the problem. For some sequences, it was possible to prove that the limit exists but that it is not a rational number. If we extend the rational numbers by all limits of such sequences, we again obtain real numbers.

Both methods of extending the rational numbers are very effective. If we repeated either of them one more time, there would be no additional new numbers. So both of these constructions extend the rational numbers by all irrational numbers and we thus obtain all real numbers.

Geometric notion

Compared to the numbers we had before, there are many more real numbers. It manifests in such a way that if we place them all on a line, we cover all its points. The rational numbers did not do that. Even though they were dense and there were infinitely many of them, they left out even more points on the line. In a geometry based on *rational line segments*, the diagonals of a square would not intersect. This cannot happen with real numbers. This is the reason why we can say that every polynomial of odd degree must have at least one real root, because its graph has to intersect the x -axis.

We again use the notation $\mathbb{R}^+, \mathbb{R}^-, \mathbb{R}_0^+, \mathbb{R}_0^-$ for positive, negative, non-positive and non-negative real numbers.

Approximations

We can approximate real numbers using fractions with various degrees of precision. It can be shown that of all fractions with single-digit denominator, $22/7$ is the best approximation of π . If we allow a three-digit denominator, the best

approximation (to six decimal places) is $355/113$. The next better fraction is $103993/33102$. Using $355/113$ is thus very effective. We can arrive at these estimates also by using the *continued fractions* of the form:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}.$$

We can expand both rational numbers (their continued fractions are finite) and irrational numbers (their fractions are infinite). If we only use the finite part of the infinite fractions, we obtain the efficient approximations.

5.1.5 Complex numbers $\mathbb{C}$

Imaginary unit and algebraic form

The complex numbers turned out to be inevitable at the beginning of the 16th century. Italian mathematician Cardano discovered formulas for calculation of roots of cubic equations and showed that to express their values, he had to work with square roots of negative numbers. In these calculations, it turned out that it is sufficient to use the square root of -1 , which was later denoted as i by Euler, and we call it an *imaginary unit*. For this number, we have a property never heard of before: $i^2 = -1$. Every complex number z now can be expressed using i and two real numbers a, b as $a + bi$. We call this an *algebraic form* of a complex number and we clearly see its *real and imaginary parts*. As expected, it was possible to solve the polynomial equations with these numbers. All of the equations now had roots.

Geometric representation

If we want to represent a complex number geometrically, we treat the x and y axes as real and imaginary. Every complex number $z = a + bi$ thus gets its place in the *complex plane*, see Figure 5.1. The place is determined by the coordinates a on the real and b on the imaginary axis. The addition of complex numbers corresponds to addition of vectors in the plane. The relation $i^2 = -1$ implies a different meaning of the multiplication (compared to real numbers). Multiplication by a real number corresponds to prolongation/shortening of the vector that represents the complex number. Multiplication by a complex number corresponds to both prolongation/shortening and rotation. The solutions of equation $z^n = 1$ correspond to vertices of regular n -sided polygons in a plane.

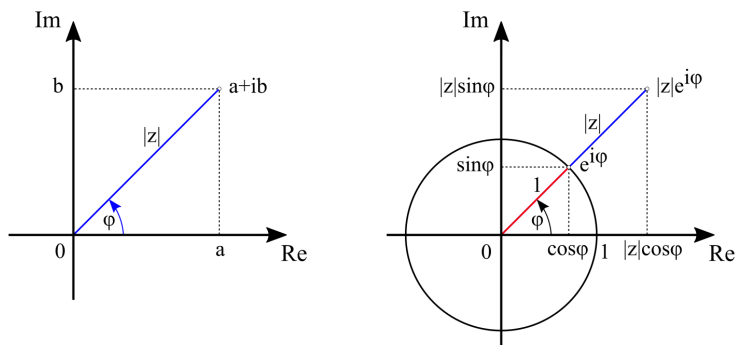


Figure 5.1: Complex numbers.

Complex conjugates

Sometimes it is useful to consider the complex number $\bar{z} = a - bi$ together with $z = a + bi$. The number $\bar{z}$ is called *complex conjugate* to z and it is its mirror image with respect to the x -axis, see Figure 5.2. For the product of a complex number and its conjugate we have $z \cdot \bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2$.

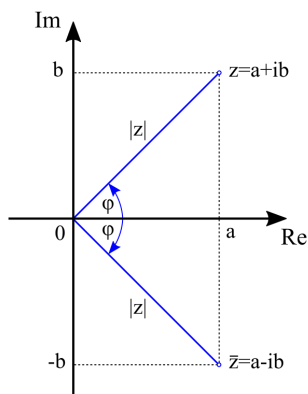


Figure 5.2: Complex conjugate numbers.

Trigonometric form

To understand the multiplication properties of complex numbers (other than $z = 0$), it is useful to write them in *trigonometric form*. It specifies the complex number by its distance r from origin $0+0i$ and by its angle φ (as measured from the positive real axis) at which we should *walk* this distance. The distance r is called the *modulus* of the number z (similar to the real axis, where the absolute value represents the distance from 0) and is denoted by $|z|$. The angle φ is also called an *argument* of the complex number. Using the unit circle, we see that a complex number z can be written as $z = a + bi = |z| \cos \varphi + i|z| \sin \varphi = |z|(\cos \varphi + i \sin \varphi)$. The last expression is the trigonometric form.

Using the Pythagorean theorem and the algebraic form $z = a + bi$, we get $|z| = \sqrt{a^2 + b^2}$ and determine φ from $\cos \varphi = \frac{a}{\sqrt{a^2 + b^2}}$ or $\sin \varphi = \frac{b}{\sqrt{a^2 + b^2}}$.

If we write two complex numbers in trigonometric form $u = |u|(\cos \varphi + i \sin \varphi)$ and $v = |v|(\cos \psi + i \sin \psi)$, their product is

$$u \cdot v = |u||v|(\cos \varphi \cdot \cos \psi - \sin \varphi \cdot \sin \psi + i(\cos \varphi \cdot \sin \psi + \sin \varphi \cdot \cos \psi)).$$

Using the trigonometric formulas, we can simplify this expression to

$$u \cdot v = |u||v|(\cos(\varphi + \psi) + i \sin(\varphi + \psi)).$$

Thus, when multiplying two complex numbers, we multiply their distances from $0+0i$ and add their angles (arguments). This allows relatively easy calculations with complex numbers and a geometrical use. The roots of the equation $z^n = 1$ represent vertices of a regular polygon. The first use of this property was the discovery of young Gauss, who determined which regular polygons with n vertices can and which cannot be constructed using a ruler and a compass.

Problem 5.5 Division of complex numbers:

Calculate $(3 + 4i) : (6 - 12i)$.

Solution: We talked about multiplication, but how do we divide complex numbers? The first thing we can try is to use what we know about multiplication. If we assume that the result of this calculation is again a complex number $a + bi$, then we can rewrite the problem as

$$3 + 4i = (6 - 12i) \cdot (a + bi)$$

and multiply everything out:

$$3 + 4i = 6a - 12ai + 6bi - 12bi^2 = (6a + 12b) + (6b - 12a)i.$$

This results in a system of two equations with two real unknowns:

$$\begin{aligned} 3 &= 6a + 12b \\ 4 &= 6b - 12a \end{aligned}$$

The unique solution of this system is $a = -1/6$ and $b = 1/3$ and thus

$$(3 + 4i) : (6 - 12i) = -1/6 + i/3.$$

An alternative way of solving this problem is to use the complex conjugate of the *denominator* and perform the following calculation which makes the denominator a real number:

$$\begin{aligned} \frac{3 + 4i}{6 - 12i} &= \frac{3 + 4i}{6 - 12i} \cdot \frac{6 + 12i}{6 + 12i} = \frac{18 + 24i + 36i + 48i^2}{36 - 144i^2} = \frac{18 + 60i - 48}{36 + 144} \\ &= \frac{-30 + 60i}{180} = -1/6 + i/3. \end{aligned}$$

Exponential form

There is one more form of a complex number that is frequently used, the *exponential form*. It results from the relation $e^{i\varphi} = \cos \varphi + i \sin \varphi$. It can be derived from functional series and is useful in many applications. A complex number $z = |z|(\cos \varphi + i \sin \psi)$ can thus be written as $z = |z|e^{i\varphi}$ and the product of two complex numbers is then $u \cdot v = |u||v|e^{i(\varphi+\psi)}$.

Ordering

We pay a price for the many useful properties of complex numbers. This is the first number set that cannot be linearly ordered, i.e. we cannot place the numbers on a suitable axis. We could order the complex numbers *alphabetically*, but then the binary operations as we are used to them would not work any more. This is one of the reasons why we display complex numbers in a plane. And even though we do not have linear ordering, it is useful to compare their magnitudes: $|z_1| < |z_2|$.

5.2 Questions

1. In which number field is mathematical induction defined and what does this concept mean?
2. Decide which of the operations (addition, subtraction, multiplication and division) may be performed without restrictions in the following number fields:
 - a) Natural numbers
 - b) Rational numbers
 - c) Real numbers
 - d) Irrational numbers
 - e) Complex numbers
3. Why is it not sufficient to use integers for mathematical calculations? Give an example of a problem that cannot be solved in integers.
4. Write as a fraction
 - a) $0.\overline{6}$
 - b) $0.\overline{9}$
 - c) $0.\overline{42}$
5. Write at least two
 - a) rational numbers between $0.\overline{7}$ and $0.\overline{8}$
 - b) rational numbers between $3/11$ and $4/12$
 - c) irrational numbers between $\sqrt{15}$ and $\sqrt{17}$
 - d) irrational numbers between 15 and 17
6. When we add the number 0.01 one thousand times in Excel, we will not get exactly 10. Why? For which numbers will this multiplication work?
7. Is there a rational number between every two rational numbers?
8. Is there an irrational number between every two irrational numbers?
9. For which natural numbers is it true that twice their square is also a square (a number raised to a second power)? Prove your answer.

10. Why is it not sufficient to use rational numbers for mathematical calculations? Give an example of a problem that cannot be solved in rational numbers.
11. Explain how to draw a square root of $2, 3, 4, 5, 6, 7, \dots$
12. Write at least 3 decimal numbers that differ from π by at most 0.002.
13. Describe a mathematical construction that can build all real numbers from rational numbers.
14. Why is it not sufficient to use real numbers for mathematical calculations? Give an example of a problem that cannot be solved in real numbers.
15. Write a quadratic equations that does not have a solution in real numbers.
16. Write a complex number with real part 1 and imaginary part 2 in
 - a) algebraic form
 - b) trigonometric form
 - c) exponential form
17. Express 360 degrees in radians. Express $\pi/2$ in degrees. What is the general formula for conversion of degrees to radians? What is the formula for conversion of radians to degrees?
18. Draw in complex plane $z_1 = 3 + 5i$, $z_2 = 5(\cos 45^\circ + i \sin 45^\circ)$, $z_3 = e^{i60^\circ}$.
19. Draw the sum of these two numbers in the complex plane: $z_1 = 3 + 5i$ and $z_2 = 2 + 3i$.
20. Draw the product of these two numbers in a complex plane $z_1 = 3e^{i60^\circ}$ and $z_2 = 3e^{i90^\circ}$.
21. Which of the complex numbers $z_1 = 3 + 5i$ and $z_2 = 2 + 3i$ is farther from $0+0i$?
22. A number in binary system is 11001101. Write the same number in decimal and in hexadecimal system.

5.3 Problem set

1. Dividing by complex numbers

Calculate

a) $\frac{3+2i}{2-3i}$

b) $\frac{(3+2i)(4-3i)}{(2-3i)}$

c) $\frac{(7-i)(-2+4i)}{(1+4i)}$

d) $\frac{(-2+5i)(2+5i)}{(5+2i)}$

e) $\frac{(6i)(5+2i)}{(-4-3i)}$

2. Complex roots

Find one complex root of the equation and verify it by plugging it into the equation:

a) $x^2 + 3x + 4 = 0$

b) $x^2 - 2x + 3 = 0$

c) $-x^2 + x - 2 = 0$

3. Trigonometric form

Write the complex numbers in trigonometric form:

a) $2 - 3i$

b) $-2i$

c) $-1 + 5i$

d) $-3 + i$

e) -4

4. Algebraic form

Write the complex numbers in algebraic form:

a) $2 \cdot (\cos 45^\circ + i \cdot \sin 45^\circ)$

b) $\frac{1}{3} \cdot (\cos 60^\circ + i \cdot \sin 60^\circ)$

c) $(-1) \cdot (\cos(-30^\circ) + i \cdot \sin(-30^\circ))$

5. Rational numbers as fractions

Write these numbers as fractions:

- a) $0.\overline{7}$
- b) $0.2\overline{7}$
- c) $0.36\overline{9}$
- d) $0.\overline{123}$
- e) $0.53\overline{42}$

6. **Continued fractions**

Write these numbers as chain (continued) fractions:

- a) 3.456
- b) $15/7$
- c) $\sqrt{2}$

7. **Conversion to decimal system**

Convert the following numbers to decimal form:

- a) From binary: 110110; 10101
- b) From ternary: 120120; 22222
- c) From hexadecimal: 0x3d39; 0xd431

8. **Conversion from decimal system**

Convert the numbers 123 and 321 from decimal to:

- a) Binary form
- b) Base-8 form
- c) Hexadecimal form

Add the converted numbers.

5.3.1 Hints and solutions

1. Dividing by complex numbers

Calculate

$$\text{b) } \frac{(3+2i)(4-3i)}{(2-3i)}$$

$$\text{c) } \frac{(7-i)(-2+4i)}{(1+4i)}$$

Hints:

In these problems, to calculate means to convert the expressions into a complex number in the algebraic form $z = a+bi$ so that the imaginary unit i does not appear in the denominator. The numerators can be multiplied out and the corresponding terms collected. To remove the imaginary part from the denominator of a fraction, we can multiply both the numerator and denominator of a fraction by a complex conjugate of the denominator.

Solution:

b)

$$\begin{aligned} & \frac{(3+2i)(4-3i)}{(2-3i)} \cdot \frac{(2+3i)}{(2+3i)} = \frac{(12-9i+8i-6i^2)(2+3i)}{(4+6i-6i-9i^2)} = \\ & = \frac{(18-i)(2+3i)}{13} = \frac{36+54i-2i-3i^2}{13} = \frac{39+52i}{13} = 3+4i \end{aligned}$$

c)

$$\begin{aligned} & \frac{(6i)(5+2i)}{(-4-3i)} \cdot \frac{(-4+3i)}{(-4+3i)} = \frac{(30i+12i^2) \cdot (-4+3i)}{16-12i+12i-9i^2} = \\ & = \frac{-120i+90i^2+48-36i}{25} = \frac{-42-156i}{25} = -\frac{42}{25} - \frac{156i}{25} \end{aligned}$$

2. Complex roots

Find one complex root of the equation and verify it by plugging it into the equation:

$$\text{a) } x^2 + 3x + 4 = 0$$

Hints:

- To calculate the roots of the quadratic equation $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$, we can use the discriminant $D = b^2 - 4ac$. The roots can be expressed as $x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$.
- What to do when the discriminant is negative? In such case, the equation does not have a real root. But now we know that the number -1 can be expressed as i^2 which lets us evaluate square roots of negative numbers, e.g. $\sqrt{-10}$ as $\sqrt{10i^2} = i\sqrt{10}$.
- To verify the result, we can plug the roots back into the original equation and check that the left-hand side equals the right-hand side.

Solution:

a) $x^2 + 3x + 4 = 0$

$$D = b^2 - 4ac = 3^2 - 4 \cdot 1 \cdot 4 = -7 = 7i^2$$

$$x_{1,2} = \frac{-b \pm \sqrt{D}}{2a} = \frac{-3 \pm \sqrt{7i^2}}{2 \cdot 1} = \frac{-3 \pm i\sqrt{7}}{2}$$

Plugging the root x_1 into the equation:

$$\begin{aligned} x_1^2 + 3x_1 + 4 &= \left(\frac{-3 + i\sqrt{7}}{2} \right)^2 + 3 \cdot \left(\frac{-3 + i\sqrt{7}}{2} \right) + 4 = \\ &= \frac{9 - 6i\sqrt{7} - 7}{4} - \frac{9 - 3i\sqrt{7}}{2} + 4 = \frac{2 - 6i\sqrt{7} - 18 + 6i\sqrt{7} + 16}{4} = 0 \end{aligned}$$

Plugging the root x_2 into the equation:

$$\begin{aligned} x_2^2 + 3x_2 + 4 &= \left(\frac{-3 - i\sqrt{7}}{2} \right)^2 + 3 \cdot \left(\frac{-3 - i\sqrt{7}}{2} \right) + 4 = \\ &= \frac{9 + 6i\sqrt{7} - 7}{4} - \frac{9 + 3i\sqrt{7}}{2} + 4 = \frac{2 + 6i\sqrt{7} - 18 - 6i\sqrt{7} + 16}{4} = 0 \end{aligned}$$

3. Trigonometric form

Write the complex numbers in trigonometric form:

a) $2 - 3i$

b) $-2i$

Hints:

- To convert a complex number to its trigonometric form we need to find out its modulus (length) and its argument (angle) φ .
- It is helpful to draw the complex number in the complex plane and look at the angle φ we are trying to find. The angle is taken in the counter-clockwise direction from the positive x -axis. If we go in the clockwise direction, the angle is negative.
- The length of the complex number can be calculated using a Pythagorean theorem and the angle using a suitable trigonometric function.

Solution:

- a) We start by drawing the complex number $2 - 3i$ in a complex plane.

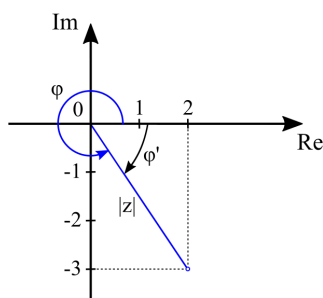


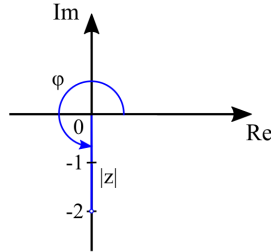
Figure 5.3: Complex number $2 - 3i$.

In Figure 5.3, we see that the angle φ can be calculated as $360^\circ - \varphi'$ and that $\tan \varphi' = \left(-\frac{3}{2}\right) \Rightarrow \varphi' = \arctan\left(-\frac{3}{2}\right) \approx -56.3^\circ$. Note that this is not an exact value. While for some angles it is possible to obtain the answer as an expression containing the irrational number π , here we give a rational approximation. Then $\varphi \approx 360^\circ - 56.3^\circ = 303.7^\circ$.

We can calculate the size of the complex number using the Pythagorean theorem as $|z| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$.

Now we express the complex number $2 - 3i$ in the trigonometric form:

$$z \approx \sqrt{13} \cdot (\cos 303.7^\circ + i \cdot \sin 303.7^\circ)$$

Figure 5.4: Complex number $-2i$.

- b) The complex number $-2i$ lies directly on the imaginary axis, see Figure 5.4, therefore its angle φ can be determined directly without any calculations as $\varphi = 270^\circ$ or as $\varphi = \frac{3}{2}\pi$ radians. The length of the complex number is $|z| = 2$, therefore the trigonometric form is

$$z = 2 \cdot \left(\cos \frac{3}{2}\pi + i \cdot \sin \frac{3}{2}\pi \right).$$

4. Algebraic form

Write the complex numbers in algebraic form:

- $2 \cdot (\cos 45^\circ + i \cdot \sin 45^\circ)$
- $\frac{1}{3} \cdot (\cos 60^\circ + i \cdot \sin 60^\circ)$
- $(-1) \cdot (\cos(-30^\circ) + i \cdot \sin(-30^\circ))$

Hints:

- For the real and imaginary components of complex numbers $a + bi$ we have: $a = |z| \cdot \cos \varphi$, $b = |z| \cdot \sin \varphi$.
- For some angles, such as 45° or 30° , the values of the trigonometric functions evaluated at them are well known and tabulated.
- For negative angles (going clockwise from the x -axis) we have: $\cos(-\varphi) = \cos \varphi$ and $\sin(-\varphi) = -\sin \varphi$. These relations follow from the facts that $\cos \varphi$ is an even function and $\sin \varphi$ is an odd function.

Solution:

$$\text{a) } 2 \cdot (\cos 45^\circ + i \cdot \sin 45^\circ) = 2 \cdot \left(\frac{\sqrt{2}}{2} + i \cdot \frac{\sqrt{2}}{2} \right) = \sqrt{2} + i\sqrt{2}$$

$$\text{b) } \frac{1}{3} \cdot (\cos 60^\circ + i \cdot \sin 60^\circ) = \frac{1}{3} \cdot \left(\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2} \right) = \frac{1}{6} + i \cdot \frac{\sqrt{3}}{6}$$

$$\text{c) } (-1) \cdot (\cos(-30^\circ) + i \cdot \sin(-30^\circ)) = (-1) \cdot \left(\frac{\sqrt{3}}{2} - i \cdot \frac{1}{2} \right) = -\frac{\sqrt{3}}{2} + i \cdot \frac{1}{2}$$

5. Rational numbers as fractions

Write these numbers as fractions:

$$\text{a) } 0.\bar{7}$$

$$\text{b) } 0.2\bar{7}$$

$$\text{c) } 0.36\bar{9}$$

$$\text{d) } 0.\overline{123}$$

Hints:

- Before we start converting decimal numbers to fractions, we can look at the problem from the other side. What are the decimal numbers for $\frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{9}$? Some of them have a finite number of digits after the decimal point, some of them are infinite. The interesting fact is that the infinite ones are periodic. To turn them back into fractions, we can observe a connection between the fractions that contain digits 9 in their denominators: $\frac{1}{9} = 0.\bar{1}$, $\frac{12}{99} = 0.\bar{12}$, $\frac{123}{999} = 0.\bar{123}$, $\frac{1234}{9999} = 0.\bar{1234}$. How much is $\frac{5}{10} + \frac{1}{10} \cdot \frac{47}{99}$?
- Using our observations, we can start converting the periodic decimal numbers into fractions, e.g. $0.\bar{6} = 6 \cdot 0.\bar{1} = 6 \cdot \frac{1}{9} = \frac{6}{9}$ or $0.0\bar{6} = 0.1 \cdot 0.\bar{6} = \frac{1}{10} \cdot \frac{6}{9} = \frac{6}{90}$. Similarly, we can write any number that contains only a periodic part after the decimal point. The situation is slightly more complicated if there is also a non-periodic part. In such case, it helps to separate the two parts and express them as separate fractions, e.g. $0.4\bar{6} = 0.4 + 0.0\bar{6} = \frac{4}{10} + \frac{6}{90} = \frac{4 \cdot 9 + 6}{90} = \frac{42}{90}$.
- An alternative method for converting a decimal number to a fraction is shown in the lecture. We represent the number as a variable, multiply both sides by a suitable number so that the periodic part of the number can be *separated* from the non-periodic. We substitute

the variable for the periodic part and express it from the equation in the form of a fraction.

- In this problem we can easily verify the results using a calculator.

Solution:

a) Let $x = 0.\overline{7}$, then
 $10x = 7.\overline{7} = 7 + 0.\overline{7}$
 $10x = 7 + x \Rightarrow 9x = 7 \Rightarrow x = \frac{7}{9}$

b) Let $z = 0.2\overline{7}$, then
 $100z = 27.\overline{7} \Rightarrow 100z = 27 + 0.\overline{7}$

Now we use the result of the previous calculation and replace $0.\overline{7}$ with $\frac{7}{9}$.

$$100z = 27 + \frac{7}{9}$$

$$100z = \frac{27 \cdot 9 + 7}{9} \Rightarrow z = \frac{27 \cdot 9 + 7}{9 \cdot 100} = \frac{27}{90}$$

c) Let $x = 0.36\overline{9}$, then
 $100x = 36.\overline{9} = 36 + 0.\overline{9}$

Now we convert $0.\overline{9}$ to a fraction separately.

Let $y = 0.\overline{9}$, then
 $10y = 9.\overline{9} = 9 + 0.\overline{9}$
 $10y = 9 + y \Rightarrow y = \frac{9}{9} = 1$

And then we insert the result into the original equation.

$$100x = 36 + 1 \Rightarrow x = \frac{37}{100}$$

d) Let $x = 0.\overline{123}$, then
 $\Rightarrow 1000x = 123.\overline{123} = 123 + 0.\overline{123} \Rightarrow 1000x = 123 + x \Rightarrow$
 $\Rightarrow 999x = 123 \Rightarrow x = \frac{123}{999}$

6. Continued fractions

Write these numbers as continued (chain) fractions:

a) 3.456

Hints: In the lecture we show the form of the chain fraction in which the term a_1 represents the integer part of the number and the digits after the decimal point are converted to a fraction, e.g. $0.5 = \frac{1}{\frac{1}{0.5}}$. The fraction that appears in the denominator should again be split into the integer part ($a_2, a_3, \dots, a_n$) and the digits after the decimal point. By repeating this algorithm we can express any real number as a chain fraction. For a rational number, the resulting fraction is *finite*, for irrational it is *infinite*.

Solution:

$$\begin{aligned}
 \text{a) } 3.456 &= 3 + 0.456 = 3 + \frac{1}{\frac{1}{0.456}} = 3 + \frac{1}{\frac{1}{2.1929..}} = 3 + \frac{1}{2 + 0.1929..} = \\
 &= 3 + \frac{1}{2 + \frac{1}{\frac{1}{0.1929..}}} = 3 + \frac{1}{2 + \frac{1}{5.1818..}} = 3 + \frac{1}{2 + \frac{1}{5 + 0.1818..}} = \\
 &= 3 + \frac{1}{2 + \frac{1}{5 + \frac{1}{\frac{1}{0.1818..}}}} = 3 + \frac{1}{2 + \frac{1}{5 + \frac{1}{5.5}}} = 3 + \frac{1}{2 + \frac{1}{5 + \frac{1}{5 + \frac{1}{2}}}}
 \end{aligned}$$

7. Conversion to decimal system

Convert the following numbers to decimal form:

- a) From binary: 110110; 10101
- b) From ternary: 120120; 22222
- c) From hexadecimal: 0x3d39; 0xd431

Hints:

- When converting a binary number to decimal, the digits 1 tell us which powers of 2 (ordered from the right) we should add.
- Conversions from other number systems work on the same principle. Every digit tells us how many times should we add the particular power of the base, e.g. $20_3 = 2 \cdot 3^1 + 0 \cdot 3^0 = 6_{10}$.

- If we do not have enough digits, e.g. in the hexadecimal system, we can use letters, e.g. a, b, c, d, e, f for *digits* 10, 11, 12, 13, 14, 15.

Solution:

$$\begin{array}{l} \text{a)} \\ \text{binary number:} \end{array} \left| \begin{array}{c} 2^5 \\ 32 \\ 1 \end{array} \right| \left| \begin{array}{c} 2^4 \\ 16 \\ 1 \end{array} \right| \left| \begin{array}{c} 2^3 \\ 8 \\ 0 \end{array} \right| \left| \begin{array}{c} 2^2 \\ 4 \\ 1 \end{array} \right| \left| \begin{array}{c} 2^1 \\ 2 \\ 1 \end{array} \right| \left| \begin{array}{c} 2^0 \\ 1 \\ 0 \end{array} \right|$$

$\Downarrow$

$$\text{decimal number: } 1 \cdot 32 + 1 \cdot 16 + 0 \cdot 8 + 1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 54_{10}$$

$$\begin{array}{l} \text{b)} \\ \text{ternary number:} \end{array} \left| \begin{array}{c} 3^5 \\ 243 \\ 1 \end{array} \right| \left| \begin{array}{c} 3^4 \\ 81 \\ 2 \end{array} \right| \left| \begin{array}{c} 3^3 \\ 27 \\ 0 \end{array} \right| \left| \begin{array}{c} 3^2 \\ 9 \\ 1 \end{array} \right| \left| \begin{array}{c} 3^1 \\ 3 \\ 2 \end{array} \right| \left| \begin{array}{c} 3^0 \\ 1 \\ 0 \end{array} \right|$$

$\Downarrow$

$$\text{decimal number: } 1 \cdot 243 + 2 \cdot 81 + 0 \cdot 27 + 1 \cdot 9 + 2 \cdot 3 + 0 \cdot 1 = 420_{10}$$

$$\begin{array}{l} \text{c)} \\ \text{hexadecimal number:} \end{array} \left| \begin{array}{c} 16^3 \\ 4096 \\ 3 \end{array} \right| \left| \begin{array}{c} 16^2 \\ 256 \\ d \end{array} \right| \left| \begin{array}{c} 16^1 \\ 16 \\ 3 \end{array} \right| \left| \begin{array}{c} 16^0 \\ 1 \\ 9 \end{array} \right|$$

$\Downarrow$

$$\text{decimal number: } 3 \cdot 4096 + d \cdot 256 + 3 \cdot 16 + 9 \cdot 1 = 15673_{10}$$

8. Conversion from decimal system

Convert the numbers 123 and 321 from decimal to:

- Binary form
- Base-8 form
- Hexadecimal form

Add the converted numbers.

Hints:

- When converting numbers from the decimal system to binary, we are looking for the largest powers of 2 that are comprised in the base-10 number. For example, the largest power of 2 comprised in 18_{10} is $2^4 = 16$. Afterwards, we check what other powers of 2 would fit. Since $18 - 16 = 2$, the powers 2^3 and 2^2 are represented by zeros, 2^1 by a one and 2^0 again by a zero. The conversion can be written as $18_{10} = 1 \cdot 16 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 10010_2$.
- Algorithmically, this method is usually performed as successive division by 2. The remainders after this division taken in reverse order represent the binary number. The same approach works for conversions to other number systems.
- If we already have a binary representation of a number, the conversion to base-8 or hexadecimal is much easier (compared to conversion from decimal system). Consider triplets (quartets) of binary digits and their relation to base-8 (hexadecimal) digits.

Solution:

a) Decimal to binary

$$123 : 2 = 61 \text{ rem } 1$$

$$61 : 2 = 30 \text{ rem } 1$$

$$30 : 2 = 15 \text{ rem } 0$$

$$15 : 2 = 7 \text{ rem } 1$$

$$7 : 2 = 3 \text{ rem } 1$$

$$3 : 2 = 1 \text{ rem } 1$$

$$1 : 2 = 0 \text{ rem } 1$$

$$\Rightarrow 123_{10} = 1111011_2$$

$$\begin{aligned}
321 : 2 &= 160 \text{ rem } \mathbf{1} \\
160 : 2 &= 80 \text{ rem } \mathbf{0} \\
80 : 2 &= 40 \text{ rem } \mathbf{0} \\
40 : 2 &= 20 \text{ rem } \mathbf{0} \\
20 : 2 &= 10 \text{ rem } \mathbf{0} \\
10 : 2 &= 5 \text{ rem } \mathbf{0} \\
5 : 2 &= 2 \text{ rem } \mathbf{1} \\
2 : 2 &= 1 \text{ rem } \mathbf{0} \\
1 : 2 &= 0 \text{ rem } \mathbf{1}
\end{aligned}$$

$$\Rightarrow 321_{10} = 101000001_2$$

$$\text{Addition in binary: } 1111011 + 101000001 = 110111100 = 444_{10}$$

b) Decimal to base-8

$$\begin{aligned}
123 : 8 &= 15 \text{ rem } \mathbf{3} \\
15 : 8 &= 1 \text{ rem } \mathbf{7} \\
7 : 8 &= 0 \text{ rem } \mathbf{7}
\end{aligned}$$

$$\Rightarrow 123_{10} = 773_8$$

c) Decimal to hexadecimal

$$\begin{aligned}
123 : 16 &= 7 \text{ rem } \mathbf{11} \\
7 : 16 &= 0 \text{ rem } \mathbf{7}
\end{aligned}$$

$$\Rightarrow 123_{10} = 0x7b_{16}$$

Alternatively, we can take the binary representation 1111011_2 and split it into quartets: 0111 and 1011. Note that we included an initial 0 in the first quartet, because the number of all digits was not divisible by 4. The first quartet 0111 represents the digit 7 in hexadecimal (and in decimal). The second quartet 1011 represents 11 in decimal and thus b in hexadecimal. In this method, it is sufficient to have a table for conversion of the 16 hexadecimal digits.

5.4 Lab problems

1. Complex numbers

- Find the commands to write and compute with complex numbers, to perform arithmetic operations with them and to convert between their arithmetic and trigonometric forms. Draw the numbers in a chart.
- Create a complex number with modulus 1 and angle $2\pi/24$. Calculate its powers and draw them on a suitable chart.
- What happens when we change the modulus slightly to 0.9 or 1.1?

2. Approximation of real numbers by fractions

Create a sheet with a single input – an irrational number r .

- Look for the most accurate approximation of this number using a fraction of the type x/n , where n can be between 1 and 1000, inclusive.
- For each fraction, determine the error of approximation.
- Indicate the rows, where the error has decreased. These correspond to the new best approximation.
- It is easy to approximate with a fraction with large denominator. Good estimate will have a small error and the smallest possible denominator.
- Find a better estimate for π than $355/113$.

3. Approximation of real numbers by continued fractions

Continued fraction is a fraction of this form:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}$$

Create a sheet that takes as input an irrational number (square root, cube root, π , ...).

- The output will be the coefficients which correspond to the continued fraction.
- Compare these fractions with the best approximations from the previous problem.

4. Conversions between number systems

- Create a simple sheet which converts a binary number to a decimal number. Use the powers of 2. Include the digits after the *decimal/binary point*.
- Create a sheet that converts a number from decimal system to binary. How do we do this for numbers that have also digits after the decimal/binary point?
- Create a sheet that converts binary numbers to hexadecimal.

5. Numerical errors in Excel

Create a sheet in which you add the same rational number 100 times (1000 times).

- Input may be a decimal number or a fraction.
- Which numbers will not give a precise sum?
- Does the error vary depending on the selected precision?
- Try using partial sums to determine when does the error occur.

5.4.1 Hints and solutions

1. Complex numbers

- Find the commands to write and compute with complex numbers, to perform arithmetic operations with them and to convert between their arithmetic and trigonometric forms. Draw the numbers in a chart.
- Create a complex number with modulus 1 and angle $2\pi/24$. Calculate its powers and draw them on a suitable chart.
- What happens when we change the modulus slightly to 0.9 or 1.1?

Solutions:

- We start by familiarising ourselves with Excel functions that work with complex numbers. The inputs are the real and imaginary parts of two complex numbers. We convert them to the algebraic form (COMPLEX), calculate their modulus (IMAB) and angle in radians (IMARGUMENT). We can convert the radians to degrees using the DEGREES function and calculate the (IMREAL) and imaginary (IMAGINARY) part of the complex number. We can use the function IMSUM to add complex numbers and IMPRODUCT to multiply them. We can also display them in the complex plane as shown in Figure 5.5.
- We start by taking the angle $2\pi/4$ and creating the algebraic form of the corresponding complex number:

mod.	angle	Real	Imaginary	algebraic form
$ z $	φ	$a = z \cos(\varphi)$	$b = z \sin(\varphi)$	$z = \text{COMPLEX}(a;b)$
1	$2\pi/4$	0	1	i

We calculate the powers of i :

power	algebraic form	Real	Imaginary
n	$z = \text{IMPOWER}(i;n)$	$\text{IMREAL}(z)$	$\text{IMAGINARY}(z)$
1	i	0	1
2	-1	-1	0
3	-i	0	-1
4	1	1	0

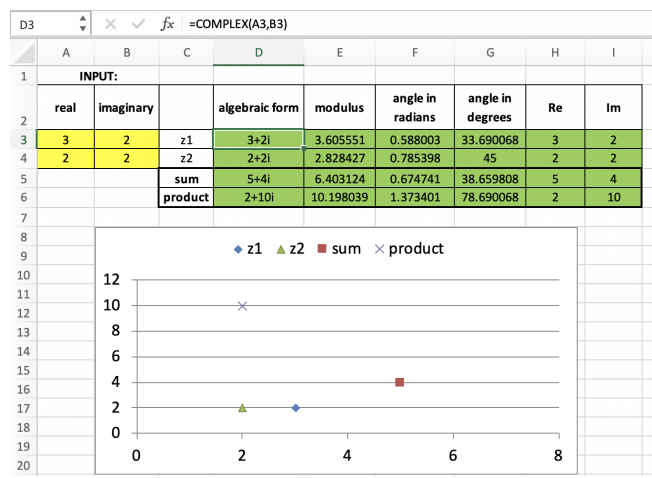


Figure 5.5: Complex numbers, operations with them and algebraic-to-trigonometric form conversions.

These are the solutions of the equation $z^4 = 1$.

If we change the angle to $2\pi/8$ and keep the modulus $|z| = 1$, we get solutions of $z^8 = 1$ at the vertices of a regular octagon, see Figure 5.6.

If we change the angle to $2\pi/24$, it means we are solving the corresponding equation $z^{24} = 1$. The solutions represented by vertices of a regular 24-sided polygon are shown in Figure 5.7 left.

- If we change the modulus of the complex number to $|z| = 1.1$, its powers create a spiral shape, see Figure 5.7 right.

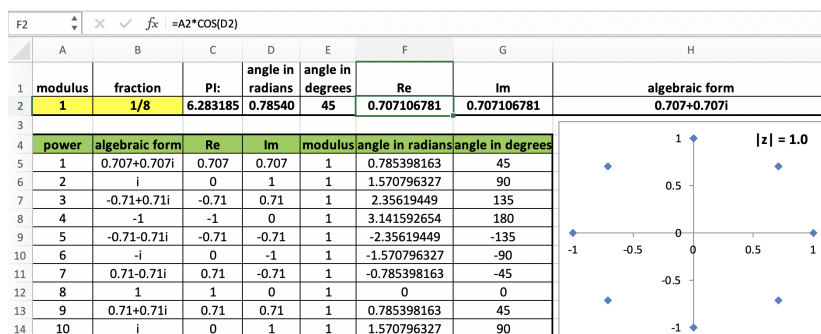
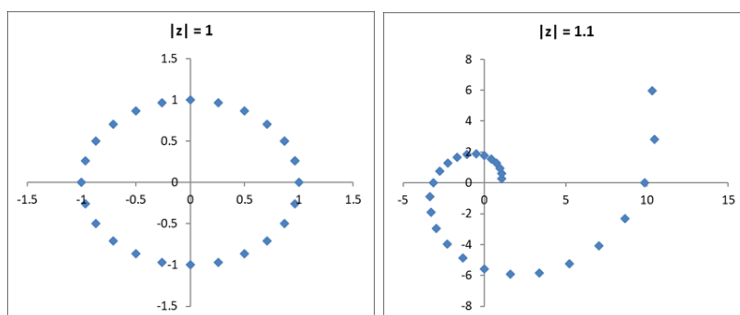
No longer a complex number

What happens to the graph if we change the modulus in the first problem to a negative number?

2. Approximation of real numbers by fractions

Create a sheet with a single input – an irrational number r .

- Look for the most accurate approximation of this number using a fraction of the type x/n , where n can be between 1 and 1000, inclusive.

Figure 5.6: Solutions of $z^8 = 1$.Figure 5.7: Powers of complex numbers for $|z| = 1.0$ and 1.1 .

- For each fraction, determine the error of approximation.
- Indicate the rows, where the error has decreased. These correspond to the new best approximation.
- It is easy to approximate with a fraction with large denominator. Good estimate will have a small error and the smallest possible denominator.
- Find a better estimate for π than $355/113$.

Solutions:

We can start with an irrational number $r = \sqrt{2} = 1.41421356237309\dots$. We want to find the best rational approximation of this number using a

fraction $\frac{n}{m}$. If we take a denominator $m = 13$, we determine the value of the numerator as

$$\frac{n}{13} = \sqrt{2} \Rightarrow n = 13 \cdot \sqrt{2} = 18.3847 \dots \doteq 18$$

This way we obtained an approximation of $\sqrt{2}$ using the fraction $18/13 = 1.3849 \dots$. The approximation is not very accurate as can be seen by calculating the error $|\sqrt{2} - 18/13| = 0.02959 \dots$.

We can write these steps in an Excel sheet as:

m	$m \cdot r$	n	n/m	error
13	18.3847	18	1.3849	0.02959

The numerator value n is calculated by rounding $n = \text{ROUND}(m \cdot r)$.

Our task is to find the the best approximation of $\sqrt{2}$ by n/m with denominators up to $m = 50$. We need to repeat the steps for $m = 1, \dots, 50$. Then we find the minimum of the approximation error (using the $\text{MIN}(:,)$ function) and determine the most accurate fraction 12/29:

m	$m \cdot r$	n	n/m	error
29	41.0121	12	1.4137	0.00041

Can we find a better approximation of $r = \pi$ than $355/113$?

m	$m \cdot r$	n	n/m	error
113	354.999	355	3.14159	0.00008

Since we do not know which denominator results in a better approximation, we need to create a check that compares the current error with the minimal error so far and notifies us in case there is a decrease, see Figure 5.8.

3. Approximation of real numbers by continued fractions

Continued fraction is a fraction of this form:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}$$

Create a sheet that takes an input – an irrational number (square root, cube root, $\pi, \dots$).

C3										
	A	B	C	D	E	F	G	H	I	J
	Denominator	Number	Numerator	Fraction	Decimal	Error	Smallest error so far	Optimal fraction	Ratio of error decrease	Error * denominator
1		3.141592654								
2										
3	1	3.141592654	3	3	3.000000000	0.1415926536	0.1415926536	TRUE		
4	2	6.283185307	6	3	3.000000000	0.1415926536	0.1415926536	FALSE	1	
5	3	9.424777961	9	3	3.000000000	0.1415926536	0.1415926536	FALSE	1	
6	4	12.56637061	13	3 1/4	3.250000000	0.1084073464	0.1415926536	TRUE	1.30611677	0.4336293856
7	5	15.70796327	16	3 1/5	3.200000000	0.0584073464	0.1084073464	TRUE	1.8560567	0.2920367321
8	6	18.84955592	19	3 1/6	3.166666667	0.0250740131	0.0584073464	TRUE	2.32939762	0.1504440785
9	7	21.99114858	22	3 1/7	3.1428571429	0.0012644893	0.0250740131	TRUE	19.8293601	0.0088514249
10	8	25.13274123	25	3 1/8	3.125000000	0.0165926536	0.0012644893	FALSE	1	
11	9	28.27433388	28	3 1/9	3.1111111111	0.0304815425	0.0012644893	FALSE	1	
12	10	31.41592654	31	3 1/10	3.100000000	0.0415926536	0.0012644893	FALSE	1	
13	11	34.55751919	35	3 2/11	3.1818181818	0.0402255282	0.0012644893	FALSE	1	
14	12	37.69911184	38	3 1/6	3.166666667	0.0250740131	0.0012644893	FALSE	1	
15	13	40.8407045	41	3 2/13	3.1538461538	0.0122535003	0.0012644893	FALSE	1	
16	14	43.98229715	44	3 1/7	3.1428571429	0.0012644893	0.0012644893	FALSE	1	
17	15	47.1238898	47	3 2/15	3.1333333333	0.0082593203	0.0012644893	FALSE	1	
18	16	50.26548246	50	3 1/8	3.125000000	0.0165926536	0.0012644893	FALSE	1	
19	17	53.40707511	53	3 2/17	3.1176470588	0.0239455948	0.0012644893	FALSE	1	
20	18	56.54866776	57	3 1/6	3.166666667	0.0250740131	0.0012644893	FALSE	1	

Figure 5.8: Approximation of π using fractions.

- The output will be the coefficients which correspond to the continued fraction.
- Compare these fractions with the best approximations from the previous problem.

Solution: Here we show the continued fraction for π .

$$\pi = 3.1415926\dots = 3 + \frac{1}{7 + \frac{1}{15 + \frac{1}{1 + \frac{1}{292 + \frac{1}{1 + \dots}}}}}$$

Using the steps shown in Figure 5.9 left, we get:

1. $\lfloor 3.1415\dots \rfloor = 3$
2. $3.1415\dots - 3 = 0.1415\dots$
3. $(0.1415\dots)^{-1} = 7.0625\dots$
1. $\lfloor 7.0625\dots \rfloor = 7$
2. $7.0625\dots - 7 = 0.0625\dots$

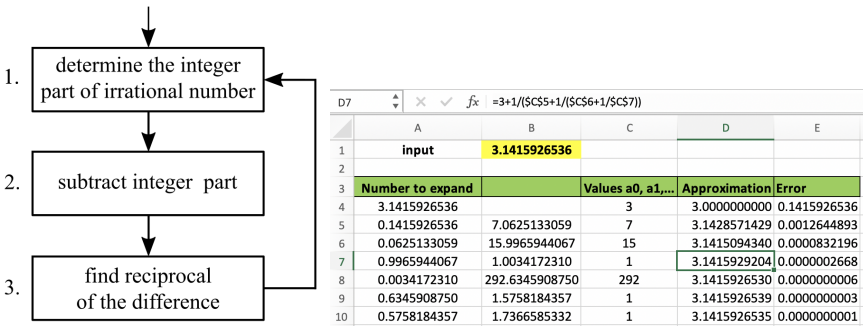


Figure 5.9: Approximation of π using continued fractions.

3. $(0.0625 \dots)^{-1} = 15.9965 \dots$
1. $\lfloor 15.9965 \dots \rfloor = 15$
- etc.

We obtained the sequence $\langle 3, 7, 15, 1, 292, 1, \dots \rangle$ which can be used to approximate π . Even if we use just the first two coefficients in the continued fraction, we get an estimate of π that is 10^{-3} accurate:

$$\pi \doteq 3 + \frac{1}{7} = \frac{22}{7} = 3.1428 \dots$$

If we include the next coefficient, the error drops to 10^{-5} :

$$\pi \doteq 3 + \frac{1}{7 + \frac{1}{15}} = 3 + \frac{15}{106} = \frac{333}{106} = 3.1415094 \dots$$

Using the continued fractions, see Figure 5.9 right, we obtain an accurate approximation of an irrational number faster than by going over many fractions as in the previous problem.

4. **Conversions between number systems**

- Create a simple sheet which converts a binary number to a decimal number. Use the powers of 2. Include the digits after the *decimal/binary point*.

- Create a sheet that converts a number from decimal system to binary. How do we do this for numbers that have also digits after the decimal/binary point?
- Create a sheet that converts binary numbers to hexadecimal.

Hints:

- Excel includes functions for conversions DEC2BIN, BIN2DEC, DEC2HEX, BIN2HEX, etc. We can use these to verify our calculations.
- The conversion from the decimal system to other number systems is based on consecutive division of the decimal number by the new base. The remainders after this division represent the digits in the new number system, see Figure 5.10 and Figure 5.11 The Excel function MOD is very helpful in these calculations. We need to read the digits of the converted number in the correct order.
- The CONCATENATE function may be helpful for joining results from several cells together.

E13						
	A	B	C	D	E	F
3	Decimal to binary:					
4	input:	453	=	011000101		
5						
6				remainder		
7		453 :2 =	226		1	
8		226 :2 =	113		0	
9		113 :2 =	56		1	
10		56 :2 =	28		0	
11		28 :2 =	14		0	
12		14 :2 =	7		0	
13		7 :2 =	3		1	
14		3 :2 =	1		1	
15		1 :2 =	0		1	
16		0 :2 =	0		0	

Figure 5.10: Conversion from decimal to binary system.

- We can also convert non-integer numbers when we use negative powers of 2 for the digits that come after the decimal point as in Figure

M19	H	I	J	K	L	M	N	O	P	Q	R
15	Decimal to binary with decimal point										
16	0.1	=				0.00011001					
17											
18				0.1 * 2 =	0.2		INT(0.2) =				0
19	FRAC(0) =	0.2 * 2 =	0.4				INT(0.4) =				0
20	FRAC(0) =	0.4 * 2 =	0.8				INT(0.8) =				0
21	FRAC(1) =	0.8 * 2 =	1.6				INT(1.6) =				1
22	FRAC(2) =	0.6 * 2 =	1.2				INT(1.2) =				1
23	FRAC(1) =	0.2 * 2 =	0.4				INT(0.4) =				0
24	FRAC(0) =	0.4 * 2 =	0.8				INT(0.8) =				0
25	FRAC(1) =	0.8 * 2 =	1.6				INT(1.6) =				1

Figure 5.13: Conversion from decimal to binary system for numbers with decimal point.

	A	B	C	D	E	F	G	H	I	J	K	L	M
46	Binary to hexadecimal:												
47	input	1	0	1	0	1	1	0	0	0	0	0	1
48		3	2	1	0	3	2	1	0	3	2	1	0
49		8	4	2	1	8	4	2	1	8	4	2	1
50		8	0	2	0	8	4	0	0	0	0	0	1
51		10				12				1			
52		a	c									1	

Figure 5.14: Conversion from binary to hexadecimal.

5. Numerical errors in Excel

Create a sheet in which you add the same rational number 100 times (1000 times).

- Input may be a decimal number or a fraction.
- Which numbers will not give a precise sum?
- Does the error vary depending on the selected precision?
- Try using partial sums to determine when does the error occur.

Hints:

Chapter 6

Sets and set theory

6.1 Lecture overview

6.1.1 Historical context

The mathematical logic is both a theoretical and a practical building block of a mathematical foundation describing exact and correct means of expressing, proving and building mathematical theories. The *set theory* is another building block of this foundation and allows us to define (basically all) objects mathematics examines. It also allows us to formulate their properties and prove them.

The set theory arose in the second half of the 19th century as a special tool for examining infinite sets related to the problems of functional series. George Cantor is the most famous mathematician who looked at the set theory as a separate research area that should become a universal tool for description of mathematical objects. After the removal of issues and paradoxes of the early set theory, this goal was reached at the beginning of the 20th century and some important questions are still being examined today. There have been attempts at new and different foundations of the set theory. In the 80s and 90s, the so-called *alternative set theory* arose in Czech Republic and Slovakia. Its author was Petr Vopěnka. Currently, the fuzzy set theory is gaining interest.

6.1.2 Current set theory

As any other theory in modern mathematics, also the set theory has its axioms. The most well known is the *Zermelo-Fraenkel axiomatic system* ZFC. Its important *axiom of extensionality* says that two sets are equal iff they have the same *elements*. This axiom carries in itself the essence of the idea of a *set*: a set is uniquely determined by its elements and by nothing else. When defining a set, we thus do not examine these elements, the relations among them, their order or function. In this sense, the idea of a set is very simple and basic. And yet, it lets us build the whole mathematics.

Another axiom declares the existence of one non-empty infinite set. The next few axioms guarantee the possibility to create new sets, e.g. by creating subsets of a set, their intersections, unions, pairs etc. Another axiom lets us create such subset of the original set that contains an element with a (correctly) defined property. The *correctness* protects us from the paradoxes of the early set theory, such as *set of all sets that do not contain themselves*. The *axiom of regularity* has a similar purpose and prevents situations such as $A \in B$ and $B \in A$ at the same time.

The final axiom is the *axiom of choice*, which guarantees the existence of a set that contains one element from each non-empty set from a group. This axiom allows some paradoxes even in the current set theory, such as: split a sphere into several parts and build two spheres from these parts that are both identical to the original sphere. Without this axiom though, it would not be possible to state and prove some important properties in very necessary parts of mathematics. The search for a suitable replacement for the axiom of choice is a current open question in the modern set theory.

Using the set theory and its axioms as a practical starting point for all areas of mathematics would be very cumbersome but some properties of sets are very useful for common mathematical notation and it is necessary and useful to know them.

6.1.3 Properties of sets

Using sets we can specify groups of objects that have certain properties. Therefore we often encounter expressions of the form $x \in \mathbb{N}$ or $\forall z \in \mathbb{R}$. The set of elements for which $x \in A$ is not true, is called the *complement* of a set A and is denoted by $\bar{A}$.

The empty set $\emptyset = \{\}$ can also be defined using a special property: $x \neq x$.

An important relation of two sets is *to be a subset*. $A \subset B$ iff all elements of

A are also elements of B . A is usually considered a subset of B even if they are equal. Otherwise we use the symbols $A \subset B$ or $A \subseteq B$.

If we have two (or more) sets $A, B, C, \dots$, we can perform basic set operations with them:

- The *intersection* $A \cap B$ ($A \cap B \cap C, \dots$) is a set that contains the elements that are in *each* of the sets A, B ($C, \dots$).
- The *union* $A \cup B$ ($A \cup B \cup C, \dots$) is a set that contains the elements that are in *at least* one of the sets A, B ($C, \dots$).
- The *difference* $A - B$ is a set that contains those elements of A which are not in B . The *symmetric difference* $A \triangle B$ contains those elements from a union of two sets that are not in their intersection.
- If we consider the set A , which is a subset of another larger set B , we can consider the complement of A in B . For example, the odd numbers are a complement of even numbers in $\mathbb{Z}$, the irrational numbers are the complement of the rational numbers in $\mathbb{R}$.)
- The *Cartesian product* $A \times B$ ($A \times B \times C, \dots$) is a set of all ordered pairs (triplets, ...) in which the first element is from the first set, the second element is from the second set (the third is from the third, ...). For example, the ordered pairs of real numbers create a set of vectors in a plane or points in a coordinate system.

Another property related to intersection and union is a *decomposition* of a set. We decompose the set A into sets $A_1, A_2, A_3, \dots$ (the decomposition may be finite or infinite), if the union of these sets is the whole A and all pairwise intersections of A_i and A_j are empty. The decomposition of a set is useful because when examining the set properties, we can examine the individual sets of the decomposition. That way we can be sure that we do not forget anything and also if we do the decomposition in a clever way, the elements of some decomposition sets may have special properties that we can use.

Venn diagrams

A useful depiction of sets and their intersections are Venn diagrams, see Figure 6.1. While easy to draw two or three sets, they get more complicated for larger numbers of sets, because all possible intersections need to be included.

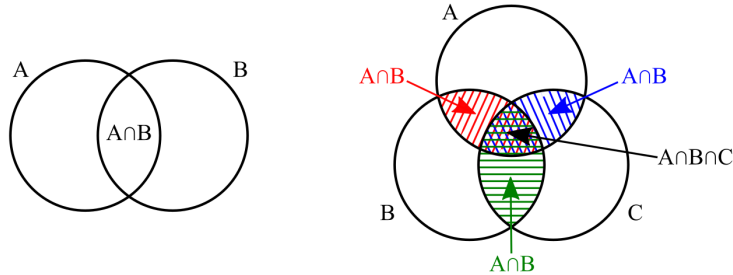


Figure 6.1: Venn diagrams for two and for three sets.

Problem 6.1 Decomposition of a set:

We have seven boys and five girls on the roster for a volleyball tournament. We need to pick a team with six members in which we need at least two boys and at least two girls. How many possible teams can we pick?

Hint: We can examine the cases with two, three and four boys separately.

6.1.4 Number of elements and cardinality**Number of elements**

For a finite set M , $|M|$ denotes the *number of elements* of the set, e.g. if $A = \{1, 2, 3, 8\}$, $|A| = 4$. If we have a decomposition of A into $A_1, A_2, A_3, \dots$, we can easily determine the number of elements of A using this decomposition: it is sufficient to determine how many elements there are in the sets $A_1, A_2, \dots$ and add these numbers.

Inclusion-exclusion principle

The decomposition of the set and the addition rule help us calculate the number of elements in a union if the two sets are disjoint. What should we do if the sets have a non-empty intersection?

- *Union of two sets* Suppose we have students who like mathematics (set M) and students who like computer science (set C). Moreover, each student in the class likes at least one of these subjects. To determine the number of students in the class, we cannot use the sum $|M| + |C|$. If we have

somebody who likes both subjects, we would count him or her twice. This is the reason why we have: $|M \cup C| = |M| + |C| - |M \cap C|$.

- *Union of three sets* If some students also like English and every student still likes at least one of the three subjects, then we have $|M \cup C \cup A| = |M| + |C| + |A| - |M \cap C| - |M \cap A| - |C \cap A| + |M \cap C \cap A|$. Check that each type of student is counted only once on the right, e.g. Susan who likes computer science and mathematics, but not English, is added twice: both in C and in M but subtracted in $|M \cap C|$. Thomas, who likes all three subjects is first added three times, then subtracted three times and finally added once.
- *Number of elements of union of n sets* $|S_1 \cup S_2 \cup \dots \cup S_n| =$
 $=$ sum of number of elements of the individual sets
 $-$ number of elements in all pairwise intersections
 $+$ number of elements of all intersections of triplets of sets
 $-$ number of elements of all intersections of all groups of four sets
 $+$ etc.

Power set

When considering the number of elements, it is also useful to mention the *power set*: $P(X)$ is the set of all subsets of X . If $X = \{1, 4, 9\}$, $P(X) = \{\emptyset, \{1\}, \{4\}, \{9\}, \{1, 4\}, \{1, 9\}, \{4, 9\}, \{1, 4, 9\}\}$. The number of elements in power set is $|P(X)| = 2^{|X|}$, i.e. in our case $|P(X)| = 2^3 = 8$.

Cardinality

In general, $|M|$ denotes the *cardinality* of the set. For finite sets, we saw that this property corresponds to the number of elements. For infinite sets, where it does not make sense to ask about a number of elements, it lets us determine how large the sets are. Two (infinite) sets A, B have the same cardinality, if we can find a mapping which assigns exactly one element of B to each element in A and covers the whole set B in the process. In other words, we can *pair the elements up* or find a one-to-one mapping. Such mapping is called a *bijection*.

It is possible to find a bijection between the natural numbers $\mathbb{N}$ and integers $\mathbb{Z}$. It is also possible to find a bijection between natural numbers and rational numbers. However, not all infinite sets have the same cardinality. For example, it is not possible to find a bijection between $\mathbb{N}$ and $\mathbb{R}$. Even the set of real numbers in the interval $(0,1)$ has a larger cardinality than $\mathbb{N}$. The power sets

$P(X)$ are important for creation of larger and larger infinite sets. For each set X it is true that $P(X)$ has larger cardinality than X , i.e. $|P(X)| > |X|$.

6.2 Questions

1. Choose three of these sets which are different from one another:
 $M_1 = \{a, a, a, b, dog\}$, $M_2 = \{dog, dog, a, b\}$, $M_3 = \{1, a, b, dog\}$, $M_4 = \{1, a, b\}$, $M_5 = \{1, 1, 1, a, a, a, b, b, b, dog\}$
2. Which of these sets are the same:
 $P_1 =$ even numbers
 $P_2 =$ multiples of 4
 $P_3 =$ multiples of 3
 $P_4 =$ odd numbers
 $P_5 =$ numbers which give remainder 1 when divided by 5
 $P_6 =$ numbers of which half is even
 $P_7 =$ numbers which give remainder 0 when divided by 2
 $P_8 =$ numbers of which half is odd
 $P_9 =$ all powers of 2
3. Using a formula with quantifiers, write a definition of two sets being equal.
4. Using a formula with quantifiers, write a definition of a subset.
5. Relate these sets using subsets: A = the set of all dogs, B = the set of all poodles, C = the set of all mammals, D = the set of all short-haired poodles, E = set of all short-haired huskies, F = the set of all long-haired poodles.
6. Relate these sets using subsets: $\mathbb{N}, \mathbb{Z}_0^+, \mathbb{R}_0^+, \mathbb{R}_0^-, \mathbb{Q}_0^+, \mathbb{Q}^+, \mathbb{Q}_0^-, \mathbb{C}$.
7. A set with which property do we postulate in the Set theory as an axiom?
8. Define the odd natural numbers by selecting them from all integers using a suitable property.
9. Define the positive numbers using some property and the set of real numbers.
10. Define a circle using an appropriate properties of complex numbers.
11. How is the empty set defined?
12. Using the sets A, B, C , an empty set and Venn diagrams, draw
 - a) the intersection of A and B

- b) the union of A , B and C
 - c) difference $B - A$
 - d) difference $A - B$
 - e) difference $(A - B) - C$
 - f) (intersection of B and C) union A
13. Let $A = \{a, b\}$, $B = \{1, 2, 3\}$, $C = \emptyset$. Write all elements of the Cartesian products
- a) $A \times B$
 - b) $B \times A$
 - c) $A \times C$
 - d) $C \times B$
14. Using the sets $A = \mathbb{N}$, $B = \mathbb{Z}_0^+$, $C = \mathbb{Q}_0^-$, describe the following sets
- a) $A \cup B$
 - b) $A \cup B \cup C$
 - c) $B - A$
 - d) $A - B$
 - e) $(A - B) - C$
 - f) $(B \cap C) \cup A$
15. Write the intersection, union, difference and symmetric difference of set $N_1 =$ integers from 10 to 30 and set $N_2 =$ integers from 20 to 40.
16. In the room, there are 60 Java programmers and 45 C++ programmers. 15 people can program in both languages. How many programmers are there in the room?
17. When is it more appropriate to say that two sets have the same cardinality and when that they have the same number of elements? What is the connection between the cardinality and the number of elements of a set?
18. Determine whether the following sets have the same cardinality and state why:
- a) $\mathbb{N}$ and $\mathbb{N}_0$

- b) $\mathbb{N}$ and even numbers from $\mathbb{N}$
 - c) $\mathbb{N}$ and integers greater than 10
19. Determine whether the following sets have the same cardinality and state why:
- a) intervals $(0, 1)$ and $(0, 4)$
 - b) intervals $(0, 1)$ and $(3, 4)$

6.3 Problem set

1. Inclusion-exclusion principle

- a) There are 40 students in the class. Every one of them speaks at least one foreign language – English or German. 27 speak English, 7 speak both. How many students speak German? How many students speak only German?
- b) Some students like to eat fruit. 22 like pears, 25 like apples and 39 like grapes. We also know that 9 like apples and grapes, 17 like pears and grapes, 20 like apples and pears. 6 students like all three kinds of fruit and 4 students do not like any fruit. How many students are there altogether?
- c) In the room, there are 57 programmers in Pascal, 48 programmers in C++ and 72 programmers in Java. 15 of them can program in both P and J, 7 can program in all three languages, 10 can program in P and C++ and 13 can program in J and C++. There are no other people in the room. How many people are there in the room?

2. Operations on sets of quadrilaterals

Let us have the following seven sets:

A: All quadrilaterals

T: Trapezoids (at least one pair of opposite sides is parallel)

P: Parallelograms (both pairs of opposite sides are parallel)

K: Kites, i.e. deltoids (two and two neighboring sides have equal length)

D: Diamonds, i.e. rhombi (all four sides have equal length)

R: Rectangles (four right angles)

S: Squares (equal sides and right angles)

In each of the subtasks, find all pairs of these sets with the desired property.

- a) The first set is a subset of the second one.
- b) The intersection of the two sets is equal to one of them.
- c) The intersection of the two sets is equal to one of the remaining five sets.
- d) The intersection of the two sets is not equal to any of the seven sets.
- e) The union of the two sets is equal to one of them.
- f) The union of the two sets is equal to one of the remaining five sets.
- g) The union of the two sets is not equal to any of the seven sets.

- h) Examine also the differences of these sets.

3. Operations on special number sets

Let us denote $C_{n,k}$ the set of natural numbers that have the digit k at the n^{th} position from the right. Let us denote $Z_{n,k}$ the set of natural numbers that give remainder k when divided by n .

- Express the set $Z_{2,0}$ using suitable sets $C_{n,k}$.
- Express the set $Z_{5,0}$ using suitable sets $C_{n,k}$.
- Express the set $Z_{5,1}$ using suitable sets $C_{n,k}$.
- Express the set $Z_{25,0}$ using suitable sets $C_{n,k}$.
- Express the set $Z_{50,1}$ using suitable sets $C_{n,k}$.
- Express single- and two-digit numbers from $Z_{3,0}$ using suitable sets $C_{n,k}$.
- Express two-digit numbers from $Z_{11,0}$ using suitable sets $C_{n,k}$.
- Express $C_{1,7}$ using suitable $Z_{n,k}$.
- Express $C_{2,0}$ using suitable $Z_{n,k}$.

4. Set difference

We have sets:

$$A = \{x: x \text{ is a natural number from 10 to 16}\}$$

$$B = \{y: y \text{ is an even number between 8 and 20}\}$$

$$C = \{7, 9, 11, 14, 18, 20\}$$

- Find $A - B$.
- Find $B - C$.
- Find $C - A$.
- Find $B - A$.

5. Operations on sets with abstract elements

We have sets:

$$K = \{a, \{a\}, b, \{c\}\}$$

$$L = \{a, b, c\}$$

$$M = \{\{a\}\}$$

- Find $K \cup L \cup M$.
- Find $K \cap L$.

- c) Find $K \cap M$.
- d) Find $K \cap L \cap M$.
- e) Find $(K \cup L) \cap M$.
- f) Find $K \cup (L \cap M)$.
- g) Find $K - L$.
- h) Find $K - M$.
- i) Find $K \times L$.
- j) Find $M \times L$.

6. Cardinality

Determine if the following pairs of sets have the same cardinality. If so, prove it by finding an appropriate pairing/bijection.

- a) $\mathbb{N}$ and numbers from $\mathbb{N}$ that give remainder 3 when divided by 7
- b) $\mathbb{N}$ and odd integers larger than 100
- c) $\mathbb{N}_0$ and $\mathbb{Z}$
- d) $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$
- e) $\mathbb{N}$ and $\mathbb{Z} \times \mathbb{N}$
- f) Intervals $(0, 1)$ and $(2, 4)$
- g) Intervals $(0, 2)$ and $(0, 1)$
- h) Intervals $(0, 2)$ and $(0, 1)$
- i) Intervals $(0, 1)$ and $(3, 6)$
- j) Intervals $(3, 5)$ and $(7, 12)$
- k) Intervals $(-2, 4)$ and $(-11, 3)$
- l) Intervals $(0, 1)$ and $(0, \infty)$
- m) Intervals $(0, 1)$ and $(-\infty, \infty)$

6.3.1 Hints and solutions

1. Inclusion-exclusion principle

There are 40 students in the class. Every one of them speaks at least one foreign language – English or German. 27 speak English, 7 speak both. How many students speak German? How many students speak only German?

Hints:

- It is helpful to draw the groups of students, e.g. in circles (Venn diagram) and note the number of students in each separate region. This is still possible with three basic groups, but gets more complicated as the number of groups grows. In such case, it is more useful to start with the inclusion-exclusion equation and look at which of its terms we know and which need to be determined.
- There is a difference between *speaks the language* (belongs to the group, may also belong to other groups) and *speaks only the language* (belongs only to this group).

Solution: If 27 students speak English and 7 speak both languages, that means there are 20 students that speak English only. Consequently, if out of 40 students 20 speak English, 7 speak both languages and there are no other students left than those who speak German, we have 13 students who speak German only. To determine how many students speak German, we have to consider both German only students and bilingual students. We get $13+7=20$, as we also see in Figure 6.2.

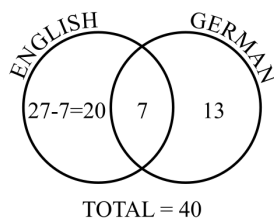


Figure 6.2: Venn diagram for English and German.

Restrictions

Determine what are the restrictions on the number of elements of the intersections in case with two sets and in case with three sets.

2. Operations on sets of quadrilaterals

Let us have the following seven sets:

A: All quadrilaterals

T: Trapezoids (at least one pair of opposite sides is parallel)

P: Parallelograms (both pairs of opposite sides are parallel)

K: Kites, i.e. deltoids (two and two neighboring sides have equal length)

D: Diamonds, i.e. rhombi (all four sides have equal length)

R: Rectangles (four right angles)

S: Squares (equal sides and right angles)

In each of the subtasks, find all pairs of these sets with the desired property.

- a) The first set is a subset of the second one.
- b) The intersection of the two sets is equal to one of them.

Hints:

- This problem does not deal with numbers but with sets of geometrical objects. And that is all right, since the elements of a set can be almost anything. Just note that the definitions of the geometrical objects (in parentheses) might be slightly different than those you are used to. This is just for the purposes of these specific problems to illustrate how we work with sets.
- Also, when we talk about intersections (or unions etc.), we mean intersections of the sets containing these elements and not intersections of the geometric shapes drawn in some plane.
- To determine which sets are subsets of other sets, it helps to consider which set has more restrictions on its elements. For example, if one set contains all blue cubes and another set contains all blue cubes that smell like oranges, the second set is a subset of the first one. Every blue cube that smells like oranges is a blue cube, so it also belongs to the first set. And this is a definition of a subset. However, only some blue cubes smell like oranges, some might smell like apples, some might not smell at all, therefore not all elements of the first set are elements of the second set. Consequently, the first set is not

a subset of the second set. We need to consider the properties of geometrical objects in this way.

- What is the intersection $A \cap B$ in case $A \subset B$?

Solution: The most general of these sets is A , therefore it cannot be a subset of any of the others. All other sets introduce restrictions on the quadrilaterals and therefore are subsets of A . If we want to form a *chain* of subsets, we need to think about which properties restrict which other properties. All squares certainly have four right angles, therefore we can write $S \subset R$. All rectangles have pairs of opposite sides parallel, therefore we can extend this chain to $S \subset R \subset P$. And having both pairs of opposite sides parallel implies having at least one pair parallel, therefore we next have $S \subset R \subset P \subset T$. Note that the definition of T here is different from what you might be used to. We allow some very peculiar-looking trapezoids. This is purely for the purposes of this problem and for the discussion of the relations among sets. Finally, since all other sets are subsets of A , we can write $S \subset R \subset P \subset T \subset A$.

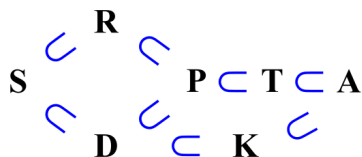


Figure 6.3: Chains of subsets.

We can start the chain differently. All squares have all sides of equal length, therefore $S \subset D$ and all rhombi are also parallelograms, therefore $S \subset D \subset P$. Using the same reasoning as above, this chain can be extended to $S \subset D \subset P \subset T \subset A$. We should now ask, what is the relationship between the sets R and D , in other words, could the two chains be combined?

On the one hand, rhombi/diamonds are not required to have right angles, so R is not a subset of D . On the other hand, rectangles are not required to have all sides of equal length, so D is not a subset of R . Consequently, we cannot merge the two chains. There are two *alternative paths* from squares to parallelograms.

$\cap$	A	T	P	K	D	R	S
A	A	T	P	K	D	R	S
T	T	T	P		D	R	S
P	P	P	P		D	R	S
K	K			K	D		S
D	D	D	D	D	D		S
R	R	R	R			R	S
S	S	S	S	S	S	S	S

Table 6.1: Intersections of sets written in the header with sets written in the first column. Note that the table is symmetric along the diagonal, since $A \cap B = B \cap A$.

So far, we considered all sets except deltoids/kites K . Where do they fit? We immediately have $D \subset K$. On the right side, there are not many options. By relaxing the kite property, we do not get T or P , therefore the best we can do here is $S \subset D \subset K \subset A$. Figure 6.3 shows all the relationships that we found.

After answering the question about subsets, we can move on to considering intersections. We know that if $A \subset B$, then we automatically have $A \cap B = A$, since all elements of A are in both sets. And if $A \subset B \subset C$, then also $A \subset C$. Therefore we can recycle the insight from chains we built earlier and collect all the information in a table such as Table 6.1. In it we also used the fact that intersection of any set with A is the set itself and that intersection of any set with S is S . In each case we also make sure that the intersection is correct with respect to the geometric properties of the set elements.

After using all the information about subsets, we are left with four empty spaces. Let us examine them one by one. Only rectangles that have all sides of equal length are squares, therefore $R \cap D = S$. Also, the only way how a rectangle can also be a kite is if it is square and we have $R \cap K = S$. The diamonds are the only kites that have the pairs of opposite sides parallel, i.e. $K \cap P = D$. And even if we relax the condition of parallel opposite sides to just one pair, we do not get any new objects, but the diamonds. This information completes the Table 6.2.

A similar table may be constructed for unions or differences of sets.

$\cap$	A	T	P	K	D	R	S
A	A	T	P	K	D	R	S
T	T	T	P	D	D	R	S
P	P	P	P	D	D	R	S
K	K	D	D	K	D	S	S
D	D	D	D	D	D	S	S
R	R	R	R	S	S	R	S
S	S	S	S	S	S	S	S

Table 6.2: Completed table of intersections of sets written in the header with sets written in the first column.

3. Operations on special number sets

Let us denote $C_{n,k}$ the set of natural numbers that have the digit k at the n^{th} position from the right. Let us denote $Z_{n,k}$ the set of natural numbers that give remainder k when divided by n .

- Express the set $Z_{2,0}$ using suitable sets $C_{n,k}$.
- Express the set $Z_{25,0}$ using suitable sets $C_{n,k}$.
- Express $C_{1,7}$ using suitable $Z_{n,k}$.

Hints:

- Before starting to express any Z sets using any C sets and vice versa, it is helpful to write out first few elements of each. For example $Z_{3,0} = \{3, 6, 9, 12, \dots\}$, $C_{2,7} = \{71, 72, 73, \dots, 171, 172, 173, \dots\}$.
- Consider if you are looking for a property that some sets share or have in common, this points towards a union, or for a property that only some portions of the sets have, which points towards an intersection.
- Consider how would the Z sets look for $n \leq k$.
- When solving this problem, it is also helpful to remember the criteria of divisibility because they are closely related to the digits of numbers.

Solution:

- $Z_{2,0} = \{2, 4, 6, 8, 10, 12, \dots\}$, since even numbers are those that give remainder 0 when divided by 2. Note that all even numbers end by one of the digits $\{2, 4, 6, 8, 0\}$. We know that all elements of $C_{1,0} =$

$\{10, 20, 30, \dots\}$, $C_{1,2} = \{2, 12, 22, 32, \dots\}$, $C_{1,4} = \{4, 14, 24, 34, \dots\}$, $C_{1,6} = \{6, 16, 26, 36, \dots\}$ and $C_{1,8} = \{8, 18, 28, 38, \dots\}$ taken together also end by one of the digits $\{2, 4, 6, 8, 0\}$. Therefore we have $Z_{2,0} = C_{1,0} \cup C_{1,2} \cup C_{1,4} \cup C_{1,6} \cup C_{1,8}$.

- d) $Z_{25,0} = \{25, 50, 75, 100, 125, 150, \dots\}$. There will be four types of numbers in this set: those that end in 25, those that end in 50, those that end in 75 and those that end in 00. We can express each of these four groups as a separate set and the answer will then be their union. To express numbers that end in 25 using the C sets, we need to consider two digits: we need the last digit to be 5 and the one before last to be 2. Since we need both of these properties at the same time, we construct the desired set using an intersection $C_{1,5} \cap C_{2,2}$. After doing this for the other three endings, we form the union $Z_{25,0} = (C_{1,5} \cap C_{2,2}) \cup (C_{1,0} \cap C_{2,5}) \cup (C_{1,5} \cap C_{2,7}) \cup (C_{1,0} \cap C_{2,0})$.
- h) $C_{1,7} = \{7, 17, 27, \dots, 137, \dots\}$. To define numbers ending in 7 using remainders after division hints at the remainder being 7. Then the question arises: by what number do we need to divide n , in order to give us remainder 7 in the case when it ends by 7? It can be seen that it is division by 10 and thus $C_{1,7} = Z_{10,7}$.

4. Set difference

We have sets:

$A = \{x: x \text{ is a natural number from 10 to 16}\}$

$B = \{y: y \text{ is an even number between 8 and 20}\}$

$C = \{7, 9, 11, 14, 18, 20\}$

Find $C - A$.

Hint: While the set C is explicitly written with all its terms, sets A and B are defined using a property. Since the property includes short ranges, it is feasible (and helpful) to write them out explicitly too. Then we can use the definition of set difference and determine which elements are in one set and are not in the other.

Solution: $A = \{10, 11, 12, 13, 14, 15, 16\}$. We are then looking for numbers that are in C but are not in A and we get $C - A = \{7, 9, 18, 20\}$.

5. Operations on sets with abstract elements

We have sets:

$K = \{a, \{a\}, b, \{c\}\}$

$$L = \{a, b, c\}$$

$$M = \{\{a\}\}$$

e) Find $(K \cup L) \cap M$.

g) Find $K - L$.

j) Find $M \times L$.

Hint: It seems strange that the set L contains both a and $\{a\}$. Are these two the same thing? No, they are not. It might be helpful to think of sets as baskets. a is an element. $\{a\}$ is a basket that contains the element a . $\{a, \{a\}\}$ is a basket that contains two things: an element a and another basket.

Empty basket

Is any of these an empty basket: $\{\emptyset\}$, $\{\}$, $\emptyset$?

Solution:

e) First, we look at the parentheses and collect all elements of K and L with each appearing only once $(K \cup L) = \{a, \{a\}, b, \{c\}, c\}$. Their order does not matter. Then we look only for those elements that appear both in this newly created set and in M to determine the intersection $(K \cup L) \cap M = \{\{a\}\} = M$.

g) To determine the difference, we look for those elements that are in K and are not in L . So we get $K - L = \{\{a\}, \{c\}\}$.

j) The Cartesian product is a set of ordered pairs. The first element of the pair is from the first set and the second element from the second set. We need to include all combinations. Thus we have $M \times L = \{[\{a\}, a], [\{a\}, b], [\{a\}, c]\}$.

6. Cardinality

Determine if the following pairs of sets have the same cardinality. If so, prove it by finding an appropriate pairing/bijection.

c) $\mathbb{N}_0$ and $\mathbb{Z}$

d) $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$

f) Intervals $(0, 1)$ and $(2, 4)$

h) Intervals $(0, 2)$ and $\langle 0, 1)$

- k) Intervals $(-2, 4)$ and $(-11, 3)$
- l) Intervals $(0, 1)$ and $(0, \infty)$
- m) Intervals $(0, 1)$ and $(-\infty, \infty)$

Hints:

- While counterintuitive, there are more kinds of infinity, i.e. not all sets that are infinite have the same cardinality. Moreover, with infinite sets it is possible to construct a bijection even between two sets one of which is a proper subset of the other, e.g. $\mathbb{N}$ and $\mathbb{N}_0$. Therefore, relying purely on intuition, can be counterproductive while solving these problems. We should rather focus on finding the bijections.
- For the discrete sets such as $\mathbb{N}$, $\mathbb{Z}$ or some combinations of them, it is useful to think in terms of ordering the elements. If we can order the elements of any set in such a way that they start with one specific element and for all others it is clear where in the row they stand, we have a bijection with $\mathbb{N}$ by simply pairing them with their index in this order.
- For intervals, the ordering is not going to work. Instead, we are looking for functions that can stretch and shift the interval as desired. Once we have found it, it always makes sense to check the endpoints, to verify that they are transformed as desired. If the function is continuous, also the inner part of the first interval will transform into the inner part of the second interval.
- Sometimes it might be useful to break the complex transformation into a sequence of simpler transformations and then compose them together.
- If an endpoint needs to be included, e.g. as problem h), we can leverage a well defined infinite sequence, whose first point gets paired up with the endpoint, the second point with the first, etc.

Solution:

- c) $\mathbb{N}_0$ and $\mathbb{Z}$:

We can order $\mathbb{Z}$ in the following way: $\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$. Now we can define the bijection in such a way that we pair 0 with 0, even numbers from $\mathbb{N}_0$ with positive numbers from $\mathbb{Z}$ and odd

numbers from $\mathbb{N}_0$ with negative numbers from $\mathbb{Z}$. Written in mathematical language:

$$n \rightarrow \begin{cases} 0 & n = 0 \\ n/2 & n = 2k, k \in \mathbb{N} \\ \frac{-n-1}{2} & \text{otherwise} \end{cases}$$

Thus the two sets have the same cardinality. This one-to-one mapping assigns different integers to different numbers from $\mathbb{N}_0$, e.g. the number 34 gets 17, the number 35 gets -18 . Moreover, for each integer there is a number in $\mathbb{N}_0$ that gets imaged on it, e.g. for 6, we find 12 and for -6 , we find 11.

The other way round

Can you find a one-to-one mapping from $\mathbb{Z}$ to $\mathbb{N}_0$?

d) $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$:

We need to order the set $\mathbb{N} \times \mathbb{N}$. Consider Figure 6.4. It shows how the ordered pairs can be numbered by traversing the table in longer and longer diagonals. Surely this way we are going to cover all elements in the table – eventually. Even though it is difficult to write a formula such as in the previous case that would explicitly state which ordered pair is assigned to which n , nevertheless, the two sets have the same cardinality because we showed that a pairing can be found.

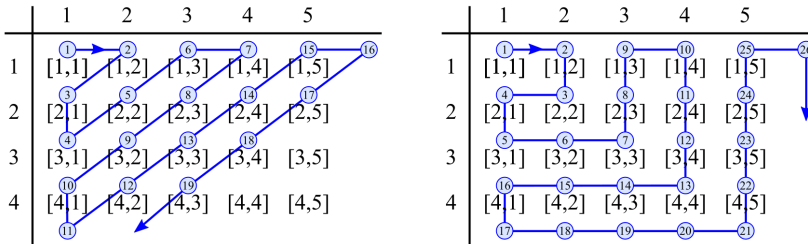


Figure 6.4: Two possible orderings of the set $\mathbb{N} \times \mathbb{N}$.

Explicit expression for mapping

Can you find an explicit expression for the one-to-one mapping between $\mathbb{N}$ and $\mathbb{N} \times \mathbb{N}$?

- f) Intervals $(0, 1)$ and $(2, 4)$:

Here we are looking for a one-to-one mapping (bijection) that stretches the interval by a factor of two (because the length of $(0, 1)$ is 1 and the length of $(2, 4)$ is 2) and shifts it by 2 units. $x \rightarrow 2x + 2$ seems to do both. We can verify this by checking the endpoints. Thus the two intervals have the same cardinality.

- h) Intervals $(0, 2)$ and $(0, 1)$:

If these were intervals $(0, 2)$ and $(0, 1)$, the answer would be simply $x/2$. However, we have the additional zero at the left endpoint of the second interval. Let us take the number 1 from the first interval and pair it with the zero from the second. That solves the issue with zero, but creates a new one with $1/2$. So let us take $1/2$ from the first interval and pair it with $1/2$ from the second, etc. This results in a formula for all numbers of the form $1/2^k$. Everything else can be scaled down as $x/2$:

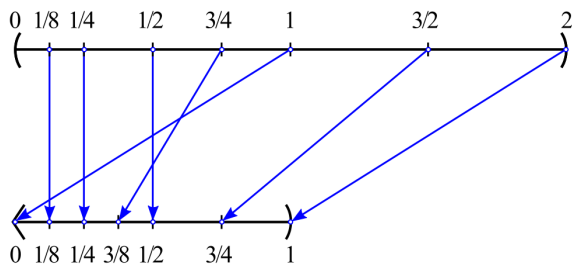
$$x \rightarrow \begin{cases} 0 & x = 1 \\ x & x = 1/2^k, k \in \mathbb{N} \\ x/2 & \text{otherwise} \end{cases}$$

This mapping is also shown in Figure 6.5. Note that the sequence $1/2^k$ is not the only one that works. We could have come up with another one that lies inside the interval. The reason why it works is that we performed a similar shift as in going from $\mathbb{N}$ to $\mathbb{N}_0$, which is possible due to the *open end*.

- k) Intervals $(-2, 4)$ and $(-11, 3)$:

This is a similar problem to f) but at the first glance it is not obvious what the bijection is. Therefore it is helpful to perform it step-by-step $(-2, 4) \rightarrow (0, 6) \rightarrow (0, 1) \rightarrow (0, 14) \rightarrow (-11, 3)$. The successive operations are $+2$, $\cdot 6$, $\cdot 14$ and -11 , so the bijection is $x \rightarrow ((x + 2)/6) \cdot 14 - 11 = \frac{7}{3}x - \frac{19}{3}$. Therefore also these intervals have the same cardinality.

A different method would be to directly look for a linear transformation $y = ax + b$. We know that the endpoint -2 is transformed to

Figure 6.5: Bijection from $(0, 2)$ to $(0, 1)$

-11 , therefore we have one equation $-11 = -2a + b$. Similarly, we know that the endpoint 4 is transformed to 3 which gives us the second equation $3 = 4a + b$. This system has a unique solution $a = 7/3$ and $b = -19/3$.

l) Intervals $(0, 1)$ and $(0, \infty)$:

Since ∞ is not a number, we cannot stretch the interval here as easily as before but we can consider other operations, such as $1/x$. Using this, we transform $(0, 1)$ to $(1, \infty)$ which is almost what we want. The numbers close to 1 in $(0, 1)$ are paired up with numbers close to 1 in $(1, \infty)$. $1/2$ is paired up with 2. $1/3$ with 3, etc. Numbers close to 0 are paired up with really large numbers. To find the bijection for the two original intervals, we shift this auxiliary result by 1 and get the bijection $x \rightarrow \frac{1}{x} - 1$. Therefore also $(0, 1)$ and $(0, \infty)$ have the same cardinality.

m) Intervals $(0, 1)$ and $(-\infty, \infty)$:

In the previous problem, we saw that a finite interval can be mapped to a *half-infinite* one. Therefore we can split $(0, 1)$ into $(0, \frac{1}{2}) \cup \{\frac{1}{2}\} \cup (\frac{1}{2}, 1)$ and $(-\infty, \infty)$ into $(-\infty, 0) \cup \{0\} \cup (0, \infty)$ and reuse what we already know.

We start by considering the intervals $(0, \frac{1}{2})$ and $(-\infty, 0)$. Mapping $x \rightarrow -\frac{1}{x}$ maps $(0, \frac{1}{2})$ to $(-\infty, -2)$. Therefore we need to add a shift.

Then we map $\frac{1}{2} \rightarrow 0$ and consider the intervals $(\frac{1}{2}, 1)$ and $(0, \infty)$. First we shift $(\frac{1}{2}, 1) \rightarrow (0, \frac{1}{2})$ and then use the same idea as before.

Combining everything together, we get the following bijection be-

tween $(0, 1)$ and $(-\infty, \infty)$:

$$x \rightarrow \begin{cases} -\frac{1}{x} + 2 & x \in (0, \frac{1}{2}) \\ 0 & x = \frac{1}{2} \\ 1/(x - \frac{1}{2}) - 2 & x \in (\frac{1}{2}, 1) \end{cases}$$

Different image for $\frac{1}{2}$

Can you find a different one-to-one mapping for this problem which maps $\frac{1}{2}$ to some number other than 0?

6.4 Lab problems

1. Inclusion and exclusion principle

- Create an Excel worksheet for three, four, (five) sets that will compute the number of elements in the union of all sets using the number of elements of individual sets and their intersections.
- Create an Excel worksheet for three, four, (five) sets that will compute the number of elements in the intersection of all sets using the number of elements of individual sets, their intersections and the union of all sets.
- Can the inputs be arbitrary integers? Add necessary checks!
- Combine the first two tasks into a sheet that will calculate either the number of elements in the intersection or the number of elements in the union, depending on the inputs.

2. Number sets

Let us denote $C_{n,k}$ the set of all natural numbers that have the digit k at the n^{th} position from the right. Let us denote $Z_{n,k}$ the set of all natural numbers that give remainder k when divided by n .

- Create an Excel sheet that will have inputs n and k . Using a column of natural numbers from 1 to 100 (or to 1000) it will then produce columns containing only the elements of sets $C_{n,k}$ and $Z_{n,k}$. Compute the number of elements in each of these sets. Then solve the following problems.
- Create the intersections of the following pairs of sets
 - $Z_{3,0}$ and $Z_{7,0}$
 - $Z_{5,0}$ and $Z_{11,0}$
 - $Z_{8,0}$ and $Z_{9,0}$
 - $Z_{4,0}$ and $Z_{12,0}$
 - $Z_{6,0}$ and $Z_{24,0}$
 - $Z_{5,0}$ and $Z_{25,0}$
 - $Z_{10,0}$ and $Z_{14,0}$
 - $Z_{15,0}$ and $Z_{24,0}$
 - $Z_{12,0}$ and $Z_{28,0}$

- Investigate the intersections of the type $Z_{n,1}$ and $Z_{m,1}$ and in general $Z_{n,k}$ and $Z_{m,j}$. When are they empty? How many elements can they have?
- Express the following sets using the sets $C_{n,k}$: $Z_{2,0}$, $Z_{2,1}$, $Z_{5,0}$, $Z_{5,2}$, $Z_{10,0}$, $Z_{10,7}$, $Z_{25,0}$, $Z_{25,24}$, $Z_{100,1}$, $Z_{100,33}, \dots$

6.4.1 Hints and solutions

1. Inclusion and exclusion principle

Hints:

- Before solving the problem for three or more sets, it might be useful to try it for two sets. What restriction do we have for the number of elements of the intersection of two sets, if we know the number of elements of the individual sets?
- When we solve the problem for three sets, consider how many inputs do we need for the inclusion and exclusion formula. Which of those need checks?

Solution: For three sets, we need seven inputs, see the yellow cells in Figure 6.6: number of elements of the sets A , B and C , number of elements in their pairwise intersections $A \cap B$, $A \cap C$ and $B \cap C$ and the intersection of all three sets $A \cap B \cap C$. In this case, the output (indicated by green in Figure 6.6) is the number of elements of the union $A \cup B \cup C$ calculated using the formula $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$. Note that if we had $|A \cup B \cup C|$ as an input, we could calculate $|A \cap B \cap C|$ using the formula $|A \cup B \cup C| - |A| - |B| - |C| + |A \cap B| + |A \cap C| + |B \cap C|$.

L4												
	A	B	C	D	E	F	G	H	I	J	K	L
1												
2		A	B	C		A∩B	A∩C	B∩C		A∩B∩C		A∪B∪C
3												
4		10	20	20		7	6	11		5		31
5						OK	OK	OK		OK		OK
6						1	1	1		1		4

Figure 6.6: Calculation of number of elements in a union of three sets.

The number of elements of any intersection of sets has to be smaller than the number of elements of either of the two sets. Therefore we should use checks such as `=IF(F4<MIN(B4;C4);"OK";"wrong input")` for the pairwise intersections. Similarly, for the intersection of three sets, we can write `=IF(J4<MIN(F4;G4;H4);"OK";"wrong input")`. To evaluate all checks together, we can use numerical values instead of *OK* and *wrong input*, with 1 denoting that the inputs are fine and 0 denoting the cases when they are not. The final check can then be performed by multiplication

IF(F6*G6*H6*J6>0;"OK";"wrong input"). This check relies on the fact that if one or more of the smaller checks fail, the final check fails too.

Generalisation

Try making a worksheet in which we can input any seven of the eight quantities involved in the inclusion-exclusion formula for three sets. The worksheet then calculates the eighth quantity. Include all necessary checks.

2. Number sets

Let us denote $C_{n,k}$ the set of all natural numbers that have the digit k at the n^{th} position from the right. Let us denote $Z_{n,k}$ the set of all natural numbers that give remainder k when divided by n .

Hints:

- See also the hints provided in Section 6.3.1.
- We can start by writing out a column of consecutive natural numbers and then checking for every number whether it belongs to $Z_{n,k}$ for a given n and k . For this check we can use the remainder after division that is calculated using the Excel function MOD().
- Constructing the set $C_{n,k}$ is slightly more complicated and we should test it for small n first.
- It is possible to construct the set $C_{n,k}$ using functions that manipulate strings, e.g. MID() combined with LEN(), or again using the remainders after division with functions MOD() and QUOTIENT().
- Before constructing the $C_{n,k}$, try taking a 5-digit number and write five formulas in Excel each of which would extract one of the digits. What do the formulas have in common?

Solution: To construct the sets $Z_{n,k}$, we write out natural numbers in one column (e.g. column K in Figure 6.7) and compare their remainders after division by n (one of the inputs highlighted in yellow) with the number k (also highlighted in yellow). We use the formula =IF(MOD(K_i;n)=k;K_i;"") to display the numbers that meet the condition and hide those which do not. We can test this by using various inputs n and k .

To construct the sets $C_{n,k}$, we first try to figure out how to extract individual digits from a given number. Suppose we have a number 12345 in

O15										
	J	K	L	M	N	O	P	Q	R	
1		Numbers	n	k				C		
2			2	1						
3		100			0	0				
4		101			1	0				
5		102			2	0				
6		103			3	0				
7		104			4	0				
8		105			5	0				
9		106			6	0				
10		107			7	0				
11		108			8	0				
12		109			9	0				
13		110			10	1		110		
14		111			11	1		111		
15		112			12	1		112		

Figure 6.7: Finding the $Z_{n,k}$ numbers for given n and k .

A1. The formula $\text{=MOD}(A1;10)$ gives the last digit 5. To get the 4, we can first calculate $\text{=MOD}(A1;100)$ to get 45 and then perform a division $\text{=QUOTIENT}(\text{MOD}(A1;100);10)$ to get 4. The QUOTIENT function performs division with integers and throws away any remainders. To get the digit 3, we first calculate $\text{=MOD}(A1;1000)$ to get 345 and then $\text{=QUOTIENT}(\text{MOD}(A1;1000);100)$. This is the same formula as before, with one zero added at two places to account for the shift in the digit. Now we can write a general formula $\text{=QUOTIENT}(\text{MOD}(A1;10^n);10^{n-1})$, which is also shown in Figure 6.8.

	number	7852		852	8,52	8
	digit from the right	3				

Figure 6.8: Picking digits from a number.

To find the intersections of pairs of Z sets, we construct two of them next to each other, each similar to the one in Figure 6.7, and use a new column

to display only those numbers that appear in both green columns. Note that for union, we would display numbers that appear in either of the two green columns.

Chapter 7

Relations and functions

7.1 Lecture overview

7.1.1 Relations

Problem 7.1 Pairs I:

What relation may the following set represent?

$$D = \{[3, 6], [3, 9], [9, 18], [1, 2], [5, 10], [,], \dots\}$$

Solution: Looking at the pairs, we might notice the rule that the first number divides the second. This relation may represent divisibility.

Problem 7.2 Pairs II:

What relation may be defined by the following?

$$L = \{[3, 6], [23, 91], [9, 8], [111, 212], [55, 16], [,], \dots\}$$

$$[7, 64] \notin L, \quad [-45, 9] \notin L, \quad [435, 19] \notin L, \quad \dots$$

Solution: Considering both pairs that belong to L and those that do not, it seems that this could be a relation of two integers that have the same number of digits.

Problem 7.3 Triplets:

In a similar fashion as in the previous problems, we can define more complex

properties. What relation may the following set represent?

$$S = \{[1, 1, 2], [1, 3, 4], [3, 4, 7], [10, 20, 30], [4, 1, 5], [,], \dots\}$$

Solution: It is a relation of triplets of natural numbers. One of the possible answers is that *the third is the sum of the first two*.

$$S = \{[a, b, c] \in \mathbb{N} \times \mathbb{N} \times \mathbb{N}, a + b = c\}$$

Relation

Any subset of the Cartesian product $A_1 \times \dots \times A_n$ is a *relation* on a set of ordered n -tuples from $A_1 \times \dots \times A_n$. For example, relation K , two lines being parallel, $K \subset L \times L$, where L is the set of all lines in a plane. Another example is a parents-child relation $C \subset P \times P \times P$, where P is the set of all people. If the relation is *binary*, i.e. involving a Cartesian product of only two sets, we can use the notation aRb instead of $[a, b] \in R$.

Properties of relations

We call the relation $R \subset A \times A$

1. *Reflexive* iff $\forall a \in A : aRa$
2. *Transitive* iff $\forall a, b, c \in A : (aRb \wedge bRc) \Rightarrow aRc$
3. *Symmetric* iff $\forall a, b \in A : aRb \Rightarrow bRa$
4. *Antisymmetric* iff $\forall a, b \in A : (aRb \wedge bRa) \Rightarrow a = b$
5. *Trichotomous* iff $\forall a, b \in A : a = b \vee aRb \vee bRa$

Table 7.1 shows several relations and their properties. A relation can also be displayed as an oriented graph and its properties imply the orientation of graph edges. Illustration is shown in Figure 7.1.

Symmetric and antisymmetric

The relations in Table 7.1 are all either symmetric or antisymmetric. Can you find a relation that is neither? And can you find a relation which is both?

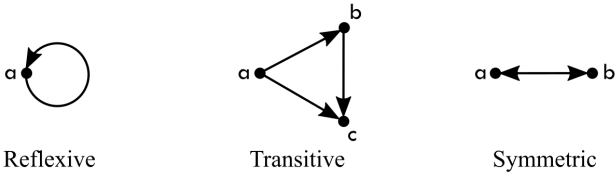


Figure 7.1: Relation is reflexive, if all vertices of its graph have a loop. It is transitive, if every $a \rightarrow b \rightarrow c$ path in its graph implies also $a \rightarrow c$ path. It is symmetric if all edges in its graph are bidirectional.

Relation	1	2	3	4	5
$\leq$ on $\mathbb{R} \times \mathbb{R}$	+	+	−	+	+
divisibility on $\mathbb{N} \times \mathbb{N}$	+	+	−	+	−
same remainder after division by 5 on $\mathbb{N} \times \mathbb{N}$	+	+	+	−	−
being parallel on $L \times L$	+	+	+	−	−
being perpendicular on $L \times L$	−	−	+	−	−
a is a descendant of b on $P \times P$	−	+	−	+	−
a lives with b on $P \times P$	+	+	+	−	−

Table 7.1: Examples of various relations. The + symbol indicates that the relation has a given property. The − symbol indicates that it does not have it.

Equivalence

A relation R on $A \times A$ is called an *equivalence* iff it is reflexive, transitive and symmetric. The equivalence relation splits the set A into *classes of equivalence*. Each of those classes contains one maximal set of elements of A that are in relation R with each other. For example, the relation *numbers that give the same remainder after division by 5* splits $\mathbb{N}$ into equivalence classes

$\{1, 6, 11, 16, \dots\}, \{2, 7, 12, 17, \dots\}, \{3, 8, 13, 18, \dots\}, \{4, 9, 14, 19, \dots\}, \{0, 5, 10, \dots\},$

which are called *residue classes*. The relation *lines being parallel* splits the set of lines into classes of parallel lines that are called *directions*. The relation *lives with* splits the people into households.

Linear ordering

A relation R on $A \times A$ is called a *linear ordering* or *total ordering* iff it is reflexive, transitive, antisymmetric and trichotomous. An example of a linear ordering is " $\leq$ " on $\mathbb{R} \times \mathbb{R}$. We draw the real numbers linearly on a line, real axis, ordered by their magnitude.

Partial ordering

A relation R on $A \times A$ is called a *partial ordering* iff it is reflexive, transitive and antisymmetric. An example of a partial ordering is *to be a divisor* on $A \times A$, where $A = \{1, 2, 3, 4, 6, 12\}$ is the set of all divisors of number 12. The graphical representation in this case is the Hasse diagram, as depicted in Figure 2.1.

Note that in case of linear ordering, we put the small numbers to the left and the large numbers to the right (on the real axis). With Hasse diagram, we put the *smaller* elements below and *larger* elements up. While these two conventions are not the same, the basic underlining property is that there is a clear progression from the small to the large, no matter what the characteristic direction is.

7.1.2 Mappings and functions

Mapping

A *mapping* is a binary relation R such that for each a we have at most one b such that $[a, b] \in R$. We say that the relation R maps the input a onto the output b . Formally, a mapping $R : A \rightarrow B$ is a relation $R \subset A \times B$ such that $([a, b] \in R \wedge c \neq b) \Rightarrow [a, c] \notin R$. Thanks to this property, we can write $R(a) = b$ or $f(a) = b$ instead of $[a, b] \in R$.

We call the set $\{a \in A : \exists b \in B, f(a) = b\}$ a *domain* and denote it by $D(f)$. We call the set $\{b \in B : \exists a \in A, f(a) = b\}$ a *range* and denote it by $H(f)$.

Function

Sometimes mappings are also called functions. We reserve the term *function* for those mappings which work with number sets, such as

$$\begin{aligned} f : \mathbb{N} &\rightarrow \mathbb{N}, f(n) = n^2 \\ f : \mathbb{N} &\rightarrow \mathbb{N}, f(n) = \sqrt{n} \\ f : \mathbb{R} &\rightarrow \mathbb{R}, f(x) = x^2 \\ f : \mathbb{R} &\rightarrow \mathbb{R}, f(x) = \sqrt{x} \end{aligned}$$

Injective, surjective and bijective mappings

A mapping $f : A \rightarrow B$ is *injective*, also called *one-to-one*, when

$$\forall a, b \in A, a \neq b \Rightarrow f(a) \neq f(b).$$

A mapping $f : A \rightarrow B$ is *surjective*, also called *onto*, iff

$$\forall b \in B \exists a \in A, f(a) = b.$$

A mapping $f : A \rightarrow B$ which is both injective and surjective is called a *bijection*.

Cardinality of a set

Now the definition of cardinality of a set in Section 6.1.4 hopefully makes more sense. We say that two sets, A and B , have the same cardinality and we write $|A| = |B|$ iff there exists a bijection between A and B . For example the set of even natural numbers E and the set of natural numbers $\mathbb{N}$ have the same cardinality, because there exists a bijection

$$f : E \rightarrow \mathbb{N}, f(n) = \frac{n}{2}.$$

Inverse mapping

For injective mapping f we always have an *inverse mapping* f^{-1} . It is a mapping which determines the input from the output, i.e. if $f(a) = b$, then $f^{-1}(b) = a$. We also have $D(f^{-1}) = H(f)$ and $H(f^{-1}) = D(f)$. Similarly, we have also inverse functions.

Composition of mappings

The *composition of mappings* is an operation which takes two mappings $f(x)$ and $g(x)$ and produces $h(x) = f(g(x))$, provided that the range of $g(x)$ is a subset of the domain of $f(x)$. Consequently, we have also composition of functions, for example, for $f(x) = 2x + 3$ and $g(x) = x^2$ we get composite functions $h(x) = f(g(x)) = 2x^2 + 3$ and $s(x) = g(f(x)) = (2x + 3)^2$.

We can also perform a composition of a mapping and its inverse and we have $\forall x \in D(f) : f^{-1}(f(x)) = x$.

7.2 Questions

1. State a few different ways how we can define relations. Include examples of each.
2. Relation defines a certain property for a group of elements. How many elements can there be in these groups?
3. List a few elements of the relation *sum of natural numbers* which is defined using sets. How does it compare to the relation *sum of real numbers*?
4. List a few elements of the relation *difference*, defined as a suitable subset of a Cartesian product.
5. List a few elements of the relation *divisibility of natural numbers* which is defined using sets.
6. When is a relation called a binary relation? Is the relation defining a sum of two real numbers a binary relation?
7. Determine if the relations $=$, $<$, *a divides b*, *a has a common divisor with b*, defined on $\mathbb{N} \times \mathbb{N}$ are reflexive, transitive, symmetric, antisymmetric, trichotomous (complete)?
8. Find a binary relation (give an example) which is not reflexive, one which is not transitive, one which is not symmetric, one which is not antisymmetric and one which is not trichotomous.
9. Is the relation of two lines *being perpendicular* (in plane, in 3D space) transitive?
10. Is the relation of two lines *being not parallel* (in plane, in 3D space) transitive?
11. Is the relation of two lines *being parallel* (in plane, in 3D space) transitive?
12. What are the properties of an equivalence?
13. What are the properties of a partial ordering?
14. What are the properties of a complete (linear) ordering?
15. State the definition of a mapping. How is it different from relation? Also give an example.

16. What is the domain of a mapping?
17. What is the range of a mapping?
18. Define injective mapping. Give an example. Also give an example of a mapping that is not injective.
19. Define surjective mapping. Give an example. Also give an example of a mapping that is not surjective.
20. Define bijective mapping. Give an example. Also give an example of a mapping that is not bijective.
21. What kind of mapping is used to determine whether two sets have the same cardinality (or number of elements)? What is the relation between the terms *cardinality* and *number of elements*?

7.3 Problem set

1. Relations on the set of people

Consider the following relations R on the set of all people in Slovakia and determine which of the typical properties of relations do they have:

- Reflexivity (xRx)
- Transitivity (if xRy and yRz then also xRz)
- Symmetry (if xRy then yRx)
- Antisymmetry (if xRy and yRx then $x = y$)
- Trichotomy (for distinct x and y we have either xRy or yRx)

If you get an equivalence, describe the equivalence classes. If you get an ordering (partial, linear), describe it in *human* characteristics. Characterise people who can and people who cannot be in these relations.

- a) x is a spouse of y .
- b) x is a child of y .
- c) x is a descendant of y .
- d) x and y have the same first two digits in their Slovak birth ID number.
- e) x has the same first name as y .
- f) x is a sibling of y .
- g) x is a sister of y .

2. Relations on numbers

Examine the relations and find their typical properties:

- a) Two integers (numbers in $\mathbb{Z}$) are in relation R_2 , if their difference is even.
- b) Two integers (numbers in $\mathbb{Z}$) are in relation R_5 , if the last digit of their difference is 0 or 5.
- c) Two integers (numbers in $\mathbb{Z}$) are in relation R_k , if their difference is a multiple of k .
- d) Two integers (numbers in $\mathbb{Z}$) are in relation R_{10} , if their difference is at most 10.
- e) Two natural numbers (from $\mathbb{N}$) are in relation D , if the first divides the second.

- f) Two real numbers are in relation C , if their difference is an integer.

3. Relations in geometry

Examine the relations and find their typical properties:

- a) Two lines are in relation R , if they cross. Solve in 2D case and then in 3D case.
- b) Two lines are in relation R , if their intersection is either an empty set or one of these two lines. Solve in 2D case and then in 3D case.
- c) Two lines are in relation R , if they are perpendicular. Solve in 2D case and then in 3D case.
- d) Two triangles are in relation R , if they have the same set of angles (*same angles*).
- e) Two triangles are in relation R , if their have the same set of side lengths (*same sides*).

4. Hasse diagram

- a) Consider the set $\{a, b\}$ and all its subsets with the relation *to be a subset*. Draw the corresponding Hasse diagram. Do this also for a set with 3 elements, with 4 elements.
- b) Draw the Hasse diagram of the relation x divides y for all divisors of number 30, 24, 48, 36, 210.
- c) Which diagrams from a) and b) are the same?

5. Functions and their properties

Check which of the following are functions $\mathbb{R} \rightarrow \mathbb{R}$. For those that are, find their $D(f)$, $H(f)$ and typical properties. Which are injective, surjective and which are bijective? Provide a justification.

- a) $f(x) = 2x + 1$
- b) $f(x) = \sqrt{x}$
- c) $f(x) = x^2 + 3x + 1$
- d) $f(x) = 3^x$
- e) $f(x) = \tan(x)$
- f) $f(x) = \sin(x)$

6. Relations vs. functions

These equalities define relations for real numbers $[x, y]$. For each case, write five pairs that belong to the relation and five that do not. Find out which of the relations are also functions $\mathbb{R} \rightarrow \mathbb{R}$. For each case, examine their typical properties.

a) $2x + 3y = 5$

b) $-x^2 + y = 4$

c) $3x - y^2 = 5$

d) $3x - y^3 = 5$

e) $\sin(x) - y = 0$

f) $y - 2^x = 1$

7.3.1 Hints and solutions

1. Relations on the set of people

Consider the following relations R on the set of all people in Slovakia and determine which of the typical properties of relations do they have.

Hints:

- When considering mathematical objects and properties in *real life examples*, we have to accept that there will be some inaccuracies and things that need clarification. If necessary, we can specify the problem more precisely and restrict the properties to e.g. citizens of particular state, adults only, etc.
- If the relation has a given property, it has to be true for all pairs of people. If not, it is sufficient to find one counterexample.
- If the relation resembles some kind of equality, it is often reflexive, symmetric and transitive. Similarly, if it does some form of comparison, it often has the properties of linear or partial ordering.
- Sometimes it is easier to imagine certain documents, e.g. birth certificates or records in databases, instead of the people. For example, x a y are spouses, if there exists a marriage certificate. A birth certificate defines the parent-child relation. We certainly do not mean to judge any form of family arrangements and preferences, this is done solely for simplification purposes when working with human relations as mathematical objects.
- The properties that are stated in the form of an implication hold also in case if the assumption is not true. Therefore it can (surprisingly) happen that they sometimes formally hold even if there is no such case that can happen.

Solution: d) Which basic properties does the relation x and y have the same first two digits in their Slovak birth ID number have?

If we restrict ourselves to the current Slovak citizens, every one of them has a unique birth ID number and we know the meaning of its digits. The first two digits correspond to the last two digits of the year when the person was born. To avoid complications, we add an assumption that we do not consider people over 100 years old. (Why?) Now we can examine the individual properties.

Is this relation reflexive? If we had two copies of an ID card of one person, both would show the same first two digits of the birth ID number. Therefore, even though it sounds somewhat peculiar, a person has the same first two digits of the birth ID number as he or she has. Thus the relation is reflexive.

To examine transitivity, consider three people: x , y and z . If x and y share the same first two digits (in the correct order of course), and also y and z have the same first two digits, then x and z have to have the same first two digits, too. Here we use the more general property that the equality of numbers is transitive. If we wanted to prove this property, we could use the fact that if $a = b$, then $a - b = 0$. Therefore

$$(a - b = 0 \wedge b - c = 0) \Rightarrow a - c = a - b + b - c = (a - b) + (b - c) = 0.$$

Similarly, we can show that the relation is symmetric. If x has the same first two digits as y , then y has to have the same first two digits as x . In other words,

$$a - b = 0 \Rightarrow b - a = 0.$$

Since the relation is reflexive, symmetric and transitive, it is an equivalence. It factors the set of people into classes. In each class, we have all the people who have the same first two digits of the birth ID number. With respect to the assumption made at the beginning, these equivalence classes split the Slovak citizens younger than 100 years into cohorts of people born in the same calendar year.

Now we check the last two properties. If I check mine and my classmate's birth certificates and find out that we have the same first two digits of the birth ID number (the relation is symmetric), this does not mean that we are the same person. Clearly, we are two distinct people and thus the relation is not antisymmetric.

This is also the underlying reason why the relation is not trichotomous. Since it is an equivalence, we can also use a different reasoning. If it was true for any pair of people that they have the same the same first two digits of the birth ID number (in any order, since the relation is symmetric), then they would belong to the same equivalence class. Then it would be true that all people were born in the same year, which is clearly not true. Thus the original assumption was false and the relation is not trichotomous.

In this case we do not have a linear or partial ordering, since the relation is not antisymmetric.

c) Which basic properties does the relation *x is a descendant of y* have?

Again, we need to clarify the problem before we start working on it. Let us say that a person *x* is a descendant of a person *y* when there exists a direct sequence of relatives of *x*, $p(x)$, $p(p(x))$, $\dots$, $p(p \dots p(x) \dots) = y$, where $p(a)$ is a parent of *a*. If necessary, we can think of this sequence as a stack of corresponding birth certificates. It is useful to notice that a parent is older than their child and after repeated application of this rule we get that a descendant is younger than his or her ancestor.

Consequently, the relation is not reflexive. Nobody (and certainly not everybody) can have a birth certificate on which they are also stated as their own parent. If the *descendancy* could be achieved indirectly, by a longer sequence of successive birth certificates, we would violate the rule that a parent is older than their child. Since the relation is not reflexive, we can immediately say that it cannot be an equivalence and it cannot be an ordering either.

It can be easily shown that this relation is transitive. If *x* is a descendant of *y* then we have a corresponding stack of birth certificates connecting the child *x* to the ancestor *y*. Similarly, we have a stack of birth certificates connecting the child *y* to great-great...parent *z*. If we put the first stack on top of the second, we get a valid sequence connecting *x* to *z*.

The relation is not symmetric. We can see this through the age property. If *x* is a descendant of *y*, he or she is younger than *y*. Then certainly *y* is not younger than *x* and thus cannot be their descendant. Therefore in any case (and consequently, certainly not for all cases) we do not have the implication if *x* is a descendant of *y* then *y* is a descendant of *x*.

As a reminder, a relation is antisymmetric iff $\forall x, y \in A : (xRy \wedge yRx) \Rightarrow x = y$. At the first glance, it might seem that the *descendancy* is not antisymmetric for the same age-related reason we used before. If *x* is a descendant of *y*, then *x* is younger than *y*. Consequently *y* certainly cannot be younger than *x* and therefore is not a descendant of *x*. When we think about this more carefully, we have to realise that this property is defined as an implication. We have shown that the premise of the implication is never true (we cannot have *x* as a descendant of *y* and *y* a descendant of *x* at the same time), therefore we consider the implication to be true even

if we cannot find any pair of people that would demonstrate this property. This relation is antisymmetric. We can come to the same conclusion if we use an equivalent definition of antisymmetric relation: *there are no x and y such that $xRy \wedge yRx \wedge x \neq y$.*

Finally, we consider the completeness of the relation. If it held for any two people x and y , it would have to be true that either x is a descendant of y or y is a descendant of x . This is clearly not true for most people, consider for example two classmates or two siblings.

Symmetry vs. antisymmetry

Once we know that a relation is not symmetric, can we say that it is antisymmetric and vice versa?

Can a relation be symmetric and antisymmetric at the same time?

Can it be neither?

2. Relations on numbers and their typical properties

Hints:

- When considering relations on numbers, we work with precisely defined properties and our thinking and answers are more precise than when talking about human relations. It is very important to carefully consider the number set on which the relation is defined and make sure that we fully understand the property or calculation which determines that a pair of numbers is in a relation.
- We can consider the mathematical implications or known properties of the relation in question. Can the property be simplified or written differently?
- Also here, if a property holds, it has to hold for all pairs of numbers. To show that it does not hold, it is sufficient to find one counterexample.
- For each relation, try to find specific pairs of numbers which belong to it and pairs which do not. Do these pairs of numbers have something in common?
- Use the specific pairs of numbers to test the properties of relations. Does the pair $(1,1)$ have the property? If the pair $(1,2)$ has the property, does also the pair $(2,1)$ have it? What pairs need to be checked for the other properties?

- We can consider the properties of basic relations on numbers, such as $=$, $<$ or $\leq$, as given. If we wanted to prove these in detail, we would use the fact that the relation $a = b$ can be replaced by $a - b = 0$ and $a < b$ by $a + x = b$, where x is a positive (real) number.

Solution: b) Which basic properties does the relation R_5 have?

When considering this relation, we are going to use the properties of divisibility as discussed in Chapter 2, but we need to extend them to negative numbers as well.

First we note that when we subtract a number from itself, we get zero. Thus the relation is reflexive and for any integer k we have kR_5k , since $k - k = 0$.

Now we consider various pairs of numbers. If one of the two numbers is a zero, we see that it is in this relation with numbers $5, 10, 15, \dots$ and also $-5, -10, -15, \dots$. When we start subtracting these numbers from one another, we see that their differences also end in 0 or 5. Thus here we have a group of numbers in which all pairs are in relation R_5 . This group contains all (integer) multiples of 5.

We can construct a similar group in which there are numbers that give a remainder 1 after division by 5. In other words, these are multiples of 5 plus 1.

Remainders after division for negative numbers

Think about how the remainders work for negative numbers. How are they evaluated in various programming languages and software tools?

Next we take numbers $-14, -9, -4, 1, 6, 11, 16, \dots$. Also here we can easily verify that their differences end in either 0 or 5 and thus they form another group of numbers bound by the relation R_5 .

If we read the definition of the relation carefully, we might realise one well-known property. We know that a number is a multiple of 5 iff it ends in 0 or 5. Thus, all numbers whose difference is divisible by 5 are in relation R_5 , i.e. $kR_5l \Rightarrow k - l = 5p$ for a suitable integer p . In Chapter 2 we also saw that if the difference of two numbers is a multiple of 5, they are congruent modulo 5, i.e. give the same remainder after division by 5. This confirms the groups of numbers that we constructed previously. With these properties in mind, we are ready to check the remaining properties.

To check for transitivity, we start by assuming kR_5l and lR_5m . This implies $k - l = 5p$ and $l - m = 5q$ for suitable p, q . Then we have $k - m = k - l + l - m = (k - l) + (l - m) = 5p + 5q = 5(p + q)$. This means that also the difference $k - m$ is a multiple of 5, therefore kR_5m . Hence, the relation is transitive.

To examine symmetry, it is sufficient to realise that if $k - l = 5p$ is a multiple of 5, also $l - k = -5p$ is a multiple of 5. The relation is thus symmetric.

Since R_5 is reflexive, symmetric and transitive, it is an equivalence. We have already discovered its equivalence classes. They are the congruence (residue) classes modulo 5.

The last two properties do not hold for this relation. For example, both differences $7 - 2$ and $2 - 7$ end in 5 but $2 \neq 7$. Therefore R_5 is not antisymmetric.

And since for example neither of the differences $2 - 1$ and $1 - 2$ is divisible by 5, the relation is not trichotomous.

e) Which basic properties does the relation D have?

In this case, we need to examine the relation of divisibility on natural numbers. We have nDm (or using a standard notation $n|m$) iff m is a multiple of n . Therefore $m = np$ for a suitable natural number p .

If we choose p to be 1, we have $n = n \cdot 1$ for any natural number n . Therefore, every number is divisible by itself (the ratio is 1) and the relation D is reflexive.

In order to consider transitivity, we start from the assumptions that lDm and mDn . Then we have $m = pl$ and $n = qm$. Consequently, $n = qm = qpl$, therefore n is a multiple of l and we have lDn . The relation D is transitive.

The relation D is not symmetric. While it is true that $2|4$, certainly it is not true that $4|2$ (2 is not an integer multiple of 4). We could find many more counterexamples like this one, but one is sufficient to disprove this property.

Conversely, the antisymmetry looks promising. If mDn and nDm , then $m = pn$ and also $n = qm$ for suitable p and q . From there we get $m = pn = pqm$. In natural numbers this is only possible if $pq = 1$ and thus $p = q = 1$. This means that $n = m$ and that the divisibility relation is antisymmetric.

The relation D is not trichotomous. For any two prime numbers, e.g. 5 and 7, we neither have $5D7$ nor $7D5$.

Considering the previously discussed properties together, we find out that D is a partial (not linear) ordering. In certain sense, it can arrange natural numbers in an order, even though this order does not place them on a *line* (like $\leq$ does). For more information, see also the solution to Problem 4 in this section.

3. Relations in geometry

Examine the relations and find their typical properties:

- a) Two lines are in relation R , if they cross. Solve in 2D case and then in 3D case.
- b) Two lines are in relation R , if their intersection is either an empty set or one of these two lines. Solve in 2D case and then in 3D case.
- c) Two lines are in relation R , if they are perpendicular. Solve in 2D case and then in 3D case.
- d) Two triangles are in relation R , if they have the same set of angles (*same angles*).
- e) Two triangles are in relation R , if they have the same set of side lengths (*same sides*).

Hints:

- We should start by looking for pairs of objects that have or do not have the given geometrical property.
- The geometric properties of pairs of objects are static and often can be drawn in a picture. Therefore we can expect that in general they are going to be symmetric, independent of the order in which we state the relation for them.
- In problem a), the intersection means that the lines have exactly one point in common. Therefore identical lines, parallel lines or skew lines in 3D are not in this relation. In general, the properties of relations of two lines in 2D and 3D are different, but for this particular relation, we get the same, albeit modest, set of properties in both spaces.
- Problem b) examines the complementary property. This relation puts together lines that are either parallel or identical. This suggests that we are dealing with equality of something, and thus an equivalence.

Note that in 3D, we also have skew lines which are not the same as parallel lines. Two lines in 3D space are parallel, if there exists a plane in which they are parallel. While skew lines are not parallel, they also do not have an intersection. Consequently, the answer to this question is going to be different in 2D and in 3D.

- *Being perpendicular* is a special case of the relation described in a). The properties of this relation should thus be *weaker*. Unlike in 3D space, in plane there is a strong connection between being perpendicular and being parallel.
- For problem d) we should recall that triangles with the same angles are similar. The *same size of angles* hints at equivalence.
- In problem e) we have the same length of sides, which points towards equivalence even more strongly than the previous case. We should remember the *sss* comparison criterion.

Solution: Here we provide the answers which you can compare to the results of your own work.

- a) In a plane, this relation is only symmetric. It is not reflexive, transitive, antisymmetric nor trichotomous. Consequently, it is neither an equivalence nor any kind of ordering. Its properties in 3D space are the same.
- b) In a plane, this relation corresponds to two lines being parallel (or identical). It is reflexive, symmetric and transitive, therefore it is also an equivalence. Its equivalence classes are groups of parallel (or identical) lines, which are sometimes called *directions*. The relation is neither antisymmetric nor trichotomous.

For lines in 3D, the situation is different. The reason is that we can also have skew lines. For those, the relation is not transitive. Therefore in 3D, it is only reflexive and symmetric.

- c) Two perpendicular lines are a special case of relation in a). In both 2D and 3D, this relation is only symmetric. In plane, we know that if $p \perp r$ and $r \perp s$, then $p \parallel s$. Being perpendicular in 3D space is a slightly more complicated property.
- d) Similarity of triangles is an equivalence (thus a reflexive, symmetric and transitive relation) and its equivalence classes are the groups of similar triangles. The relation is neither symmetric nor trichotomous.

- e) In this case, we have congruence of triangles. The properties of the relation (except the equivalence classes) are the same as in case of the similarity of triangles.

4. Hasse diagram

Hints: Hasse diagram is a useful depiction of elements of a set that are in partial ordering. (In case of linear ordering, the elements of the set can be drawn on a line, where we draw the smaller elements to the left of the larger elements. The properties of linear ordering guarantee that it is always possible.) There are few simple rules for drawing a Hasse diagram of a relation R :

- Before drawing the diagram, we write out all the elements that belong to this relation. If xRy , the vertex representing x is drawn below the vertex representing y . We draw the *small* elements in the lower part of the diagram, the *large* elements in the upper part of the diagram and the rest *in the middle*.
- If xRy , there is a connection between the vertex x with the vertex y . This connection might, but does not have to correspond to an edge connecting these two vertices.
- The diagram respects transitivity. Therefore, there are no redundant edges. If xRy and yRz , we do not draw an xz edge. The edges xy and yz are sufficient for capturing the fact that x is also in relation with z . More precisely, we only draw the edge xy that denotes the xRy relation, if it cannot be replaced by a pair of edges xz and zy , i.e. relations xRz and zRy . If we have inadvertently already placed such an edge in the diagram, we should later erase it and replace it with the relevant pair of edges that go through the element z .

Solution:

- a) Consider all subsets of the set $\{a, b, c\}$. We know that the subset defines a partial ordering, it should thus be possible to draw the Hasse diagram.

There are eight subsets of the set $\{a, b, c\}$: $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$. The *smallest* one is the $\emptyset$, which we draw at the bottom of the diagram. Immediately larger, without any *middlemen* are one-element subsets $\{a\}, \{b\}, \{c\}$. The edges connecting them to the $\emptyset$ cannot be replaced by any other pairs. The next *layer*

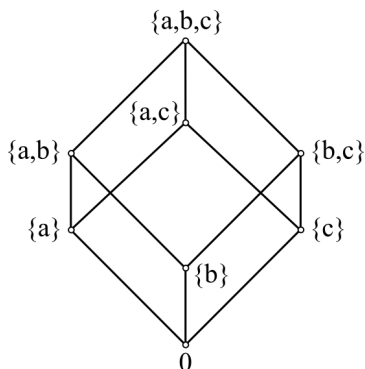


Figure 7.2: Hasse diagram for subsets of the set $\{a, b, c\}$.

comprises the two-element sets $\{a, b\}$, $\{a, c\}$, $\{b, c\}$ and the top layer is the *self-subset* $\{a, b, c\}$.

Try drawing your own diagram and comparing it with Figure 7.2. If there are differences, find out which edges are redundant and which are missing. If this figure reminds you of another picture, it is not a coincidence.

- b) To draw the diagram of relation $x|y$ for x and y from the set of divisors of the number 30, we start by a prime factorisation $30 = 2 \cdot 3 \cdot 5$. From Section 2.1 we know that the divisors of the number 30 are 1, 2, 3, 5, 6, 10, 15 and 30. In terms of divisibility, the *smallest* of these is the number 1 and the *largest* is the number 30. The rest of the divisors can be split into a group of primes 2, 3, 5 and their pairwise products 6, 10, 15. These two groups correspond to the layers and vertices of the relation D in the Hasse diagram.

At this point, draw your own diagram, compare it with Figure 7.3 and explain any differences that might appear.

Now compare the two diagrams in this problem. Up to the vertex notation, we see the same picture. This demonstrates one of the advantages of using the Hasse diagram. Partial orderings from various mathematical topics in fact have the same structure. Many properties obtained by examining subsets apply to divisibility and vice versa.

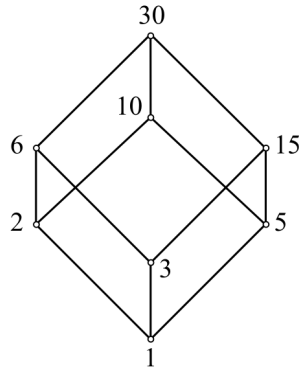


Figure 7.3: Hasse diagram for divisors of the number 30.

5. Functions and their properties

Hints:

- In case of mappings, the typical properties of relations (reflexivity, transitivity, etc.) are not so important. The only reflexive mapping on $\mathbb{R} \times \mathbb{R}$ is the function $f(x) = x$. While it is unique because it is the only one, it is neither very useful nor interesting.
- Much more important is for example the ability of mappings to uniquely map inputs to outputs and therefore in the case of mappings, we typically investigate whether they are injective, surjective and bijective.
- We can recognise and observe the properties of mappings from real numbers to real numbers from their graphs in Cartesian coordinate system. We return to this topic again in Chapters 10 and 11 but already now we should start thinking about where in the graph do we see $D(f)$ and $H(f)$ and how does the fact that f is injective, surjective, bijective manifest in the graph.
- To distinguish a mapping from a relation, we need to carefully consider whether we get a unique output $f(x)$ for every (reasonable) input x . In order to do that, we should focus on the formula of $f(x)$. Can it give the same answer for different values of x ? Certainly x^2 has the same value as $(-x)^2$, $|x|$ is the same as $|-x|$, etc. Therefore

we should suspect formulas that contain even powers of x , absolute values, etc.

- Similarly, we investigate the formulas of $f(x)$ when we want to determine the domain and the range of f . If we do not *artificially* restrict the range of arguments x (e.g. possible range of income for calculation of income tax or non-destructive acceleration values for calculation of rocket fuel consumption, etc.) it is important to find which values of x cannot enter the formula. These are the values for which the expression is not defined. The suspicious places are denominators (they cannot be zero), even roots (for which x ?), logarithms and exponential functions (why?), undefined values of tangent and cotangent (what are they?), etc.
- Determining $D(f)$ and $H(f)$, although in a slightly different sense, is also important in computer science and in writing source codes. The domain is very often related to the declaration of data types and admissible inputs for algorithms and the range to values used for saving and output. If we do not define them correctly, we might end up with inappropriate data types, which may lead to surprising outcomes for the results of calculations, e.g. the number -1 cannot be stored in an *unsigned int* variable and an attempt to do so will instead produce an unexpectedly large positive number. It is a serious problem when a program formally performs a calculation and returns a value even in case when the solution does not make sense, e.g. Heron's formula returns some value of triangle area even if the inputs are negative lengths of sides. Another problem is incorrect estimate of possible range of outputs, which might lead to overflow of used variables or display of absurd reports.

Solution:

- a) To find the domain of $f(x) = 2x + 1$ we need to think about which real values of the input (argument) x would make it impossible to calculate the output value $2x + 1$. We can perform doubling and adding 1 for all real numbers, therefore $D(f) = \mathbb{R}$.

We get the same answer for the range $H(f) = \mathbb{R}$ but the reasoning is slightly more complicated here. How do find out whether all real numbers can be the outputs of calculation $2x + 1$ for some x ? We can try a few cases. Can the answer be 123? If we decrement it by 1, we get 122 and the half of that is 61. Therefore we know that

$f(61)$ is 123. We can repeat these steps for different values of x or generalise it instead. Simple transformation of the original function gives us $x = (y - 1)/2$. Since this calculation can also be performed without problems, we see that for every real y we can find an x . This also gives us the answer to the question whether $f(x)$ is surjective.

For a function to be injective, it has to meet another, in this particular case a very intuitive, criterion: when we substitute two different values of x into $2x + 1$, we need to get two different answers. The simplicity of the formula suggests that this could be true. We prove it by assuming that for two real numbers u and v we get the same output values $2u + 1 = 2v + 1$. If we subtract 1 from this equation and divide by 2, we get $u = v$. Since this results in two equal numbers, they had to be equal already at the time of input. This is the same as saying (see Chapter 4) that we get different outputs for different inputs. The function is thus injective.

Since f is both injective and surjective, it means it is also a bijection.

The steps and analysis are very similar for other functions, therefore we only show brief results here:

- b) For function $f(x) = \sqrt{x}$ we have $D(f) = H(f) = \mathbb{R}_0^+$. This means it is not surjective and consequently not bijective either. It is injective though, because square roots of distinct non-negative real numbers are distinct non-negative numbers. With respect to this problem and working with square roots in general, it is useful to remember that there is a difference between the solution of the equation $x = \sqrt{9}$ and the solution of the equation $x^2 = 9$.
- c) The function $f(x) = x^2 + 3x + 1$ has $D(f) = \mathbb{R}$ and $H(f) = \langle -5/4, \infty \rangle$. The point $[-3/2, -5/4]$ is a vertex of the parabola representing this mapping. $H(f) \neq \mathbb{R}$, therefore this function cannot be surjective and consequently cannot be bijective. Like all other quadratic functions, it is also not injective, since for example $f(-2) = f(-1)$. The reason for this is the symmetric shape of the parabola, which is the graph of this function.
- d) The function $f(x) = 3^x$ is exponential and we return to it in Chapter 11. At this point it is sufficient to draw it in a suitable software and deduce the properties from the graph. We should see $D(f) = \mathbb{R}$ and $H(f) = \mathbb{R}^+$. The function is injective (why?). It is not surjective ($H(f) \neq \mathbb{R}$) and therefore it is not bijective.

- e) We will also return to the function $f(x) = \tan(x)$ later, but now we can analyse its properties from its graph or from its definition $\tan(x) = \frac{\sin(x)}{\cos(x)}$. Since the expression is a fraction, this implies a restriction on the domain, because we cannot calculate the values of $\tan(x)$ for zero points of $\cos(x)$ in the denominator. $D(f)$ is therefore the set of real numbers except the odd multiples of $\pi/2$, i.e. $D(f) = \mathbb{R} - \{(2k+1)\pi/2; k \in \mathbb{Z}\}$. The range is the whole $\mathbb{R}$, because close to the asymptotes in undefined points the graph of the function $\tan(x)$ *escapes* towards infinity. The function is periodic which implies that it is not injective (why?) and therefore it is not bijective.
- f) For $f(x) = \sin(x)$ we have $D(f) = \mathbb{R}$, $H(f) = \langle -1, 1 \rangle$, the function is neither surjective nor injective (for each multiple of π we get the same value $\sin(x) = 0$), therefore it is not a bijection either.

6. Relations vs. functions

Hints:

- In the first step we want to distinguish mappings from general relations that are not mappings. This is done using the criterion whether for each value of the input variable, here denoted by x , exists only one or more output values, here denoted by y . To make this distinction, we first transform the expression into the form $y = \text{formula}(x)$. If there is only one such y , we have a mapping, if there are more, we have a general relation.
- Once we distinguish the general relations and mappings, we can decide which properties we are going to examine. For relations that are not mappings, we investigate whether they are reflexive, symmetric, etc. For mapping we focus on whether they are injective, surjective and bijective.
- Similarly to the previous problems, it is useful to first look at examples and counterexamples of specific pairs of real numbers that satisfy or do not satisfy the given formula.

Solution:

- a) $2x + 3y = 5$

In this case we have a mapping. For example for $x = 3$ there exists a unique $y = -1/3$, which satisfies the equality. One example is not sufficient to claim that we have a mapping, but after expressing y as

$y = (5 - 2x)/3$ we get a one-to-one formula for calculation of output y for any input x . We can either draw the graph or manipulate the expression into $x = (5 - 3y)/2$ to realise that the mapping is injective, surjective and bijective. $D(f) = H(f) = \mathbb{R}$.

b) $-x^2 + y = 4$

Here we could again start with testing a few $[x, y]$ pairs. Then manipulate the equation into $y = 4 + x^2$ and realise that for every x we get one output value y . Therefore we again have a mapping, its $D(f) = \mathbb{R}$, but $H(f)$ is only $\langle 4, \infty \rangle$. This means the mapping is not surjective. Neither it is injective since the pairs $+1$ and -1 , $+2$ and -2 (and x and $-x$ in general) result in the same value of y .

c) $3x - y^2 = 5$

In this case we have $y^2 = 3x - 5$. For $x = 7$ this means $y^2 = 3 \cdot 7 - 5 = 16$. There are two values that satisfy this: $y = 4$ and $y = -4$. Therefore this relation contains both $[7, 4]$ and $[7, -4]$ and thus it is not a mapping. It does not have any of the typical properties of the relations, which can be verified by finding specific counterexamples in each case.

d) $3x - y^3 = 5$

Compared to the previous problem, we are looking at a different root now. The cube function does not have any issues with the sign and distinguishes between positive and negative numbers. It is injective and its inverse – a cube root – is defined on the whole set $\mathbb{R}$. After few steps we get $y = \sqrt[3]{3x - 5}$ which gives a unique output y for every input x . Therefore this is a mapping with $D(f) = H(f) = \mathbb{R}$. It is both surjective and injective (and therefore also bijective) as can also be seen from the graph.

e) $\sin(x) - y = 0$

We can easily change the expression into $y = \sin(x)$. Therefore we now analyse the properties of the mapping $\sin(x)$. We know that it is well defined on the whole $\mathbb{R}$. However, the range is only the interval $\langle -1, 1 \rangle$, therefore the mapping is not surjective and consequently it is not bijective. Moreover, it is not even injective, since it is periodic and for example all integer multiples of π give the same output value $\forall k \in \mathbb{Z} : \sin(k\pi) = 0$.

f) $y - 2^x = 1$

In this case we are working with an exponential function $y = 2^x + 1$.

We return to it in more detail in Chapter 11. Here we just state that $D(f) = \mathbb{R}$. $H(f) = (1, \infty)$. Look at the graph to determine that the function is injective, but not surjective (and therefore not bijective).

7.4 Lab problems

1. Infinite hotel

Suppose we have an infinite hotel and the rooms are numbered $1, 2, 3, 4, \dots$. All the rooms are single-occupancy and if there is a guest staying in the room, he or she remembers the room number. At the beginning, infinitely many of the rooms are occupied, but some may still be free.

At the reception, there are new guests waiting to check-in. Sometimes they come from a next-door infinite hotel and they remember the room number they had there. The task is to accommodate everyone in such a way that both the existing and the new guests have rooms and there are no rooms left empty.

Give the guests who already stay in the hotel an instruction (a suitable function) they can use to calculate their new room number (they will then move into this room). Also give an instruction (a suitable function) to the people waiting at the front desk to calculate the new room number from their old room number.

Create an Excel sheet in which one column corresponds to the infinite hotel and another column to a line at the front desk. In the next column, apply the functions that tell them where to move. Finally, check and show that in every room there is exactly one guest.

Consider these situations:

current guests	new guests
$1, 2, 3, 4, 5, \dots$	$1, 2, 3, \dots, 9, 10$
$1, 2, 3, 4, 5, \dots$	$1, 2, 3, \dots, 999, 1000$
$11, 12, 13, 14, 15, \dots$	$201, 202, 203, \dots, 299, 300$
$1, 2, 3, 4, 5, \dots$	$1, 2, 3, 4, 5, \dots$
$10, 11, 12, 13, 14, 15, \dots$	$100, 101, 102, 103, 104, 105, \dots$
$2, 4, 6, 8, 10, \dots$	$11, 13, 15, 17, 19, \dots$
$10, 20, 30, 40, 50, \dots$	$3, 6, 9, 12, 15, \dots$
$1, 101, 201, 301, 401, \dots$	$-2, -4, -6, -8, -10, \dots$

How did you take into account the fact that the hotel is infinite? How is the infinite hotel different from a finite one?

2. Bijection between intervals

Even though it seems counterintuitive, any two finite intervals have the

same cardinality – there is a bijection between them. If we think of them as segments on a number axis, the segments have the *same amount of points*.

Create a sheet that will demonstrate this. The input are the endpoints of two closed intervals.

Using suitable formulas and the input values (over multiple steps, if needed) transform the points of one interval to the points of the second interval. The intervals should be tabulated uniformly using 10 (or 20) steps. Find an appropriate method (graph) for displaying the gradual transformations of the interval.

3. Mappings vs. relations

- For an independent variable x from -10 to 10 calculate the corresponding values y using a relation $y = ax^2 + b$, where a, b are parameters. Draw the graph of this relation in a plane. Is it a function of variable x ? Is it a function for all admissible values of parameters a, b ?
- Solve the same problem for *switched* values of x and y : For y from -10 to 10 calculate the corresponding values x with respect to relation $x = ay^2 + b$, where a, b are parameters. Draw the graph of this relation in a plane. Is it a function of variable x ? Is it a function for all admissible values of parameters a, b ?
- Compare the pairs $[x, y]$, the two relations and their graphs for the same values of parameters a, b .

7.4.1 Hints and solutions

1. Infinite hotel

Hints:

- Of course, we are not going to model an infinite hotel. It is sufficient to restrict ourselves to the first few, e.g. 100, rooms. However, we should carefully consider, whether the solution checked using this range also works if we extended the hotel to infinity.
- It is useful to designate separate columns to contain:
 - the list of room numbers 1, 2, 3, ... This is the basic *control* column that guides the other columns in the worksheet.
 - only the occupied room numbers.
 - the room numbers of the new guests waiting at the front desk. This list is going to be more complex than 1, 2, 3 ... In all specific situations, it is sufficient to input the first few values and then fill them down either by dragging or by double-clicking the lower right corner of highlighted cell(s).
- For output, it might be helpful to have columns that contain:
 - new rooms for the current guests. The room numbers in this column should be obtained using a formula into which a guest inputs their current room number.
 - new rooms for waiting guests who obtain their new rooms numbers by plugging their old room number into a second suitable formula.
- The goal is to occupy each room by exactly one guest. How can we check that we have achieved this using the two formulas mentioned previously? What numbers do we expect in the two output columns?
- We can create a few verification columns. For clarity and easier debugging, we recommend to perform the verification separately for the current guests and for the new guests.
 - The verification columns should correspond to the control column.
 - For each hotel room, we can count how many original guests and how many new guests have moved into it. The function COUNTIF is a good choice for this task if we use the room numbers as parameters.

- The final check is done by combining the two previous results. Then we can visually highlight *empty* rooms, *correctly occupied* rooms and *overcrowded* rooms.

Solution: We describe the solution shown in Figure 7.4. The column A contains the list of all (although in fact only just the first 100) hotel rooms.

D12									
	A	B	C	D	E	F	G	H	I
1	Rooms	Hotel cust.	Desk client	Hotel formula	Desk formula	Cust. occup.	Client occup.	Room occup.	Check
2									
3	1		2		1	0	1	1	OK
4	2		4		3	0	0	0	Wrong
5	3		6		5	0	1	1	OK
6	4		8		7	0	0	0	Wrong
7	5		10		9	0	1	1	OK
8	6		12		11	0	0	0	Wrong
9	7		14		13	0	1	1	OK
10	8		16		15	0	0	0	Wrong
11	9		18		17	0	1	1	OK
12	10	10	20	20	19	0	0	0	Wrong
13	11	11	22	22	21	0	1	1	OK
14	12	12	24	24	23	0	0	0	Wrong
15	13	13	26	26	25	0	1	1	OK
16	14	14	28	28	27	0	0	0	Wrong
17	15	15	30	30	29	0	1	1	OK
18	16	16	32	32	31	0	0	0	Wrong

Figure 7.4: Infinite hotel and new guests at the front desk.

Columns B and C contain the particular situation described in the problem. In this case, the current guests occupy the rooms starting with number 10 and new guests come from another infinite hotel where they stayed in all the rooms with even numbers.

First we move the current guests to room numbers that are double their original room numbers. The formula in column D calculates these values only for the occupied rooms. The current guests thus move to rooms with even numbers starting with 20.

The new guests waiting at the front desk with their old numbers $2k$ then get keys to rooms $2k - 1$. Once they check in, all odd room numbers are occupied and we see the result in column E.

We notice that the formulas were not completely correct. There are some empty rooms left: the rooms with even numbers smaller than 20. We need a check that can detect this kind of situation.

Using the formula `=COUNTIF(D:D,A3)` in column F, we calculate the number of current guests staying in the room number 1, etc. Similarly, using the formula `=COUNTIF(E:E,A3)` in column G we calculate the number of newly checked-in guests in room number 1, etc.

The column H adds the values in F and G and thus determines the number of people now staying in the rooms with numbers 1, 2, 3, ... There should be exactly one guest in each of them. If that is the case, the column I displays the value *OK*. If not, it shows the value *Wrong*.

In our problem, we see that the Excel sheet detected the empty rooms with numbers 2, 4, 8, ..., 18. If we want to correct this mistake, the current guests should use the formula $2k - 18 = 2(k - 9)$ instead of $2k$.

This Excel sheet can check your solution for all situations given in this problem once you input the formulas in the columns D and E. In all cases it is possible to find a solution that moves the current guests to suitable even numbers and new guests in suitable odd numbers.

2. Bijection between intervals

Hints:

- We can start by tabulating the first (input) interval into a column in the worksheet. In order to do that, we need to use the interval endpoints and decide on the number of tabulated values. Once we have that, we can calculate and use a *step* to get from one tabulated value to the next.
- The neighboring columns should represent the consecutive solution steps, each performing a certain transformation of the interval, e.g. shift right, scale down, etc. Once we apply the whole sequence of steps, we can compare the resulting interval (in terms of its endpoints) with the expected answer (the endpoints of the second interval).
- The key to solving this problem lies in a correct choice of the transformation functions that map the numbers from one interval to another. We recommend working in simple, clear steps. Suppose we want to map the input interval $\langle 2, 5 \rangle$ to output interval $\langle 8, 12 \rangle$. Our steps could be as follows: $\langle 2, 5 \rangle \rightarrow \langle 0, 3 \rangle \rightarrow \langle 0, 1 \rangle \rightarrow \langle 0, 4 \rangle \rightarrow \langle 8, 12 \rangle$. We first shift the interval three units to the left, then scale it down three times, scale it up four times and shift it eight units to the right.

- The parameters of the changes are determined by the endpoints of the intervals and their differences. We need to calculate them in general from the endpoints of the two intervals. In any case, we *go through* the universal *middle* interval $\langle 0, 1 \rangle$.
- Of course, it is possible to combine the steps into one single formula. Such approach might not be as clear as performing the transformation in successive steps, is more error-prone and more difficult to debug. Moreover, when using the composite formula, we cannot display and observe the individual transformation steps. Still, computationally, it is more efficient.
- In order to display the sequence of intervals in a chart, we need to add a second coordinate to the tabulated values. The second coordinate can be a different constant for each interval, which then determines the position of the interval in an x - y chart.

Solution:

In Figure 7.5, we show one possible solution.

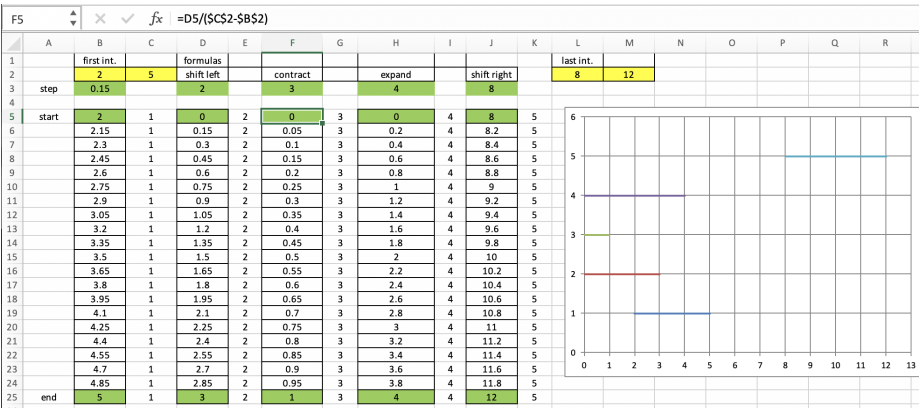


Figure 7.5: Bijective transformation of one interval into another.

The values representing the bounds of the intervals $\langle 2, 5 \rangle$ and $\langle 8, 12 \rangle$ mentioned in *Hints* are entered as input in the yellow cells.

The bounds of the first interval are used to determine the *step* for tabulating it. All the bounds are used to calculate the shifts and scaling needed for the bijection.

In the rows 5 to 25, the columns B, D, F, H and J contain the values of the individual intervals. The endpoints are highlighted in orange. The column B corresponds to the initial interval and the column J to the resulting final interval.

The formula shown in the cell F5 is used in the whole column F. This column recalculates the tabulated values from the interval $\langle 0, 3 \rangle$ shown in column D by dividing it by 3 to the values from the interval $\langle 0, 1 \rangle$.

The constant values in columns C, E, G, I and K are used as the y -coordinates for displaying the sequence of intervals in a suitable x - y chart.

3. Mappings vs. relations

Hints:

- In this problem we want to display and compare two relations. At the first glance they might appear to be the same, but there is a significant difference between them. The second relation switches the first value x with the second value y in the formula that defines it.
- In practice, we can start with two tables showing the x - y dependence and drawing their graphs. In both cases we use the same parameters a and b . How are the tables going to differ? Which values are going to be the input and which are going to be calculated using the given formulas?
- What format can we use to display these relations? How are the two graphs going to differ and what will they have in common?
- In the first case, the values x tabulated from $x = -10$ to $x = 10$ are the input. Even if we tabulate with a *step* = 1, we get a reasonably good graph. The values in the output column y are calculated using the formula with parameters a, b .
- In the second case, we can create a table with the same two columns in the same order, but this time, it is the y column that is tabulated with values from -10 to 10 with the step 1 and the values of x are calculated using a formula.
- If both of the tables have the same format, we can use the same x - y chart to display them. One of them corresponds to a graph of a mapping that is single-valued for each input x . The other is a relation in which a given x can be with more than one y . Which graph is which?

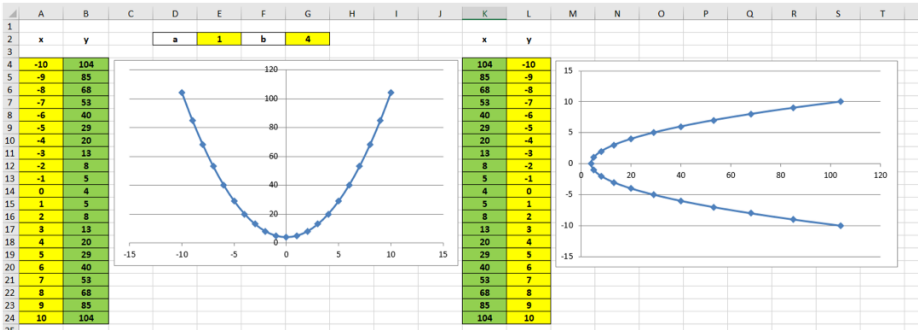


Figure 7.6: A mapping vs. a relation.

Solution: In Figure 7.6, we show one possible solution.

The values of input parameters a , b and tabulated values x and y are highlighted in yellow. The calculated values are shown in green.

The first table and its graph show a mapping. For each tabulated value x we have one corresponding output value y . We see the output values *above* the input values x in the graph.

In the second table and graph we see that for example the value $x = 104$ is in relation $x = 1 \cdot y^2 + 4$ with $y = -10$ and also with $y = 10$. Therefore this is *only* a relation and not a mapping. Unlike the previous case, when we take a line parallel with the y -axis here, e.g. at $x = 104$, it intersects the graph twice, giving us two related values of y .

A more careful observation reveals that the two graphs are very similar (up to the scaling of the axes). We can transform one into the other by rotating it by 90 degrees, i.e. by switching the variables x a y .

Chapter 8

Sequences

8.1 Lecture overview

8.1.1 Introduction and definition

Examples of sequences around us

- Time series: stock prices, exchange rates, measurements of physical variables (temperature, pressure) in time, sampling of sound.
- Recurrent processes: adding interest to savings or loans, calculating populations based on data from previous time intervals.
- Description of (potentially) infinite processes or constructions: inscribing triangles into triangles, calculation of approximate values.
- Definition, investigation and manipulation of sequentially given data, e.g. elements of arrays.
- Random processes: number of devices connected to a network at a given time, number of e-mails received in an hour, interval between e-mails, times when the e-mails were received (how are these related?), numbers obtained by rolling dice (what we got, how many rolls did it take to get a six).

These values may be finite, discrete, continuous. In the following we assume they are real numbers.

Sequence

The basic idea is to define an ordered sequence of (real) numbers in which we assume a certain connection, a common meaning. It is clear which number is the first and how they are ordered. Each number has its immediate successor and (except for the first one) also a predecessor. That way we can clearly determine, which is the first, the second, the third etc. number in the sequence.

A *sequence* is a function $a : \mathbb{N} \rightarrow \mathbb{R}$. The sequence consists of values $a(n)$ denoted a_n and called *sequence terms*. The numbers n are called their *indices*. The infinite ordered list $\langle a_1, a_2, a_3, a_4, \dots \rangle$ is denoted $\{a_n\}_{n=1}^{\infty}$ or $\{a_n\}_{n \in \mathbb{N}}$. If we restrict the domain from $\mathbb{N}$ to a finite subset of $\mathbb{N}$, we can formally define a *finite sequence*.

Some ways to define a sequence

A sequence can be given by:

- A formula for the term a_n , e.g. $a_n = \frac{3n-4}{7n-1}$ or $b_n = \frac{1-\cos(n\pi)}{2}$. We can also write $\{\frac{3n+4}{7n-1}\}_{n \in \mathbb{N}}$, $\{\frac{1-\cos(n\pi)}{2}\}_{n \in \mathbb{N}}$ or $\{\frac{3n+4}{7n-1}\}_{n=1}^{\infty}$, $\{\frac{1-\cos(n\pi)}{2}\}_{n=1}^{\infty}$. Two well known sequences are given by formulas:

$a_n = k + (n-1) \cdot d$, (The first term $a_1 = k$ and the difference d are real parameters.) – *arithmetic sequence*

$g_n = k \cdot q^{n-1}$, (The first term $g_1 = k$ and the quotient q are non-zero real parameters) – *geometric sequence*

- Stating the first terms and a recurrent formula how to calculate next terms. The calculation is then performed *recursively*.

$$a_1 = 1, a_n = n \cdot a_{n-1}$$

$$a_1 = 1, a_2 = 1, a_n = a_{n-1} + a_{n-2}$$

For an arithmetic sequence, it is $a_1 = k, a_n = a_{n-1} + d$

For a geometric sequence, it is $g_1 = k, g_n = g_{n-1} \cdot q$

- Listing the first few terms $\langle 1, 3, 9, 27, 81, \dots \rangle$, $\langle 1, 2, 1, -1, -2, -1, 1, 2, \dots \rangle$. Here it is expected that we discover the rule.

- A precise definition but without knowledge of the underlying formula or the recurrent relation:

p_n is the n^{th} prime

t_n is the n^{th} digit of π after the decimal point

To understand the sequences and their properties better, it is useful to draw them in a graph.

Notation

The definition of a sequence strictly requires that the set of indices is $\mathbb{N}$. Clearly, this can always be done, if we denote the first term of the sequence a_1 , the next a_2 etc. Sometimes, we can use a more clever indexing, e.g. the arithmetic and geometric sequences have simpler formulas, if the indices start at 0. Therefore we also use the following notation:

$$\{a_n\}_{n=0}^{\infty}, \quad \{a_n\}_{n=4}^{\infty}, \quad \{a_n\}_{n=-6}^{\infty}, \quad \{k + n \cdot d\}_{n=0}^{\infty}, \quad \{k \cdot q^n\}_{n=0}^{\infty}$$

If we define the set of indices I , we can also write $\{a_n\}_{n \in I}$. It is also possible to consider the sequence $\{a_n\}_{n \in \mathbb{Z}} = \{a_n\}_{n=-\infty}^{\infty}$. Another exception is a finite sequence, where I is a finite set. Note that not all of the following properties apply to finite sequences.

Subsequence

Sometimes, we only need to examine some terms of a given sequence. We may for example wish to skip the first 5 or 50 000 or every other term. The selection of a subsequence has to maintain order of the terms of the original sequence, i.e. if we select a term, the next selected term has to come after it. We say that a sequence $\{b_n\}_{n \in \mathbb{N}}$ is a *subsequence* of sequence $\{a_n\}_{n \in \mathbb{N}}$, iff $\forall n \in \mathbb{N} \exists m \in \mathbb{N} : b_n = a_m$, moreover $\forall n \in \mathbb{N}$ if $(b_n = a_k) \wedge (b_{n+1} = a_l)$ for some $k, l \in \mathbb{N}$ then $k < l$.

8.1.2 Basic properties

We say that the sequence $\{a_n\}_{n \in \mathbb{N}}$ is

- *Constant* iff there exists a real number c such that $\forall n \in \mathbb{N} : a_n = c$

$$\langle 1, 1, 1, 1, 1, \dots \rangle, \quad \langle -\pi, -\pi, -\pi, -\pi, \dots \rangle$$

- *Increasing* iff $\forall i, j \in \mathbb{N} : i < j \Rightarrow a_i < a_j$ (it is also sufficient to have $\forall i \in \mathbb{N} : a_i < a_{i+1}$), for example:

$$\langle 1, 2, 3, 4, 5, \dots \rangle, \quad \langle \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \rangle$$

- *Decreasing* iff $\forall i, j \in \mathbb{N} : i < j \Rightarrow a_i > a_j$ (or $\forall i \in \mathbb{N} : a_i > a_{i+1}$)
- *Non-decreasing* iff $\forall i, j \in \mathbb{N} : i < j \Rightarrow a_i \leq a_j$ (or $\forall i \in \mathbb{N} : a_i \leq a_{i+1}$)
- *Non-increasing* iff $\forall i, j \in \mathbb{N} : i < j \Rightarrow a_i \geq a_j$ (or $\forall i \in \mathbb{N} : a_i \geq a_{i+1}$)
- *Bounded from above* iff there exists a real number H such that $\forall i \in \mathbb{N} : a_i \leq H$
- *Bounded from below* iff there exists a real number D such that $\forall i \in \mathbb{N} : a_i \geq D$
- *Bounded* iff it is bounded both from below and from above.

Properties of sequences

Which of the properties of sequences imply other properties?

Which preclude other properties?

8.1.3 Limits of sequences

Another important set of properties is related to the fact that sequence is infinite. One might ask, why do we care about sequence behavior in infinity. In real world, we only encounter finite sequences (which may be parts of the theoretically infinite ones). There are two basic reasons: the infinite sequences are an important tool for developing parts of mathematics and many properties of sequences following from the sequence behavior in infinity are useful (and otherwise undetectable) for determining the sequence behavior in its finite initial parts. Most often we would like to know whether the sequence values *stabilise* at a certain value. It could mean a steady stream of e-mails per hour or a numerical value of π or e .

The key property of the sequence, which makes it a useful term and mathematical tool, is thus its ability to tend to a certain value. It is important that the sequence approaches this value (or even reaches it, as is the case of a constant sequence) arbitrarily closely and *almost the whole sequence* achieves this closeness. There is only a finite number of exceptions. All subsequent terms after the initial part containing exceptions will be close. We call the value, which they are approaching, a *limit*.

Limit

The closeness of sequence to a limit value L can be understood by its terms belonging to a (small) neighborhood $(L - \varepsilon, L + \varepsilon)$. We can think of the neighborhood as a *tunnel* around the limit value. Even when we make the tunnel arbitrarily narrow, starting at some sequence term, the whole rest of the sequence will be inside the tunnel. If we denote the width of the tunnel by a real number ε and distance using an absolute value, we can define the limit as follows.

We say that a sequence $\{a_n\}_{n \in \mathbb{N}}$ has a *limit* $L \in \mathbb{R}$ iff

$$(\forall \varepsilon \in \mathbb{R}^+)(\exists m \in \mathbb{N})(\forall n \geq m, n \in \mathbb{N})(|a_n - L| < \varepsilon)$$

Since the limit L is a real number, we also say that the sequence has a *proper limit*. When a sequence has a limit (L), we say that the sequence *converges* (to L). We write $\lim_{n \rightarrow \infty} a_n = L$ or $\lim a_n = L$, if it is clear which index was meant. In such case, we can also write $a_n \rightarrow L$.

Improper limit

We now state a special case when a sequence increases or decreases without a bound. Again, we require that it has only finitely many exceptions to this property in some early part of the sequence. Then we say that the sequence $\{a_n\}_{n \in \mathbb{N}}$ has an *improper limit* ∞ iff

$$(\forall B \in \mathbb{R}), (\exists m \in \mathbb{N})(\forall n \geq m, n \in \mathbb{N})(a_n > B).$$

We say that the sequence $\{a_n\}_{n \in \mathbb{N}}$ has an *improper limit* $-\infty$ iff

$$(\forall B \in \mathbb{R}), (\exists m \in \mathbb{N})(\forall n \geq m, n \in \mathbb{N})(a_n < B).$$

8.1.4 Properties and calculations of limits

Calculation of limits from definition

If we want to calculate a limit of a sequence based on what we know so far, we need to perform two steps. First we need to guess a (proper or improper) limit and then prove that it meets the *tunnel property*. These steps are difficult and impractical but necessary for calculation of few basic limits. Once we have these, we can use the following properties for calculation of more limits. The properties hold for proper limits and some of them can also be extended for improper limits.

Intuitive calculation of limits

In order to determine the limit of a sequence, we can also use an estimate obtained by plugging in increasingly large values of n .

Problem 8.1 Intuitive limit:

Calculate $\lim_{n \rightarrow \infty} \frac{4-n}{2n+5}$.

Solution: We are looking for an L such that

$$\frac{4-n}{2n+5} \rightarrow L.$$

We start by plugging in $n = 100$

$$\frac{-96}{205} \doteq -0.468292683;$$

then $n = 1000$

$$\frac{-996}{2005} \doteq -0.49675;$$

and then $n = 1\,000\,000$

$$\frac{-999996}{2000005} \doteq -0.49999675.$$

At this point we guess that $L = -0.5 = -\frac{1}{2}$ and try to prove it.

To prove it means to show that the following difference tends to zero:

$$\frac{4-n}{2n+5} - L \rightarrow 0.$$

After plugging in, we get

$$\frac{4-n}{2n+5} + \frac{1}{2} = \frac{8-2n+2n+5}{2 \cdot (2n+5)} = \frac{13}{2 \cdot (2n+5)} = \frac{\mathcal{O}(n)}{\mathcal{O}(n^2)} \rightarrow 0.$$

We use the $\mathcal{O}$ notation to indicate the behavior of the sequence in infinity. We return to this in more detail in Section 12.1.1.

Problem 8.2 Another intuitive limit:

Calculate the limit of the sequence $\{\frac{2n^2-3n+5}{3n^2-2n+5}\}_{n \in \mathbb{N}}$.

Solution: By plugging in increasingly large values of n we estimate L and then show that the sequence decreased by L tends to zero.

$$\begin{aligned} \frac{2n^2 - 3n + 5}{3n^2 - 2n + 5} - L &\rightarrow 0 \\ \frac{2n^2 - 3n + 5}{3n^2 - 2n + 5} - \frac{2}{3} &= \frac{6n^2 - 9n + 15 - 6n^2 + 4n - 10}{3 \cdot (3n^2 - 2n + 5)} = \\ &= \frac{-5n + 5}{9n^2 - 6n + 15} = \frac{\mathcal{O}(n)}{\mathcal{O}(n^2)} \rightarrow 0 \end{aligned}$$

Multiplication by a constant

Suppose $a_n \rightarrow A$, $A \in \mathbb{R}$. Then

1. $\forall c \in \mathbb{R}$ it is true that $ca_n \rightarrow cA$;
2. $|a_n| \rightarrow |A|$.

The limit of a sequence whose terms are constant multiples (or absolute values) of terms of a convergent sequence, is the same multiple (or absolute value) of the original limit.

Multiplication by a constant for improper limits

Suppose $a_n \rightarrow \pm\infty$. Then

1. $\forall c \in \mathbb{R}^+$ it is true that $ca_n \rightarrow \pm\infty$;
2. $\forall c \in \mathbb{R}^-$ it is true that $ca_n \rightarrow \mp\infty$ (the sign changes!);
3. $|a_n| \rightarrow \infty$.

The limit of a sequence whose terms are constant multiples (or absolute values) of terms of a sequence that converges to an improper limit, is also an improper limit (possibly with opposite sign).

Basic operations with two sequences

Suppose $a_n \rightarrow A, b_n \rightarrow B$; $A, B \in \mathbb{R}$. We consider sequences whose terms are obtained by arithmetic operations on terms a_n and b_n .

1. $a_n + b_n \rightarrow A + B$;

2. $a_n - b_n \rightarrow A - B$;
3. $a_n \cdot b_n \rightarrow A \cdot B$;
4. if $B \neq 0$ then $a_n/b_n \rightarrow A/B$.

Sequences whose terms are created from two convergent sequences using an arithmetic operation converge to the limits obtained by the same operation from the original limits.

Ratio of bounded and infinitely growing sequences

Suppose a_n is a bounded sequence and $b_n \rightarrow \pm\infty$. Then $a_n/b_n \rightarrow 0$.

If the numerator is bounded and denominator grows or decreases unboundedly, their ratio tends to zero. Note that every sequence that has a finite limit, i.e. $a_n \rightarrow A, A \in \mathbb{R}$, is bounded and thus this theorem applies to it.

Ratio of a sequence with a proper limit and a sequence tending to zero

Suppose $a_n \rightarrow A \neq 0$, $b_n \rightarrow 0$ and all b_n are positive. Denote the sign of A by $\text{sign}(A)$. Then $a_n/b_n \rightarrow \text{sign}(A)\infty$. If all b_n are negative, then $a_n/b_n \rightarrow -\text{sign}(A)\infty$.

If the denominator tends to zero, the ratio unboundedly grows or decreases, depending on the sign of A and the sign of terms b_n . The tunnel property does not work, if b_n changes signs.

Ratio of a bounded sequence and a sequence tending to zero?

Could we just use a bounded sequence in the previous statement instead of a sequence that has a proper limit? If yes, why? If not, why not?

Composition of continuous functions

Suppose $a_n \rightarrow A$, $A \in \mathbb{R}$. Suppose b_n is a sequence, whose terms are generated from a_n using the same *continuous* function (e.g. square root, exponential, power, trigonometric function, etc.). Suppose all a_n and the limit value A are in the domain of this function. Then also b_n converges and we can obtain its limit by applying the same function to the original limit A , i.e.:

$$a_n \rightarrow A, b_n = f(a_n) \Rightarrow b_n \rightarrow f(A)$$

Some examples:

$$\frac{2n}{n+1} \rightarrow 2 \Rightarrow \sqrt{\frac{2n}{n+1}} \rightarrow \sqrt{2}, \quad 3^{\frac{2n}{n+1}} \rightarrow 3^2, \quad \cos\left(\frac{2n}{n+1}\right) \rightarrow \cos(2)$$

Therefore if we are calculating a limit of a sequence, whose terms are given by a formula that enters an outer function, in some (i.e. continuous) cases, we can first calculate the limit of the inner part and then apply the function to the result.

The squeeze theorem

This property uses the inequalities between pairs of sequence terms. It is also unofficially called a *Two policemen theorem* or *Two friends theorem*, since in both cases the two of them will lead the middle person where they should go.

Suppose we have three sequences $\{a_n\}_{n \in \mathbb{N}}$, $\{b_n\}_{n \in \mathbb{N}}$ and $\{c_n\}_{n \in \mathbb{N}}$. Suppose $\forall n \in \mathbb{N}$ we have $a_n \leq b_n \leq c_n$, and let $a_n \rightarrow L, c_n \rightarrow L$. The sequences $\{a_n\}_{n \in \mathbb{N}}$ and $\{c_n\}_{n \in \mathbb{N}}$ thus have the same limit, either proper or improper. Then it follows that $\{b_n\}_{n \in \mathbb{N}}$ has the same limit, i.e. $b_n \rightarrow L$.

The two good friends (or cops) who are headed to the same limit and *guard* the third sequence between them, lead the third sequence to the same limit as well. The inequalities allow us to deal with situation in which it is complicated or unfeasible to use the arithmetic operations from previous properties. Some examples of sequences which lead the *middle* sequence to zero are:

$$\frac{-1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}, \quad 0 \leq \frac{1}{n^2} \leq \frac{1}{n} \quad 0 \leq \frac{1}{2^n} \leq \frac{1}{n}$$

In cases where we used a constant zero (the second and the third example), we meant a constant sequence.

Uniqueness of the limit

Suppose $a_n \rightarrow A$, $A \in \mathbb{R}$. Then A is the only limit of this sequence. Remember that the interval $(A - \varepsilon, A + \varepsilon)$ is the *tunnel of the limit*. Since every *neighborhood* (tunnel) around the limit value A contains all sequence terms starting at some index, it is not possible for infinitely many of them to appear in another tunnel $(B - \varepsilon, B + \varepsilon)$, around the limit value B . To make sure that these two tunnels do not overlap, we can sufficiently decrease ε .

Monotonicity with boundedness implies proper limit

Suppose a sequence is non-decreasing and bounded from above. Then it has a proper limit. Similarly if it is non-increasing and bounded from below, then it has a proper limit.

To understand this property, it is helpful to realise, what is the limit of such sequence. In the case of non-decreasing sequence bounded from above, it is the smallest upper bound. If we make an *arbitrarily small tunnel* around it, the sequence eventually crosses its lower edge (otherwise, this lower edge would be a smaller upper bound). And once the sequence crosses this edge, it will remain in the tunnel due to its monotonicity. (For non-increasing sequence bounded from below, the greatest lower bound serves a similar purpose.)

This property is very useful both for proving that the sequence has a limit and for finding this limit. Once we know that the limit exist, we can use heavy caliber weapons, such as subsequences, for finding it and then proving monotonicity and boundedness of the sequence, which is much easier than examining complicated limits. Moreover, the monotonicity implies that the sequence is getting closer and closer to the limit, so its terms themselves can serve as a self-improving estimate of the limit value.

The most famous application of this property is analysis of the sequence

$$\left\{ \left(1 + \frac{1}{n} \right)^n \right\}_{n \in \mathbb{N}}.$$

Using the binomial theorem it is possible to show that this sequence is increasing and all its terms are smaller than three. Therefore this sequence has to have a proper limit. This limit is the well-known and important number $e \cong 2.718281828\dots$

Limit of subsequence

Suppose that a sequence $\{a_n\}_{n \in \mathbb{N}}$ has a proper or improper limit. Next suppose that $\{b_n\}_{n \in \mathbb{N}}$ is any subsequence of $\{a_n\}_{n \in \mathbb{N}}$. Then $\{b_n\}_{n \in \mathbb{N}}$ has the same limit. This can be seen directly from the definition of both proper and improper limit.

The main use of this property lies in proving that a sequence does not have a limit. For example a sequence $\{-1^n\}_{n \in \mathbb{N}} = \langle -1, 1, -1, 1, \dots \rangle$ has two special subsequences. One of them has limit 1 and the other -1 . If two subsequences have two different limits, it is clear that the original sequence does not have a limit. Without this property, it would be very difficult to prove that a sequence

does not have a limit. We would need to negate the definition and show that every real number (as well as $\mp\infty$) meets the negation.

The property also gives us new possibilities for determining the value of the limit, especially if we already know (using the previous property) that a limit exists.

Problem 8.3 n^{th} square root of 2:

Show that $\lim_{n \rightarrow \infty} \sqrt[n]{2} = 1$.

Solution: We first show that the sequence decreasing and bounded from below, i.e. for all n we have $\sqrt[n]{2} > \sqrt[n+1]{2}$ and $\sqrt[n]{2} > 1$. From that we know that a limit exists. We can denote it A . Therefore we have $\lim_{n \rightarrow \infty} \sqrt[n]{2} = A$. But how much is A ? For that we use a subsequence of $\{\sqrt[n]{2}\}_{n \in \mathbb{N}}$. We only take the even terms: $\{\sqrt[2n]{2}\}_{n \in \mathbb{N}}$. What is its limit? Since it is a subsequence of a sequence with limit A , it must be true that $\lim_{n \rightarrow \infty} \sqrt[2n]{2} = A$. Using the properties we mentioned, we can write:

$$\lim_{n \rightarrow \infty} \sqrt[2n]{2} = \lim_{n \rightarrow \infty} \sqrt{\sqrt[n]{2}} = \sqrt{\lim_{n \rightarrow \infty} \sqrt[n]{2}} = \sqrt{A}.$$

So we see that $A = \sqrt{A}$. There are only two such numbers: 1 and 0. Since all sequence terms are greater than 1, we can disregard 0 and we have:

$$\lim_{n \rightarrow \infty} \sqrt[n]{2} = \lim_{n \rightarrow \infty} \sqrt[2n]{2} = 1.$$

Other important limits can be determined similarly: $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$, $\lim_{n \rightarrow \infty} q^n = 0$ for $q \in (0, 1)$. The limit is also 0 for $q \in (-1, 0)$, but the proof is different.

Ratio of successive terms

Consider a sequence with the property that each new term q^{n+1} is created from the previous term q^n by multiplying it by a number less than 1:

$$\frac{a_{n+1}}{a_n} = \frac{q^{n+1}}{q^n}, q \in (-1, 1).$$

Then it is true that $\lim_{n \rightarrow \infty} a_n = 0$. The idea is that if the sequence stabilises to the extent that each successive term is less-than-a-1-multiple of the previous term, the sequence decreases in such a way that it tends to 0. This property is very useful in examining the zero-limit sequences and it can be shown that this property also works if the ratio is not fixed but tends to a limit.

Problem 8.4 Ratio of successive terms:

Evaluate $\lim_{n \rightarrow \infty} \frac{4n}{6^n}$.

Solution: We suspect that the numerator tends to infinity much slower than the denominator and the limit could be zero. We evaluate the auxiliary limit:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{4(n+1)}{6^{n+1}}}{\frac{4n}{6^n}} \right| = \lim_{n \rightarrow \infty} \frac{4(n+1) \cdot 6^n}{4n \cdot 6^{n+1}} = \lim_{n \rightarrow \infty} \frac{n+1}{6n} = \frac{1}{6}.$$

Since $1/6 < 1$, the original limit is 0.

8.2 Questions

1. Write the first six terms of the sequence

a) $\left\{ 2 - \frac{\sin(\pi \cdot n)}{2} \right\}_{n=1}^{\infty}$,

b) $\left\{ \frac{4n+7}{3n-2} \right\}_{n=4}^{\infty}$.

2. Write the first seven terms of the arithmetic sequence with the first term 8 and difference 3.
3. Write the first six terms of the geometric sequence with the first term 2 and quotient 4.
4. Write the first five terms of the recurrent sequence $a_0 = 3$; $a_1 = 2$; $a_n = 3 \cdot a_{n-1} + a_{n-2}$.
5. Write a simple formula for a sequence that starts with terms

a) $\langle 1, 3, 9, 27, 81, \dots \rangle$

b) $\langle 1, 4, 9, 16, 25, \dots \rangle$

c) $\langle 1, 2, 4, 6, 10, 12, 16, \dots \rangle$

6. Let $a_n = \langle 1, 4, 9, 16, 25, 36, \dots \rangle$, $b_n = \langle 4, 16, 36, \dots \rangle$ and $c_n = \langle 4, 36, 16, \dots \rangle$. What is the relation between a_n , b_n and c_n ?

7. Give an example of a constant sequence.

8. Give an example of

a) an increasing sequence,

b) a non-decreasing sequence which is not increasing,

c) a non-increasing sequence which is not decreasing,

d) a sequence which is bounded from above,

e) a sequence which is bounded from below and decreasing.

9. Which of these properties does the sequence $\left\{ \frac{4n+1}{3n+1} \right\}_{n=1}^{\infty}$ have: increasing, non-increasing, bounded from below, bounded from above, unbounded, decreasing, non-decreasing?

10. Find the smallest upper bound for the sequence $\left\{2 - \frac{1}{3n+1}\right\}_{n=1}^{\infty}$.
11. Find the first such n , that the distance of the n^{th} term of the sequence $\left\{2 - \frac{1}{3n}\right\}_{n=1}^{\infty}$ from 2 is at most
 - a) 0.1,
 - b) 0.01.
1. Give examples of at least three sequences which
 - a) have limit 5.
 - b) do not have a limit.
 - c) have a limit ∞ .
 - d) have a limit $-\infty$.
2. Find a q such that $\lim_{n \rightarrow \infty} q^n$
 - a) equals 0,
 - b) equals 1,
 - c) is ∞ ,
 - d) does not exist.
3. Let $\lim_{n \rightarrow \infty} a_n = 7$ and let $c = 3$. How much is $\lim_{n \rightarrow \infty} c \cdot a_n$?
4. Let $\lim_{n \rightarrow \infty} a_n = -7$. How much is $\lim_{n \rightarrow \infty} |a_n|$?
5. Let $\lim_{n \rightarrow \infty} a_n = -5$ and let $\lim_{n \rightarrow \infty} b_n = 5$. How much is $\lim_{n \rightarrow \infty} (a_n + b_n)$?
6. Let $\lim_{n \rightarrow \infty} a_n = -5$ and let $\lim_{n \rightarrow \infty} b_n = 5$. How much is $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$?
7. Let $\lim_{n \rightarrow \infty} a_n = 17$. How much is $\lim_{n \rightarrow \infty} \sin(a_n)$?
8. Let the sequence $\{a_n\}_{n=1}^{\infty}$ be increasing and let it have an upper bound 4. Can the limit of this sequence be -5 ?
9. Let the sequence $\{a_n\}_{n=1}^{\infty}$ be decreasing and let it have a lower bound 4. Can the limit of this sequence be -5 ?
10. Give an example of a sequence that has two different limits.
11. Give an example of a sequence that has two subsequences with different limits.

8.3 Problem set

1. Sequences given by a closed form formula

Write the first few terms of sequences given by the formula.

- a) $\{5n\}_{n=1}^{\infty}$; $\{5n\}_{n=0}^{\infty}$; $\{5n\}_{n=3}^{\infty}$
- b) $\{n^2\}_{n=1}^{\infty}$; $\{n^2 - 1\}_{n=0}^{\infty}$; $\{n^2 + 5\}_{n=-2}^{\infty}$; $\{3n^2\}_{n=1}^{\infty}$
- c) $\{2^n\}_{n=1}^{\infty}$; $\{2^n\}_{n=0}^{\infty}$; $\{2^n + 2\}_{n=-2}^{\infty}$; $\{2^{n+2}\}_{n=-2}^{\infty}$
- d) $\{(-1)^n\}_{n=1}^{\infty}$; $\{(-1)^n\}_{n=0}^{\infty}$; $\{(-1)^{n+1}\}_{n=1}^{\infty}$; $\{(-1)^{n+1} - (-1)^n\}_{n=1}^{\infty}$
- e) $\{(-1)^n + 3\}_{n=1}^{\infty}$; $\{3 \cdot (-1)^n\}_{n=1}^{\infty}$
- f) $\left\{\frac{(-1)^{n+1} - (-1)^n}{2}\right\}_{n=1}^{\infty}$; $\left\{\frac{(-1)^{n+1} - (-1)^n}{-2}\right\}_{n=1}^{\infty}$
- g) $\{\frac{1}{2^n}\}_{n=1}^{\infty}$; $\{\frac{1}{2^n}\}_{n=0}^{\infty}$; $\{\frac{1}{n^2}\}_{n=1}^{\infty}$

2. Sequences given by their terms

For each sequence write a formula for the n^{th} term. If possible, write also a recurrent formula, which can be used to calculate a term using some previous terms.

- a) $\langle 3, 6, 9, 12, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=0}^{\infty} = \{c_n\}_{n=3}^{\infty}$
 $a_n = ?$, $b_n = ?$, $c_n = ?$
- b) $\langle 3, 7, 11, 15, 19, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty} = \{c_n\}_{n=-5}^{\infty}$
 $a_n = ?$, $b_n = ?$, $c_n = ?$
- c) $\langle 0, 3, 8, 15, 24, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=5}^{\infty}$
 $a_n = ?$, $b_n = ?$
- d) $\langle 1, 2, 4, 8, 16, 32, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = ?$, $b_n = ?$
- e) $\langle 3, -3, 3, -3, 3, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = ?$, $b_n = ?$
- f) $\langle 2, 3, 5, 9, 17, 33, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = ?$, $b_n = ?$
- g) $\langle 5, 6, 5, 6, 5, 6, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = ?$, $b_n = ?$

3. Sequences given by a recurrent formula

The following sequences are given by a recurrent formula. Use it to write out at least the first five terms. Try determining the closed formula for the n^{th} term (formula that uses just the index n and not the previous terms).

- a) $a_1 = 4, \quad a_n = a_{n-1} + 2$
- b) $a_1 = 2, \quad a_n = 2 - a_{n-1}$
- c) $a_1 = 2, \quad a_n = -a_{n-1}$
- d) $a_1 = 1, \quad a_n = 1 + 2a_{n-1}$
- e) $a_1 = 2, \quad a_2 = 1, \quad a_n = a_{n-1} - a_{n-2}$

4. Properties of sequences

Write at least the first five terms of these sequences. Are the sequences increasing, decreasing, bounded from above, bounded from below? Prove your answers.

- a) $\left\{ \frac{2n+1}{5n-3} \right\}_{n=1}^{\infty}$
- b) $\left\{ \frac{4-m}{2m+5} \right\}_{m=1}^{\infty}$
- c) $\left\{ \frac{n+4}{n+3} \right\}_{n=1}^{\infty}$
- d) $\left\{ \frac{k+1}{2k-11} \right\}_{k=1}^{\infty}$

5. Limits

Find the limits of sequences:

a)

$$\left\{ \frac{4-n}{2n+5} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{t+4}{t+3} \right\}_{t=1}^{\infty}, \quad \left\{ \frac{k+1}{3k-11} \right\}_{k=1}^{\infty}, \quad \left\{ \frac{3r^2+2}{3-4r^2} \right\}_{r=1}^{\infty}$$

$$\left\{ \frac{n^2+2n-1}{1-2n+5n^2} \right\}_{n=1}^{\infty}$$

b)

$$\left\{ \frac{4n^5 - 9n^3 + 148}{4n^5 + 7n^2 - n} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{6l^4 + 5l + 4}{7l^3 + 8l + 9} \right\}_{l=1}^{\infty}, \quad \left\{ \frac{m^2 + 10m}{2m^2 + 5m^3} \right\}_{m=1}^{\infty}$$

c)

$$\left\{ \frac{3 + 5 \cdot 4^n}{5 + 7 \cdot 4^n} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{1 + 7 \cdot 5^j}{2 - 5 \cdot 7^j} \right\}_{j=1}^{\infty}$$

d)

$$\left\{s - \sqrt{s^2 + 2}\right\}_{s=1}^{\infty}, \quad \left\{\sqrt{n+3} - \sqrt{n-3}\right\}_{s=1}^{\infty}$$

e)

$$\left\{\frac{1}{n^2}\right\}_{n=1}^{\infty}, \quad \left\{\frac{1}{n^3+1}\right\}_{n=1}^{\infty}, \quad \left\{\frac{-2}{n^2+n}\right\}_{n=1}^{\infty}, \quad \left\{\frac{(-1)^n}{n^2}\right\}_{n=1}^{\infty}$$
$$\left\{\frac{\sin n}{n^2}\right\}_{n=1}^{\infty}$$

8.3.1 Hints and solutions

1. Sequences given by a closed form formula

Write the first few terms of sequences given by the formula.

- a) $\{5n\}_{n=1}^{\infty}; \quad \{5n\}_{n=0}^{\infty}; \quad \{5n\}_{n=3}^{\infty}$
- b) $\{n^2\}_{n=1}^{\infty}; \quad \{n^2 - 1\}_{n=0}^{\infty}; \quad \{n^2 + 5\}_{n=-2}^{\infty}; \quad \{3n^2\}_{n=1}^{\infty}$
- c) $\{2^n\}_{n=1}^{\infty}; \quad \{2^n\}_{n=0}^{\infty}; \quad \{2^n + 2\}_{n=-2}^{\infty}; \quad \{2^{n+2}\}_{n=-2}^{\infty}$
- d) $\{(-1)^n\}_{n=1}^{\infty}; \quad \{(-1)^n\}_{n=0}^{\infty}; \quad \{(-1)^{n+1}\}_{n=1}^{\infty}; \quad \{(-1)^{n+1} - (-1)^n\}_{n=1}^{\infty}$
- e) $\{(-1)^n + 3\}_{n=1}^{\infty}; \quad \{3 \cdot (-1)^n\}_{n=1}^{\infty}$
- f) $\left\{\frac{(-1)^{n+1} - (-1)^n}{2}\right\}_{n=1}^{\infty}; \quad \left\{\frac{(-1)^{n+1} - (-1)^n}{-2}\right\}_{n=1}^{\infty}$
- g) $\left\{\frac{1}{2^n}\right\}_{n=1}^{\infty}; \quad \left\{\frac{1}{2^n}\right\}_{n=0}^{\infty}; \quad \left\{\frac{1}{n^2}\right\}_{n=1}^{\infty}$

Hint: In computer science and coding, the sequences are related to cycles and indexing fields. We often need to input different values for a first few terms and then starting from some term, the sequence observes a certain rule. We should be able to capture such behavior using indices. In this problem, we consecutively insert the natural numbers into the variable n and express the individual sequence terms. The mathematicians say that the sequence is given explicitly, the computer scientists say that the terms are given in the closed form.

Solution:

- a) $\langle 5, 10, 15, 20, \dots \rangle; \quad \langle 0, 5, 10, 15, 20, \dots \rangle; \quad \langle 15, 20, 25, 30, \dots \rangle$
- b) $\langle 1, 4, 9, 16, 25, \dots \rangle; \quad \langle -1, 0, 3, 8, 15, 24, \dots \rangle; \quad \langle 9; 6; 5; 6; 9, 18, 21, \dots \rangle; \quad \langle 3, 12, 27, 48, 75, \dots \rangle$
- c) $\langle 2, 4, 8, 16, \dots \rangle; \quad \langle 1, 2, 4, 8, 16, \dots \rangle; \quad \langle 2\frac{1}{4}, 2\frac{1}{2}, 3, 4, 6, 10, 18, \dots \rangle; \quad \langle 1, 2, 4, 8, 16, \dots \rangle$
- d) $\langle -1, 1, -1, 1, -1, \dots \rangle; \quad \langle 1, -1, 1, -1, 1, \dots \rangle; \quad \{1, -1, 1, -1, 1, \dots\}; \quad \langle 2, -2, 2, -2, 2, -2, \dots \rangle$
- e) $\langle 2, 4, 2, 4, 2, 4, \dots \rangle; \quad \langle -3, 3, -3, 3, -3, 3, \dots \rangle$
- f) $\langle 1, -1, 1, -1, 1, -1, \dots \rangle; \quad \langle -1, 1, -1, 1, -1, 1, -1, \dots \rangle$
- g) $\langle \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots \rangle; \quad \langle 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots \rangle; \quad \langle 1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \frac{1}{25}, \frac{1}{36}, \dots \rangle$

2. Sequences given by their terms

For each sequence write a formula for the n^{th} term. If possible, write also a recurrent formula, which can be used to calculate a term using some previous terms.

- a) $\langle 3, 6, 9, 12, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=0}^{\infty} = \{c_n\}_{n=3}^{\infty}$
 $a_n = ?, \quad b_n = ?, \quad c_n = ?$
- b) $\langle 3, 7, 11, 15, 19, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty} = \{c_n\}_{n=-5}^{\infty}$
 $a_n = ?, \quad b_n = ?, \quad c_n = ?$
- c) $\langle 0, 3, 8, 15, 24, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=5}^{\infty}$
 $a_n = ?, \quad b_n = ?$
- d) $\langle 1, 2, 4, 8, 16, 32, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = ?, \quad b_n = ?$
- e) $\langle 3, -3, 3, -3, 3, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = ?, \quad b_n = ?$
- f) $\langle 2, 3, 5, 9, 17, 33, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = ?, \quad b_n = ?$
- g) $\langle 5, 6, 5, 6, 5, 6, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = ?, \quad b_n = ?$

Hints:

- While writing out the individual terms of a sequence given in a closed form is relatively simple, the inverse problem – finding the closed form of the sequence given by its terms – is more difficult.
- We need to look for regularity for example in differences or in ratios of the consecutive terms of the sequence.
- The expression for the n^{th} term of the sequence will depend on the first value of n .
- The recurrent relations may be easy (multiplying the previous term or adding a constant), may be impossible to find (prime numbers) and sometimes just difficult (using the powers of -1 or multiple sequences).
- For oscillating sequences, such as the one in g), we can use the fact that the sequence $\langle 1, -1, 1, -1, 1, -1, \dots \rangle$ oscillates around 0, with amplitude 1. It can be shifted to oscillate around a different number and stretched to oscillate with a different amplitude.

Solution:

- a) We notice that the terms are multiples of 3.
 $\langle 3, 6, 9, 12, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=0}^{\infty} = \{c_n\}_{n=3}^{\infty}$
 $a_n = 3n, \quad b_n = 3(n+1), \quad c_n = 3(n-2)$
- b) We notice that the terms are multiples of 4 decremented by 1.
 $\langle 3, 7, 11, 15, 19, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty} = \{c_n\}_{n=-5}^{\infty}$
 $a_n = 3 + 4n, \quad b_n = 3 + (n-2) \cdot 4, \quad c_n = 3 + (n+5) \cdot 4$
- c) $\langle 0, 3, 8, 15, 24, \dots \rangle = \{a_n\}_{n=1}^{\infty} = \{b_n\}_{n=5}^{\infty}$
 $a_n = n^2 - 1, \quad b_n = (n-4)^2 - 1$
- d) $\langle 1, 2, 4, 8, 16, 32, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = 2^n, \quad b_n = 2^{n-2}$
- e) $\langle 3, -3, 3, -3, 3, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = (-1)^n \cdot 3, \quad b_n = (-1)^{n+1} \cdot 3$
- f) $\langle 2, 3, 5, 9, 17, 33, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=2}^{\infty}$
 $a_n = 2^n + 1, \quad b_n = 2^{n-2} + 1$
- g) $\langle 5, 6, 5, 6, 5, 6, \dots \rangle = \{a_n\}_{n=0}^{\infty} = \{b_n\}_{n=1}^{\infty}$
 $a_n = 5.5 + 0.5 \cdot (-1)^{n+1}, \quad b_n = 5.5 + 0.5 \cdot (-1)^n$

3. Sequences given by a recurrent formula

The following sequences are given using a recurrent formula. Use it to write out at least the first five terms. Try determining the closed formula for the n^{th} term (formula that uses just the index n and not the previous terms).

- a) $a_1 = 4, \quad a_n = a_{n-1} + 2$
- b) $a_1 = 2, \quad a_n = 2 - a_{n-1}$
- c) $a_1 = 2, \quad a_n = -a_{n-1}$
- d) $a_1 = 1, \quad a_n = 1 + 2a_{n-1}$
- e) $a_1 = 2, \quad a_2 = 1, \quad a_n = a_{n-1} - a_{n-2}$

Hint: The expression for the n^{th} term of the sequence depends on the initial value of n . It is useful to start with writing the first few terms and looking at possible patterns that might emerge. Why do we use recurrent formulas then at all? Sometimes they are more convenient for calculating the sequence terms, e.g. calculating a binomial coefficient from an explicit formula is computationally intensive. Evaluating it as a sum

of two suitable smaller binomial coefficients is much easier. In cases where we cannot detect a pattern in the values, the closed form formula can be obtained using generating functions (these are beyond the scope of this text).

Solution:

- a) $a_1 = 4, \quad a_n = a_{n-1} + 2, \quad \langle 4, 6, 8, 10, \dots \rangle, \quad \{4 + 2(n-1)\}_{n=1}^{\infty}$
- b) $a_1 = 2, \quad a_n = 2 - a_{n-1}, \quad \langle 2, 0, 2, 0, 2 \dots \rangle, \quad \{(-1)^{n+1} - (-1)^n\}_{n=1}^{\infty}$
- c) $a_1 = 2, \quad a_n = -a_{n-1}, \quad \langle 2, -2, 2, -2, 2 \dots \rangle, \quad \{(-1)^{n+1} \cdot 2\}_{n=1}^{\infty}$
- d) $a_1 = 1, \quad a_n = 1 + 2a_{n-1}, \quad \langle 1, 3, 7, 15, 31 \dots \rangle, \quad \{2^n - 1\}_{n=1}^{\infty}$
- e) $a_1 = 2, \quad a_2 = 1, \quad a_n = a_{n-1} - a_{n-2}, \quad \langle 2, 1, -1, -2, -1, 1, 2, 1, -1, \dots \rangle$

4. Properties of sequences

Write at least the first five terms of these sequences. Are the sequences increasing, decreasing, bounded from above, bounded from below? Prove your answers.

- a) $\left\{ \frac{2n+1}{5n-3} \right\}_{n=1}^{\infty}$
- b) $\left\{ \frac{4-m}{2m+5} \right\}_{m=1}^{\infty}$
- c) $\left\{ \frac{n+4}{n+3} \right\}_{n=1}^{\infty}$
- d) $\left\{ \frac{k+1}{2k-11} \right\}_{k=1}^{\infty}$

Hint: When we try to answer the questions about sequences and prove their properties, it is useful to have a graphic representation of them.

Looking at Figure 8.1 we see that the sequence is decreasing and possibly bounded. For a decreasing sequence, it is true that $\forall k \in \mathbb{N} : a_k > a_{k+1}$. For our sequence $\left\{ \frac{2n+1}{5n-3} \right\}_{n=1}^{\infty}$ we should then have $\forall k \in \mathbb{N} :$

$$\frac{2 \cdot k + 1}{5 \cdot k - 3} > \frac{2 \cdot (k+1) + 1}{5 \cdot (k+1) - 3}.$$

We prove this statement by starting from a true statement, e.g. $2 > -9$, and modifying it by admissible (equivalent) operations to get the desired statement. How do we know where to start? That is, of course, difficult

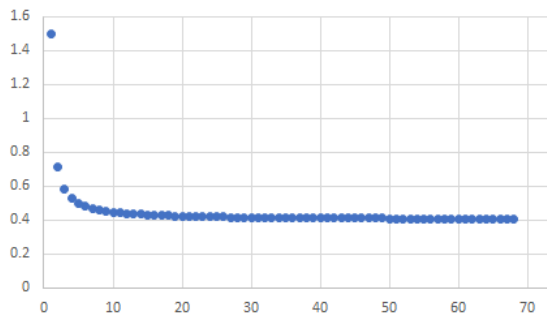


Figure 8.1: Graph of sequence $\left\{ \frac{2n+1}{5n-3} \right\}_{n=1}^{\infty}$.

to guess. Instead of guessing, we can take the inequality $\frac{2k+1}{5k-3} > \frac{2k+3}{5k+2}$ and modify it by equivalent operations until we get to an obviously true statement. We then take this as a starting point and reverse the order of the equivalent operations. In Chapter 12 we discuss this approach (and proofs in general) in more detail.

$$\begin{aligned}
 2 &> -9 \\
 2 + 10k^2 &> -9 + 10k^2 \\
 2 + 10k^2 + 9k &> -9 + 10k^2 + 9k \\
 (2k+1)(5k+2) &> (2k+3)(5k-3) \\
 \frac{2k+1}{5k-3} &> \frac{2k+3}{5k+2}
 \end{aligned}$$

Alternatively, we can calculate the difference

$$a_{k+1} - a_k = \frac{2(k+1)+1}{5(k+1)-3} - \frac{2k+1}{5k-3} = \frac{2k+3}{5k+2} - \frac{2k+1}{5k-3} = \frac{-11}{(5k+2)(5k-3)} < 0.$$

Since the difference is always negative, each next term is smaller than the previous one, hence the sequence is decreasing.

Regarding monotonicity, we should realise that if a sequence is decreasing, it cannot be increasing and vice versa. Consider also the difference between a decreasing and non-increasing sequence.

Solution:

- a) The terms of the sequence $\left\{\frac{2n+1}{5n-3}\right\}_{n=1}^{\infty}$ are: $\langle \frac{3}{2}, \frac{5}{7}, \frac{7}{12}, \frac{9}{17}, \frac{11}{22}, \dots \rangle$. The sequence is decreasing and bounded both from below and above.
- b) The terms of the sequence $\left\{\frac{4-m}{2m+5}\right\}_{m=1}^{\infty}$ are: $\langle \frac{3}{7}, \frac{2}{9}, \frac{1}{11}, 0, \frac{-1}{15}, \dots \rangle$. The sequence is decreasing and bounded both from below and above.
- c) The terms of the sequence $\left\{\frac{n+4}{n+3}\right\}_{n=1}^{\infty}$ are: $\langle \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{9}{8}, \dots \rangle$. The sequence is decreasing and bounded both from below and above.
- d) The terms of the sequence $\left\{\frac{k+1}{2k-11}\right\}_{k=1}^{\infty}$ are: $\langle \frac{-2}{9}, \frac{-3}{7}, \frac{-4}{5}, \frac{-5}{3}, \frac{-6}{1}, 7, \dots \rangle$. The sequence is neither decreasing nor increasing. It is bounded both from below and above.

5. Limits

Find the limits of sequences:

a)

$$\left\{\frac{4-n}{2n+5}\right\}_{n=1}^{\infty}, \quad \left\{\frac{t+4}{t+3}\right\}_{t=1}^{\infty}, \quad \left\{\frac{k+1}{3k-11}\right\}_{k=1}^{\infty}, \quad \left\{\frac{3r^2+2}{3-4r^2}\right\}_{r=1}^{\infty}$$

$$\left\{\frac{n^2+2n-1}{1-2n+5n^2}\right\}_{n=1}^{\infty}$$

b)

$$\left\{\frac{4n^5-9n^3+148}{4n^5+7n^2-n}\right\}_{n=1}^{\infty}, \quad \left\{\frac{6l^4+5l+4}{7l^3+8l+9}\right\}_{l=1}^{\infty}, \quad \left\{\frac{m^2+10m}{2m^2+5m^3}\right\}_{m=1}^{\infty}$$

c)

$$\left\{\frac{3+5 \cdot 4^n}{5+7 \cdot 4^n}\right\}_{n=1}^{\infty}, \quad \left\{\frac{1+7 \cdot 5^j}{2-5 \cdot 7^j}\right\}_{j=1}^{\infty}$$

d)

$$\left\{s - \sqrt{s^2+2}\right\}_{s=1}^{\infty}, \quad \left\{\sqrt{n+3} - \sqrt{n-3}\right\}_{s=1}^{\infty}$$

e)

$$\left\{\frac{1}{n^2}\right\}_{n=1}^{\infty}, \quad \left\{\frac{1}{n^3+1}\right\}_{n=1}^{\infty}, \quad \left\{\frac{-2}{n^2+n}\right\}_{n=1}^{\infty}, \quad \left\{\frac{(-1)^n}{n^2}\right\}_{n=1}^{\infty}$$

$$\left\{\frac{\sin n}{n^2}\right\}_{n=1}^{\infty}$$

Hints:

a)

$$\left\{ \frac{4-n}{2n+5} \right\}_{n=1}^{\infty}$$

A limit is a value to which the terms of the sequence tend to with increasing n . One way to find out the value of the limit is to draw the sequence terms, see Figure 8.2.

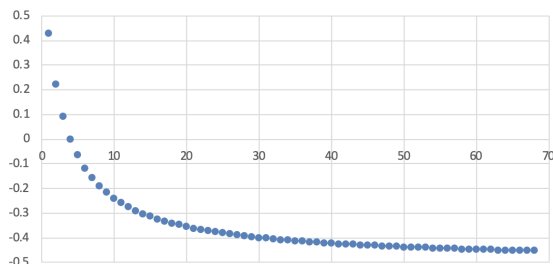


Figure 8.2: Graph of the sequence $\left\{ \frac{4-n}{2n+5} \right\}_{n=1}^{\infty}$.

Historically, limits have also been calculated by substituting infinitely large values for n and working with them as if they were real values, e.g.:

$$\frac{4-\infty}{2\infty+5} \quad \text{or} \quad \frac{-\infty}{\infty}.$$

This kind of calculation does not always lead to a correct answer. Therefore we should rather perform suitable algebraic operations with the expression in the limit that do not change the answer. In this particular case, we can multiply both the numerator and denominator by $\frac{1}{n}$.

$$\left\{ \frac{4-n}{2n+5} \cdot \frac{\frac{1}{n}}{\frac{1}{n}} \right\}_{n=1}^{\infty} = \left\{ \frac{4 \cdot \frac{1}{n} - n \cdot \frac{1}{n}}{2n \cdot \frac{1}{n} + 5 \cdot \frac{1}{n}} \right\}_{n=1}^{\infty} = \left\{ \frac{\frac{4}{n} - 1}{2 + \frac{5}{n}} \right\}_{n=1}^{\infty}$$

Since each of the four expressions has a finite limit, we can calculate the limits separately and insert the results into the original fraction.

Then we get

$$\lim_{n \rightarrow \infty} \frac{4-n}{2n+5} = \lim_{n \rightarrow \infty} \frac{\frac{4}{n} - 1}{2 + \frac{5}{n}} = \frac{\lim_{n \rightarrow \infty} \frac{4}{n} - \lim_{n \rightarrow \infty} 1}{\lim_{n \rightarrow \infty} 2 + \lim_{n \rightarrow \infty} \frac{5}{n}} = \frac{0-1}{2+0} = -\frac{1}{2}.$$

b)

$$\left\{ \frac{4n^5 - 9n^3 + 148}{4n^5 + 7n^2 - n} \right\}_{n=1}^{\infty}$$

Similarly to the previous problem, we multiply both the numerator and denominator by $\frac{1}{n^5}$.

For the problem

$$\left\{ \frac{6l^4 + 5l + 4}{7l^3 + 8l + 9} \right\}_{l=1}^{\infty}$$

we multiply both numerator and denominator by $\frac{1}{l^4}$ and get

$$\left\{ \frac{6 + \frac{5}{l^3} + \frac{4}{l^4}}{\frac{7}{l} + \frac{8}{l^3} + \frac{9}{l^4}} \right\}_{l=1}^{\infty}.$$

In this expression we see that the limit of the numerator is 6 and the limit of the denominator is 0. According to the Theorem *Ratio of a sequence with a proper limit and a sequence tending to zero* in Section 8.1.3 we get

$$\lim_{n \rightarrow \infty} \frac{6 + \frac{5}{l^3} + \frac{4}{l^4}}{\frac{7}{l} + \frac{8}{l^3} + \frac{9}{l^4}} = \infty.$$

c)

$$\left\{ \frac{3 + 5 \cdot 4^n}{5 + 7 \cdot 4^n} \right\}_{n=1}^{\infty}$$

In this case, we multiply both the numerator and denominator by $\frac{1}{4^n}$.

d)

$$\left\{ s - \sqrt{s^2 + 2} \right\}_{s=1}^{\infty}$$

In this case, for *infinite* s , we symbolically get $\infty - \infty$, which can evaluate to $-\infty$, ∞ or anything in between. To calculate the value of this kind of limit, we use the formula

$$(a-b)(a+b) = a^2 - b^2.$$

We can treat $s - \sqrt{s^2 + 2}$ as a fraction

$$\frac{s - \sqrt{s^2 + 2}}{1}$$

and multiply both its numerator and denominator by $s + \sqrt{s^2 + 2}$.

- e) Since $-1 \leq \sin n \leq 1$, we can find a lower bound $\{\frac{-1}{n^2}\}_{n=1}^{\infty}$ and an upper bound $\{\frac{1}{n^2}\}_{n=1}^{\infty}$ for the sequence $\{\frac{\sin n}{n^2}\}_{n=1}^{\infty}$. These two bounding sequences have the same limit, which is 0, so it is possible to deduce the limit of the sequence bounded by them using the *Squeeze theorem* from Section 8.1.3.

Solution: For the rest of the limits we provide the answers so that you can check your work:

a)

$$\lim_{n \rightarrow \infty} \frac{4-n}{2n+5} = -\frac{1}{2}, \quad \lim_{n \rightarrow \infty} \frac{t+4}{t+3} = 1, \quad \lim_{n \rightarrow \infty} \frac{k+1}{3k-11} = \frac{1}{3},$$

$$\lim_{n \rightarrow \infty} \frac{3r^2+2}{3-4r^2} = -\frac{3}{4}, \quad \lim_{n \rightarrow \infty} \frac{n^2+2n-1}{1-2n+5n^2} = \frac{1}{5}$$

b)

$$\lim_{n \rightarrow \infty} \frac{4n^5 - 9n^3 + 148}{4n^5 + 7n^2 - n} = 1, \quad \lim_{l \rightarrow \infty} \frac{6l^4 + 5l + 4}{7l^3 + 8l + 9} = \infty,$$

$$\lim_{m \rightarrow \infty} \frac{m^2 + 10m}{2m^2 + 5m^3} = \frac{1}{2}$$

c)

$$\lim_{n \rightarrow \infty} \frac{3 + 5 \cdot 4^n}{5 + 7 \cdot 4^n} = \frac{5}{7}, \quad \lim_{j \rightarrow \infty} \frac{1 + 7 \cdot 5^j}{2 - 5 \cdot 7^j} = 0$$

d)

$$\lim_{s \rightarrow \infty} s - \sqrt{s^2 + 2} = 0, \quad \lim_{n \rightarrow \infty} \sqrt{n+3} - \sqrt{n-3} = 0$$

e)

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0, \quad \lim_{n \rightarrow \infty} \frac{1}{n^3 + 1} = 0, \quad \lim_{n \rightarrow \infty} \frac{-2}{n^2 + n} = 0,$$

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n^2} = 0, \quad \lim_{n \rightarrow \infty} \frac{\sin n}{n^2} = 0$$

8.4 Lab problems

1. Drawing sequences

- a) Calculate the first 1000 terms of a linear fraction sequence $\{a_n\}_{n=1}^{\infty}$ and draw them in a chart. Using the chart, estimate the properties of the sequence (boundedness, monotonicity, limit). Prove these properties also theoretically. Use can use the sequences:

$$\left\{ \frac{3n+2}{4n-3} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{5-n}{2n+4} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{n+3}{n+2} \right\}_{n=1}^{\infty},$$

$$\left\{ \frac{2n+1}{3n-13} \right\}_{n=1}^{\infty}, \quad \left\{ \frac{2n-3}{3n+2} \right\}_{n=1}^{\infty}.$$

- b) Calculate the first 1000 terms of an arithmetic sequence $\{a_n\}_{n=1}^{\infty}$ using parameters a_1 , d and n , where a_1 is the first term of this sequence, d is the difference and n is the index of the n^{th} term. Find also a recurrent formula for this sequence.
- c) Calculate the first 1000 terms of a geometric sequence $\{a_n\}_{n=1}^{\infty}$ using parameters a_1 , q and n , where a_1 is the first term of this sequence, q is its quotient and n is the index of the n^{th} term. Find also a recurrent formula for this sequence.
- d) Calculate the first 1000 terms of the Fibonacci sequence given by the recurrent formula $a_n = a_{n-1} + a_{n-2}$, $a_1 = a_2 = 1$.
- e) Create a new sequence from the Fibonacci sequence using a formula: $\left\{ \frac{f_n}{f_{n+1}} \right\}_{n=1}^{\infty}$, where f_n are the terms of the Fibonacci sequence. What are the properties of this new sequence?

2. Properties of sequences

- a) Calculate the terms and draw a chart of the sequence $\left\{ \frac{a \cdot n + b}{c \cdot n + d} \right\}_{n=1}^{\infty}$ given by the input parameters a, b, c, d . Observe how the properties of the sequence vary with respect to the values of parameters a, b, c, d .
- b) For which values of the input parameters a, b, c, d is the sequence $\left\{ \frac{a \cdot n + b}{c \cdot n + d} \right\}_{n=1}^{\infty}$ increasing? For which values is it decreasing? When is it (is not) bounded from above/below?

- c) For which values does the sequence have a limit? What is the value of the limit? How does the sequence approach this limit?
- d) Propose a *typology* of functions according to how their properties depend on the parameters a, b, c, d .

3. Logistic map

Calculate the terms and draw a chart of the sequence $\{x_n\}_{n=1}^{\infty}$ given by the recurrent relation:

$$x_n = a \cdot x_{n-1} \cdot (1 - x_{n-1})$$

The values of the parameter a are from the interval $(0, 4)$ and the values of x_1 from the interval $(0, 1)$. Calculate at least the first 5 000 terms of the sequence and display them in a chart. Examine the behavior of the sequence in response to changes in parameter a .

4. Operations with sequences

Calculate the first 100 terms of a sequence $\{a_n\}_{n=1}^{\infty}$.

- a) Calculate the terms of the sequence of partial sums of the original sequence $\{a_n\}_{n=1}^{\infty}$, i.e. the sum of terms from the first to the n^{th} term.
- b) Calculate the differences of two consecutive terms.
- c) Calculate the ratios of two consecutive terms.
- d) Calculate the ratio of a term and its index.
- e) Calculate the ratio of the sum of the first n terms and the index n .

Display the original sequence and the new sequences together in a chart. Try this for various sequences $\{a_n\}_{n=1}^{\infty}$ and observe the properties of sequences a) to e). How do the partial sums look for the sequence $(\frac{1}{2})^n$? How do they look for the sequence $(\frac{1}{10})^n$? And how do they look for a geometric sequence in general?

Consider the sequence

$$\left(1 + \frac{1}{n}\right)^n.$$

Is it bounded? Does it have a limit?

Examine the sequence

$$\left(\frac{1}{n!}\right)$$

and the sequence of its partial sums.

5. Iterative calculation of π and e

- a) Iteratively calculate approximate values of π as a ratio of the circumference and diameter of a regular n -sided polygon for increasing n .
- b) Create a similar Excel sheet for calculation of e using compound interest rates, when we add the yearly 100% interest in two 50% sums, three 33.3333% sums, four 25% sums, etc. Transform this problem into a problem with power function. What is the connection of this problem to previous problems?

8.4.1 Hints and solutions

1. Drawing sequences

Hints:

- a) To get an idea of how the terms of a sequence look like, we draw it in a chart. It is useful to create an Excel sheet for a sequence $\left\{ \frac{a \cdot n + b}{c \cdot n + d} \right\}_{n=1}^{\infty}$ so that we have the sequence parametrised by four values a , b , c and d . There should be infinitely many values of the independent variable n , but of course, we will only use a (large) finite number, e.g. $1, 2, 3, \dots, 1000$. Then we calculate the value of a general a_n . Note that even if we draw the sequence as individual points, if there are sufficiently many of them, they will look like a continuous line. The graphs of the sequences will be different for different values of parameters a , b , c and d .

If we wanted to determine the values of parameters for which the sequences would like we see in Figure 8.3 we could evaluate the expression $\frac{a \cdot n + b}{c \cdot n + d}$ for $n = 1, n = 2, n = 3$ and $n = 4$. This would give us a system of four equations with four unknown parameters. We can also look at the ratio of parameters a and c and observe an interesting rule. Similarly, we can try different values of parameters to determine their impact on monotonicity.

- b) First, we should designate two cells as input cells for the parameters a_1 and d . Then we can calculate the values of the sequence $\{a_n\}_{n=1}^{\infty}$ by using the closed form formula involving the current value of n and fixed values of a_1 and d . Next to it, we can calculate the terms of the sequence using a recurrent formula, i.e. using the previous term of the sequence and the parameter d . If we do not make an error, the two columns with the sequence terms should be identical.
- c) Again, we can first designate the input cells for the parameters a_1 and q . Then we can calculate the values of the sequence by using the closed form formula involving the current value of n and fixed values of a_1 and q . Next to it, we can calculate the terms of the sequence using a recurrent formula. Also in this case, we should obtain two identical columns.
- d) The graph of the Fibonacci sequence is shown in Figure 8.4.

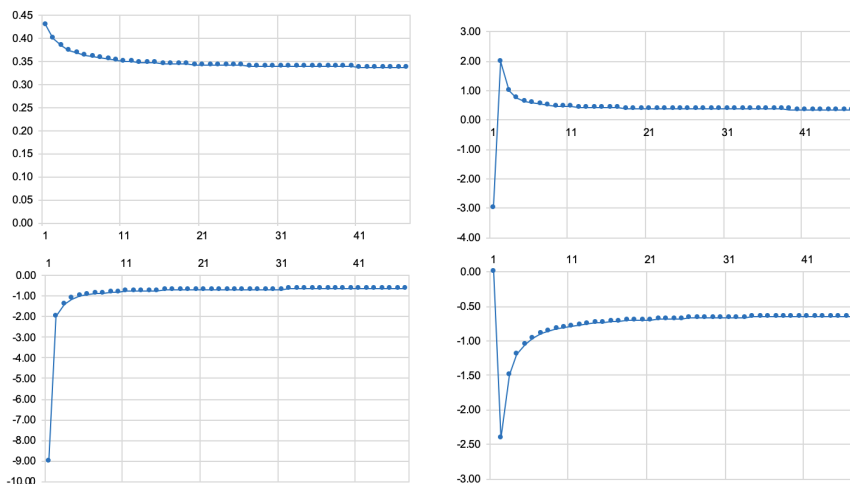


Figure 8.3: Graphs of the sequence $\left\{ \frac{a \cdot n + b}{c \cdot n + d} \right\}_{n=1}^{\infty}$ for different values of parameters a, b, c, d .

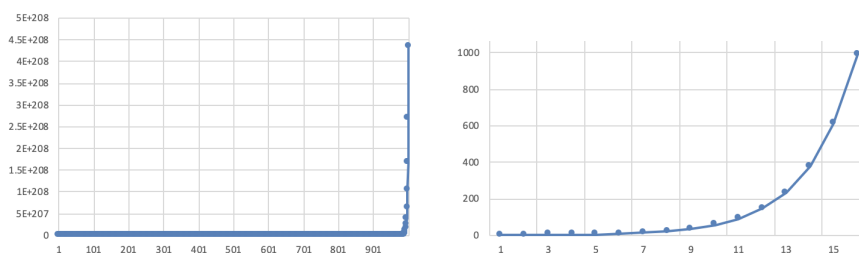


Figure 8.4: Fibonacci sequence, first 1000 terms and first 16 terms.

e) Using the Fibonacci sequence, we create a new sequence

$$\left\{ \frac{f_n}{f_{n+1}} \right\}_{n=1}^{\infty},$$

where f_n are the original Fibonacci terms. From a graph, Figure 8.5, we see that this sequence is neither increasing, nor decreasing. It has a limit $\frac{1-\sqrt{5}}{2}$, which is known as the *golden ratio*.

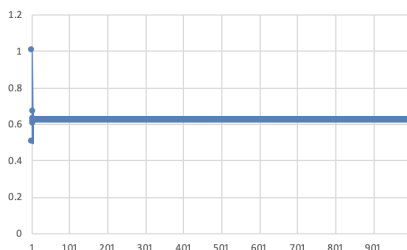


Figure 8.5: Golden ratio.

2. Properties of sequences

Hints:

- Similarly to the previous problem, we start with parameters a, b, c, d that can vary and create and draw a sequence $\left\{ \frac{a \cdot n + b}{c \cdot n + d} \right\}_{n=1}^{\infty}$ using an independent variable n . First, try varying only one of the four parameters at a time and observe the resulting changes in the graph. Then try changing two parameters at the same time. Examine which parameters have impact on which properties of the sequence.
- If we try for example $a = 2, b = 1, c = 0, d = 1$, we see that the sequence is not bounded from above and has an improper limit ∞ . Using the same parameters $a = 2, b = 1, c = 0$ and changing d to $d = -1$, we get a sequence that is not bounded from below with improper limit $-\infty$. We get the same properties for parameters $a = 4, b = 1, c = 0, d = -2$.

We see that not only the specific values of parameters impact the sorting of sequences into categories, but (and perhaps even more so) the ratios between them.

3. Logistic map

Hints: We create a sheet in which we calculate the values of a sequence given by a recurrent formula $x_n = a \cdot x_{n-1} \cdot (1 - x_{n-1})$ for parameters $a \in (0, 4)$ and $x_1 \in (0, 1)$. The change of sequence behavior is much more interesting when we change the parameter a compared to changes in x_1 .

Solution: As a increases, the sequence loses its limit but has a growing number of subsequences that have limits (we call these numbers *accumu-*

lation points). Figure 8.6 shows the sequences with two and four accumulation points.

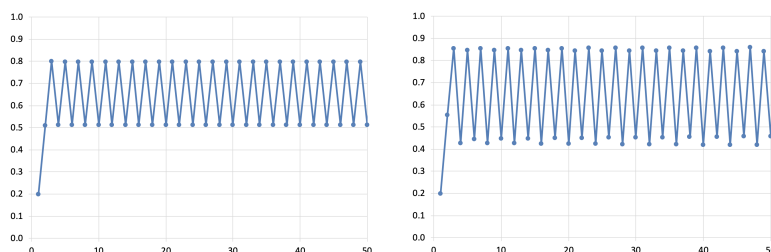


Figure 8.6: Logistic map with parameters $x_1 = 0.2$, $a = 3.1$ (left) has two accumulation points, with parameters $x_1 = 0.2$, $a = 3.46$ (right) has four accumulation points.

If we wanted to investigate the accumulation points, we could split the sequence into subsequences by picking every other term (into the next column and separate chart) in case of two accumulation points. Similarly, we could split the terms into four subsequences in case of four accumulation points.

For even larger values of a we observe *chaos*, Figure 8.7. For almost all $a > 3.6$ we no longer see regular oscillations and even a slight change in x_1 yields dramatically different results.

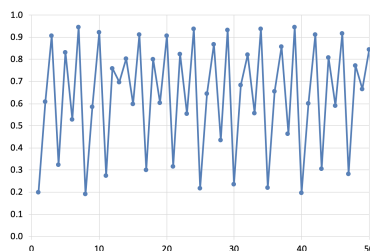


Figure 8.7: Logistic map with parameters $x_1 = 0.2$, $a = 3.8$.

4. Operations with sequences

Hints:

- It is useful to calculate each new sequence (difference, ratio, partial sums, etc.) in a separate column.
- Differences and ratios are primarily interesting for the arithmetic and geometric sequences.
- The sequence of partial sums grows rapidly and thus forces an increase in the range displayed on the y -axis. The behavior of the other sequences cannot be clearly seen then. Therefore it might be useful to remove it from the graph when observing the other sequences or perhaps create a separate graph for it.

Solution: An example sequence and the new sequences created from it are shown in Figure 8.8.

5. Iterative calculation of π and e

- a) *Solution:* Since we can approximate a circle using a regular polygon with n -sides, we can approximate the value of π using either the area or circumference of this polygon. If we use polygons inscribed into a circle, we get an approximation from below, if we use circumscribed polygons, we get an approximation from the above, Figure 8.9. The larger the n , the better the approximation.

To calculate the circumference of a unit circle ($r = 1$), we use the formula $O = 2\pi \cdot r$. The circumference of an equilateral triangle approximates (albeit badly) this quantity. To find out the circumference of the triangle, we can draw and measure it. This is what Archimedes of Syracuse did approximately 250 years AD. He inscribed and circumscribed a regular 96-sided polygon to the circle.

German-Dutch mathematician Ludolph van Ceulen repeated this calculation in 1596 with higher precision and obtained an approximation of π which is correct up to the first 20 digits. Is it possible to find out what polygon he used? And what was the error of Archimedes' approximation?

In Excel, we calculate the circumference of a regular n -sided polygon inscribed in a circle for n from 3 to 100 (or more if necessary). For that we use the sine function, inner angle of the regular polygon and the radius of the circle $r = 1$. The calculation also uses the

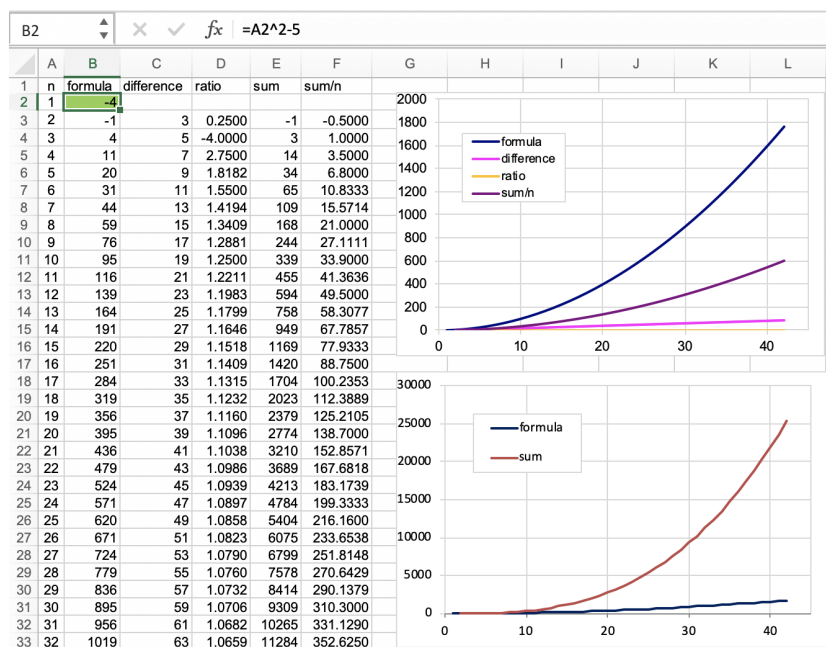


Figure 8.8: The sequence $\{n^2 - 5\}_{n=1}^{\infty}$, differences and ratios of its successive terms, partial sum and a ratio of the partial sum to the index.

Excel value of π which, of course, was not available for Archimedes or Ludolph, but we are not looking for the value of π , but rather how well can we approximate this value by a precise drawing.

The size of the angle α , see Figure 8.10, in a regular polygon is $\alpha = \frac{2\pi}{2n}$ and the sine of this angle is $\sin(\alpha) = \frac{x}{r}$. The circumference of the polygon is then

$$O_n = n \cdot 2 \cdot x = n \cdot 2 \cdot \sin\left(\frac{2\pi}{2n}\right) = 2 \cdot n \cdot \sin\left(\frac{\pi}{n}\right).$$

With growing n , the circumference of the polygon O_n approaches the circumference of the circle O_c .

$$O_n = 2 \cdot n \cdot \sin\left(\frac{\pi}{n}\right), \quad O_c = 2 \cdot \pi \cdot r$$

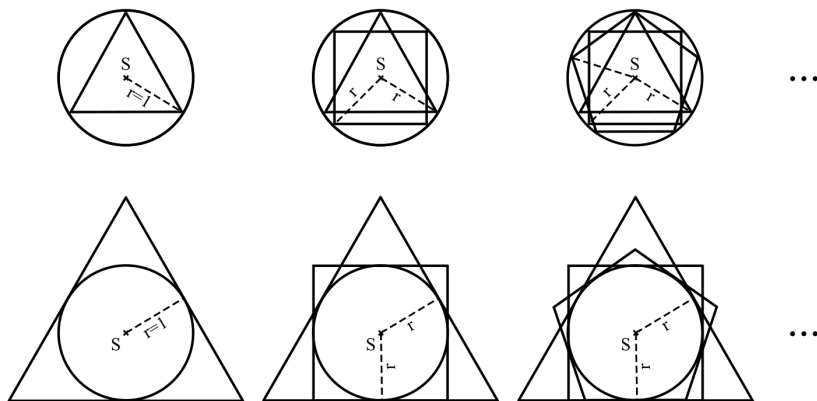


Figure 8.9: Iterative approximation of π using a regular polygon with n sides. The inscribed polygons lead to an approximation from below, the circumscribed polygons lead to the approximation from above.

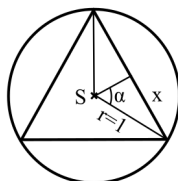


Figure 8.10: Relationship of the side length of a regular n -sided polygon and a unit circle circumscribed to this polygon.

The approximation of π can thus be calculated from

$$\pi = \frac{O_n}{2}.$$

The difference of the approximation and π is the approximation error. (In Excel, π can be evaluated using the function =PI().)

Using a table shown in Figure 8.11, it can be shown that the approximation error in case of regular 22-sided polygon is 0.010666212. Regular 96-sided polygon gives the error 0.000560702, which means the first four digits of the approximation are correct.

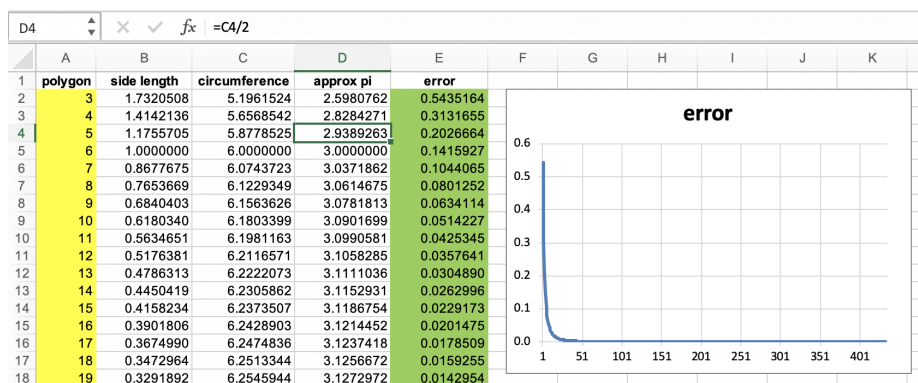


Figure 8.11: The side length of a regular polygon with n sides, which is inscribed in a circle, calculated using a sine function. The table also shows the iterative approximations of the value of π and the approximation error.

- b) *Solution:* Figure 8.12 shows how we can use the cumulative interest rates to iteratively approximate the value of the Euler number e .

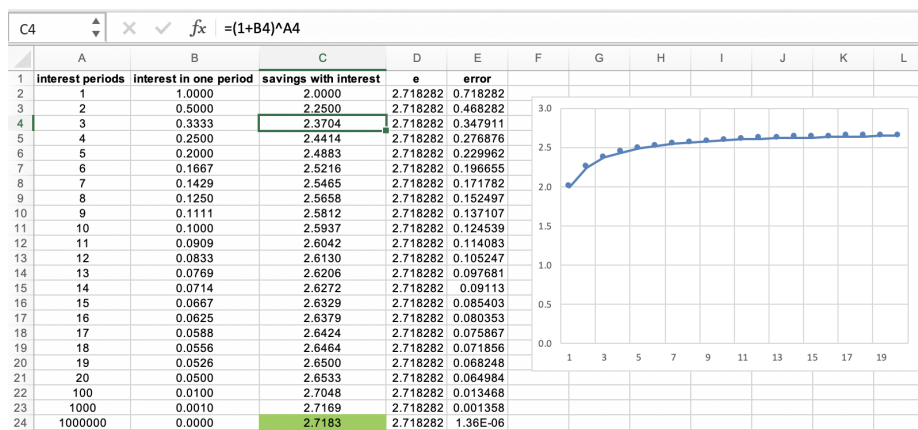


Figure 8.12: Calculating the final balance for more frequent addition of the compound interest.

Chapter 9

Sums and products

9.1 Lecture overview

9.1.1 Sums

Notation for sums

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$, we then write

$$a_1 + a_2 + \dots + a_n = \sum_{i=1}^n a_i$$

More generally: If $a_1, a_2, \dots, a_n \in \mathbb{R}$, we then write

$$a_k + a_{k+1} + a_{k+2} + \dots + a_l = \sum_{i=k}^l a_i$$

Some examples of using this notation are:

- $1 + 2 + 3 + 4 + 5 + 6 = \sum_{i=1}^6 i$
- $1 + 3 + 5 + 7 + 9 = \sum_{i=1}^5 (2i - 1) = \sum_{j=0}^4 (2j + 1) = 25 = 5^2$
- $\sum_{i=1}^7 (2i - 1) = 1 + 3 + 5 + 7 + 9 + 11 + 13 = 49 = 7^2$

Sums of first n numbers with given property

Using mathematical induction (Section 12.1.2), it can be proven that the *sum of the first n odd numbers* is

$$\sum_{i=1}^n (2i - 1) = n^2$$

The formula for the *sum of the first n natural numbers* can be proven using the Pythagorean method with little pebbles arranged in a triangle or a Gaussian method by adding two opposite rows of numbers:

$$\sum_{i=1}^n i = \frac{n \cdot (n + 1)}{2}$$

The following formula for the *sum of the first n squares* can also be proven by induction:

$$\sum_{i=1}^n i^2 = 1 + 4 + 9 + 16 + 25 + \cdots + n^2 = \frac{n \cdot (n + 1) \cdot (2n + 1)}{6}$$

Problem 9.1 Sum of the first 100 numbers:

Calculate $1 + 2 + 3 + 4 + \cdots + 100 =$

$$\text{Solution: } 1 + 2 + 3 + 4 + \cdots + 100 = \sum_{i=1}^{100} i = \frac{100 \cdot 101}{2} = 5050$$

Problem 9.2 Sum of even numbers up to 100:

Calculate $2 + 4 + 6 + 8 + \cdots + 100 =$

$$\text{Solution: } \sum_{i=1}^{50} 2i = 2 + 4 + 6 + 8 + \cdots + 100 = 2 \cdot (1 + 2 + 3 + 4 + \cdots + 50) =$$

$$2 \sum_{i=1}^{50} i = 2 \frac{50 \cdot 51}{2} = 2550. \text{ Here we showed that}$$

$$\sum_{i=1}^{50} 2i = 2 \sum_{i=1}^{50} i.$$

Problem 9.3 Sum that does not start at the beginning:

Evaluate $51 + 52 + 53 + 54 + \cdots + 100 =$

Solution: $51 + 52 + 53 + 54 + \dots + 100 = \sum_{i=51}^{100} i = (50 + 1) + (50 + 2) + (50 + 3) + (50 + 4) + \dots + (50 + 50) =$

$$= \sum_{i=1}^{50} (50 + i) = 50 \cdot 50 + \sum_{i=1}^{50} i = 50 \cdot 50 + \frac{50 \cdot 51}{2} = 2500 + 1275 = 3775$$

Another solution:

$$\sum_{i=1}^{100} i = \sum_{i=1}^{50} i + \sum_{i=51}^{100} i$$

$$\sum_{i=51}^{100} i = \sum_{i=1}^{100} i - \sum_{i=1}^{50} i = \frac{100 \cdot 101}{2} - \frac{50 \cdot 51}{2} = 5050 - 1275 = 3775$$

Rules for working with sums

$$1. \sum_{i=1}^n c = n \cdot c$$

$$2. \sum_{i=k}^n c = (n - k + 1) \cdot c$$

$$3. \sum_{i=1}^n c \cdot a_i = c \cdot \sum_{i=1}^n a_i$$

$$4. \sum_{i=k}^n c \cdot a_i = c \cdot \sum_{i=k}^n a_i$$

$$5. \sum_{i=1}^n a_i = \sum_{i=1}^k a_i + \sum_{i=k+1}^n a_i$$

$$6. \sum_{i=1}^n a_i = \sum_{i=1+k}^{n+k} a_{i-k}$$

$$7. \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

9.1.2 Products

Notation for products

$$1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = \prod_{i=1}^6 i = 6!$$

Some examples of using this notation are:

- $2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 = \prod_{i=1}^6 2i = 2^6 \cdot 6!$

- One possible explanation for $0! = 1$

$$4! = 4 \cdot 3! \Rightarrow 3! = \frac{4!}{4} \Rightarrow 2! = \frac{3!}{3} \Rightarrow 1! = \frac{2!}{2} \Rightarrow 0! = \frac{1!}{1} \Rightarrow 0! = 1$$

- $\prod_{i=4}^7 i = 4 \cdot 5 \cdot 6 \cdot 7 = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7}{1 \cdot 2 \cdot 3} = \frac{\prod_{i=1}^7 i}{\prod_{i=1}^3 i} = \frac{7!}{3!}$

Rules for working with products

1. $\prod_{i=1}^n c = c^n$

2. $\prod_{i=1}^n c \cdot i = c^n \prod_{i=1}^n i = c^n \cdot n!$

3. $\prod_{i=1}^n i = \prod_{i=1+k}^{n+k} (i-k)$

4. $\prod_{i=1}^k i \cdot \prod_{i=k+1}^n i = \prod_{i=1}^n i$

5. $\prod_{i=1}^k (a_i b_i) = \prod_{i=1}^k a_i \cdot \prod_{i=1}^k b_i$

9.1.3 Double sums/products

In the following problems, we use these two finite sequences:

$$\{a_n\}_{n=1}^5 = \{2, 5, 8, 11, 14\}, \quad \{b_m\}_{m=1}^3 = \{3, 6, 12\}$$

Problem 9.4 Double sum:

Evaluate $\sum_{n=1}^5 \sum_{m=1}^3 a_n b_m$ in two different ways.

Solution:

$$\begin{aligned} \sum_{n=1}^5 \sum_{m=1}^3 a_n b_m &= \left(\sum_{n=1}^5 a_n \right) \left(\sum_{m=1}^3 b_m \right) = (2 + 5 + 8 + 11 + 14)(3 + 6 + 12) \\ \sum_{n=1}^5 \sum_{m=1}^3 a_n b_m &= \sum_{n=1}^5 \sum_{m=1}^3 (a_n b_m) = \sum_{n=1}^5 (3a_n + 6a_n + 12a_n) \\ &= 3 \cdot 2 + 6 \cdot 2 + 12 \cdot 2 + 3 \cdot 5 + 6 \cdot 5 + 12 \cdot 5 + \dots \end{aligned}$$

Problem 9.5 Sum of products:

Only one of the following calculations is correct. Which one? How can be the incorrect one corrected?

$$\begin{aligned} \sum_{n=1}^5 \prod_{m=1}^3 a_n b_m &= \left(\sum_{n=1}^5 a_n \right) \left(\prod_{m=1}^3 b_m \right) = (2 + 5 + 8 + 11 + 14)(3 \cdot 6 \cdot 12) \\ \sum_{n=1}^5 \prod_{m=1}^3 a_n b_m &= \sum_{n=1}^5 \prod_{m=1}^3 (a_n b_m) = \sum_{n=1}^5 (3a_n \cdot 6a_n \cdot 12a_n) \\ &= 3 \cdot 2 \cdot 6 \cdot 2 \cdot 12 \cdot 2 + 3 \cdot 5 \cdot 6 \cdot 5 \cdot 12 \cdot 5 + \dots \end{aligned}$$

Problem 9.6 Product of sums:

Only one of the following calculations is correct. Which one?

$$\begin{aligned} \prod_{n=1}^5 \sum_{m=1}^3 a_n b_m &= \left(\prod_{n=1}^5 a_n \right) \left(\sum_{m=1}^3 b_m \right) = (2 \cdot 5 \cdot 8 \cdot 11 \cdot 14)(3 + 6 + 12) \\ \prod_{n=1}^5 \sum_{m=1}^3 a_n b_m &= \prod_{n=1}^5 \sum_{m=1}^3 (a_n b_m) = \prod_{n=1}^5 (3a_n + 6a_n + 12a_n) \\ &= (3 \cdot 2 + 6 \cdot 2 + 12 \cdot 2)(3 \cdot 5 + 6 \cdot 5 + 12 \cdot 5) \cdot \dots \end{aligned}$$

Problem 9.7 Double product:

Only one of the following calculations is correct. Which one?

$$\begin{aligned} \prod_{n=1}^5 \prod_{m=1}^3 a_n b_m &= \left(\prod_{n=1}^5 a_n \right) \left(\prod_{m=1}^3 b_m \right) = (2 \cdot 5 \cdot 8 \cdot 11 \cdot 14)(3 \cdot 6 \cdot 12) \\ \prod_{n=1}^5 \prod_{m=1}^3 a_n b_m &= \prod_{n=1}^5 \prod_{m=1}^3 (a_n b_m) = \prod_{n=1}^5 (3a_n \cdot 6a_n \cdot 12a_n) \\ &= (3 \cdot 2 \cdot 6 \cdot 2 \cdot 12 \cdot 2)(3 \cdot 5 \cdot 6 \cdot 5 \cdot 12 \cdot 5) \dots \end{aligned}$$

9.1.4 Geometric sequence and series**Problem 9.8 Finite chocolate:**

Five friends eat chocolate in such a way that the first one eats half of it and each next friend eats half of what was left. How much chocolate remains after they all had their shares?

Solution: In the end there is $\frac{1}{32}$ of the chocolate and they ate $\frac{31}{32}$:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = 1 - \frac{1}{32} = \frac{31}{32}$$

More generally for n friends:

$$\sum_{i=1}^n \frac{1}{2^i} = \frac{2^n - 1}{2^n}$$

Sum of finite geometric series

Suppose the first term of the series is 1 and quotient is q . The sum of the series is then

$$\begin{aligned} S_n &= 1 + q + q^2 + q^3 + \dots + q^n \\ q \cdot S_n &= q + q^2 + q^3 + \dots + q^n + q^{n+1} \\ S_n - q \cdot S_n &= 1 - q^{n+1} \\ S_n &= \frac{1 - q^{n+1}}{1 - q}, \quad q \neq 1 \end{aligned}$$

Problem 9.9 Infinite chocolate:

What happens when we extend the finite chocolate series from Problem 9.8 and have infinitely many friends?

Solution: We get the sum

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots = 1.$$

Problem 9.10 Achilles and turtle:

The ancient Greek hero Achilles runs 10 times faster than a turtle. The turtle gets a head start of 1km. Every time Achilles gets to the place where the turtle was in the previous step, the turtle covers $\frac{1}{10}$ of the distance Achilles does. Will he catch the turtle or not? (This paradox belongs to the Zenon's aporias.)

Solution: The ancient Greeks considered this problem long and hard. The crucial idea is that a finite length can be expressed as a sum of an infinite number of shorter and shorter parts. Even though we can keep on telling the race story indefinitely, Achilles catches the turtle after $10/9$ km. This can be calculated using the sum of geometric series with the first term 1 and a quotient $\frac{1}{10}$.

$$\begin{aligned} 1 + \frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \dots &= S \\ 10 + 1 + \frac{1}{10} + \frac{1}{100} + \dots &= 10S \\ 10 + S &= 10S \\ 10 &= 9S \\ S &= \frac{10}{9} \end{aligned}$$

Sum of infinite geometric series

This sum depends on the limit of the corresponding sequence:

$$\begin{aligned} \lim_{n \rightarrow \infty} q^n &= 0 && \text{for } |q| < 1 \\ &= 1 && \text{for } q = 1 \\ &= \infty && \text{for } q > 1 \\ &= \text{does not exist} && \text{for } q = -1 \\ &= \text{does not exist} && \text{for } q < -1 \end{aligned}$$

When the first term of the series is 1 and quotient is $|q| < 1$:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{i=0}^n q^i = \lim_{n \rightarrow \infty} \frac{1 - q^{n+1}}{1 - q} = \frac{1}{1 - q}$$

Similarly when the first term is a_0 and quotient is $|q| < 1$:

$$\lim_{n \rightarrow \infty} a_0 \cdot S_n = \lim_{n \rightarrow \infty} \sum_{i=0}^n a_0 \cdot q^i = \lim_{n \rightarrow \infty} a_0 \cdot \frac{1 - q^{n+1}}{1 - q} = a_0 \cdot \frac{1}{1 - q}$$

If we use this approach with unsuitable q (e.g. with $q = 10$, $q = 20$, $q = 100$ or other cases when the series do not have a finite sum), we get an incorrect answer.

Problem 9.11 Periodic numbers:

How much is $0.\bar{9}$?

Solution:

$$\begin{aligned} 0.\bar{9} &= 3 \cdot 0.\bar{3} = 3 \cdot \frac{1}{3} \\ &= 0.9 + 0.09 + 0.009 + \dots = \frac{9}{10} \cdot \left(1 + \frac{1}{10} + \frac{1}{100} + \dots\right) \\ &= \frac{9}{10} \cdot \frac{1}{1 - \frac{1}{10}} = \frac{9}{10} \cdot \frac{10}{9} = 1 \end{aligned}$$

Geometric series we can now evaluate

1.

$$\sum_{n=0}^{\infty} q^n = (1 + q + q^2 + q^3 + \dots) = \frac{1}{1 - q}$$

2.

$$\sum_{n=0}^{\infty} a_0 \cdot q^n = a_0 \cdot (1 + q + q^2 + q^3 + \dots) = a_0 \cdot \frac{1}{1 - q}$$

3.

$$\sum_{n=k}^{\infty} q^n = (q^k + q^{k+1} + q^{k+2} + \dots) = q^k (1 + q + q^2 + \dots) = \frac{q^k}{1 - q}$$

4.

$$\sum_{n=0}^{\infty} (-q)^n = (1 - q + q^2 - q^3 + q^4 - \dots) = \frac{1}{1 - (-q)} = \frac{1}{1 + q}$$

5.

$$\sum_{n=0}^{\infty} q^{2n} = (1 + q^2 + q^4 + q^6 + \dots) = \frac{1}{1 - q^2}$$

9.2 Questions

1. State the formula for the sum of first n natural numbers. How much is $1 + 2 + 3 + 4 + \dots + 98 + 99$?
2. State the formula for the sum of first n odd numbers. How much is $1 + 3 + 5 + 7 + \dots + 33$?
3. State the formula for the sum of first n squares. How much is $1 + 4 + 9 + 16 + \dots + 64$?
4. Express $\sum_{i=1}^5 (a_{i+1} - a_i)$ using two numbers.
5. Let a_k be the k^{th} term of the geometric sequence, where $a_1 = 7$ and $q = 1/7$. Calculate $\sum_{k=1}^7 a_k$
6. Calculate $4! =$, $3! + 3! =$, $4! \cdot 4! =$, $0! =$, $(2 \cdot 3)! =$, $1! =$
7. Calculate
 - a) $\prod_{k=1}^5 2k$
 - b) $\prod_{k=1}^5 k^2$
8. Express the following using factorials
 - a) $\prod_{k=3}^1 5k$
 - b) $\prod_{k=1}^5 3k$
9. Suppose we divide a chocolate in two halves, eat one and leave the other for next day. The next day we repeat the process with the half and so on. The last piece we eat was $1/128$ of the original chocolate. How many pieces have we eaten? What part of original chocolate was it? How many days do we need until the remaining piece of chocolate is one molecule?
10. Calculate the sum of geometric series with quotient $q = 1/6$. Use the assumption that the sum S exists.

11. Calculate the sum of geometric series with quotient $q = -1/6$. Use the assumption that the sum S exists.
12. Calculate the sum of geometric series with quotient $q = 6$. Use the assumption that the sum S exists.
13. Express as infinite series and evaluate
 - a) $1.111\bar{1}$
 - b) $9.99999\bar{9}$
 - c) $3.333\bar{3}$
14. How do we calculate $\sum_{k=0}^{\infty} q^k$?

9.3 Problem set

Sums

1. Write the following expressions using the $\sum$ notation:

a) $1 + 2 + 3 + \dots + 20$

b) $7 + 14 + 21 + \dots + 70$

c) $6 + 8 + 10 + \dots + 6n$

d) $12n + 12n + 6 + 12n + 12 + 12n + 18 + \dots + 24n$

2. Using $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ calculate

a) $\sum_{i=1}^{10} 4i, \quad \sum_{j=1}^n 4j$

b) $\sum_{k=1}^{100} 3k, \quad \sum_{k=1}^{2n} 3k$

c) $\sum_{t=1}^{25} 5, \quad \sum_{i=5}^{3n} 2i$

d) $\sum_{i=8}^{16} i, \quad \sum_{j=n}^{2n} j$

e) $\sum_{i=2n}^{3n} 2k, \quad \sum_{t=2n}^{4n} 6t$

3. Using $\sum_{i=1}^n (2i-1) = n^2$ calculate

a) $\sum_{j=1}^{3n} (2j-1)$

b) $\sum_{i=0}^n 4(2i+1)$

c) $\sum_{k=2n}^{4n} (2k-1)$

4. Split into two sums and calculate

a) $\sum_{i=1}^n (4i + 3)$

b) $\sum_{j=1}^{3n} (2j - 1)$

c) $\sum_{k=1}^{2n} (-1)^k 3k$

5. Calculate

a) $\sum_{k=1}^{20} (k + 2)$

b) $\sum_{i=1}^{100} (-1)^i i$

c) $\sum_{k=1}^n (4k + 1)$

d) $\sum_{k=4}^n 2k$

e) $\sum_{k=n}^{2n} (2k - 1)$

Products

6. Denote the product $\prod_{i=1}^n i$ as $n!$ Using this notation, multiplication, division, etc., simplify the following expressions:

a) $\prod_{i=1}^n (4i)$

b) $\prod_{j=2n}^{4n} j$

c) $\prod_{i=1}^n \frac{4+i}{i+2}$

d) $\prod_{j=1}^n \frac{j-5}{j+2}$

e) $\prod_{k=n}^{3n} \frac{k+6}{k}$

$$\text{f) } \prod_{i=1}^n (3i+2)(3i+1)$$

$$\text{g) } \prod_{j=1}^n (4j-1)(4j+1)$$

$$\text{h) } \prod_{i=1}^n \frac{(4i-3)(4i+3)}{4i}$$

Geometric series

7. Using $\sum_{i=0}^n 2^i = 2^{n+1} - 1$ calculate

$$\text{a) } \sum_{j=10}^{20} 2^j$$

$$\text{b) } \sum_{k=n}^{2n} 2^k$$

$$\text{c) } \sum_{i=3}^{3n} (2^i \cdot 3)$$

8. Calculate the sums of finite geometric series:

$$\text{a) } \sum_{i=0}^{10} 3^i, \quad \sum_{j=1}^{20} \left(\frac{1}{3}\right)^j, \quad \sum_{i=1}^{99} 7^i, \quad \sum_{j=20}^n \left(\frac{2}{3}\right)^j$$

$$\text{b) } \sum_{i=0}^{10} (-3)^i, \quad \sum_{j=1}^{20} \left(-\frac{1}{3}\right)^j, \quad \sum_{i=1}^{99} (-7)^i, \quad \sum_{j=20}^n \left(-\frac{2}{3}\right)^j$$

$$\text{c) } \sum_{i=4}^6 10^i, \quad \sum_{j=1}^{10} \left(\frac{1}{2}\right)^j, \quad \sum_{i=k}^{2k} 10^i, \quad \sum_{j=2n}^{3n} \left(\frac{1}{2}\right)^j$$

$$\text{d) } \sum_{i=4}^6 (-10)^i, \quad \sum_{j=1}^{10} \left(-\frac{1}{2}\right)^j, \quad \sum_{i=k}^{2k} (-10)^i, \quad \sum_{j=2n}^{3n} \left(-\frac{1}{2}\right)^j$$

9. Calculate the sums of infinite geometric series:

$$\text{a) } \sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i, \quad \sum_{j=1}^{\infty} \left(-\frac{2}{5}\right)^j, \quad \sum_{i=4}^{\infty} \left(\frac{5}{7}\right)^i, \quad \sum_{j=30}^{\infty} \left(-\frac{2}{3}\right)^j,$$

$$\text{b) } \sum_{i=2}^{\infty} q^i, \quad \sum_{j=1}^{\infty} (a^2)^j, \quad \sum_{j=k}^{\infty} (-b)^j,$$

$$\text{c) } \sum_{j=2n}^{\infty} \left(\frac{1}{2}\right)^j, \quad \sum_{j=1}^{\infty} (-b^3)^j, \quad \sum_{j=k^2}^{\infty} \left(-\frac{2}{7}\right)^j$$

Double sums and products

10. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$.
Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$.
Check whether the following equality holds:

$$\sum_{n=1}^3 \sum_{k=1}^2 (a_n \cdot b_k) = \left(\sum_{n=1}^3 a_n\right) \cdot \left(\sum_{k=1}^2 b_k\right)$$

11. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$.
Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$.
Check whether the following equality holds:

$$\sum_{n=1}^3 \sum_{k=1}^2 (a_n / b_k) = \left(\sum_{n=1}^3 a_n\right) / \left(\sum_{k=1}^2 b_k\right)$$

12. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$.
Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$.
Check whether the following equality holds:

$$\sum_{n=1}^3 \sum_{k=1}^2 (a_n + b_k) = \sum_{n=1}^3 a_n + \sum_{k=1}^2 b_k$$

13. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$.
Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$.
Check whether the following equality holds:

$$\prod_{k=1}^2 \sum_{n=1}^3 (a_n + b_k) = \sum_{n=1}^3 \prod_{k=1}^2 (a_n + b_k)$$

14. Write out the terms of the double sums and calculate their values. Which results are the same and why?

$$\sum_{i=1}^5 \sum_{j=1}^4 i \cdot j, \quad \sum_{i=2}^5 \sum_{j=3}^6 (i+1)(j+2), \quad \sum_{i=2}^6 (i-1) \sum_{j=0}^3 (j+1), \quad \sum_{i=0}^3 (i+3) \sum_{j=5}^8 (j-1)$$

15. Write out and compare the terms of the two expressions. Are they the same?

$$\begin{aligned} \text{a) } \sum_{i=1}^5 \left[\sum_{j=3}^6 (a_i \cdot b_j) \right] \quad & \text{and} \quad \sum_{i=1}^5 a_i \sum_{j=3}^6 b_j \\ \text{b) } \sum_{i=3}^5 \left[\sum_{j=7}^{10} (c_i \cdot d_j) \right] \quad & \text{and} \quad \sum_{i=6}^{10} d_{i+1} \sum_{j=4}^6 c_{j-1} \end{aligned}$$

16. Write out the terms of these expressions and transform them into a product of two sums:

$$\text{a) } \sum_{i=4}^8 \sum_{j=0}^3 c_i \cdot d_j, \quad \text{b) } \sum_{i=2}^6 \sum_{j=3}^4 u_i \cdot v_j, \quad \text{c) } \sum_{k=6}^{10} (k-2) \sum_{j=1}^4 (j+2),$$

17. Write out the terms, calculate and compare

$$\sum_{i=5}^8 \prod_{j=3}^5 (i+2)j \quad \text{and} \quad \sum_{i=5}^8 (i+2) \prod_{j=3}^5 j$$

9.3.1 Hints and solutions

Sums

1. Write the following expressions using the $\sum$ notation:

a) $1 + 2 + 3 + \dots + 20$

b) $7 + 14 + 21 + \dots + 70$

c) $6 + 8 + 10 + \dots + 6n$

d) $12n + 12n + 6 + 12n + 12 + 12n + 18 + \dots + 24n$

Hints:

- It is straightforward to use the $\sum$ notation when the two successive terms in the expression differ by +1.
- If the +1 difference is not apparent, we can consider whether the terms have a common divisor. If so, we can factor the divisor out of all terms.
- In some cases it might be possible to write the sum several different ways.
- Sometimes, e.g. in d), it is helpful to rearrange the order of terms.

Solution:

a) $1 + 2 + 3 + \dots + 20 = \sum_{i=1}^{20} i$

b) $7 + 14 + 21 + \dots + 70 = 7 \cdot (1 + 2 + 3 + \dots + 10) = \sum_{i=1}^{10} 7i$

c) $6 + 8 + 10 + \dots + 6n = 2 \cdot (3 + 4 + 5 + \dots + 3n) = \sum_{i=3}^{3n} 2i$

d) $12n + 12n + 6 + 12n + 12 + 12n + 18 + \dots + 24n =$
 $= 6 \cdot (2n + 2n + 1 + 2n + 2 + 2n + 3 + \dots + 4n) =$
 $= 6 \cdot (2n + [2n + 1] + [2n + 2] + [2n + 3] + \dots + 4n) = \sum_{i=2n}^{4n} 6i$

2. Using $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ calculate

$$\begin{aligned}
\text{a) } & \sum_{i=1}^{10} 4i, & \sum_{j=1}^n 4j \\
\text{b) } & \sum_{k=1}^{100} 3k, & \sum_{k=1}^{2n} 3k \\
\text{c) } & \sum_{i=1}^{25} 5, & \sum_{i=5}^{3n} 2i \\
\text{d) } & \sum_{i=8}^{16} i, & \sum_{j=n}^{2n} j \\
\text{e) } & \sum_{i=2n}^{3n} 2k, & \sum_{t=2n}^{4n} 6t
\end{aligned}$$

Hints:

- A constant can be factored out from the sum, e.g. $\sum_{i=1}^4 3i = 3 + 6 + 9 + 12 = 3 \cdot (1 + 2 + 3 + 4)$, which can be written as $3 \cdot \sum_{i=1}^4 i$.
- The formula for addition of the first n natural numbers starts at 1. If the sum we are trying to calculate does not start at 1, we can either shift the bounds using the rule $\sum_{i=1}^n a_i = \sum_{i=1+k}^{n+k} a_{i-k}$ or use the fact that $\sum_{i=1}^n a_i = \sum_{i=1}^{k-1} a_i + \sum_{i=k}^n a_i$.
- When shifting the bounds, we need to also adjust the index in the sum. For example if the original sum was from $3n$ to $6n$, then by subtracting $3n$ and adding 1 to both bounds, we adjust the lower bound to 1. To adjust the iteration index inside the sum accordingly, we need to add $3n$ and subtract 1. This way we are still adding the same terms as before. The whole transformation is $\sum_{i=3n}^{6n} 3i = \sum_{i=1}^{3n+1} 3(i + 3n - 1)$.

For verification, it is always a good idea to check the first few terms before and after the transformation. In both cases we have to get the same results. In our example, $\sum_{i=3n}^{6n} 3i$, this would be $a_1 = 3 \cdot 3n = 9n$

and after the transformation to $\sum_{i=1}^{3n+1} 3(i+3n-1)$ we get $a_1 = 3(1+3n-1) = 9n$.

- In case the iteration variable in the sum does not correspond to the variable indices, e.g. the first problem in part e), it is constant with respect to the sum and can be factored out.

Solution:

$$\begin{aligned} \text{a) } \sum_{i=1}^{10} 4i &= 4 + 8 + 12 + \dots + 40 = 4 \cdot (1 + 2 + 3 + \dots + 10) = \\ &= 4 \cdot \frac{10(10+1)}{2} = 220 \\ \sum_{j=1}^n 4j &= 4 + 8 + 12 + \dots + 4n = 4 \cdot \frac{n(n+1)}{2} = 2n(n+1) \end{aligned}$$

$$\begin{aligned} \text{b) } \sum_{k=1}^{100} 3k &= 3 \cdot \sum_{k=1}^{100} k = 3 \cdot \frac{100 \cdot 101}{2} \\ \sum_{k=1}^{2n} 3k &= 3 \cdot (1 + 2 + 3 + \dots + 2n) = 3 \cdot \frac{2n(2n+1)}{2} \end{aligned}$$

$$\text{c) } \sum_{t=1}^{25} 5 = 5 \cdot 25 = 125$$

In $\sum_{i=5}^{3n} 2i$ the sum does not start from 1. We show two ways how we can deal with that.

First method: We add the missing finite series and then we subtract it, i.e. we add a cleverly written 0 to the whole series.

$$\begin{aligned} \sum_{i=5}^{3n} 2i &= \sum_{i=5}^{3n} 2i + \sum_{i=1}^4 2i - \sum_{i=1}^4 2i = \sum_{i=1}^{3n} 2i - \sum_{i=1}^4 2i = \\ &= 2 \cdot \frac{3n(3n+1)}{2} - 2 \cdot \frac{4(4+1)}{2} = 3n(3n+1) - 20 = 9n^2 + 3n - 20 \end{aligned}$$

Second method: We shift the lower bound to $i = 1$. With that we also have to shift the upper bound and apply the corresponding change to the iteration index. In the next step, we split the sum into two sums.

$$\begin{aligned}\sum_{i=5}^{3n} 2i &= \sum_{i=1}^{3n-4} 2(i+4) = \sum_{i=1}^{3n-4} 2i + \sum_{i=1}^{3n-4} 8 = \\ &= 2 \frac{(3n-4)(3n-3)}{2} + 8(3n-4) = 9n^2 + 3n - 20\end{aligned}$$

$$\text{d) } \sum_{j=n}^{2n} j = \sum_{j=1}^{2n} j - \sum_{j=1}^{n-1} j = \frac{2n(2n+1)}{2} - \frac{(n-1)n}{2} = \frac{3n}{2}(n+1)$$

$$\begin{aligned}\text{e) } \sum_{i=2n}^{3n} 2k &= 2k \sum_{i=2n}^{3n} 1 = 2k(n+1) \\ \sum_{t=2n}^{4n} 6t &= \sum_{t=1}^{2n+1} 6(t+2n-1) = \sum_{t=1}^{2n+1} 6t + \sum_{t=1}^{2n+1} 6(2n-1) = \\ &= 6 \frac{(2n+1)(2n+2)}{2} + 6(2n-1)(2n+1) = 18n(2n+1)\end{aligned}$$

3. Using $\sum_{i=1}^n (2i-1) = n^2$ calculate

$$\text{a) } \sum_{j=1}^{3n} (2j-1)$$

$$\text{b) } \sum_{i=0}^n 4(2i+1)$$

$$\text{c) } \sum_{k=2n}^{4n} (2k-1)$$

Hints:

- The given formula (sum of the first n odd numbers) starts with 1. If we want to use it for other problems that involve odd numbers or their multiples, we need to be careful about the indices.
- The problems can also be solved without using the formula, just by splitting the sum into two sums.

Solution:

a) First method: Using the formula for addition of the first n odd numbers.

$$\sum_{j=1}^{3n} (2j-1) = (3n)^2$$

Second method: Splitting the sum.

$$\sum_{j=1}^{3n} (2j-1) = \sum_{j=1}^{3n} 2j - \sum_{j=1}^{3n} 1 = 3n(3n+1) - 3n = 9n^2$$

$$\text{b) } \sum_{i=0}^n 4(2i+1) = \sum_{i=1}^{n+1} 4(2(i-1)+1) = \sum_{i=1}^{n+1} 4(2i-1) = 4(n+1)^2$$

$$\begin{aligned} \text{c) } \sum_{k=2n}^{4n} (2k-1) &= \sum_{k=1}^{2n+1} (2(k+2n-1)-1) = \sum_{k=1}^{2n+1} 2k + \sum_{k=1}^{2n+1} (4n-3) = \\ &= (2n+1)(2n+2) + (4n-3)(2n+1) = (2n+1)(6n-1) \end{aligned}$$

4. Split into two sums and calculate

$$\text{a) } \sum_{i=1}^n (4i+3)$$

$$\text{b) } \sum_{j=1}^{3n} (2j-1)$$

$$\text{c) } \sum_{k=1}^{2n} (-1)^k 3k$$

Hints:

- We can split a sum into two sums using the rule

$$\sum_{i=1}^n (a_i \pm b_i) = \sum_{i=1}^n a_i \pm \sum_{i=1}^n b_i$$
- If it is not clear at the first glance how to split the sum into two sums, e.g. in problem c), it might help to write out the first few terms.

Solution:

$$\text{a) } \sum_{i=1}^n (4i+3) = \sum_{i=1}^n 4i + \sum_{i=1}^n 3 = 4 \frac{1}{2} n(n+1) + 3n = 2n^2 + 5n$$

$$\text{b) } \sum_{j=1}^{3n} (2j-1) = \sum_{j=1}^{3n} 2j - \sum_{j=1}^{3n} 1 = 3n(3n+1) - 3n = 9n^2$$

$$\begin{aligned} \text{c) } \sum_{k=1}^{2n} (-1)^k 3k &= 3 [-1 + 2 - 3 + 4 - 5 + \dots + (-1)^{2n} 2n] = \\ &= 3 [(-1 - 3 - 5 - \dots - (2n-1)) + (2 + 4 + 6 + \dots + 2n)] = \end{aligned}$$

$$= -3n^2 + 3 \cdot 2 \frac{1}{2} n(n+1) = 3n$$

5. Calculate the sums

Solution:

$$\text{a) } \sum_{k=1}^{20} (k+2) = \sum_{k=1}^{20} k + 2 \sum_{k=1}^{20} 1 = \frac{20 \cdot 21}{2} + 2 \cdot 20$$

$$\begin{aligned} \text{b) } \sum_{i=1}^{100} (-1)^i i &= -1 + 2 - 3 + 4 - 5 + 6 + \dots + 100 = \\ &= (-1 - 3 - 5 - \dots - 99) + (2 + 4 + 6 + \dots + 100) = \\ &= -1(1 + 3 + 5 + \dots + 99) + 2(1 + 2 + 3 + \dots + 50) = \\ &= -(50)^2 + 2 \frac{50 \cdot 51}{2} = 100 \end{aligned}$$

$$\text{c) } \sum_{k=1}^n (4k+1) = \sum_{k=1}^n 4k + \sum_{k=1}^n 1 = 4 \frac{n(n+1)}{2} + n = 2n^2 + 3n$$

$$\text{d) } \sum_{k=4}^n 2k = 2 \sum_{k=1}^{n-3} (k+3) = 2 \frac{(n-3)(n-2)}{2} + 6(n-3) = (n-3)(n+4)$$

$$\begin{aligned} \text{e) } \sum_{k=n}^{2n} (2k-1) &= \sum_{k=1}^{2n-n+1} (2(k+n-1)-1) = \sum_{k=1}^{n+1} (2k+2n-3) = \\ &= 2 \frac{(n+1)(n+2)}{2} + (2n-3)(n+1) = (n+1)(3n-1) \end{aligned}$$

Products

6. Denote the product $\prod_{i=1}^n i$ as $n!$ Using this notation, multiplication, division, etc, simplify the following expressions

$$\text{a) } \prod_{i=1}^n (4i)$$

$$\text{b) } \prod_{j=2n}^{4n} j$$

$$\text{c) } \prod_{i=1}^n \frac{4+i}{i+2}$$

$$\text{d) } \prod_{j=1}^n \frac{j-5}{j+2}$$

$$\begin{aligned} \text{e)} \quad & \prod_{k=n}^{3n} \frac{k+6}{k} \\ \text{f)} \quad & \prod_{i=1}^n (3i+2)(3i+1) \\ \text{g)} \quad & \prod_{j=1}^n (4j-1)(4j+1) \\ \text{h)} \quad & \prod_{i=1}^n \frac{(4i-3)(4i+3)}{4i} \end{aligned}$$

Hints:

- With the products, it is very useful to write out the individual terms (few from the beginning and few from the end), check if we see any factorials and perhaps multiply by suitable terms to get a factorial if anything is missing. For example, if we write out the product $\prod_{i=5}^{10} i = 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10$, we see that we could have $10!$ if we also had $1 \cdot 2 \cdot 3 \cdot 4 = 4!$. Therefore we can multiply the product by $\frac{4!}{4!}$ to get $\frac{10!}{4!}$.
- An expression can be simplified by factoring out a common divisor, but we have to remember its multiplicity, e.g. in the product of the first n even numbers we have:

$$\prod_{k=1}^n (2i) = 2 \cdot 4 \cdot 6 \cdot 8 \cdot \dots \cdot 2n = 2^n (1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot n) = 2^n \prod_{k=1}^n i = 2^n n!$$

- If we wanted to calculate the product of the first n odd numbers, we can use the result for the even numbers

$$\prod_{k=1}^n (2i-1) = 1 \cdot 3 \cdot 5 \cdot 7 \cdot \dots \cdot (2n-1) = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot 2n}{2 \cdot 4 \cdot 6 \cdot 8 \cdot \dots \cdot 2n} = \frac{(2n)!}{2^n n!}$$

Solution:

$$\begin{aligned} \text{a)} \quad & \prod_{i=1}^n 4i = 4 \cdot 1 \cdot 4 \cdot 2 \cdot 4 \cdot 3 \cdot 4 \cdot 4 \cdot \dots \cdot 4 \cdot (n-1) \cdot 4 \cdot n = 4^n n! \\ \text{b)} \quad & \prod_{j=2n}^{4n} j = 2n \cdot (2n+1) \cdot (2n+1) \cdot (2n+1) \cdot \dots \cdot (4n-1) \cdot 4n = \frac{(4n)!}{(2n-1)!} \end{aligned}$$

- c) When we write out this product, we see that the numerator is almost $(4+n)!$ What is missing is $4!$ therefore we multiply the whole fraction by a cleverly written 1 in the form of $\frac{4!}{4!}$. Similarly, in the denominator, we almost have $(n+2)!$ To get it, we multiply the fraction also by $\frac{2!}{2!}$.

$$\prod_{i=1}^n \frac{4+i}{i+2} = \frac{5}{3} \cdot \frac{6}{4} \cdot \frac{7}{5} \cdots \frac{4+n-1}{n+2-1} \cdot \frac{4+n}{n+2} \cdot \frac{4!}{4!} \cdot \frac{2!}{2!} = \frac{(4+n)!}{4!} \cdot \frac{2!}{(n+2)!} \\ = \frac{(n+3)(n+4)}{12}$$

$$\text{d) } \prod_{j=1}^n \frac{j-5}{j+2} = \frac{-4}{3} \cdot \frac{-3}{4} \cdot \frac{-2}{5} \cdot \frac{-1}{6} \cdot \frac{0}{7} \cdot \frac{1}{8} \cdots \frac{n-6}{n+1} \cdot \frac{n-5}{n+2} = 0$$

$$\text{e) } \prod_{k=n}^{3n} \frac{k+6}{k} = \frac{n+6}{6} \cdot \frac{n+7}{7} \cdot \frac{n+8}{8} \cdots \frac{3n+5}{3n-1} \cdot \frac{3n+6}{3n} = \\ = \frac{(3n+6)!}{(n+5)!} \cdot \frac{5!}{(3n)!}$$

$$\text{f) } \prod_{i=1}^n (3i+2)(3i+1) = 5 \cdot 4 \cdot 8 \cdot 7 \cdot 11 \cdot 10 \cdots (3n+2)(3n+1) = \\ = \frac{5 \cdot 4 \cdot 3 \cdot 8 \cdot 7 \cdot 6 \cdot 11 \cdot 10 \cdot 9 \cdots (3n+2)(3n+1)3n}{3 \cdot 6 \cdot 9 \cdots 3n} = \frac{(3n+2)!}{2!3^n n!}$$

$$\text{g) } \prod_{j=1}^n (4j-1)(4j+1) = 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 \cdots (4n-1) \cdot (4n+1) = \\ = \frac{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 6 \cdots (4n-1)4n(4n+1)}{2 \cdot 4 \cdot 6 \cdots 4n} = \frac{(4n+1)!}{2^{2n}(2n)!}$$

$$\text{h) } \prod_{i=1}^n \frac{(4i-3)(4i+3)}{4i} = \frac{1 \cdot 7}{4} \cdot \frac{5 \cdot 11}{8} \cdot \frac{9 \cdot 15}{12} \cdots \frac{(4n-3)(4n+3)}{4n} = \\ = \frac{(4n+3)!}{(4n+1) \cdot 3 \cdot 4^n n! \cdot 2^n n!}$$

Geometric series

7. Using $\sum_{i=0}^n 2^i = 2^{n+1} - 1$ calculate

$$\text{a) } \sum_{j=10}^{20} 2^j$$

$$\text{b) } \sum_{k=n}^{2n} 2^k$$

$$c) \sum_{i=3}^{3n} (2^i \cdot 3)$$

Hints:

- The formula is derived from a general formula for sum of finite geometric series using a quotient $q = 2$.
- When changing the bounds of the index, we need to also change the exponent. We can use the properties $2^{a+b} = 2^a \cdot 2^b$, $2^{a-b} = 2^a \cdot 2^{-b} = \frac{2^a}{2^b}$.
- We can also consider the connection of these sums to a binary system. For example for $n = 4$ we can write the sum $\sum_{i=0}^4 2^i = 1+2+4+8+16 = 2^5 - 1$ in binary as:

$$1_2 + 10_2 + 100_2 + 1000_2 + 10000_2 = 11111_2 = 100000_2 - 1_2$$

This gives us a simple rule for consecutive addition of powers of two 2^i in binary.

Solution:

- a) First method: When we write out the first few terms, we see that we can factor out 2^{10} .
- $$\sum_{j=10}^{20} 2^j = 2^{10} + 2^{11} + \dots + 2^{20} = 2^{10} [1 + 2 + 2^2 + \dots + 2^{10}] = 2^{10} [2^{11} - 1]$$
- Second method: We can shift the lower bound to zero and then use the formula to calculate the sum
- $$\sum_{j=10}^{20} 2^j = \sum_{j=0}^{10} 2^{j+10} = 2^{21} - 2^{10}$$
- b) $\sum_{k=n}^{2n} 2^k = \sum_{k=0}^n 2^{k+n} = 2^n [2^{n+1} - 1]$
- c) $\sum_{i=3}^{3n} (2^i \cdot 3) = 3 \cdot 2^3 \sum_{i=3}^{3n} 2^{i-3} = 3 \cdot 2^3 \sum_{i=0}^{3n-3} 2^i = 3 \cdot 2^3 [2^{3n-2} - 1]$

8. Calculate the sums of finite geometric series.

$$a) \sum_{i=0}^{10} 3^i, \quad \sum_{j=1}^{20} \left(\frac{1}{3}\right)^j, \quad \sum_{i=1}^{99} 7^i, \quad \sum_{j=20}^n \left(\frac{2}{3}\right)^j$$

$$\begin{aligned}
\text{b) } & \sum_{i=0}^{10} (-3)^i, & \sum_{j=1}^{20} \left(-\frac{1}{3}\right)^j, & \sum_{i=1}^{99} (-7)^i, & \sum_{j=20}^n \left(-\frac{2}{3}\right)^j \\
\text{c) } & \sum_{i=4}^6 10^i, & \sum_{j=1}^{10} \left(\frac{1}{2}\right)^j, & \sum_{i=k}^{2k} 10^i, & \sum_{j=2n}^{3n} \left(\frac{1}{2}\right)^j \\
\text{d) } & \sum_{i=4}^6 (-10)^i, & \sum_{j=1}^{10} \left(-\frac{1}{2}\right)^j, & \sum_{i=k}^{2k} (-10)^i, & \sum_{j=2n}^{3n} \left(-\frac{1}{2}\right)^j
\end{aligned}$$

Hints:

- The sum of finite geometric series is $S_n = \sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$. This series S_n starts at $k=0$, i.e. S_n has $n+1$ terms.
- If we want to use this formula directly, the lower bound in the iteration index has to be zero. We can achieve this by shifting the summation indices directly or by factoring out a suitable power of the quotient q in front of the sum.
- When calculating powers of negative numbers, we need to be careful about the signs, e.g. in $(-1)^k$ we get a positive result for even powers.

Solution:

$$\begin{aligned}
\text{a) } & \sum_{i=0}^{10} 3^i = \frac{1-3^{11}}{1-3}, & \sum_{j=1}^{20} \left(\frac{1}{3}\right)^j &= \sum_{j=0}^{19} \left(\frac{1}{3}\right)^{j+1} = \frac{1}{3} \frac{1-\left(\frac{1}{3}\right)^{20}}{1-\frac{1}{3}} = \frac{1}{2} \left[1 - \left(\frac{1}{3}\right)^{20}\right] \\
\text{b) } & \sum_{j=1}^{20} \left(-\frac{1}{3}\right)^j = \sum_{j=0}^{19} \left(-\frac{1}{3}\right)^{j+1} = -\frac{1}{3} \frac{1-\left(-\frac{1}{3}\right)^{20}}{1+\frac{1}{3}} = -\frac{1}{4} \left[1 - \left(\frac{1}{3}\right)^{20}\right], \\
\text{c) } & \sum_{i=4}^6 10^i = \sum_{i=0}^2 10^{i+4} = 10^4 \cdot \frac{1-10^3}{1-10} = 10^4 [1 + 10 + 100] = 111000, \\
& \sum_{j=2n}^{3n} \left(\frac{1}{2}\right)^j = \sum_{j=0}^n \left(\frac{1}{2}\right)^{j+2n} = \left(\frac{1}{2}\right)^{2n} \cdot \frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}} = 2 \left(\frac{1}{2}\right)^{2n} \cdot \left[1 - \left(\frac{1}{2}\right)^{n+1}\right] \\
\text{d) } & \sum_{i=4}^6 (-10)^i = 10^4 - 10^5 + 10^6, & \sum_{j=2n}^{3n} \left(-\frac{1}{2}\right)^j &= \frac{2}{3} \left(\frac{1}{2}\right)^{2n} \cdot \left[1 - \left(-\frac{1}{2}\right)^{n+1}\right]
\end{aligned}$$

9. Calculate the sums of infinite geometric series:

$$\text{a) } \sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i, \quad \sum_{j=1}^{\infty} \left(-\frac{2}{5}\right)^j, \quad \sum_{i=4}^{\infty} \left(\frac{5}{7}\right)^i, \quad \sum_{j=30}^{\infty} \left(-\frac{2}{3}\right)^j,$$

$$\begin{aligned} \text{b) } & \sum_{i=2}^{\infty} q^i, \quad \sum_{j=1}^{\infty} (a^2)^j, \quad \sum_{j=k}^{\infty} (-b)^j, \\ \text{c) } & \sum_{j=2n}^{\infty} \left(\frac{1}{2}\right)^j, \quad \sum_{j=1}^{\infty} (-b^3)^j, \quad \sum_{j=k^2}^{\infty} \left(-\frac{2}{7}\right)^j \end{aligned}$$

Hints:

- The sum of the infinite geometric series is $S_n = \frac{1}{1-q}$.
- If we need to shift the indices, the upper bound remains ∞ .

Solution:

$$\begin{aligned} \text{a) } & \sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i = \frac{1}{1-\frac{1}{3}} = \frac{3}{2} \\ & \sum_{j=1}^{\infty} \left(-\frac{2}{5}\right)^j = \sum_{j=0}^{\infty} \left(-\frac{2}{5}\right)^{j+1} = \sum_{j=1}^{\infty} \left(-\frac{2}{5}\right)^j \cdot \left(-\frac{2}{5}\right)^1 = -\frac{2}{5} \cdot \frac{1}{1+\frac{2}{5}} \\ & \sum_{i=4}^{\infty} \left(\frac{5}{7}\right)^i = \left(\frac{5}{7}\right)^4 \frac{1}{1-\frac{5}{7}} \\ \text{b) } & \sum_{i=2}^{\infty} q^i = q^2 \frac{1}{1-q} \\ & \sum_{j=1}^{\infty} (a^2)^j = a^2 \frac{1}{1-a^2} \\ & \sum_{j=k}^{\infty} (-b)^j = (-b)^k \frac{1}{1+b} \\ \text{c) } & \sum_{j=2n}^{\infty} \left(\frac{1}{2}\right)^j = \left(\frac{1}{2}\right)^{2n} \cdot \frac{1}{1-\frac{1}{2}} = 2 \left(\frac{1}{2}\right)^{2n} \\ & \sum_{j=k^2}^{\infty} \left(-\frac{2}{7}\right)^j = \left(-\frac{2}{7}\right)^{k^2} \cdot \frac{1}{1-\frac{2}{7}} \end{aligned}$$

Double sums and products

Hints:

- A double sum $\sum_n \sum_k$ or a combination of a sum and a product $\sum_n \prod_k$ can be understood as a double cycle in a source code which is supposed to perform a binary operation on terms of two sequences. The

index n can be the outer index and the index k can be the inner index.

- First we set the outer index to its lowest value, e.g. $n = 1$. We also set the inner index to its lowest value, e.g. $k = 1$ and then subsequently increment the inner index with step $+1$ until it covers its whole range. When it does, we increment the outer index to $n = 2$ and *reset* the inner index to $k = 1$. We then cycle through the whole range of k again and then reset it together with incrementing n . We repeat these steps until the outer index reaches its upper limit.
11. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$. Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$. Check whether the following equality holds:

$$\sum_{n=1}^3 \sum_{k=1}^2 (a_n/b_k) = (\sum_{n=1}^3 a_n) / (\sum_{k=1}^2 b_k)$$

Solution:

$$\begin{aligned} \sum_{n=1}^3 \sum_{k=1}^2 (a_n/b_k) &= \frac{a_1}{b_1} + \frac{a_1}{b_2} + \frac{a_2}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_1} + \frac{a_3}{b_2} \\ (\sum_{n=1}^3 a_n) / (\sum_{k=1}^2 b_k) &= \frac{a_1 + a_2 + a_3}{b_1 + b_2} \\ &\Rightarrow \text{The equality does not hold.} \end{aligned}$$

13. Let a_n be the n^{th} term of the geometric sequence, where $a_1 = 4$ and $q = 2$. Let b_k be the k^{th} term of arithmetic sequence, where $b_1 = 3$ and $d = 1$. Check whether the following equality holds:

$$\prod_{k=1}^2 \sum_{n=1}^3 (a_n + b_k) = \sum_{n=1}^3 \prod_{k=1}^2 (a_n + b_k)$$

Solution:

$$\prod_{k=1}^2 \sum_{n=1}^3 (a_n + b_k) = (a_1 + a_2 + a_3 + 3b_1)(a_1 + a_2 + a_3 + 3b_2)$$

$$\sum_{n=1}^3 \prod_{k=1}^2 (a_n + b_k) = (a_1 + b_1)(a_1 + b_2) + (a_2 + b_1)(a_2 + b_2) + (a_3 + b_1)(a_3 + b_2)$$

$\Rightarrow$ The equality does not hold.

17. Write out the terms, calculate and compare

$$\sum_{i=5}^8 \prod_{j=3}^5 (i+2)j \quad \text{and} \quad \sum_{i=5}^8 (i+2) \prod_{j=3}^5 j$$

Solution:

$$\sum_{i=5}^8 \prod_{j=3}^5 (i+2)j = 7 \cdot 3 \cdot 7 \cdot 4 \cdot 7 \cdot 5 + \dots + 10 \cdot 3 \cdot 10 \cdot 4 \cdot 10 \cdot 5 = 3 \cdot 4 \cdot 5 \cdot (7^3 + \dots + 10^3)$$

$$\sum_{i=5}^8 (i+2) \prod_{j=3}^5 j = 7 \cdot 3 \cdot 4 \cdot 5 + \dots + 10 \cdot 3 \cdot 4 \cdot 5 = 3 \cdot 4 \cdot 5 \cdot (7 + \dots + 10)$$

$\Rightarrow$ The equality does not hold.

4. Distance to Mars

Create an Excel sheet which shows the first 1000 steps of the journey to Mars. The length of the first step is equal to a paper width. The second step is the thickness of a paper folded in half. The next is the paper folded in half again and so on. How many times do we need to fold the paper in half to get the thickness equal to the distance from Earth to Mars?

A regular package of 500 sheets of paper is 5.3cm thick.

The distance to Mars from Earth varies from 55 to 400 million kilometres.

9.4.1 Hints and solutions

1. Sums and products

Create an Excel sheet with the first 15 terms of an arithmetic sequence $\{a_n\}_{n=1}^{15}$ and first 10 terms of a geometric sequence $\{b_m\}_{m=1}^{10}$. Calculate:

- a) $\sum_{n=1}^{15} a_n \cdot \sum_{m=1}^{10} b_m$
- b) $\sum_{m=1}^{10} \left(\sum_{n=1}^{15} a_n b_m \right)$
- c) $\sum_{n=1}^{15} a_n \cdot \prod_{m=1}^{10} b_m$
- d) $\sum_{n=1}^{15} \left(\prod_{m=1}^{10} (a_n + b_m) \right)$
- e) $\prod_{m=1}^{10} \left(\sum_{n=1}^{15} (a_n + b_m) \right)$

Find out which of these expressions give the same result.

Hints:

- The double sums/products resemble a double cycle used in source code. The iteration variable of the outer cycle is set to the first value and the whole inner cycle is run. Then, the iteration variable of the outer cycle is set to the next value and again the whole inner cycle is run and so on. We can work with double sums the same way.
- In the problems where we need to add or multiply the terms of sequences pairwise, it is useful to create a table for that. We can use the same table for problems a) and b) and another table for problems c), d) and e).
- The problems do not specify the values of the sequence terms, so we can make our own choice. We can prepare the Excel sheets in such a way that there are four inputs (the first term of arithmetic sequence, the difference, the first term of geometric sequence and the quotient). When these are changed, the expressions with sums and products are recalculated.
- The indices help us determine whether we are working with rows or columns in the individual cases.

Solution: Since there are no specific sequences given in the problem and we only know that one of them is arithmetic and the other geometric, we can select our own. We are going to use the same sequences for all five problems. The arithmetic sequence is defined by $d = 1$ and $a_1 = 1$ and the geometric sequence is defined by $q = 2$ and $b_1 = 1$. The terms of the sequences are:

$$a_n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15$$

$$b_m = 1, 2, 4, 8, 16, 32, 64, 128, 256, 512$$

We use the same table for problems a) and b). The terms in the individual table cells are the products of every a_n with every b_m term. In our case, the sequence a_n is in a column and the sequence b_m in a row, see Figure 9.1, but the results would be the same if a_n was in the row and b_m in the column.

For problem a), we add the terms a_n and b_m separately and then multiply the results. In problem b), we first calculate the inner sums, i.e. we add the terms for each column and then add the results.

We find out that $\sum_{m=1}^{10} (\sum_{n=1}^{15} a_n b_m) = \sum_{n=1}^{15} (\sum_{m=1}^{10} a_n b_m)$ is true.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	q	2	b1	b2	b3	b4	b5	b6	b7	b8	b9	b10		
2	d	1	1	2	4	8	16	32	64	128	256	512	sum b:	1023
3	a1	1	1	2	4	8	16	32	64	128	256	512		
4	a2	2	2	4	8	16	32	64	128	256	512	1024		
5	a3	3	3	6	12	24	48	96	192	384	768	1536		
6	a4	4	4	8	16	32	64	128	256	512	1024	2048		
7	a5	5	5	10	20	40	80	160	320	640	1280	2560		
8	a6	6	6	12	24	48	96	192	384	768	1536	3072		
9	a7	7	7	14	28	56	112	224	448	896	1792	3584		
10	a8	8	8	16	32	64	128	256	512	1024	2048	4096		
11	a9	9	9	18	36	72	144	288	576	1152	2304	4608		
12	a10	10	10	20	40	80	160	320	640	1280	2560	5120		
13	a11	11	11	22	44	88	176	352	704	1408	2816	5632		
14	a12	12	12	24	48	96	192	384	768	1536	3072	6144		
15	a13	13	13	26	52	104	208	416	832	1664	3328	6656		
16	a14	14	14	28	56	112	224	448	896	1792	3584	7168		
17	a15	15	15	30	60	120	240	480	960	1920	3840	7680	problem b)	
18	sum a:	120											sum of sums	
19	inner sums:		120	240	480	960	1920	3840	7680	15360	30720	61440	122760	
20														
21													product:	122760
22														

Figure 9.1: Sums in a) and b).

We create a similar table for problems c), d) and e) that contains a product of each term a_n with each term b_m , see Figure 9.2.

In problem c), we add the terms of the arithmetic sequence and calculate the product of the terms of the geometric sequence. Then we calculate the product of these two partial calculations.

Problems very similar to d) and e) are discussed in Section 9.3, problem 13. Practically, here we need to first perform a product of sums (in the rows) and then a sum of these products. In problem e), we first perform a sum of sums (in the columns) and then a product of these sums. We do not get the same results, since the order of operations matters here.

E5	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	q	2	b1	b2	b3	b4	b5	b6	b7	b8	b9	b10			problem d)
2	d	1	1	2	4	8	16	32	64	128	256	512			product of sums
3	a1	1	2	3	5	9	17	33	65	129	257	513			1.674848E+14
4	a2	2	3	4	6	10	18	34	66	130	258	514			5.013856E+15
5	a3	3	4	5	7	11	19	35	67	131	259	515			1.198930E+15
6	a4	4	5	6	8	12	20	36	68	132	260	516			2.497070E+15
7	a5	5	6	7	9	13	21	37	69	133	261	517			4.728115E+15
8	a6	6	7	8	10	14	22	38	70	134	262	518			8.348560E+15
9	a7	7	8	9	11	15	23	39	71	135	263	519			1.3942E+16
10	a8	8	9	10	12	16	24	40	72	136	264	520			2.229496E+16
11	a9	9	10	11	13	17	25	41	73	137	265	521			3.604401E+16
12	a10	10	11	12	14	18	26	42	74	138	266	522			5.150836E+16
13	a11	11	12	13	15	19	27	43	75	139	267	523			7.051345E+16
14	a12	12	13	14	16	20	28	44	76	140	268	524			1.072113E+17
15	a13	13	14	15	17	21	29	45	77	141	269	525			1.5001E+17
16	a14	14	15	16	18	22	30	46	78	142	270	526			2.065097E+17
17	a15	15	16	17	19	23	31	47	79	143	271	527			2.794517E+17
18															sum of products
19															9.57881E+17
20	problem e)														
21	sum of sums	135	150	180	240	360	600	1080	2040	3960	7800			product of sums	1.2859E+28
22															
23	sum a	120	sum b	1023											
24	product a	1.30767E+12	product b	3.51844E+13		problem c)	4.22212E+15								

Figure 9.2: Calculations for problems c), d) and e).

2. Geometric series

Prepare an Excel sheet which will contain the first 100 terms of a geometric sequence. Do it in such a way that you can vary the quotient of geometric sequence in one cell. Calculate also the partial sums that correspond to the terms of this sequence. Find the limit of these partial sums. In the next column calculate the sum of the first n terms using the formula. Compare the results.

For geometric series with quotient less than 1 examine how the partial sums converge to the limit of the infinite sum.

For geometric series with a negative quotient examine the behavior of the partial sums. How does the behavior change if the absolute value of the quotient is larger/smaller than 1?

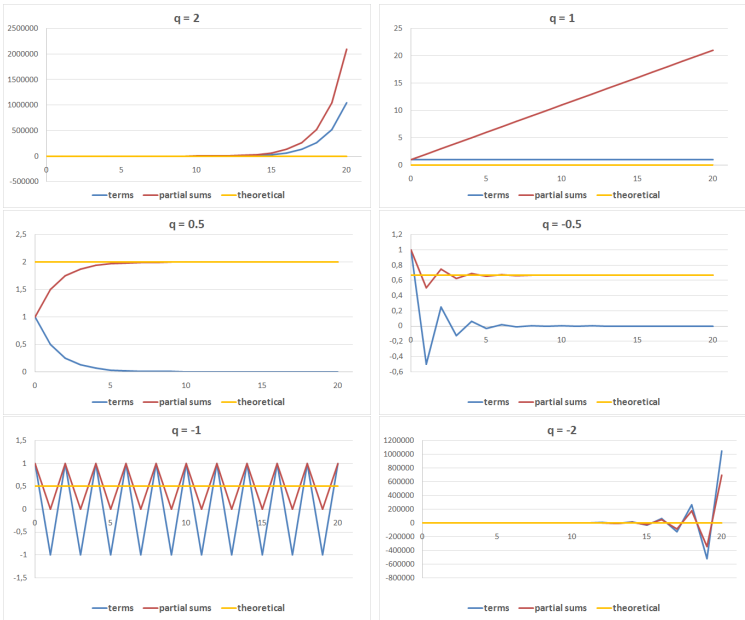


Figure 9.4: Geometric sequences and their partial sums for $q = 2, 1, 0.5, -0.5, -1, -2$.

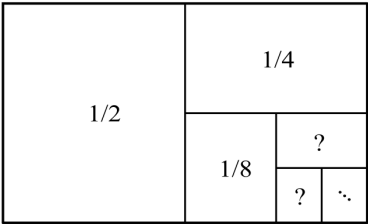


Figure 9.5: Successive splitting of chocolate in half.

- Using the recurrent formula, keep splitting the 100 g chocolate until the mass of the smallest piece is approximately 10^{-26} kg . Make sure that everything is calculated in the same units.
- To make the search for the right number of splits easier, we can write

a check that would inform us once we reach the mass of an atom.

Solution:

- The input data are the chocolate mass ($100g$) and an approximate atom mass ($1 \cdot 10^{-23}g$). These parameters can serve as an input and perhaps be also changed later on.
- We can designate one column (Num), see Figure 9.6, for tracking the numbers of splits needed.
- In another column (Geom. sequence), we write the splitting sequence. This is a geometric sequence with $q = \frac{1}{2}$. If we calculate it using a recurrent formula, it is sufficient to divide the previous cell by two, with the first term being 1. We can also directly use the formula $=1*POWER((1/2),E3)$ for calculation of the n^{th} term.
- The next column (Mass) calculates the mass of each successive piece of chocolate as $=100*F3$.
- The final column contains a check whether the new piece has smaller mass than an atom using the formula $=IF(G3<C3,"TRUE", "")$.

F7 ▲▼ ✕ ✓ f_x =F6/2									
	A	B	C	D	E	F	G	H	I
1	INPUT				Num	Geom. sequence	Mass [g]	Check	
2	Mass of chocolate:	100 g				1	100		
3	Mass of atom:	1.00E-23 g			1	0.5	50		
4					2	0.25	25		
5					3	0.125	12.5		
6					4	0.0625	6.25		
7					5	0.03125	3.125		
8					6	0.015625	1.5625		
:					:	:	:	:	
85					83	1.034E-25	1.03E-23		
86					84	5.17E-26	5.17E-24	TRUE	
87					85	2.585E-26	2.58E-24	TRUE	
88					86	1.292E-26	1.29E-24	TRUE	
89					87	6.462E-27	6.46E-25	TRUE	
90					88	3.231E-27	3.23E-25	TRUE	
91					89	1.616E-27	1.62E-25	TRUE	
92					90	8.078E-28	8.08E-26	TRUE	

Figure 9.6: Chocolate mass after successive splitting in half.

4. Distance to Mars

Create an Excel sheet which shows the first 1000 steps of the journey to Mars. The length of the first step is equal to a paper width. The second step is the thickness of a paper folded in half. The next is the paper folded in half again and so on. How many times do we need to fold the paper in half to get the thickness equal to the distance from Earth to Mars?

A regular package of 500 sheets of paper is 5.3cm thick.

The distance from Earth to Mars varies from 55 to 400 millions kilometres.

Hints:

- We can start by assuming that we have an infinitely large paper. We are going to fold it in half until the stack is thick enough to reach Mars. We can calculate the thickness of one paper from the data given in the problem.
- The distance from Earth to Mars varies depending on their mutual positions as they travel around the sun, however, once we calculate the number of folds needed for the minimum distance, we can easily check how this number changes for the maximum distance (and the distances between them).
- What type of sequence do we have here? Looking at the first few terms, we can find a recurrent formula for the n^{th} term. We can use it to determine the number of layers we have after each fold.
- Try folding a paper a few times yourself. Can you fold it more than eight times? What is the height of your tallest stack?

Solution:

As in the previous problem, we are dealing with geometric series here, with $q = 2$. The recurrent formula $a_n = 2a_{n-1}$ helps us calculate the n^{th} term $a_n = a_0 \cdot q^n$. We can use either of them to determine the thickness of each new layer and then add them up. The thickness of one paper can be calculated from the thickness information about one 500-paper package.

As we see in Figure 9.7, we need to fold the paper 49 to 52 times to create a stack that would reach Mars.

Paper dimensions

Suppose we start with an A4 paper with dimensions $210\text{ mm} \times 297\text{ mm}$. What dimensions does the folded paper have, once it has been folded enough times to reach Mars?

C8						
1		thickness of package [cm]:	5.3			
2		thickness of 1 paper [km]:	0.000000106			
3		Mars-Earth MIN [km]:	55000000			
4		Mars-Earth MAX [km]:	400000000			
5						
6		folds	layers	thickness [km]		
7		0	1	0.000000106		
8		1	2	0.000000212		
9		2	4	0.000000424		
10		3	8	0.000000848		
11		4	16	0.000001696		
:		:	:	:		
55		48	2.81475E+14	29836347.53		
56		49	5.6295E+14	59672695.06	min Mars	
57		50	1.1259E+15	119345390.1		
58		51	2.2518E+15	238690780.3		
59		52	4.5036E+15	477381560.5	max Mars	
60		53	9.0072E+15	954763121		
61		54	1.80144E+16	1909526242		
62		55	3.60288E+16	3819052484		

Figure 9.7: Calculating the number of paper folds needed to get from Earth to Mars.

Chapter 10

Real functions – part 1

10.1 Lecture overview

Problem 10.1 Velocity with respect to time:

Draw a graph of a function that describes the dependence of velocity on time for a 100 km long stage of a bicycle race in which Peter Sagan went 50 km uphill at 25 km/hr and then 50 km downhill at 100 km/hr.

Solution: The graph is shown in Figure 10.1 a). In practice, there would not be a *jump* (discontinuity) between the two horizontal lines. Similarly, there would be a gradual increase from 0 to 25 as the race starts and decrease from 100 to 0 as Sagan breaks after the finish line.

Problem 10.2 Distance with respect to time:

Draw a graph of a function that describes the dependence of distance (from the starting point) on time for a 100 km long stage of a bicycle race in which Peter Sagan went 50 km uphill at 25 km/hr and then 50 km downhill at 100 km/hr. Using this graph, determine, how fast he went, if there were any breaks, when did he accelerate and when did he slow down.

Solution: The graph is shown in Figure 10.1 b). We can see that in the first hour he covered 25 km, in two hours 50 km. This corresponds to the velocity 25 km/hr for the first two hours. Then the distance grows faster, he covers additional 50 km in just 1/2 hour. Therefore the velocity is 100 km/hr during this half hour. There were no breaks, since they would appear as flat horizontal

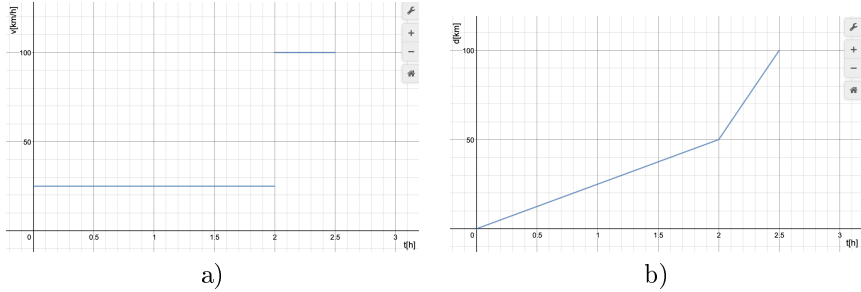


Figure 10.1: One stage of a bicycle race: a) velocity dependence on time, b) distance dependent on time. Note that this is an idealised situation, since obviously velocity cannot instantaneously increase from 25 to 100 km/h.

lines in the distance graph. Also, there was no acceleration/deceleration - that would appear as non-straight lines in this graph.

Problem 10.3 Water in the tub:

From a graph of water tap flow rate depending on time, Figure 10.2, find out how much water is in the tub.

Solution: We see that in the first five minutes, the water was flowing at constant rate 10 liters per minute (l/min), therefore after five minutes there were $5 \cdot 10$ liters of water in the bathtub. This corresponds to the area under the first line segment in the graph. In the next five minutes the water flew in faster – at 20 l/min, therefore during this time, we gained $5 \cdot 20$ liters. Again this number corresponds to the area under the second segment. The total area (including the third rectangle) thus corresponds to the total amount of water in the bathtub after 15 minutes: $50 + 100 + 15 \cdot 5 = 225$ liters.

Functions – reminder

Functions express the dependence of numerical output on a numerical input. The input can be any number in $\mathbb{R}$ or in its relevant subset. Examples of functions include: relations between physical quantities (e.g. dependence of atmospheric pressure on the altitude), dependence of time needed for an algorithm to finish on the size of the inputs, server load dependence on time, interest amount as a function of time, profit as a function of costs, etc.

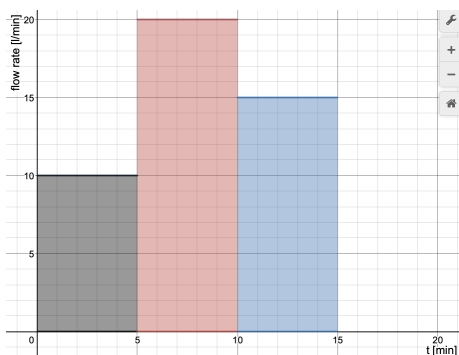


Figure 10.2: Water flow rate into a tub.

Domain

The set of all $x \in \mathbb{R}$ for which there exists a value $f(x)$ is called a *domain* $D(f)$.

$$D(f) = \{x \in \mathbb{R}; \exists f(x)\}$$

Range

The function $f(x)$ assigns each $x \in D(f)$ exactly one value $f(x)$. The set of all values $f(x)$ is called a *range* $H(f)$.

$$H(f) = \{y \in \mathbb{R}, \exists x \in \mathbb{R}; f(x) = y\}$$

Ways to define a function

- by a table
- by a graph
- by a formula or algebraic expression which lets us calculate the value of the function, e.g. $3x^6 - 2x^2 + 5$
- by a known notation, e.g. $\sin(x)$, $\cos(x)$, $\sinh(x)$
- by instructions, e.g. find functions $f_A(x)$ and $g_A(x)$ which are the areas *under the constant* $f(x) = 1$ or *under the line* $g(x) = x$ from 0 to x . (The functions are $f_A(x) = x$ and $g_A(x) = x^2/2$.)

Basic properties of functions

A function $f(x)$

- is *increasing on an interval* I iff $\forall x, y \in I, x < y \Rightarrow f(x) < f(y)$.
- is *decreasing on an interval* I iff $\forall x, y \in I, x < y \Rightarrow f(x) > f(y)$.
- is *non-decreasing on an interval* I iff $\forall x, y \in I, x < y \Rightarrow f(x) \leq f(y)$.
- is *non-increasing on an interval* I iff $\forall x, y \in I, x < y \Rightarrow f(x) \geq f(y)$.
- has a *minimum* m on I iff $\forall x \in I : (f(x) \geq m \wedge \exists x_m \in I : f(x_m) = m)$.
- has a *maximum* M on I iff $\forall x \in I : (f(x) \leq M \wedge \exists x_M \in I : f(x_M) = M)$.

10.1.1 Linear functions

Problem 10.4 Values of a linear function:

Calculate selected values of the function $f(x) = 2x + 3$.

Solution:

x	1	2	3	0	10	-1	-2	-10	0.1	0.3	$\sqrt{2}$	π
$f(x)$	5	7	9	3	23	1	-1	-17	3.2	3.6	$2\sqrt{2} + 3$	$2\pi + 3$

Graph of a function

A *graph of a function* $f(x)$ is a set of points $G(f)$ in a plane, such that $G(f) = \{[x, f(x)]; x \in D(f)\}$.

Problem 10.5 Graphs of linear functions:

Draw graphs of $f_1(x) = x$, $f_2(x) = x + 2$, $f_3(x) = x - 3$, $f_4(x) = 2x$, $f_5(x) = \frac{x}{2}$, $f_6(x) = 2x + 3$, $f_7(x) = -x$, $f_8(x) = 2x + 3$, $f_9(x) = -2x + 3$.

Solution: Some of these functions are shown in Figure 10.3. The graphs of $f_1(x) = x$, $f_2(x) = x + 2$ and $f_3(x) = x - 3$ are parallel lines. $f_1(x)$ passes through zero and $f_2(x)$ and $f_3(x)$ are shifted with respect to it. $f_7(x)$ has a completely different slope.

Discovering properties of linear functions

The graph of a function $f(x) = 2x + 3$ is a line. Where in the graph do we see the value 2 and where do we see the value 3? What do the graphs $f_1(x) = 2x + 1$, $f_2(x) = 2x - 2$ and $f_3(x) = 2x$ have in common? What

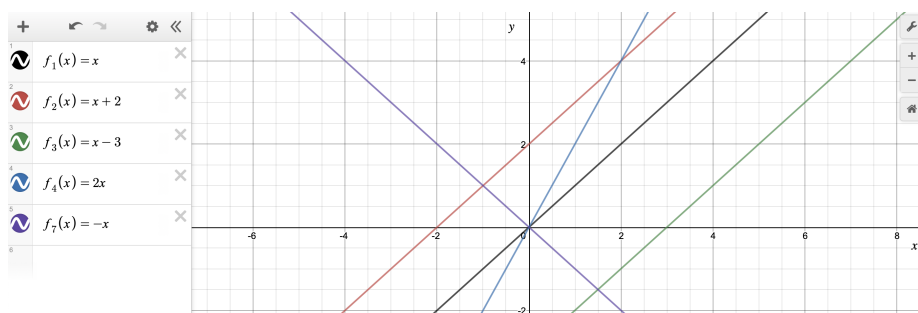


Figure 10.3: Linear functions.

do the graphs $f_1(x) = 2x + 1$, $f_2(x) = 5x + 1$ and $f_3(x) = \frac{1}{2}x + 1$ have in common?

Linear function

A *linear function* is a function of the form:

$$f(x) = ax + b$$

Depending on the value of a , a linear function $f(x) = ax + b$ is either increasing, constant or decreasing. Its domain is $\mathbb{R}$. Its range is either $\mathbb{R}$ or just one real number.

Scaling and shifts in graphs

- The coefficient a in front of variable x scales the graph. For $a > 1$ we get a *faster* behavior (as if we squeezed the x -axis from both sides towards the center). For $0 < a < 1$ we get a *slower* behavior (as if we stretched the x -axis from the center to both sides). The graph is mirrored about the y -axis for $a < 0$.
- The coefficient b shifts the graph. For $b > 0$ we have a shift up and for $b < 0$ down.
- The graphs can also be shifted left and right (this is useful when drawing more complicated functions). If we have $x - k$ instead of x , the graph is shifted k units to the right. For $x + k$, it is shifted k units to the left.

left. Compare the graphs of $f(x) = 2x + 3$, $g(x) = 2(x - 1) + 3$ and $h(x) = 2(x + 1) + 3$ shown in Figure 10.4.

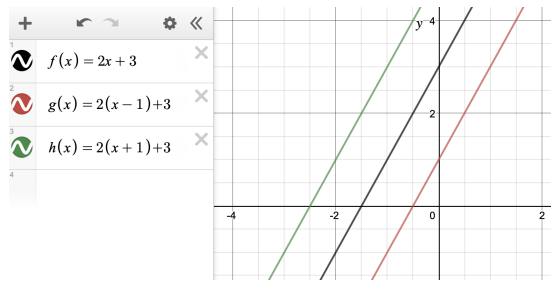


Figure 10.4: Shifting a linear function.

Inverse function

A function $f(x)$ is *inverse* to function $g(y)$ iff the following is true: $\forall x \in D(f), \forall y \in D(g) : y = f(x)$ and $x = g(y)$. More precisely, if we have:

$$g(f(x)) = x$$

We can only look for the inverse function if the original function $f(x)$ was injective. Some examples of inverse functions:

- $2x$ and $x/2$ are mutually inverse.
- We can derive the inverse function $g(y)$ to $f(x) = 2x + 3$. First we denote $y = 2x + 3$. Then we have $2x = y - 3$ and then $x = g(y) = \frac{y-3}{2}$. So we can write $g(x) = \frac{x-3}{2}$. And it is true that $f(\frac{x-3}{2}) = 2(\frac{x-3}{2}) + 3 = x - 3 + 3 = x$.

10.1.2 Quadratic functions

Problem 10.6 Values of a quadratic function:

Calculate selected values of the function $f(x) = x^2$.

Solution:

x	0	1	-1	2	-2	$\frac{1}{2}$	$-\frac{1}{2}$	$\sqrt{2}$	$-\sqrt{2}$	3	-3
$f(x)$	0	1	1	4	4	$\frac{1}{4}$	$\frac{1}{4}$	2	2	9	9

The graph of the function $f(x) = x^2$ is a *parabola*. The minimum of this function is 0 and it is achieved at $x = 0$.

Quadratic function

A *quadratic function* is defined as:

$$f(x) = ax^2 + bx + c$$

The domain of a quadratic function is $\mathbb{R}$. The range is a half-closed interval (for $a \neq 0$).

Problem 10.7 Graphs of quadratic functions:

Draw graphs of $f_1(x) = x^2 + 1$, $f_2(x) = x^2 - 1$, $f_3(x) = 2x^2$, $f_4(x) = \frac{x^2}{2}$, $f_5(x) = (x - 1)^2$, $f_6(x) = (x + 2)^2$, $f_7(x) = x^2 + 4x + 4 = (x + 2)^2$, $f_8(x) = (x - 3)^2$, $f_9(x) = (x + 10)^2$. What is the effect of coefficients a, b, c on the graph of $f(x) = ax^2 + bx + c$? For $f_5(x)$ determine the intervals on which it is increasing and decreasing.

Solution: Some of these functions are shown in Figure 10.5.

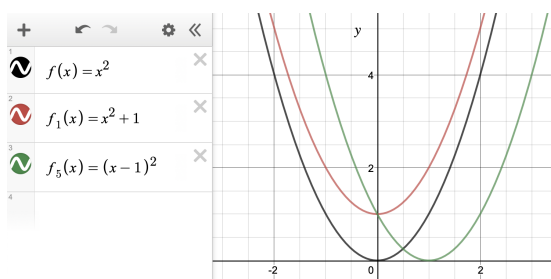


Figure 10.5: Quadratic functions.

Zero of a function

Let the function $f(x)$ have a domain $D(f)$. We call the point $x_0 \in D(f)$ a *zero (or root) of the function $f(x)$* iff $f(x_0) = 0$. This means that the zeros of a function $f(x)$ are the solutions of the equation $f(x) = 0$.

- The function $f(x) = x^2 + 4x + 1$ has two zeros.
- The function $f(x) = x^2 + 4x + 4$ has one zero point.
- The function $f(x) = x^2 + 4x + 6$ does not have any zeros.

- Using a graph of a function, we can learn about the solutions of an equation or inequality, e.g. $x^2 + 4x + 1 \geq 0$. We find the zeros of the function $f(x) = x^2 + 4x + 1$:

$$x^2 + 4x + 4 - 3 = 0$$

$$(x + 2)^2 - 3 = 0$$

$$(x + 2)^2 = 3$$

$$x + 2 = \pm\sqrt{3}$$

$$x = -2 \pm \sqrt{3}$$

The solutions of inequality $x^2 + 4x + 1 \geq 0$ are $\forall x \in (-\infty, -2 - \sqrt{3}) \cup (-2 + \sqrt{3}, \infty)$.

- The solutions of $x^2 + 4x + 1 < 0$ are $\forall x \in (-2 - \sqrt{3}, -2 + \sqrt{3})$.

10.1.3 Absolute value function

Problem 10.8 Values of an absolute value function:

Calculate selected values of the function $f(x) = |x|$.

Solution:

x	1	2	3	0	10	-1	-2	-10	0.1	0.2	-0.3	$-\sqrt{2}$	π
$f(x)$	1	2	3	0	10	1	2	10	0.1	0.2	0.3	$\sqrt{2}$	π

The graph of $f(x) = |x|$ is a piecewise linear line. The domain of $f(x) = |ax + b|$ is $\mathbb{R}$. The range is a half-open interval.

Problem 10.9 Graphs of absolute value functions:

Draw graphs of

- $g(x) = |x - 3|$, $h(x) = |x + 1|$.
- $g(x) = |x - 3| - 3$, $h(x) = |x + 1| + 5$. How do we determine the zeros of these functions?
- $g_1(x) = |2x - 3|$ (more bent, sharper angle), $h_1(x) = |\frac{1}{2}x + 1|$ (less bent, less sharp angle).
- $g(x) + h(x) = |x - 3| + |x + 1|$.
- $g(x) - h(x) = |x - 3| - |x + 1|$.

10.2 Questions

1. Draw a graph of a function that describes your journey from home to school on Monday morning. Independent variable is time and dependent variable is your velocity.
2. Draw a graph of a function that describes your journey from home to school on Monday morning. Independent variable is time and dependent variable is your distance from home.
3. Real function of a real variable is a mapping from A to B . What are A and B ?
4. Give an example of a function for which $D(f) = H(f)$.
5. Give an example of a function for which $D(f) \cap H(f) = \emptyset$.
6. Find a linear function $g(x)$ which grows faster than $f(x) = 2x + 3$.
7. Find a linear function $g(x)$ whose graph intersects the x -axis to the right of the function $f(x) = 2x + 3$.
8. Find a linear function $g(x)$ whose graph is parallel with the graph of the function $f(x) = -2x + 3$.
9. Find a linear function $g(x)$ which is neither increasing nor decreasing.
10. Find a quadratic function $g(x)$ whose graph is completely above the graph of $f(x) = x^2 + 2x + 1$.
11. Find a quadratic function $g(x)$ whose roots are between the roots of the function $f(x) = x^2 - 5x + 6$.
12. Find a quadratic function $g(x)$ whose graph touches the graph of $f(x) = x^2 + 2x + 1$ in just one point.
13. Find a quadratic function $g(x)$ whose graph does not cross the x -axis.
14. Find a quadratic function $g(x)$ whose graph does not cross the y -axis.
15. For numbers from which interval is the following statement true: if $x > y$ then $x^2 > y^2$?
16. For numbers from which interval is the following statement true: if $x > y$ then $x^2 < y^2$?

17. For which number x is $f(x) = x^2 + 2x + 1$ minimal? What is the minimum value of $f(x)$?
18. Draw the graphs $f(x) = |2x|$, $g(x) = |x + 1|$, $h(x) = |2x + 1|$ and $z(x) = |2x - 1|$.

10.3 Problem set

1. Equations and inequalities with absolute values

State the set of solutions, draw the graph of the function and display the method of solution and the result. Draw the solution on a real axis where the absolute value can be interpreted as a distance.

a) $|x + 2| = 3$

b) $|x - 1| \geq 7$

c) $|x + 2| + |x - 2| = 4$

d) $|x + 3| + |x - 4| \leq 7$

e) $|x - 1| + |x + 8| > 7$

2. Equations and inequalities with quadratic functions

Solve the equations and inequalities with quadratic functions. State the set of solutions, draw the graph of the function and display the method and result.

a) $x^2 - x - 6 = -6$

b) $-x^2 + 4x + 2 = 5$

c) $x^2 + 4x - 5 \leq 14$

d) $x^2 + 6x - 4 \geq 9$

e) $-x^2 + 5x - 8 \geq -5$

3. Domains and graphs of rational functions

Draw the graphs and state the domains of the following functions:

a) $\frac{1}{x}$

b) $\frac{1}{x-3}$

c) $\frac{1}{x+3}$

d) $\frac{3}{2x+3}$

e) $\frac{3x-3}{x+2}$

f) $\frac{5x+3}{2-x}$

g) $\frac{1}{(x-2)^2}$

h) $\frac{1}{(x+4)(x-1)}$

4. Domains, ranges, roots and intervals of monotonicity

State the domains, ranges, roots and intervals of monotonicity of the following functions:

a) $x^2 - x - 6$

b) $|x + 1| + |x - 3|$

c) $\frac{3x-4}{5-x}$

d) $\frac{1}{(x+1)(x-3)}$

5. Zeroes of quadratic functions

How many zero points can a function $f(x) = ax^2 + bx + c$ have? Give an example for each case.

10.3.1 Hints and solutions

1. Equations and inequalities with absolute values

- a) $|x + 2| = 3$
d) $|x + 3| + |x - 4| \leq 7$

Hints:

- Think about what the absolute value does. We evaluate whatever is inside it. If it is positive or zero, the absolute value does not do anything at all. On the other hand, if the result is negative, it simply removes the negative sign. With respect to the graph of a function, we can think about the absolute value as a mirror that lies on the x -axis and flips everything from below the axis to the above.
- The problems can be solved analytically by replacing the absolute value(s) with a set of suitable functions over different intervals.
- The absolute value can also be considered as distance. In this case, remember that $|x - 2|$ represents the distance of x from 2, while $|x + 2|$ represents the distance of x from -2 .
- Alternatively, these problems can also be solved graphically by sketching the left-hand side and right-hand side separately and then considering where the graphs intersect (in case of equations) or regions where the larger side is above the smaller side (in case of inequalities).

Solution:

- a) One way to approach this problem is to remove the absolute values. We assume that $x \in \mathbb{R}$ and we solve this problem separately two times. Once on the interval $(-\infty, -2)$, where the absolute value changes the sign of what is inside it and therefore we have the equation $-x - 2 = 3$. The solution is $x_1 = -5$.

And we also solve the equation $x + 2 = 3$ on the interval $(2, \infty)$, where the absolute value does not change the sign of the expression within. Here we get a solution $x_2 = 1$. The set of solutions is then $S = \{-5, 1\}$. This can be interpreted as a set of numbers that have a distance 3 from the number -2 , as illustrated in Figure 10.6 a).

Another way to look at this problem is to draw the graph of the left-hand side of the equation, a V as shown in red in Figure 10.6 b)

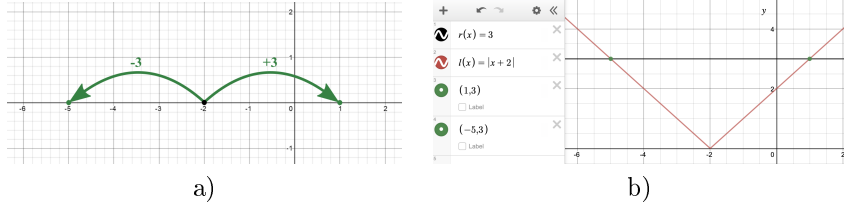


Figure 10.6: Absolute value represents distance as in a), or we can solve the problem using the graphs of both sides of the equation as in b).

and the constant as a horizontal line (green). The solutions are then the x -coordinates of the intersections of these two graphs.

- d) We again solve this problems in two ways: once by splitting the domain and removing the absolute values and the second time by plotting both sides of the inequality.

The domain is split by the zero points of each absolute value, i.e. into $(-\infty, -3) \cup (-3, 4) \cup (4, \infty)$. Note that it does not matter into which interval we include the zero points, we could use e.g. $(-\infty, -3) \cup (-3, 4) \cup (4, \infty)$ instead.

On $(-\infty, -3)$ we have the inequality

$$\begin{aligned} -x - 3 - x + 4 &\leq 7 \\ -2x + 1 &\leq 7 \\ -6 &\leq 2x \\ -3 &\leq x \end{aligned}$$

We have one solution here $S_1 = \{-3\}$, since -3 is the only $x \in (-\infty, -3)$ that is greater than (or equal to) negative three.

So we look at the second interval $(-3, 4)$. Only one of the absolute values changes the sign of its inner expression there (we can verify that by plugging in any value from the interval):

$$\begin{aligned} x + 3 - x + 4 &\leq 7 \\ 7 &\leq 7 \end{aligned}$$

This is true for all x in this interval, so $S_2 = (-3, 4)$.

And then we look at the third interval $\langle 4, \infty)$. The absolute value does not change sign here, so we have

$$\begin{aligned} x + 3 + x - 4 &\leq 7 \\ 2x - 1 &\leq 7 \\ x &\leq 4 \end{aligned}$$

We have one solution here $S_3 = \{4\}$.

Combining them all, we get $S = S_1 \cup S_2 \cup S_3 = \langle -3, 4 \rangle$. The same answer can be obtained by looking at Figure 10.7. We see the individual absolute values in black, their sum in red and the right-hand side in green. Coincidentally, the intersection of the red and green graphs is the horizontal part of the left-hand side. In general, this is not true. The green line may intersect the red one at two points or at none at all. In the latter case there are no solutions, in the former, the solutions are the x -coordinates of all points on the red line that lie below or on the green line.

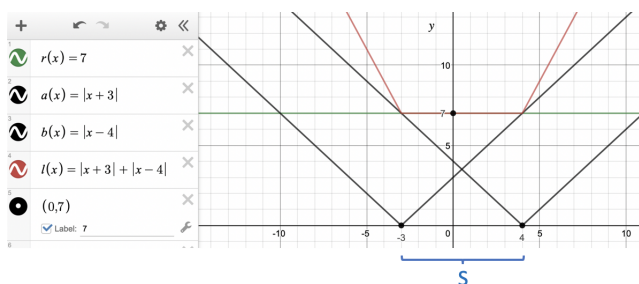


Figure 10.7: A graphic solution to inequality with two absolute values.

2. Equations and inequalities with quadratic functions

a) $x^2 - x - 6 = -6$

d) $x^2 + 6x - 4 \geq 9$

Hints:

- It is helpful to find the roots of the quadratic function, e.g. using discriminant, and the vertex of the parabola that lies between them.

- The problems can be solved analytically by manipulating the equations/inequalities or graphically by drawing both sides and looking for intersections.

Solution:

- a) When looking for the zeros of the quadratic function, we first write it in the form $ax^2 + bx + c = 0$ by adding 6 to both sides of the equation (this is one of the equivalent modifications that we can perform on an equation):

$$x^2 - x = x(x - 1) = 0$$

The zeros are $x_1 = 0$ and $x_2 = 1$.

We can also look at the graphs of both sides of the original equation, see Figure 10.8. The solutions are then the x -coordinates of their intersections.

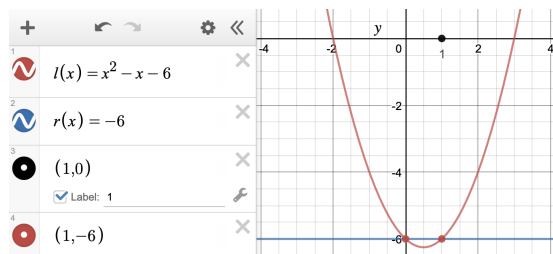


Figure 10.8: Left-hand side (red) and right-hand side (blue) of the equation $x^2 - x - 6 = -6$.

- d) Here we also start by writing it in the form $ax^2 + bx + c \geq 0$:

$$x^2 + 6x - 13 \geq 0.$$

It is not immediately clear how to factor the left-hand side into a product of linear terms. Therefore, we use discriminant $D = b^2 - 4ac = 36 - 4 \cdot (-13) = 88$ to determine $(x_1 = -6 + \sqrt{88})/2 = -3 + \sqrt{22}$ and $x_1 = (-6 - \sqrt{88})/2 = -3 - \sqrt{22}$. The inequality can now be written as

$$(x - (-3 + \sqrt{22})) \cdot (x - (-3 - \sqrt{22})) \geq 0.$$

Now we might remember the fact that $a > 0$ means that we have a parabola with branches pointed upwards and thus the solution to the inequality is a union of two intervals where these branches are above the x -axis: $(-\infty, -3 - \sqrt{22}) \cup (-3 + \sqrt{22}, \infty)$.

If we do not know this, we can examine a number between the two roots, e.g. $x = -3$ and check whether the left-hand side is greater than 0. Since $(-3)^2 + 6 \cdot (-3) - 13 = 9 - 18 - 13 < 0$, we know that the whole part of the parabola that lies between the zeros has to be below the x -axis (if it crossed the axis somewhere, we would have more zeros, so it cannot do that) and consequently the branches have to be above it. Then we can make the same conclusion about the solution as before.

To draw the graph of a quadratic function, it is certainly helpful to know the roots (if there are any), but there is another point of interest: the vertex. This is either the maximum (if the branches of the parabola point downward, i.e. for $a < 0$) or minimum (if the branches point upward, i.e. for $a > 0$). We can find it by *completing the square*:

$$x^2 + 6x - 13 = (x + 3)^2 - 9 - 13 = (x + 3)^2 - 22.$$

Here we are using the fact that $(x^2 + ax)$ differs from $(x + a/2)^2$ only by a constant. The expression $x^2 + 6x - 13$ is the smallest, when $(x + 3)^2 = 0$, because it cannot be negative. This happens for $x_m = -3 = -\frac{b}{2a}$ and this is a point of minimum. The actual value of minimum is $m = f(x_m)$ and can be obtained either by evaluating $f(x_m)$ or by looking at expression $(x_m + 3)^2 - 22 = 0 - 22 = -22$.

3. Domains and graphs of rational functions

- a) $\frac{1}{x}$
- c) $\frac{1}{x+3}$
- f) $\frac{5x+3}{2-x}$
- g) $\frac{1}{(x-2)^2}$
- h) $\frac{1}{(x+4)(x-1)}$

Hints:

- What do these functions have in common is that when drawing their graphs, we can start from a hyperbola and shift it to the desired position.
- If the rational function (or portion of a function between two vertical asymptotes) does not have a zero point, then it has to stay either above the x -axis or below it.

Solution:

- a) The domain of $\frac{1}{x}$ is $\mathbb{R} - \{0\}$ and the graph – a hyperbola – is shown in Figure 10.9 a).
- c) Similarly, $\frac{1}{x+3}$ is not defined at the zero point of the denominator $x = -3$ and its domain is $\mathbb{R} - \{-3\}$. The graph is shown in Figure 10.9 a).
- f) The domain of $\frac{5x+3}{2-x}$ is $\mathbb{R} - \{2\}$. To draw the graph, we express the function in the form $\frac{5x+3}{2-x} = \frac{5x-10+13}{2-x} = -5 + \frac{13}{2-x}$. We start by the graph of $\frac{1}{2-x}$, which is a hyperbola in the second and the fourth quadrant (due to the negative sign in front of x). Multiplication by a constant 13 stretches both the branches away from the x -axis and -5 shifts the whole graph down, as can be seen in Figure 10.9 b).
- g) The domain of $\frac{1}{(x-2)^2}$ is also $\mathbb{R} - \{2\}$. The graph is a hyperbola with both branches above the x -axis and closer the axes (due to the second power) as shown in Figure 10.9 c).
- h) The domain of $\frac{1}{(x+4)(x-1)}$ is $\mathbb{R} - \{-4, 1\}$. The graph is shown in Figure 10.9 d). It is helpful to start by examining the behavior of the function immediately to the left and to the right of the zeros of denominator.

Inverse graph problem

Can you find a function, whose graph is shown in Figure 10.10?

4. Domains, ranges, roots and intervals of monotonicity

- b) $f(x) = |x + 1| + |x - 3|$
- d) $g(x) = \frac{1}{(x+1)(x-3)}$

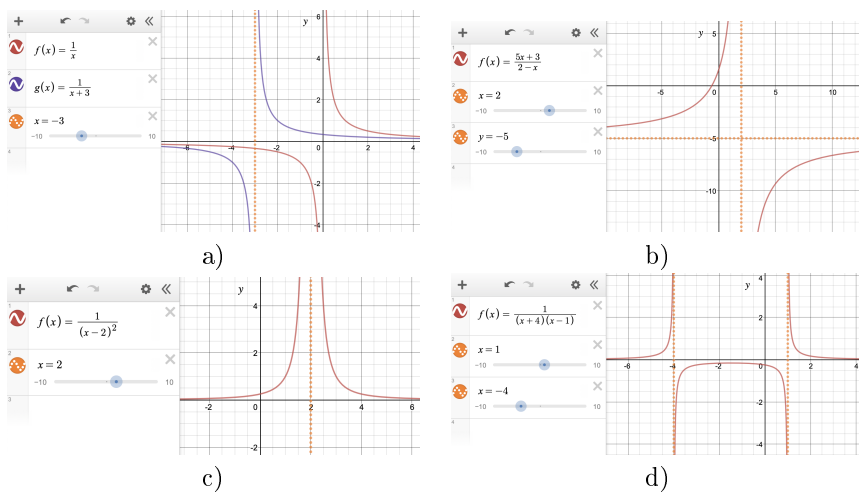


Figure 10.9: Graphs of rational functions.

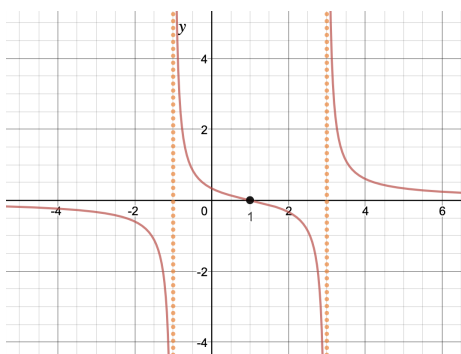


Figure 10.10: Can you write the formula of the function shown in this graph?

Hints:

- When trying to determine a domain, we should look for *problematic spots* in the formula of the function. These are typically square roots or denominators of fractions.
- A graph of the function can be very useful in determining the range.

In the graph we can easily see all the values of y that are images of some $x \in D(f)$. The check can be performed by sliding a horizontal line over the graph up and down – each position corresponding to a value y . All the values y for which the horizontal line intersects the graph of the function belong to the range.

- Roots/zeros are the intersections of the graph with the x -axis. They can be found by solving the equation $f(x) = 0$.
- While there are analytical methods (which use derivatives) to determine the intervals of monotonicity of functions, they are beyond the scope of this material. To determine monotonicity without them, we have to visually inspect the graph of the function and/or use the definition.

Solution:

- b) Looking at the function $f(x) = |x + 1| + |x - 3|$, we see that there are no problematic values of x at which this function would not be defined. Therefore $D(f) = \mathbb{R}$. Clearly $f(x) \geq 0$, since it is a sum of two non-negative terms, however this is not sufficient to state the range. For that we draw a graph, Figure 10.11. From the graph we see that the range is $H(f) = \langle 4, \infty \rangle$.

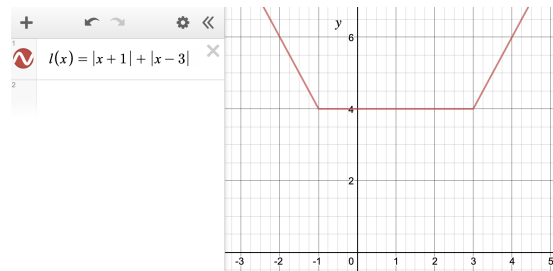


Figure 10.11: A graph of the function $f(x) = |x + 1| + |x - 3|$.

Since the graph does not intersect the x -axis, this function has no roots and we can also directly determine the intervals of monotonicity: $f(x)$ is decreasing on $(-\infty, -1)$, constant on $\langle -1, 3 \rangle$ and increasing on $\langle 3, \infty \rangle$.

We can also check monotonicity using the definition for example on the interval $(-\infty, -1)$. On this interval $f(x) = -x - 1 - x + 3 =$

$-2x + 2$. For $y > x$ we have $-2y < -2x$ and $f(y) = -2y + 2 < -2x + 2 = f(x)$. Since for any $y > x$ on the interval $(-\infty, -1)$ we have $f(y) < f(x)$, this means the function is decreasing on this interval.

- d) The function $g(x) = \frac{1}{(x+1)(x-3)}$ has two zero points in the denominator: $x = -1$ and $x = 3$. Therefore the domain is $D(g) = \mathbb{R} - \{-1, 3\}$. Looking at the graph, Figure 10.12, we see that the range is $H(g) = \mathbb{R} - (-\frac{1}{4}, 0)$. Clearly $g(x) \neq 0$, there are no zeros of this function. How can we be sure that it is sufficient to exclude 0 and nothing else? We can calculate the limits $\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow -\infty} g(x) = 0$ and from this we know that the functional values have to get arbitrarily close to 0 and thus nothing else needs to be excluded from the range.

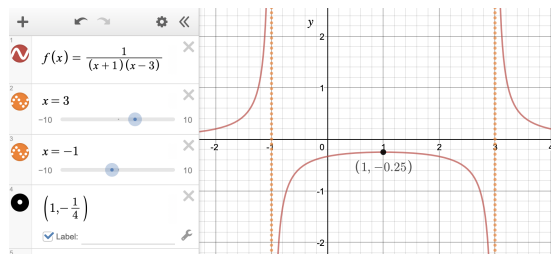


Figure 10.12: A graph of the function $g(x) = \frac{1}{(x+1)(x-3)}$.

The local minimum in the middle part of the graph lies in the middle between the two vertical asymptotes at $x = \frac{-1+3}{2} = 1$.

The graph in Figure 10.12 also helps with investigating monotonicity. We see that the function is increasing on $(-\infty, -1)$ and on $(-1, 1)$. It is decreasing on $(1, 3)$ and on $(3, \infty)$.

Monotonicity on two intervals

Can we write that $g(x)$ is increasing on $(-\infty, -1) \cup (-1, 1)$? Why? (Or why not?)

5. Zeroes of quadratic functions

Hints:

- The fundamental theorem of algebra tells us at most how many zeroes can a polynomial function have. Once we know this number k for a

quadratic function, we can try to construct different functions, that will have $0, 1, \dots, k$ roots.

- It is also useful to think about how many parameters there are in a quadratic function that can be tuned (three) and what implications this has for the number of roots.
- If a polynomial $f(x)$ has a root r , then it can be written as $f(x) = (x-r)g(x)$, where the degree of $g(x)$ (largest power of x) is the degree of $f(x)$ minus 1.

Solution: The fundamental theorem of algebra says that a polynomial of degree n can have at most n roots. Therefore a quadratic function can have at most two. Suppose we want the roots to be $x_1 = 2$ and $x_2 = -3$. We can construct the corresponding quadratic function as $f_2(x) = (x-2)(x+3) = x^2 + x - 6$. Note that this is not the only quadratic function with these roots, $g_2(x) = 2x^2 + 2x - 12 = 2(x-2)(x+3)$ is another one and any other constant multiple of $f_2(x)$ would work. Of course, we could have picked any other two real numbers for roots.

Quadratic function can also have just one double real root. This happens for a suitable combination of coefficients when the discriminant $D = 0$. Thus we can construct $f_1(x) = (x-1)^2 = x^2 - 2x + 1$. Again we can take any constant multiple of this function. The parabola touches the x -axis at precisely one point.

For a quadratic function to have no roots, it cannot cross or even touch the x -axis. We can start with x^2 and shift it upward, i.e. $f_0(x) = x^2 + 3$.

10.4 Lab problems

1. Absolute value

- Draw the graph of $|x - a|$ for a variable x and a parameter a . Try shifts to the right, left, up, down and change of slope.
- Write the equations of individual lines (removal of absolute value).
- Use the graph to analyse equations and inequalities and look for their solution.
- Solve similar problems for a sum of two absolute values, examine the shape and shifts. Write the equations of individual lines.
- Solve similar problems for a difference of two absolute values.
- Create the most peculiar graph with a given number of $|x - a|$ expressions.
- Create dragon teeth: iterative drawing of $||x - a| - b|$, $||x - a| - b| - c|$. Examine the properties, shift up and down, draw a required number of teeth, at required locations, etc.

2. Quadratic function

Calculate and draw a quadratic function $f(x) = ax^2 + bx + c$ for x in the range $\langle -10, 10 \rangle$. Keep a , b , c as adjustable parameters. Observe what is the impact of individual parameters on the shape of the graph. Calculate the roots (output them only if real roots exist), number of roots, x -coordinate of the vertex and the value of minimum/maximum.

3. Cubic and bi-quadratic functions

Draw graphs of cubic and bi-quadratic ($f(x) = ax^4 + bx^2 + c$) functions. What shapes can they have? How many roots can they have? Can you set the parameters in such a way that you get specific number and values of roots? Can you set the parameters in such a way that you get specific intervals of monotonicity?

4. Area under the curve

Tabulate a function $f(x)$ (constant, linear with only parameter a) for nonnegative real numbers and areas under the graph $f(x)$ from 0 to x using a suitable formula for calculation of area. Draw the graph of the function $f(x)$ and area $A_f(x)$. What relationship between $f(x)$ and $A_f(x)$ do you observe?

10.4.1 Hints and solutions

1. Absolute value

Hints:

- We can start in Excel by tabulating the values of $f(x) = |x|$. The absolute value changes the sign of everything inside it that is below the x -axis. For a linear function this means that the left half is reflected above and the resulting graph is a V -shape.
- When we parametrise the expression as $|x - a|$ with a parameter a , we soon realise that the resulting shifts due to different inputs for a might cause our graph to *escape* to the right or to the left. Of course, we could tabulate more values or new domains, but that might distract us from the actual task – investigating the effects of parameter changes. Therefore, we recommend using a different software tool for this problem, such as the versatile webpage wolframAlpha [3] or some other webpage dedicated to graphing, such as desmos.com [1] or perhaps a standalone application like Geogebra [5].
- It is useful to change the parameters one at a time. If you change more of them at once, it is difficult to realise which effects were due to which changes.
- To solve the equations or inequalities graphically, draw both sides and examine their intersections.
- To create the dragon teeth, try using the absolute values iteratively: $||x - a| - b|$, $|||x - a| - b| - c|$. Which parameters determine the size of individual teeth? How to make the teeth all the same size? How to make the desired number of them? If needed, add more parameters.

Solution:

Putting the parameter inside the absolute value causes horizontal shift in the graph. $|x - 2|$ shifts the basic absolute value function two units to the right, $|x + 2| = |x - (-2)|$ shifts it two units to the left (or -2 units to the right), as can be seen in Figure 10.13. To achieve a vertical shift, we need to add something *once the value of the function is already calculated*, i.e. outside the absolute value. $|x| + 3$ shifts the absolute value up by 3 units and $|x| - 3$ shifts it down.

To change the slope, we need a multiplicative constant with x , similarly to the basic linear function. For constant $k > 1$, the V in the graph is steeper (also seen in Figure 10.13), for $k < 1$ the V is more shallow.

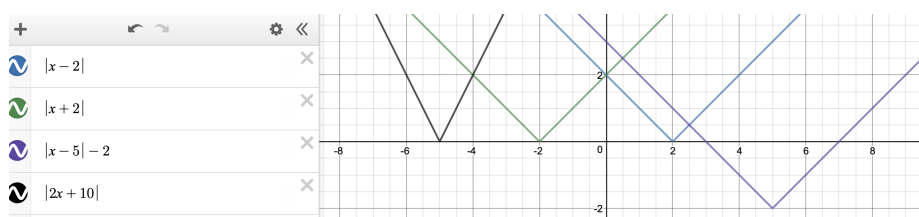


Figure 10.13: Manipulations with absolute value function: the blue one is shifted to the right by 2, the green one is shifted to the left by 2, the purple one is shifted to the right by 5 and down by 2. The black one is shifted to the left by 5 and its slope is increased by a factor of 2.

Various graphing software lets us experiment with parameters in a very user-friendly way, as shown in Figure 10.14.

Inverse problems

Can you solve an inverse problem, i.e. find the parameters k and a given a set of solutions of an equation $|kx - a| = c$ or inequality $|kx - a| < c$?

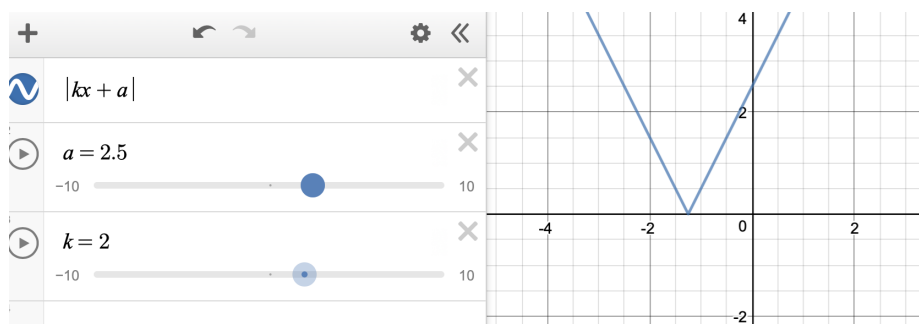


Figure 10.14: In graphing softwares it is possible to parametrise the function and easily observe the influence of parameters on the graph of the function – in this case on the slope (parameter k) and horizontal shift (parameter a). Performed in Desmos.

To draw the dragon teeth, we apply the absolute values repeatedly. Every time we shift the function down, the next absolute value mirrors the part

of V that is below the x -axis up and results in a new tooth. If we want all teeth be of the same size, the downward shifts should be the same too. Once we have the desired number of teeth we can display a second function, which only differs from the original one by a vertical shift, i.e. a constant added or subtracted at the end. This results in a *mouth* with two rows of teeth, Figure 10.15, which can even be animated.

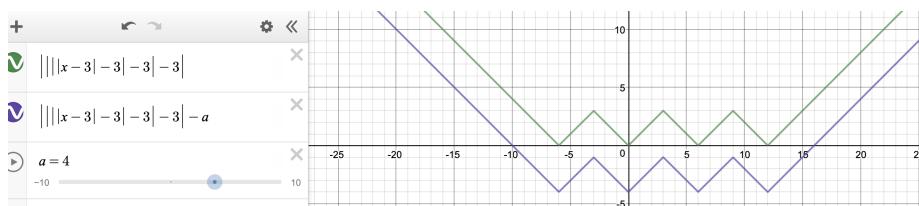


Figure 10.15: By applying the absolute value repeatedly on function that is repeatedly shifted downward, we can get a *teeth* function. We can even take two of such functions, one with parametric vertical shift (parameter a) and either by manual or automatic changes in the parameter animate the chewing motion.

Blunt dragon teeth

Can you create dragon teeth that are not sharp like those shown in Figure 10.15, but rather round (*semicircular* or *parabolic*)?

2. Quadratic function

Calculate and draw a quadratic function $f(x) = ax^2 + bx + c$ for x in the range $\langle -10, 10 \rangle$. Keep a , b , c as adjustable parameters. Observe what is the impact of individual parameters on the shape of the graph. Calculate the roots (output them only if real roots exist), number of roots, x -coordinate of the vertex and the value of minimum/maximum.

Hints: If we only wanted a graph, we could experiment with the parameters in a graphing software as in the previous problem, but since we should also calculate some characteristics of the parabola, it is useful to use Excel again here. The roots can be determined using the discriminant and conditions that check if it is positive, negative or equal to zero. The x -coordinate of the vertex can either be found in the middle between the roots or again using a suitable formula that contains some of the parameters a , b , c . The y -coordinate of the vertex (global extreme of the

quadratic function) can be evaluated as the value of f at the x -coordinate of the vertex.

Solution: An example worksheet is shown in Figure 10.16. We can vary the input parameters a , b , c and observe their impact on the graph – for example changing the sign of a flips the branches of the parabola, changes in c impact the number of roots, etc. Which parameter(s) influence the location of the roots?

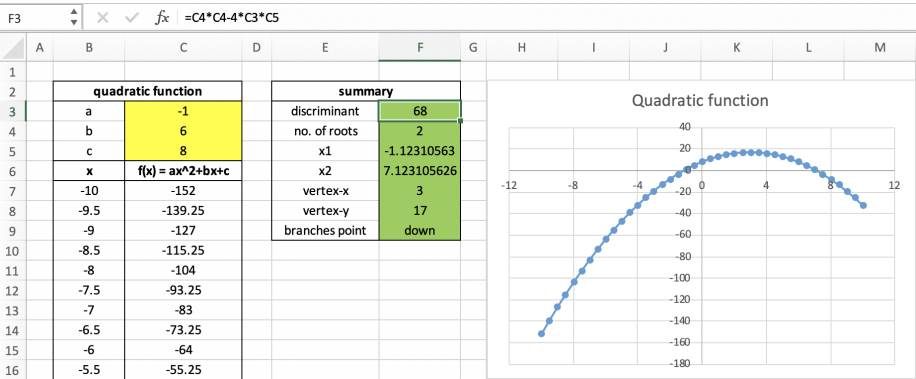


Figure 10.16: Quadratic function displayed in Excel along with information on its discriminant, roots and vertex.

Inverse problem
Create an Excel sheet in which the inputs are roots and coordinates of the vertex. The sheet checks that the input is consistent, draws the parabola and outputs the values a , b , c .

3. Cubic and bi-quadratic functions
- Draw graphs of cubic and bi-quadratic functions. What shapes can they have? How many roots can they have? Can you set the parameters in such a way that you get specific number and values of roots? Can you set the parameters in such a way that you get specific intervals of monotonicity?
- Hints:*

- While we could deal with this problem similarly to the previous one and consider the cubic function as $f(x) = ax^3 + bx^2 + cx + d$, with

four input parameters a, b, c and d , it would be difficult to get the right number of roots at the right positions. For that, it is useful to consider the formula in a different form. More details can be found in the next chapter – both in the problem section and in the lab section.

- The bi-quadratic function has the form $f(x) = ax^4 + bx^2 + c$. It is a fourth degree polynomial function with no odd terms. Therefore it is possible to make a change of variables $y = x^2$ to get the well known $f(y) = ay^2 + by + c$. Once know the roots of $f(y) = 0$, we can go back and find the roots of $f(x) = 0$.
- When thinking about the number of roots, we have three cases that can happen to y and then for each of them there are a few cases for x .
- To get specific values of the roots, we can *work backwards* and reverse-engineer them.

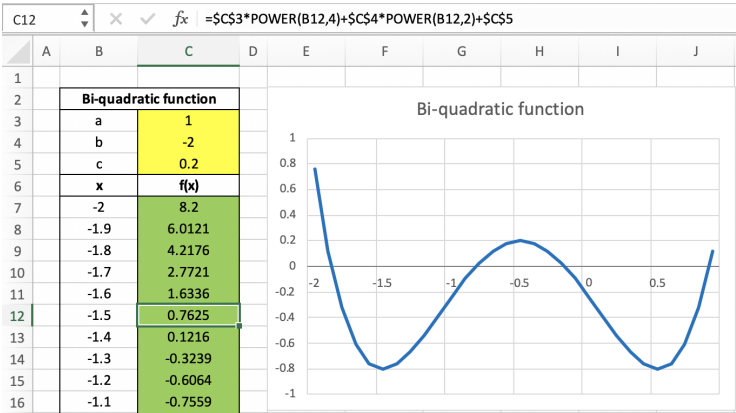


Figure 10.17: Tabulated values and graph of a bi-quadratic function.

Solution: An example of a bi-quadratic function is shown in Figure 10.17. There are the following possibilities:

- No real roots in case $f(y) = 0$ had no real roots (the whole graph lies above or below the x -axis).
- One real root, for example in case $f(y) = 0$ has one root equal to 0 and one negative root.

- Two real roots, for example in case $f(y) = 0$ has one positive and one negative root.
- Three real roots, in case $f(y) = 0$ has one root equal to 0 and one positive root.
- Four real roots in case $f(y) = 0$ has two positive roots.

4. Area under the curve

Tabulate a function $f(x)$ (constant, linear with only parameter a) for nonnegative real numbers and areas under the graph $f(x)$ from 0 to x using a suitable formula for calculation of area. Draw the graph of the function $f(x)$ and area $A_f(x)$. What relationship between $f(x)$ and $A_f(x)$ do you observe?

Hints:

- When calculating the area under a (positive) constant function, we are calculating the area of rectangles. One side of the rectangle is the constant $f(x) = c$ and the other is the independent variable x . In this case, $A_f(x)$ grows as x grows.
- When calculating the area under a linear function, it is easier to start with $f(x) = ax$. In this case, we are calculating areas of triangles. It is of course possible to use a general linear function $f(x) = ax + b$ for positive b and the qualitative results would not change much, but we would have to calculate with trapezoids instead of triangles or split the problem into triangle and rectangle.
- We could extend this problem and calculate area under different curves, e.g. parabola, but in this case, we would not have exact area calculations. We could approximate the area by cutting it into slices (rectangles and trapezoids) and adding the areas of those.

Solution: In Figure 10.18 we show a constant $f(x)$ and its area function $A_f(x)$, which turns out to be linear. We also show a linear $f(x)$ and its area function $A_f(x)$, which turns out to be a parabola. This is not a coincidence. If we repeated this procedure for higher order polynomials the area function is always a one-order-higher polynomial.

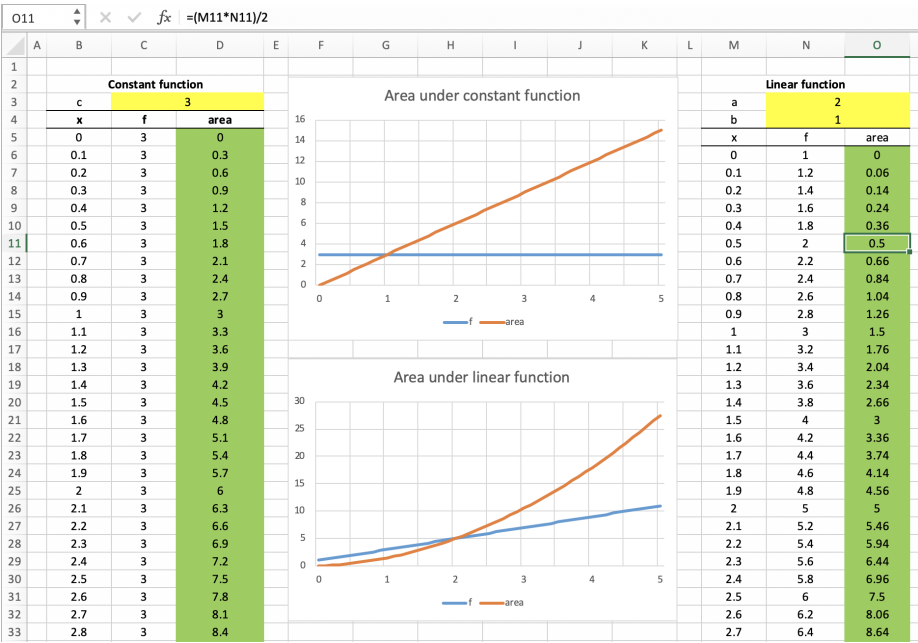


Figure 10.18: Area under the graph: for a constant and for a linear function.

Area under a parabola

Create a sheet which calculates and displays the values of $A_f(x)$ for a quadratic function $f(x)$. Consider different options for calculating the area.

Chapter 11

Real functions – part 2

11.1 Lecture overview

11.1.1 Cubic, polynomial and rational functions

Cubic function

A *cubic function* has the form $f(x) = a \cdot x^3 + b \cdot x^2 + c \cdot x + d$, where $a \neq 0, b, c, d$ are real parameters.

Problem 11.1 Zeros of cubic function:

Draw graphs of three cubic functions: one that has one zero point, one that has two zero points and one that has three zero points.

Hint: The graphs can be very similar, it is the shift in y -coordinate that makes a difference.

Problem 11.2 Monotonicity of cubic function:

Draw graphs of three cubic functions: one that has one interval of monotonicity, one that has two intervals of monotonicity and one that has three intervals of monotonicity.

Problem 11.3 Extremes of cubic function:

Find the extremes of $f(x) = 2x^3 - 3x^2 + x$.

Solution: The function $f(x)$ has three zero points: 0, $1/2$ and 1. It has a *local maximum* at approximately 0.211325 and *local minimum* at approximately 0.788675. It does not have a (global) minimum or maximum.

Polynomial functions

A *polynomial function* has the form $f(x) = a_n \cdot x^n + a_{n-1} \cdot x^{n-1} + \cdots + a_2 \cdot x^2 + a_1 \cdot x + a_0$.

11.1.2 Rational functions

Rational functions are defined as a ratio of polynomials, e.g. $f(x) = \frac{1}{x}$. The graph of this function is a *hyperbola*, the domain is $D(f) = \mathbb{R} - \{0\}$ and range $H(f) = \mathbb{R} - \{0\}$. The graph of the function never intersects the lines $x = 0$ and $y = 0$, but gets arbitrarily close to them. We call such lines *asymptotes*.

Problem 11.4 Asymptotes:

What are the asymptotes of

$$f(x) = \frac{3}{x-4}?$$

Solution: $y = 0$ and $x = 4$.

11.1.3 Exponential functions

Starting point

We start from a sequence $\{2^n\}_{n=0}^\infty$. By generalising this sequence, we get a function that is defined not only for natural numbers but also for all real numbers: $f(x) = 2^x$.

Problem 11.5 Interest rate:

How do we calculate interest at times which are not natural numbers?

Hint: We have already looked at this for rational values of exponent in Section 8.4.1.

Problem 11.6 Bacteria population:

How does the population of bacteria grow in time, not only in discrete time intervals such as minutes?

Solution: If we had z bacteria at the beginning and this number grows 10% per minute, in 5 minutes we have: $z \cdot 1.1 \cdot 1.1 \cdot 1.1 \cdot 1.1 \cdot 1.1 = z \cdot 1.1^5$ bacteria. In 20 seconds, we have $z \cdot 1.1^{\frac{1}{3}}$ bacteria.

If there were z bacteria at the beginning, and $z \cdot 1.1^{10}$ after 10 minutes, how many were there in the middle?

It is not $\frac{z+z \cdot 1.1^{10}}{2}$ but rather:

$$\sqrt{z \cdot z \cdot 1.1^{10}} = z \cdot 1.1^5.$$

Desired properties

How do we define $f(x) = 2^x$ for $x = 1$, $x = 0$, $x = -1$ and $x = \frac{1}{2}$, i.e. what are the values $f(1)$, $f(0)$, $f(-1)$, $f(\frac{1}{2})$?

We want to keep the rule:

$$2^2 \cdot 2^7 = 2^{2+7} = 2^9$$

We are looking for a function for which:

$$f(1) = 2, \quad f(x+y) = f(x) \cdot f(y)$$

We start by examining specific values of x and y :

$$x = 0, \quad y = 0, \quad \text{then} \quad f(0+0) = f(0) \cdot f(0), \quad \text{then} \quad f(0) = f(0)^2$$

This is only true for $f(0) = 0$ and for $f(0) = 1$.

If we go with $f(0) = 0$, we get a constant zero function. Therefore we select $f(0) = 1$.

$$f(2) = f(1+1) = f(1) \cdot f(1)$$

$$f(0) = f(1-1) = f(1) \cdot f(-1), \quad \text{then} \quad 1 = 2 \cdot f(-1), \quad \text{then} \quad f(-1) = \frac{1}{2}$$

$$2 = f(1) = f\left(\frac{1}{2} + \frac{1}{2}\right) = f\left(\frac{1}{2}\right) \cdot f\left(\frac{1}{2}\right), \quad \text{then} \quad 2 = f\left(\frac{1}{2}\right)^2$$

$$\text{then} \quad f\left(\frac{1}{2}\right) = \pm\sqrt{2}$$

Here we only use $+$, therefore $f(\frac{1}{2}) = \sqrt{2}$. So:

$$f\left(\frac{a}{b}\right) = 2^{\frac{a}{b}} = \sqrt[b]{2^a}$$

To find the value $2^{\sqrt{2}}$, i.e. to calculate $f(x)$ for an irrational x , $f(\sqrt{2})$, we use fractions. $\sqrt{2} \neq \frac{a}{b}$ but there are many fractions that are close to $\sqrt{2}$. $f(\sqrt{2})$ is the smallest upper bound of the set

$$M = \left\{ 2^{\frac{a}{b}}; \frac{a}{b} \in \mathbb{Q}, 0 < \frac{a}{b} < \sqrt{2} \right\}.$$

If we complete the definition of exponential function in this way, its graph is going to be continuous for all rational and irrational numbers.

Exponential function

Exponential function is denoted by $a^x, a \in \mathbb{R}^+$. The domain is $D(f) = \mathbb{R}$ and the range is $H(f) = \mathbb{R}^+$. A special case of exponential function is $f(x) = e^x$.

It is sufficient to calculate e^x :

$$2^x = e^{x \cdot \ln 2}$$

$$3^x = e^{x \cdot \ln 3}$$

$$a^x = e^{x \cdot \ln a} = e^{c \cdot x}$$

The function e^x can be evaluated using infinite series:

$$e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

11.1.4 Logarithmic functions

Logarithms and their properties

We derived the exponential function from the rule $f(x + y) = f(x) \cdot f(y)$. For a logarithmic function, we use the rule $f(x \cdot y) = f(x) + f(y)$. *Logarithm* $\log_2 x$ is such an exponent of 2 which *hits* x , e.g. $\log_2 16 = 4$.

value	x	2	4	8	16	32	64	128	256	512
exponent	$\log_2 x$	1	2	3	4	5	6	7	8	9

Multiplication using logarithms

If we want to multiply 4 by 16, we can first change the product of 4 and 16 to product of powers of 2, then add the exponents and look up the result in the table above.

$$4 \cdot 16 = 2^2 \cdot 2^4 = 2^{2+4} = 2^6 = 64$$

If we want to multiply $7 \cdot 33$, we choose approximate values, e.g.

$$7 \cdot 33 \doteq 2^3 \cdot 2^5 = 2^{3+5} = 2^8 = 256.$$

The correct result is 231, so the calculation was just an approximation. We would get an exact result, if we knew the powers of 2 which give values 7 and 33. We call such exponents binary logarithms, i.e. $\log_2 7$ a $\log_2 33$. In the past, logarithms have indeed been used for multiplication of large numbers and logarithmic tables were a desired possession of medieval navigators.

Base of the logarithm

Depending on the base we use, we call the logarithm *binary* (2), *decimal* (10) and *natural* (e). We denote them $\log_2 x$, $\log_{10} x = \log x$ and $\log_e x = \ln x$.

Logarithm and exponential

The functions $\ln x$ and e^x are mutually inverse. Thus it is true that:

$$\ln e^x = x \ln e = x \quad \text{and} \quad e^{\ln x} = x.$$

This also holds for other bases, e.g. $\log x$ and 10^x are also mutually inverse.

Stirling formula

After taking a logarithm of a factorial $n! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot \dots \cdot n$, we get:

$$\ln(n!) = \ln 1 + \ln 2 + \ln 3 + \ln 4 + \dots + \ln n.$$

This equality is used to derive an approximation of a factorial, that is called *Stirling formula*:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

You may encounter this formula for example when working with binomial distribution, trying to estimate expressions containing factorials.

For $n = 30$, compare the exact value of expression $\binom{2n}{n} \frac{1}{2^{2n}}$ with its approx-

imation using the Stirling formula. Show that for smaller n , e.g. $n = 5$, the approximation is less precise.

11.1.5 Trigonometric functions

Where do they appear?

The most often used trigonometric functions are *sine* and *cosine*, denoted by $f(x) = \sin(x)$ and $g(x) = \cos(x)$ respectively. We encounter them for example when calculating with angles in a right triangle or when using the law of cosines $c^2 = a^2 + b^2 - 2ab \cos \gamma$ in a general triangle. (What is its connection to Pythagorean theorem?) The functions sine and cosine are used in analysis of periodic oscillations, e.g. in sound or signal processing.

Sine

The function $\sin x$ is defined for all real numbers. Its range is the closed interval $\langle -1, 1 \rangle$. Its zero points are integer multiples of π . $\sin x$ is *periodic* with period 2π . Thus it is true that $\sin(x) = \sin(x + 2\pi)$. Shifting the sine function by $\frac{\pi}{2}$, we get a cosine.

Problem 11.7 Determining the period:

What is the period of these functions? $\sin(2x), \sin\left(\frac{x}{2}\right), \sin\left(\frac{x}{2\pi}\right), 2\sin x, 2 + \sin x, \sin(x - 2)$?

Cosine

The function $\cos x$ is defined for all real numbers. Its range is the closed interval $\langle -1, 1 \rangle$. Its zero points are $\frac{\pi}{2} + k\pi$, where $k \in \mathbb{Z}$. $\cos x$ is periodic with period 2π .

Sine and cosine as infinite series

The functions $\sin x$ and $\cos x$ can be expanded into infinite series:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

As a consequence, we can relate the functions e^x , $\sin x$ and $\cos x$ (This can be verified using the corresponding infinite series.):

$$e^{ix} = \cos x + i \sin x$$

From this follows the beauty queen of mathematical formulas:

$$e^{i\pi} + 1 = 0.$$

Tangent

Another trigonometric function is called a *tangent*, denoted by $f(x) = \tan(x)$ or sometimes $f(x) = \operatorname{tg}(x)$. Its properties can be derived directly from its definition

$$\tan(x) = \frac{\sin(x)}{\cos(x)}.$$

It has the same zero points as $\sin x$. The domain are all real numbers except those for which $\cos x = 0$. Its asymptotes are defined by $x = \frac{\pi}{2}$, $x = \frac{\pi}{2} + \pi$ etc., in general, $x = \frac{\pi}{2} + k\pi$, where $k \in \mathbb{Z}$. The range of $\tan x$ is $\mathbb{R}$ and it is a periodic function.

Cotangent

A *twin* function to tangent is the *cotangent* function. It is denoted by $g(x) = \cot(x)$ or sometimes $g(x) = \operatorname{cotg}(x)$. Its properties also follow from its definition

$$\cot(x) = \frac{\cos(x)}{\sin(x)}.$$

It has the same zero points as the function $\cos x$. The domain are all real numbers except those for which $\sin x = 0$. Its asymptotes are defined by $x = \pi$, $x = 2\pi$ etc., in general, $x = k\pi$, where $k \in \mathbb{Z}$. The range of $\cot x$ is $\mathbb{R}$ and it is a periodic function.

Problem 11.8 Measuring function change over short interval:

Suppose we select several points $x_1, x_2, \dots, x_n, \dots, x_N$ in the domain of a function f . They might or might not be regularly spaced. And then we look at the value of the following ratio for different $n \in \{1, 2, \dots, N\}$:

$$g(x) = \frac{f(x_{n+1}) - f(x_n)}{x_{n+1} - x_n}.$$

Consider the properties of this ratio, $g(x)$, depending on the function $f(x)$. How does $g(x)$ look at parts of the domain where $f(x)$ is increasing, decreasing? How does it look at an extremal point of $f(x)$?

11.2 Questions

1. Write how much is
 - a) $2^3 \cdot 2^5 =$
 - b) $2^{-3} \cdot 2^5 =$
 - c) $2^{-3} \cdot 2^{-5} =$
 - d) $(-2)^3 \cdot 2^5 =$
 - e) $(-2)^3 \cdot 2^{-5} =$
 - f) $2^{\frac{1}{2}} \cdot 2^4 =$
2. Find the function $f(x)$ for which $f(1) = 3$ and $f(x + y) = f(x) \cdot f(y)$.
3. Explain how do we get the value $2^{\sqrt{3}}$.
4. Why is it useful to use the function $f(x) = e^x$?
5. Write the function $f(x) = e^x$ as the sum of infinite series.
6. Find the function $f(x)$ for which $f(x \cdot y) = f(x) + f(y)$.
7. Find the sum of infinite series $\sum_{n=0}^{\infty} \frac{5^n}{n!}$.
8. How much is $14^{\log_{14} 5}$ and how much is $5^{\log_5 14}$?
9. What is the relation between Pythagorean theorem and cosine theorem?
10. What kind of phenomena are modeled by the sine function?
11. How are the functions $\sin x$, $\cos x$, $\tan x$, $\cot x$ related?
12. Find $f(x)$ and $g(x)$ such that $f(x) = g(x + \frac{\pi}{2})$.
13. Find a function for which $f(x + y) = f(x) \cdot f(\pi/2 - y) + f(y) \cdot f(\pi/2 - x)$.
14. Find a function for which $f(x) = x - x^3/3! + x^5/5! - x^7/7! + \dots$?
15. Find a function for which $f(x) = 1 - x^2/2! + x^4/4! - x^6/6! + \dots$?
16. Fix the equality $e^x = \sin x + \cos x$.
17. Draw the functions in one graph: $\sin x$, $\sin 2x$, $2 \sin x$, $\sin(x + \pi)$.

11.3 Problem set

1. Cubic functions

Sketch the graphs of $f : \mathbb{R} \rightarrow \mathbb{R}$. What do these graphs have in common and how do they differ?

- a) $f(x) = (x-3)(x-1)(x+2)$
- b) $f(x) = 3(x-3)(x-1)(x+2)$
- c) $f(x) = -2(x-3)(x-1)(x+2)$
- d) $f(x) = (x-3)(x-1)(x+2) + 4$

2. Roots and extremes

Find cubic functions f that have 0, 1, 2, 3, 4 zeros, i.e. roots of the equations $f(x) = 0$. How many and what kinds of extremes do they have? Can you estimate the location of the extreme values?

3. Hyperbolae

Sketch the graphs of $f : \mathbb{R} \rightarrow \mathbb{R}$.

- a) $f(x) = \frac{1}{x-1}$
- b) $f(x) = \frac{1}{(x-1)(x+2)}$
- c) $f(x) = \frac{-1}{(x-1)(x+2)}$
- d) $f(x) = \frac{1}{x-1} + 2$
- e) $f(x) = \frac{1}{(x-1)+2}$
- f) $f(x) = \frac{-1}{(x-1)(x+2)+1}$
- g) $f(x) = \frac{1}{(x-1)(x+2)} + 1$

4. Periodic functions

Write the function f which meets the following criteria:

- a) Period = 10π , range = $\langle 0, 3 \rangle$, $f(0) = 0$
- b) Period = 10π , range = $\langle 0, 3 \rangle$, $f(0) = 1.5$
- c) Period = 4π , range = $\langle 2, 14 \rangle$, $f(0) = 14$
- d) Period = $\pi/2$, range = $\langle -3, 3 \rangle$, $f(0) = -3$
- e) Period = 8, range = $\langle \pi, 5\pi \rangle$, $f(0) = 4\pi$

5. Addition to product

Find the function f which meets the following criteria:

- a) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(1) = 2$
- b) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(1) = 1/2$
- c) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(2) = 9$

6. Product to addition

Find the function f which meets the following criteria:

- a) $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(2) = 1$
- b) $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(2) = 1/2$
- c) $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(9) = 2$

11.3.1 Hints and solutions

1. Cubic functions

Sketch the graphs of $f : \mathbb{R} \rightarrow \mathbb{R}$. What do these graphs have in common and how do they differ?

- a) $f(x) = (x - 3)(x - 1)(x + 2)$
- b) $f(x) = 3(x - 3)(x - 1)(x + 2)$
- c) $f(x) = -2(x - 3)(x - 1)(x + 2)$
- d) $f(x) = (x - 3)(x - 1)(x + 2) + 4$

Hints:

- In each case we can sketch the graph by evaluating the function in a reasonable number of points tabulated with a reasonable step, for example values from -5 to 5 with the step 0.5 . We then draw these points in a coordinate system and connect them. For these functions though, there exists a more efficient method.
- If we multiply the individual factors, we find out that all of them are cubic functions. In general, a graph of a cubic function is a *twice-bent curve*. There are some special cases, when the graph is a simpler curve of the x^3 type. Which of these curves do we have in this problem?
- Let us look at $f(x) = (x - 3)(x - 1)(x + 2)$ first. The function is expressed as a product, which allows us to evaluate it very easily for one type of arguments x . For which x can we easily evaluate this function? This is also an answer to a simpler question: Which three real values can be substituted for x and give us the functional value practically without any calculation?

Once we realise we are talking about roots of the function, we are half done. The graph of a function crosses the x -axis exactly at the roots. From which direction? Does it come to the first root from above or from below of the x -axis? And which way does it leave from the last root? To answer these questions, we can think about what values does the function have for inputs close to ∞ and $-\infty$. In practice, it is sufficient to consider the signs of $f(-10)$ and $f(10)$.

If we answered all these questions, we should be able to draw the correct curve passing the roots of the function. We can verify the

sketch by checking some other functional values, such as $f(0)$ or $f(2)$, etc. The exact values might not lie precisely on the graph, but they have to have the correct sign.

- To sketch the other graphs, we can start by checking how do they differ from the function in a). The function in b) has three times larger values than the function in a), therefore its graph is *stretched* three times in the vertical direction and keeps the same zeros.
- The function in c) behaves similarly, but is multiplied by a negative number. This means that all function values (except the roots) change their sign. How does this manifest in a graph?
- The function in d) is shifted up by 4. In this case, the shape of the graph remains the same, but the roots change. A careful examination of function values between the second and the third root of the original function reveals whether the number of roots remains the same or decreases.

Solution: To check the solution, we can copy the formula for the function into any of the freely available tools, webpages or softwares that can draw graphs, such as WolframAlpha or Desmos.

2. Roots and extremes

Find cubic functions f that have 0, 1, 2, 3, 4 zeros, i.e. roots of the equations $f(x) = 0$. How many and what kinds of extremes do they have? Can you estimate the the location of the extreme values?

Hints:

- We have already worked with a cubic function in the previous problem. This experience should give us a hint that two of the five cases do not have a solution. The remaining three cases can be answered using a multiplicative form of the function formula, which is very convenient when working with roots.
- Every cubic function with a positive coefficient multiplying x^3 takes on negative values for sufficiently large negative inputs and positive values for sufficiently large positive inputs. Its graph thus *comes from* the negative values of y -coordinates and at some point has to *leave* to the positive values. The situation is reversed (with respect to the x -axis) for a negative coefficient by x^3 . Clearly, in both cases the graph of the function has to cross the x -axis at some point (and this

can be proven by the tools of mathematical analysis). This crossing is a zero of the function and therefore every cubic function has at least one root. This means there is no cubic function that would have zero roots.

- A function that has four zeros has to *wiggle* (switch from growth to decrease or vice versa) at least three times. No cubic function can do that. If we consider the product form of the formula for $f(x)$ which has four roots a, b, c, d , it would have to contain (at least) the terms $(x - a)(x - b)(x - c)(x - d)$. This implies a polynomial of at least fourth degree. That is another reason why there is no cubic function with four roots.
- We saw examples of a cubic function with three roots in the previous problem. Suppose we want the roots $-2, 1$ and 3 . The function is then $f(x) = c(x - (-2))(x - 1)(x - 3) = c(x + 2)(x - 1)(x - 3)$, where c is a real non-zero constant. We see that we have three roots and thus a polynomial of third degree.
- If we want a cubic function with fewer roots, we can shift its graph up or down. For one root we can also use various multiples and shifts of a *simple* cubic function x^3 .
- There is also a simpler solution. If we want a cubic function to have one root, its formula in the product form has to contain exactly three root terms of the form $(x - a)$. The number a is the solution of equation $f(x) = 0$ and thus a zero or root of the function. When we do not need three different roots, we can repeat some of them.
- How do we get a cubic function with exactly two roots, e.g. 2 and 4 ? We need to use the root terms $(x - 2)$ and $(x - 4)$ and add one more term that does not result in a new solution of $f(x) = 0$. We can repeat one of the already used terms. We recommend checking both of them and comparing the resulting graphs. To get a cubic function with just a one root, we use the same root term three times (possibly multiplied by a constant).
- We are not going to give an exact answer to the question about extremes but rather estimate their positions from the graphs of the function. The answer depends on the number of roots. The location of the extremes changes with the roots. Vertical shifts of the function change the value of the extremes. Which cubic functions do not have any extremes?

- Graphs of the proposed functions can again be displayed and checked in suitable available graphical tools.

Solution: Here we provide a few examples of functions that meet the criteria. Many more can be obtained by small modifications.

In case of three roots, we have for example $f(x) = (x-2)(x-4)(x-8)$ with the roots 2, 4 and 8, Figure 11.1. (Functions in a), b) or c) in the previous problem work too.) One of the local extremes lies between the first and the second root, i.e. in the interval $(2, 4)$ and the other between the second and the third root, i.e. in the interval $(4, 8)$. Note and check that they do not lie exactly in the middle of these intervals.

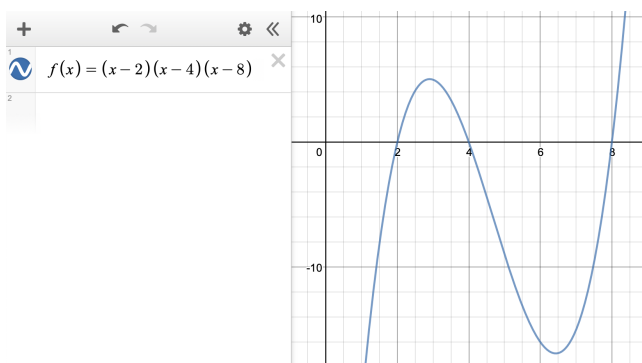


Figure 11.1: Cubic function.

To obtain a function with two roots, we modify it to $f(x) = (x-2)^2(x-4)$ or $f(x) = (x-2)(x-4)^2$. One of the local extremes lies in the interval $(2, 4)$ and the other is exactly at the double root.

To obtain a function for which the equation $f(x) = 0$ has a unique solution, we need a multiple root, such as $f(x) = (x-2)^3$ or $f(x) = (x-4)^3$. These functions do not have any extremes. Alternatively, we can use a vertical shift and we get for example $f(x) = (x-2)(x-4)(x-8) + 10$ or $f(x) = (x-2)(x-4)(x-8) - 5$. These functions have local extremes at the same points x as the original function $f(x) = (x-2)(x-4)(x-8)$.

3. Hyperbolae

Sketch the graphs of $f : \mathbb{R} \rightarrow \mathbb{R}$.

- a) $f(x) = \frac{1}{x-1}$
- b) $f(x) = \frac{1}{(x-1)(x+2)}$
- c) $f(x) = \frac{-1}{(x-1)(x+2)}$
- d) $f(x) = \frac{1}{x-1} + 2$
- e) $f(x) = \frac{1}{(x-1)+2}$
- f) $f(x) = \frac{-1}{(x-1)(x+2)+1}$
- g) $f(x) = \frac{1}{(x-1)(x+2)} + 1$

Hints:

- The functions in this problem have the form of fractions with the variable x in the denominator. Therefore the zero points of the denominators are points which are excluded from the domain. It is important to examine the behavior of the function in the immediate vicinity of these points. Therefore, to sketch their graphs, it is not sufficient to tabulate the function values with a standard step 1 or 0.5, but rather we need to determine the asymptotes – lines to which the graph tends for *infinite* values of arguments x or function values $f(x)$ – and the function behavior close to these asymptotes.
- We start with function $f(x) = \frac{1}{(x-1)(x+2)}$. First, we realise what its values are for very large negative or very large positive x . The denominator grows with the multiplication of terms. The value of $f(x)$ is its reciprocal, therefore a very small number close to zero. We still need to consider its sign. In both cases, the function will tend to zero both for $x \rightarrow \infty$ and $x \rightarrow -\infty$. This means that the line $y = 0$, i.e. the x -axis, is the horizontal asymptote in this case.
- The second type of asymptotes corresponds to the zero points of the denominator in which the function is not defined. In this case they are $x = -2$ and $x = 1$. What are the values of the function in the immediate vicinity of $x = 1$? We can try to calculate the values $f(x)$ for $x = 0.9$, $x = 0.95$, $x = 0.99$, $x = 0.999$. Similarly from the other side for values $x = 1.1$, $x = 1.05$, $x = 1.01$, $x = 1.001$. Can you guess and draw the graph for values of x even closer to 1? Similar consideration should be given to the values of x close to number -2 .

- We can also evaluate the behavior of the function near these points just qualitatively, e.g. the term $(x-1)$ is very small for values of x very close to 1. The whole denominator remains small when multiplied by the term $(x+2)$, which is about 3 in this case. The value $f(x)$ is the reciprocal of the denominator and therefore a large number here. Since x can get arbitrarily close to 1, the value of $f(x)$ will tend to infinity. Here we also need to be careful about the sign of the denominator, which determines for which values of x we get $+\infty$ and for which $-\infty$. In all cases, the vertical lines $x = -2$ and $x = 1$ are also the asymptotes of the function.
- The other cases in this problem are variations on the one we just discussed. We should carefully consider the impact of their differences on the graphs. A negative value in the numerator flips the graph symmetrically about the x -axis. Adding a constant to the denominator changes its zero points, i.e. points in which the function is not defined and thus the location of the vertical asymptotes. Adding a constant to the whole fraction shifts the graph up or down, which results in the shift of the horizontal asymptote.

Solution:

Once we have a sketch, we should compare it with a graph obtained by a suitable software tool. If they do not match, we need to check the differences and look for what might have caused them.

Primarily, we should check the asymptotes. In a) they are the x -axis and the line $x = 1$. In the case d), the function $f(x)$ is increased by 2. The graph thus shifts 2 units up. The asymptote $x = 1$ remains unchanged by this shift, but the asymptote that was at the x -axis is now moved to $y = 2$.

In e), we can rewrite the function as $f(x) = 1/(x+1)$. The horizontal asymptote is the x -axis, the vertical is $x = -1$.

In b), we have the horizontal asymptote x -axis and two vertical asymptotes: $x = -2$ and $x = 1$. Case c) is a mirror image of this, but the asymptotes remain the same. In g), the graph is shifted one unit up and the horizontal asymptote is moved to $y = 1$. In case f), there is a new quadratic expression in the denominator with new roots and thus new vertical asymptotes.

4. Periodic functions

Write the function f which meets the following criteria:

- a) Period = 10π , range = $\langle 0, 3 \rangle$, $f(0) = 0$
- b) Period = 10π , range = $\langle 0, 3 \rangle$, $f(0) = 1.5$
- c) Period = 4π , range = $\langle 2, 14 \rangle$, $f(0) = 14$
- d) Period = $\pi/2$, range = $\langle -3, 3 \rangle$, $f(0) = -3$
- e) Period = 8 , range = $\langle \pi, 5\pi \rangle$, $f(0) = 4\pi$

Hints:

- The core of this problems is to fine-tune a correct periodic function so that it meets three properties: period length, range $H(f)$ and the function value at $x = 0$, which in fact represents a correct horizontal shift.
- In the following we use the function $\sin(x)$ and its modifications but we could work with $\cos(x)$ as well. The period of $\sin(x)$ is 2π and its range is $\langle -1, 1 \rangle$. The function values at the interesting points $x = 0$, $x = \pi/2$ and $x = 3\pi/2$ are 0, 1 and -1 respectively. It is a good idea to draw this function in a software tool and observe the changes resulting from the following modifications.
- First we are going to change the period. To draw one period of the basic function $\sin(x)$ we need the independent variable x to run over all values from the interval $\langle 0, 2\pi \rangle$. If we change the function to $f(x) = \sin(\pi x)$, the constant π helps the variable x cover the domain faster. To draw one period of the function $\sin(\pi x)$ it is sufficient for the independent variable x to run over π -times shorter interval $\langle 0, 2 \rangle$. The period of the function $\sin(\pi x)$ is thus π -times shorter and equal to 2.

Conversely, the division by 2 in the formula $f(x) = \sin(x/2)$ slows down the growth of x by a factor of 2. The independent variable thus needs to cover twice as big interval $\langle 0, 4\pi \rangle$ in order to draw one period. The period of $\sin(x/2)$ is therefore 4π .

With this in mind, we can correctly determine the value of parameter c in $f(x) = \sin(cx)$ to achieve the correct period in problems a) to e). Note that these changes do not change the range of the functions.

- We have already discussed transformation of intervals into other intervals in Section 7.4. Here we need to transform the range $\langle -1, 1 \rangle$ of $\sin(x)$ into another interval. First, we make sure that the size of the range interval is correct. For example, if $H(f)$ is supposed to be $\langle 2, 14 \rangle$, we need to increase the original size of range six-fold from 2 to 12. For that we modify the function to $6\sin(x)$. However, the range of this function is $\langle -6, 6 \rangle$, so we still need to shift it into the correct position $\langle 2, 14 \rangle$. A modification to $6\sin(x) + 8$ does precisely that.
- The last property we need to ensure is the function value at $x = 0$. Note that in all the cases we want $f(0)$ to be either the minimum, maximum or middle of the range. For the original $\sin(x)$ those would be values $-1, 1$ or 0 . Since $\sin(0) = 0$, we do not need any changes to achieve the middle value. If we want $\sin(x) = 1$, we need to remember that the maximum of the sine function is reached at, e.g. $x = \pi/2$. Therefore the graph of the function needs to be shifted to the left by $\pi/2$, which can be done by modifying the function to $\sin(x + \pi/2)$.
- Since $\sin(3\pi/2) = -1$ (or $\sin(-\pi/2) = -1$), if we want to achieve the minimal value at $x = 0$, we need to shift the graph $3\pi/2$ units to the left (or $\pi/2$ units to the right). This can be done by modifying the function to $\sin(x + 3\pi/2)$ or $\sin(x - \pi/2)$.
- To solve the problem, it is now sufficient to perform all these modifications one after another. The solution can be verified by graphing the individual steps as well as the final result in an appropriate software tool.

Solution:

Here we show one of the possible answers for each case:

- $\frac{3}{2} \sin(\frac{x-5\pi/2}{5}) + \frac{3}{2}$, Figure 11.2.
- $\frac{3}{2} \sin(x/5) + \frac{3}{2}$
- $6 \sin(\frac{x+\pi}{2}) + 8$
- $3 \sin(4(x - \pi/8))$
- $2\pi \sin(\frac{\pi(x+2)}{4}) + 3\pi$

5. Addition to product

Find the function f which meets the following criteria:

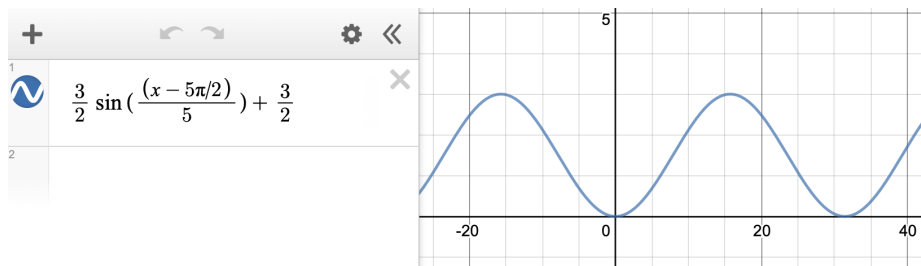


Figure 11.2: Sine function.

- a) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(1) = 2$
- b) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(1) = 1/2$
- c) $f(x + y) = f(x) \cdot f(y)$, $f(0) = 1$ and $f(2) = 9$

Hints:

- In this problem, we have two types of information that we can use to find the functions: a general relation – the so-called functional equation, which our functions satisfies – and its values at two specific points. The functional equation is the same in all three cases and is mentioned in the lecture. From there we know that the only functions that satisfy it are exponential functions.
- Once we know we are dealing with exponentials, the only thing that remains open is to figure out the base of the function in each case. The functional values should help with that. One of the two is more helpful than the other. Which one?
- We can also use a software tool for drawing the exponential functions, *guesstimate* the base and check that the function has the desired properties.

Solution: For an exponential function, the equality $f(0) = 1$ means $a^0 = 1$. Since this is true for all admissible bases $a \in \mathbb{R}^+$, we do not get any new information about the function from here. We need to focus on the other two equalities.

- a) We need $a^1 = 2$. This is true for $a = 2$ which gives us the function 2^x .

- b) We need $a^1 = \frac{1}{2}$. The solution is thus a decreasing exponential $(\frac{1}{2})^x$, Figure 11.3.
- c) Here we need $a^2 = 9$. There are two values that satisfy this equation: $a = -3$ and $a = 3$. The base of the exponential has to be a real positive number, therefore we get just one solution: the function 3^x .

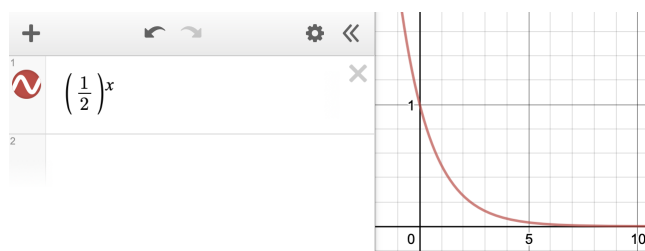


Figure 11.3: Exponential function.

6. Product to addition

Find the function f which meets the following criteria:

- $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(2) = 1$
- $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(2) = 1/2$
- $f(x \cdot y) = f(x) + f(y)$, $f(1) = 0$ and $f(9) = 2$

Hints:

- Similarly to the previous problem, also here we are looking for three functions that satisfy the same functional equation. The rule it describes is unique and typical for logarithmic functions. The functions differ by their base which we need to determine. For each function, we are given two functional values. We can use one of them to determine the base. The other serves as confirmation that we have the correct type of the function. Can you tell which is which?
- Transforming the functional equation into $x \cdot y = f^{-1}(f(x) + f(y))$ not only helps us to re-discover it, but also reveals the historical reason for discovery of logarithmic functions. In 16th and 17th centuries, people were trying to simplify multiplications of two large numbers. Instead of multiplying them, they could find the logarithms of x and y

in tables, add these (smaller) values and use inverse tables to lookup the result to get the product $x \cdot y$.

- In order to determine the base of the logarithm, we should realise what does the logarithm tell us: If we are looking for $\log_b x$, we are essentially asking, to what power do we need to raise the base b to get the value x ? The relation $f(1) = 0$ tells us that $\log_b 1 = 0$. This is not very helpful, since for every positive real number b we have $b^0 = 1$.

Therefore, the crucial piece of information is in the final relations. For example, in a) we are looking for such logarithm base b that gives us 2 when raised to 1. For which base do we get $b^1 = 2$?

Solution: Since the last hint is quite explicit, we assume that the bases of the logarithms can now be easily determined. Therefore, here we only provide the answers without additional explanation:

- We are looking for a base which gives 2 when raised to the power 1. Therefore $b = 2$ and the function is $\log_2 x$.
- Now we are looking for a base which gives 2 when raised to power $\frac{1}{2}$, i.e. when we take a square root. Since $2 = \sqrt{4}$, we have $b = 4$ and the function is $\log_4 x$.
- Finally, we have $9 = b^2$. The base is 3 and the function is $\log_3 x$, Figure 11.4.

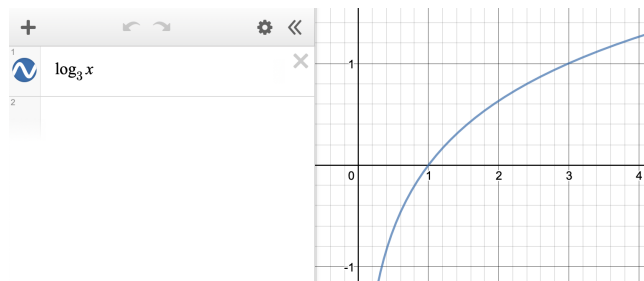


Figure 11.4: Logarithmic function.

11.4 Lab problems

1. Trigonometric functions

Draw the graph of a sine function on the interval approximately $(-20, 20)$ and save it for later comparison.

Then observe how the graph changes if the function changes to $a \sin(x)$, $\sin(x) + a$, $\sin(ax)$, $\sin(x - a)$ for various values of parameter a and generalise to $a \sin(bx + c) + d$. Next, look for parameters that change the range of sine function to $(-2, 2)$, $(-4, 4)$, $(0, 4)$, $(-3, 7)$, $(0.5, 1.5)$, $(3, 3.1)$, $(1/2, 1/3)$, $(2/5, 4/7)$. Can the parameters be determined from the end-points of the range interval?

Change the period of the sine function. Try to double it, triple it, quadruple it. Make it half, quarter, etc. Which parameter causes this? How to cause the period to be 2, 1, 5, 10?

Shift the sine curve to the left and to the right. Which parameter needs to be changed and how? Can you make the graph intersect the y -axis at the point $1/2$, $\sqrt{2}/2$, $-\sqrt{3}/2$?

Using the examples above, find the following sine curves:

- Period = $\pi/2$, range = $\langle -2, 2 \rangle$, $f(0) = 1$
- Period = 2, range = $\langle \pi, 2\pi \rangle$, $f(0) = \pi$
- Period = 3π , range = $\langle 3, 4 \rangle$, $f(0) = 3.25$
- Period = 5π , range = $\langle 4, 14 \rangle$, $f(0) = 4$
- Period = 10, range = $\langle \pi, 5\pi \rangle$, $f(0) = 4\pi$
- Period = 10π , range = $\langle 0, 2 \rangle$, $f(0) = 1 + \sqrt{3}/2$
- Period = 2, range = $\langle -1, 1 \rangle$, $f(0) = -\sqrt{2}/2$

2. Increase and decrease of a function, intuitive derivative

Tabulate a quadratic function $ax^2 + bx + c$ at 20 points, using input parameters a , b , c and lower and upper bounds of the desired range. In the next column watch the rate of change of the function relative to the change in x , i.e.

$$\frac{f(x_{n+1}) - f(x_n)}{x_{n+1} - x_n}.$$

Draw the graph of the original function and the graph of the function describing its rate of change. How are they related?

Which changes of the quadratic function change and which do not change

the rate of change function? How are the important points and properties of the two functions related?

Do the same for a cubic function? What kind of function describes its rate of change? How are they related?

Do the same also for sine, cosine and the exponential function.

3. Value of exponential function calculated at irrational points

Calculate the value of an exponential function a^x for an irrational point x , e.g. $x = \sqrt{2}$. The base a and the irrational number x are given as input parameters.

For denominators $1, 2, 3, 4, 5, \dots, 100$ find the best lower estimates of the irrational number by a fraction with the given denominator. For these fractions k/l calculate the values $a^{k/l}$. Display them in a chart and compare them with a line drawn at a^x (calculated by Excel). Check whether the values $a^{k/l}$ are getting closer to the exact value. How does the behavior change if we sort the table according to the value k/l ? How does the behavior change for different bases and different irrational exponents?

11.4.1 Hints and solutions

1. Trigonometric functions

Hints:

- This problem is very suitable for experimentation. As in previous chapter, we again recommend Desmos [1]. Figure 11.5 illustrates this environment.
- The Problem 4 from Section 11.3.1 is very similar to this problem and its hints and solutions can be used here.

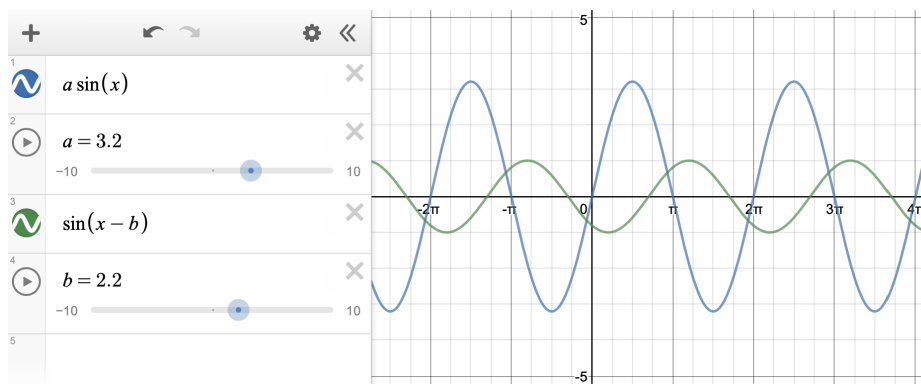


Figure 11.5: Changing amplitude and performing a horizontal shift of a sine function.

2. Increase and decrease of a function, intuitive derivative

Hints:

- First, we tabulate the quadratic function at 20 values of the independent variable x from the given range, similarly to Problem 2 in Section 10.4.
- We need a column for calculation of the increase (decrease) of the function values between two successive tabulated steps. In general, this is an increase (decrease) of the function values $f(x)$ divided by the increase in independent variable x . The increase in the independent variable is always the same in our case, therefore the calculation can refer to a constant value of this step.

- We should draw the function values and the column with rate of change into a common graph. The graph of the function is a parabola. What is the graph of the rate of change? Try changing the lower and upper bounds of the range for the independent variable. Also try changing the parameters of the quadratic function and observe their impact on the function representing the rate of change.
- To modify this worksheet for another function, all we need to do is use a different formula for calculation of $f(x)$. Everything else remains the same.
- For an exponential function, we see that the graph of changes is very similar to the original graph. Can you find such a base of the exponential function that the graphs are the same? For an exact answer, we recommend using an interval $(1, 2)$ or smaller. Another interesting connection is between the functions $\sin(x)$ and $\cos(x)$. The rate of change of a cubic function corresponds to another (simpler) polynomial.
- The rate of change of a quadratic function is captured by a graph of a linear function. Similarly, the rate of change of a cubic function is captured by a one-degree smaller, i.e. quadratic, polynomial. The exponential function that is identical to its own rate of change is the e^x which is one of the reasons behind its special status among the exponential functions. The rate of change of $\sin(x)$ corresponds to $\cos(x)$ and the rate of change of $\cos(x)$ corresponds to $-\sin(x)$.

Solution: An example solution is shown in Figure 11.6.

First we calculate the tabulation step in I5. Then we determine the values of the arguments x in the column B and values of the quadratic function $f(x) = ax^2 + bx + c$ given by the input parameters a , b , c in the column D. We calculate the differences in function values $f(x)$ in the column E and divide them by the constant step in the column F.

The values of $f(x)$ and its rate of change $\frac{f(x_{n+1}) - f(x_n)}{x_{n+1} - x_n}$ can be shown using a suitable x - y graph. We can observe that the rate of change is a linear function. This linear function remains unchanged if we change the parameter c of the quadratic function. Observe the changes caused by changing the input parameters a and b .

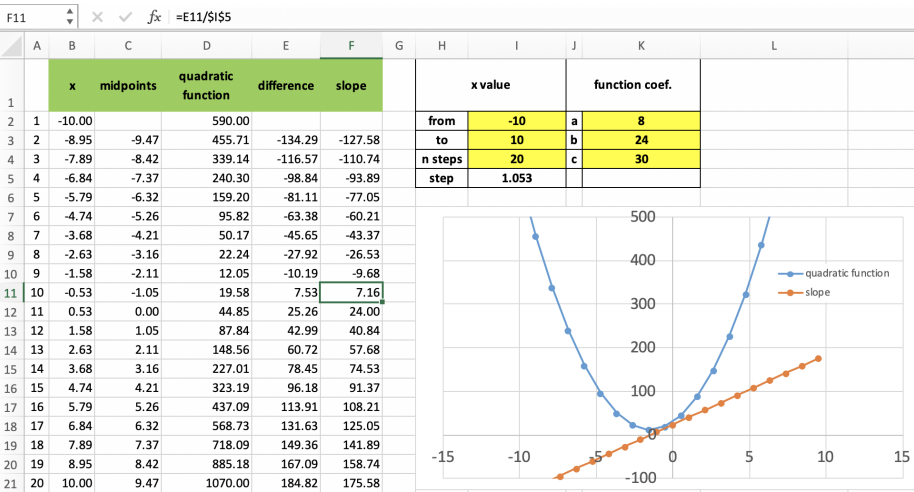


Figure 11.6: A quadratic function and its rate of change.

3. Value of exponential function calculated at irrational points

Hints:

- In this problem we follow a very similar approach to Problem 2 in Section 5.4. For example, when we are looking for a fraction of the form $k/5$, which approximates the value $\sqrt{3}$ best from below, the number k can be obtained by rounding $5\sqrt{3}$ down. In Excel, this is calculated using the formula `=ROUNDDOWN(SQRT(3)*5;0)`.
- One column should contain the denominators l from 1 to 100. Another column should calculate the numerators k and the next one fractions k/l . Then we can use the `POWER()` function to calculate the values $a^{k/l}$.
- We should also create columns for the *exact* values (as the Excel calculates them) and display them together with our approximations in an appropriate chart.

Solution:

An example solution is shown in Figure 11.7. The input parameters are entered in column K. Those shown correspond to calculation of $2^{\sqrt{3}}$.

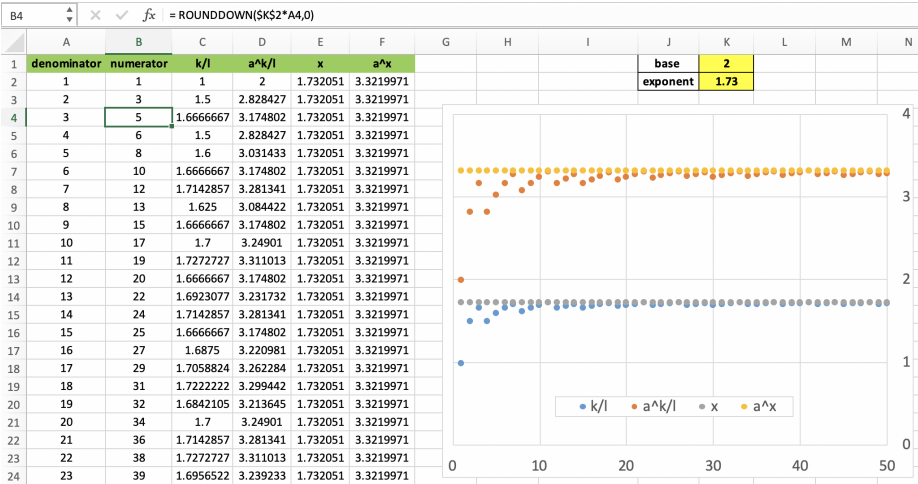


Figure 11.7: Exponential function at an irrational point approximated using fractions.

In column A, we have tabulated the denominators from 1 to 50. In column B, we use the formula $\text{ROUNDDOWN}(K2 \cdot A4, 0)$ to find the optimal numerator. In column D, we calculate the values $2^{k/l}$ using $K1^{C3}$. The constant values $\sqrt{3}$ in column E and $2^{\sqrt{3}}$ in column F are used for drawing the exact values.

It is highly recommended to experiment with the input base and exponent and observe how the fractions k/l and powers $a^{k/l}$ approximate the desired values. Also try changing the function ROUNDDOWN to regular ROUND . What happens?

Chapter 12

Asymptotic approximations and proofs

12.1 Lecture overview

12.1.1 Asymptotic approximations

Where do we see $\mathfrak{o}$ and $\mathcal{O}$?

When evaluating *computational complexity of algorithms*, it is worth considering asymptotic approximations of sequences and functions. For example, if we look at sorting algorithms, we find out that BubbleSort is $\mathcal{O}(n^2)$ and QuickSort is $\mathcal{O}(n \log n)$. What does the big $\mathcal{O}$ mean and which of the algorithms is better?

When considering the efficiency of the algorithms, two things need to be considered: How long does the computation last for a given size of input? And how fast does this time grow with growing inputs? If we can write a functional dependence of computational time on the size of the input, then these questions can be answered using analytical methods for asymptotic comparison of functions with the help of $\mathcal{O}$ notation.

Little $\mathfrak{o}$ is useful for approximations. In Section 11.1.3, we saw that for x close to 0 the value of an exponential function (and similarly also values of trigonometric functions) can be calculated using infinite series:

$$e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

In practice, for technical or physics applications it is sufficient to obtain the value of e^x or $\sin(x)$ with certain precision therefore we can stop the calculation at some point (we sum only finite series). In doing this, we make an approximation error, which can be estimated if we know what we had neglected. For example, if we take only the first three terms when evaluating e^x , we can say that the approximation error is of the order $\mathcal{O}(x^3)$, i.e. negligible when compared to x^2 , which is denoted by $\mathfrak{o}(x^2)$:

$$e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \mathfrak{o}(x^2).$$

If we take only the first three terms when evaluating $\sin(x)$, we can say that the approximation error is of the order $\mathcal{O}(x^7)$, which is negligible when compared to x^5 :

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \mathfrak{o}(x^5).$$

We formulate the following definitions of asymptotic approximations for functions (sequences) from $\mathbb{N}$ to $\mathbb{R}$ and their limits as $n \rightarrow \infty$. However, they can be extended to functions from $\mathbb{R}$ to $\mathbb{R}$ and limits in other accumulation points, e.g. $x \rightarrow 0$.

Little $\mathfrak{o}$ notation

For two functions (sequences) $f(n)$ and $g(n)$ from $\mathbb{N}$ to $\mathbb{R}$, we say that $f \in \mathfrak{o}(g)$, (sometimes also denoted $f = \mathfrak{o}(g)$) if and only if

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0.$$

We say that the function f is *asymptotically smaller* than g . This means that the function f is significantly smaller than g and we can neglect it. $\mathfrak{o}(g)$ is a set of functions.

Big $\mathcal{O}$ notation

For two functions $f(n)$ and $g(n)$ from $\mathbb{N}$ to $\mathbb{R}$, we say that $f \in \mathcal{O}(g)$, (sometimes also denoted $f = \mathcal{O}(g)$) if and only if there exists a $c \in \mathbb{R}$ and an $n_0 \in \mathbb{N}$, such that $\forall n \geq n_0$ we have:

$$|f(n)| \leq c \cdot |g(n)| \quad \text{or} \quad -c \cdot |g(n)| \leq f(n) \leq c \cdot |g(n)|$$

The function f is asymptotically at most a constant multiple of g . This includes all cases where f is asymptotically smaller than g . $\mathcal{O}(g)$ is a set of functions.

Theta notation

For two functions $f(n)$ and $g(n)$ from $\mathbb{N}$ to $\mathbb{R}$ we say that $f = \Theta(g)$ if and only if $f \in \mathcal{O}(g)$ and $g \in \mathcal{O}(f)$. We say that the function f differs from g at most by a constant multiple.

Commentary

$f \in \mathcal{O}(g)$ essentially says: *There is at least one constant c for which the inequality $f(n) < c \cdot g(n)$ holds for all n .*

$f \in \mathfrak{o}(g)$ says that: *For any constant $c > 0$ we can find a number n_0 such that the inequality $f(n) < c \cdot g(n)$ holds for all $n > n_0$.*

The difference between these two criteria is whether the functions can be asymptotically *equal*. Intuitively: big $\mathcal{O}$ says that f *does not grow faster than* g (it can grow at the same rate or slower) and little $\mathfrak{o}$ says f *grows strictly slower than* g . In some sense, the big $\mathcal{O}$ is like $\leq$ and little $\mathfrak{o}$ like $<$. $f \in \Theta(g)$ means that the growth rate of f is the same as the growth rate of g .

So for example the following statements are true for big $\mathcal{O}$ and not for little $\mathfrak{o}$:

$$n^2 \in \mathcal{O}(n^2)$$

$$n^2 \in \mathcal{O}(n^2 + n)$$

$$n^2 \in \mathcal{O}(200 \cdot n^2)$$

$$200 \cdot n^2 \in \mathcal{O}(n^2)$$

And the following statements are true for little $\mathfrak{o}$ (and therefore also for big $\mathcal{O}$):

$$n^2 \in \mathfrak{o}(n^3)$$

$$n^2 \in \mathfrak{o}(n!)$$

$$\ln(n) \in \mathfrak{o}(n)$$

While the Θ relation is symmetric, the $\mathcal{O}$ and $\mathfrak{o}$ clearly are not.

For many functions we can also use limits to determine $\mathcal{O}$. If the $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$ exists and is final (including the possibility that it is equal to 0), then $f \in \mathcal{O}(g)$. Similarly, if the limit exists and is non-zero, then $f \in \Theta(g)$.

$\mathcal{O}$ and Θ notation for complexity of algorithms

Formally, we should use the Θ notation when talking about the computational complexity of algorithms. Exactly speaking, when we say that *Algorithm A has computational complexity $\mathcal{O}(n^2 + 3n)$* , it means that the *complexity of A is asymptotically at most $n^2 + 3n$* . This does not exclude the possibility that it could actually be better. Can you give such an example?

However, in practice, people tend to use $\mathcal{O}$ in place of Θ and we adopt the same approach when talking about computational complexity.

Problem 12.1 Asymptotic comparison:

For the functions $f(n) = 2n^2$ and $g(n) = 4n$ determine whether they are in any of the following relations $f \in \mathfrak{o}(g)$, $g \in \mathfrak{o}(f)$, $f \in \mathcal{O}(g)$, $g \in \mathcal{O}(f)$.

Solution: We first examine whether $f \in \mathfrak{o}(g)$.

$$\lim_{n \rightarrow \infty} \frac{2n^2}{4n} = \lim_{n \rightarrow \infty} \frac{n}{2} = \infty \neq 0.$$

Thus f is not asymptotically smaller than g . Next we consider the $f \in \mathcal{O}(g)$ relation and so we look for a constant c such that $|2n^2| < c \cdot |4n|$ for all n starting at n_0 . That would imply $|n| < 2c$, which is a contradiction, because for any fixed c we can find an $n_0 > c$. Therefore $f \notin \mathcal{O}(g)$.

Now we try the reverse order

$$\lim_{n \rightarrow \infty} \frac{4n}{2n^2} = \lim_{n \rightarrow \infty} \frac{2}{n} = 0.$$

Therefore g is asymptotically smaller than f , i.e. $g \in \mathfrak{o}(f)$. When considering $g \in \mathcal{O}(f)$, we are looking for a constant c such that $|4n| < c \cdot |2n^2|$ for all n starting at n_0 . Suppose we take $n_0 = 2$ and $c = 2$. We get the relation:

$$\begin{aligned} |4n| &< 2 \cdot |2n^2| \\ |n| &< |n^2| \end{aligned}$$

The last inequality holds for all $n > n_0$, so we have $g \in \mathcal{O}(f)$.

Related to the previous problemCan we take a smaller c ?**Problem 12.2 Asymptotic comparison:**Find the asymptotic relation(s) between $f(n) = \frac{2n-3}{n+5}$ and $g(n) = 54$.*Solution:*

$$\lim_{n \rightarrow \infty} \frac{\frac{2n-3}{n+5}}{54} = \dots = \frac{1}{27}$$

$$\lim_{n \rightarrow \infty} \frac{54}{\frac{2n-3}{n+5}} = \dots = 27$$

So $f \notin \mathfrak{o}(g)$, $g \notin \mathfrak{o}(f)$. The inequality

$$\left| \frac{2n-3}{n+5} \right| < c \cdot |54|$$

holds for e.g. $c = 1$ a $n_0 = 1$. Therefore $f \in \mathcal{O}(g)$. Moreover,

$$|54| < c \cdot \left| \frac{2n-3}{n+5} \right|$$

holds for e.g. $c = 54$ a $n_0 = 10$, so we also have $g \in \mathcal{O}(f)$. The last two relations together tell us that $g \in \Theta(f)$ and vice versa. Note that we could have made this conclusion also from either of the non-zero finite limits that we calculated at the beginning.

Problem 12.3 Asymptotic comparison:Find the asymptotic relation(s) between $f(n) = 1 + (n \cdot \sin(n\frac{\pi}{3}))^2$ and $g(n) = 4n$.*Solution:* The limit

$$\lim_{n \rightarrow \infty} \frac{1 + (n \cdot \sin(n\frac{\pi}{3}))^2}{4n}$$

and the limit

$$\lim_{n \rightarrow \infty} \frac{4n}{1 + (n \cdot \sin(n\frac{\pi}{3}))^2}$$

do not exist, so $f \notin \mathfrak{o}(g)$, $g \notin \mathfrak{o}(f)$. The $\mathcal{O}$ relations also do not hold because f oscillates between zero and a quadratic function and therefore (starting at a suitably chosen n_0) is alternatively above and below the linear function $4n$.

Problem 12.4 Asymptotic comparison:

Find the asymptotic relation(s) between $f(n) = 2^n$ and $g(n) = 5^n$.

Solution:

$$\lim_{n \rightarrow \infty} \frac{2^n}{5^n} = \dots = 0$$

$$\lim_{n \rightarrow \infty} \frac{5^n}{2^n} = \dots = \infty$$

So we have $f \in \mathfrak{o}(g)$ (and consequently also $f \in \mathcal{O}(g)$), $g \notin \mathfrak{o}(f)$. The inequality

$$\begin{aligned} |5^n| &< c \cdot |2^n| \\ \left(\frac{5}{2}\right)^n &< c \end{aligned}$$

is not true for any c , because we can always find an n_0 such that for $n > n_0$ is the left-hand side greater than the right-hand side. So $g \notin \mathcal{O}(f)$.

Problem 12.5 Asymptotic comparison:

Find the asymptotic relation(s) between $f(n) = n^{11}$ and $g(n) = e^n$.

Solution:

$$\lim_{n \rightarrow \infty} \frac{n^{11}}{e^n} = \dots = 0$$

$$\lim_{n \rightarrow \infty} \frac{e^n}{n^{11}} = \dots = \infty$$

So we have $f \in \mathfrak{o}(g)$ (and consequently also $f \in \mathcal{O}(g)$), $g \notin \mathfrak{o}(f)$. The inequality

$$\begin{aligned} |e^n| &< c \cdot |n^{11}| \\ \frac{e^n}{n^{11}} &< c \end{aligned}$$

is not true for any c , because we can always find an n_0 such that for $n > n_0$ is the left-hand side greater than the right-hand side, e.g. for $c = 1$, $n_0 = 50$. n_0 can be found using the graph of e^n/n^{11} . So $g \notin \mathcal{O}(f)$.

Problem 12.6 Asymptotic comparison:

Find the asymptotic relation(s) between $f(n) = (\log(n))^k$ and $g(n) = n^k$.

Solution: Assuming k is a natural number

$$\lim_{n \rightarrow \infty} \frac{(\log(n))^k}{n^k} = \dots = 0$$

$$\lim_{n \rightarrow \infty} \frac{n^k}{(\log(n))^k} = \dots = \infty$$

So we have $f \in \mathfrak{o}(g)$ (and consequently also $f \in \mathcal{O}(g)$), $g \notin \mathfrak{o}(f)$. The inequality

$$|n^k| < c \cdot |(\log(n))^k|$$

$$\left| \left(\frac{n}{\log(n)} \right)^k \right| < c$$

is not true for any c , because we can always find an n_0 such that for $n > n_0$ is the left-hand side greater than the right-hand side, e.g. for $c = 1$, $n_0 = 5$. So $g \notin \mathcal{O}(f)$.

Redundant calculations

In the previous problems, we considered both limits for f/g and g/f . Could we tell the value of one of them once we calculate the value of the other? How? If so, this would save us half of the work when considering $\mathfrak{o}$. Moreover, could we also deduce something for $\mathcal{O}$ from this one limit?

12.1.2 Proofs

In general, a mathematical theorem is stated in the form *Suppose X is an object that has properties $A, B, C, \dots$. Then something else has to be true (about this object or about some other object)*. A *proof* of such statement is a sequence of logical steps which shows the validity of the statement without a doubt and meets the necessary mathematical criteria. The proof typically:

- clears the doubts about the general validity of the statement,
- shows how the statement is related to other mathematical statements,
- helps us understand the statement and the meaning and importance of terms used in it.

Sometimes it is useful to reformulate the original statement, because it might be easier to prove the reformulated version.

There are several types of proofs some of which are listed here:

Direct proof

Using a finite number of steps, we derive the statement from already proven statements and/or axioms. When proving an implication $A \Rightarrow B$ directly, we add A to the assumptions and then show that B has to be true.

Problem 12.7 Inequality:

Prove: $\frac{1}{3-2\sqrt{2}} < 6$

Solution: $\frac{1}{3-2\sqrt{2}} \cdot \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = \frac{3+2\sqrt{2}}{3^2-(2\sqrt{2})^2} = \frac{3+2\sqrt{2}}{9-8} = 3+2\sqrt{2} < 3+2 \cdot 3/2 = 6$

To prove the inequality, we used the relation $\sqrt{2} < 3/2$.

Problem 12.8 Property of divisibility by 2:

Prove: $\forall n \in \mathbb{N} \quad 2|n \Rightarrow 2|n^2$

Solution: $2|n \Rightarrow \exists k \in \mathbb{N} : n = 2k$

$\Rightarrow n^2 = 2k \cdot 2k$

$\Rightarrow \exists m \in \mathbb{N} : n^2 = 2m$, it is $m = 2k^2$

$\Rightarrow 2|n^2$

Indirect proof

It is mostly used for proving implications $A \Rightarrow B$. We start with a *converse* $B' \Rightarrow A'$ and prove it directly. We are using the fact that an implication and its converse always have the same truth value.

Problem 12.9 Another property of divisibility by 2:

Prove: $\forall n \in \mathbb{N} \quad 2|n^2 \Rightarrow 2|n$

Solution: Converse: $\forall n \in \mathbb{N} \quad 2 \text{ does not divide } n \Rightarrow 2 \text{ does not divide } n^2$.

Proof: $2 \text{ does not divide } n \Rightarrow \exists k \in \mathbb{N}_0 : n = 2k + 1$

$\Rightarrow n^2 = 4k^2 + 4k + 1$

$\Rightarrow \exists m \in \mathbb{N} : n^2 = 2m + 1$, it is $m = 2k^2 + 2k$

$\Rightarrow 2 \text{ does not divide } n^2$.

Proof by contradiction

First we make a negation of the statement we are trying to prove and then we perform a sequence of logical steps until we get to a contradiction. The

contradiction means that the starting point was incorrect and as implied by the law of excluded third (see Chapter 4), the original statement must have been true. We have already seen an example of this in Section 2.1.2, where we proved that there are infinitely many prime numbers.

Problem 12.10 Injective function:

Prove that $f(x) = \frac{x}{3-x}$ is an injective function.

Solution: Suppose that the given statement is not true, i.e. that its negation holds: $f(x)$ is not an injective function and thus $\exists x_1, x_2 \in \mathbb{R} - \{3\} : x_1 \neq x_2 \wedge f(x_1) = f(x_2)$. So we have

$$\begin{aligned}\frac{x_1}{3-x_1} &= \frac{x_2}{3-x_2} \\ x_1(3-x_2) &= x_2(3-x_1) \\ 3x_1 - x_1x_2 &= 3x_2 - x_1x_2 \\ 3x_1 &= 3x_2 \\ x_1 &= x_2\end{aligned}$$

And this is a contradiction because we assumed $x_1 \neq x_2$.

Proof of equivalence

When proving a statement $A \Leftrightarrow B$, it is necessary to prove both $A \Rightarrow B$ and $B \Rightarrow A$.

Problem 12.11 Divisibility by 6:

Prove: A natural number is divisible by 6 $\Leftrightarrow$ it is divisible by 3 and by 2.

Solution: First we prove that $A \Rightarrow B$:

When a number n is divisible by 6 $\Rightarrow \exists k \in \mathbb{N} : n = 6k = 2 \cdot 3k = 3 \cdot 2k \Rightarrow 2|n \wedge 3|n$

Next we prove $A \Leftarrow B$:

When a number is divisible by 3 and by 2 $\Rightarrow \exists k \in \mathbb{N} : n = 3k \wedge 2|3k \Rightarrow 2|k \Rightarrow \exists m \in \mathbb{N} : k = 2m \Rightarrow n = 6m \Rightarrow 6|m$.

Mathematical induction

The mathematical induction is used for proving statements of the type: *For each natural number $n \in \mathbb{N}$ a statement $V(n)$ is true.* The proof is done in two steps:

1. We prove that the statement holds for $n = 1$.
2. We prove that $\forall k \in \mathbb{N} : V(k) \Rightarrow V(k + 1)$, i.e. that if V holds for k then it also holds for $k + 1$. $V(k)$ is called an *induction hypothesis*.

Problem 12.12 Sum of odd numbers:

Prove $\sum_{i=1}^n (2i - 1) = 1 + 3 + 5 + \dots + (2n - 1) = n^2$.

Solution: First, we verify that the statement is true for $n = 1$. By plugging into both sides of the equation we get $1 = 1^2$, so this is ok.

Next we assume that the statement is true for $n = k$ and we check whether it implies that the statement holds also for $n = k + 1$. For $n = k + 1$, we have on the left-hand side

$$\begin{aligned} \sum_{i=1}^{k+1} (2i - 1) &= 1 + 3 + 5 + \dots + (2k - 1) + (2(k + 1) - 1) = \\ &= \sum_{i=1}^k (2i - 1) + (2(k + 1) - 1) = k^2 + 2k + 1. \end{aligned}$$

The last equality uses the induction hypothesis. And finally: $k^2 + 2k + 1 = (k + 1)^2$. This is exactly what we needed.

12.2 Questions

1. What is the computational complexity of an algorithm and why should we know it?
2. Which of the $\mathcal{O}(f)$ or $\mathfrak{o}(f)$ is used when expressing the computational complexity of an algorithm and why?
3. Which asymptotic comparisons are used when approximating functions using infinite series?
4. The computational complexity of BubbleSort is $\mathcal{O}(n^2)$ and the computational complexity of QuickSort is $\mathcal{O}(n \log n)$. Which of them is better?
5. Algorithm A has computational complexity $\mathcal{O}(n^2 + 3n)$ and algorithm B has computational complexity $\mathcal{O}(n^2 + 24)$. Which of them is better?
6. Algorithm C has computational complexity $\mathcal{O}(n \log n^2)$ and algorithm D has computational complexity $\mathcal{O}(n^2 \log n)$. Which of them is better?
7. A function $f(n)$ is asymptotically smaller than $g(n)$, i.e. $f \in \mathfrak{o}(g)$, iff:

a)

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$$

or iff

b) there exists a $c \in \mathbb{R}$ and $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0 : |f(n)| \leq c|g(n)|$?

8. A function f is either asymptotically smaller or at most larger by a constant multiple than g , iff:

a)

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$$

or iff

b) there exists a $c \in \mathbb{R}$ and $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0 : |f(n)| \leq c|g(n)|$?

9. Find two functions $f(n)$ and $g(n)$ from $\mathbb{N}$ to $\mathbb{R}$ for which $f(n)$ differs from $g(n)$ at most by a constant multiple and $f \neq \Theta(g)$.

10. Give both direct and indirect proof of the same theorem. Choose your own theorem or try to prove: *If a natural number is divisible by 6, then it is even.*
11. Prove by contradiction: *There are infinitely many prime numbers.*
12. Prove: *The sum of the first n natural numbers is*

$$\frac{n \cdot (n + 1)}{2}$$

- a) by contradiction,
- b) by mathematical induction.

12.3 Problem set

Asymptotic approximations

1. Which functions belong to $\mathfrak{o}(n^2)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
2. Which functions belong to $\mathfrak{o}(n)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
3. Which functions belong to $\mathfrak{o}(1)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
4. Which functions belong to $\mathfrak{o}(n \log n)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
5. Which functions belong to $\mathcal{O}(n^3)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
6. Which functions belong to $\mathcal{O}(n^2)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
7. Which functions belong to $\mathcal{O}(n)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
8. Which functions belong to $\mathcal{O}(1)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
9. Which functions belong to $\mathcal{O}(n \log n)$ and which do not:
 $n, n^4, n^3+n, 5n, n+2, 7, n^3+n+4, \log(n^3), \log n, n \log n$
10. Which of the functions in the previous problems are $f = \Theta(g)$?
11. Find out which of the following pairs of functions $f(n)$ and $g(n)$ are in relations $f \in \mathfrak{o}(g), g \in \mathfrak{o}(f), f \in \mathcal{O}(g), g \in \mathcal{O}(f), f = \Theta(g)$. In the case of $\mathcal{O}$, find the closest bounding constant c .
 - a) $f(n) = 2n^2, g(n) = 4n$
 - b) $f(n) = \frac{2n-3}{n+5}, g(n) = 54$
 - c) $f(n) = 1 + (n \sin(n\frac{\pi}{3}))^2, g(n) = 4n$
 - d) $f(n) = 2^n, g(n) = 5^n$
 - e) $f(n) = n^2, g(n) = e^n$

$$\text{f) } f(n) = (\log(n))^k, \quad g(n) = n^k$$

Proofs

12. Prove that $\sqrt{2}$ is irrational, i.e. $\forall a, b \in \mathbb{N} : \frac{a}{b} \neq \sqrt{2}$. What kind of proof can we use? Can we also use it to prove that $\sqrt{3}$ is irrational? How will the proof break down if we attempt to prove that also $\sqrt{4}$ is irrational?
13. Prove the criterion of divisibility by 9, i.e. prove *A natural number is divisible by 9 iff the sum of its digits is divisible by 9*. What kind of proof can we use? How did we use this criterion when drawing the spirolaterals in Section 2.4? Can you modify the proof and use it for proving other divisibility criteria?
14. Prove that the following statements define the same set of quadrilaterals, i.e. that the statements are equivalent.
 - a) The set (*A*) of quadrilaterals that have all sides of equal length and all angles 90° .
 - b) The set (*B*) of quadrilaterals that have all sides of equal length and all angles of the same size.
 - c) The set (*C*) of quadrilaterals that are both rectangles and diamonds.
 - d) The set (*D*) of quadrilaterals that have diagonals of equal length. The diagonals are perpendicular to each other and cut each other in half.
15. Prove by mathematical induction $\forall n \in \mathbb{N}$:

a)

$$\sum_{k=1}^n k = \frac{n \cdot (n+1)}{2}$$

b)

$$\sum_{k=1}^n k^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$$

16. Suppose we have a few lines in a plane. No two of them are parallel, i.e. every pair of lines has an intersection. Moreover, no three lines intersect at a single point. How many intersections are there? Express this number as a function of number of lines and prove it using mathematical induction.

17. In the lecture, there is an indirect proof of the statement $2|n^2 \Rightarrow 2|n$. Repeat this proof for divisibility by 3, i.e. prove $3|n^2 \Rightarrow 3|n$. Then, try to prove $4|n^2 \Rightarrow 4|n$.

12.3.1 Hints and solutions

Asymptotic approximations

1. Which functions belong to $\mathfrak{o}(n^2)$ and which do not:
 n , n^4 , n^3+n , $5n$, $n+2$, 7 , n^3+n+4 , $\log(n^3)$, $\log n$, $n \log n$

Hints: The function f for which it is true that $f \in \mathfrak{o}(g)$ has to be significantly smaller than g . This is expressed by the little $\mathfrak{o}$. We use the definition: $f \in \mathfrak{o}(g)$ if and only if

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0,$$

i.e. for functions $f(n) = n$ and $g(n) = n^2$, we need to check if the limit

$$\lim_{n \rightarrow \infty} \frac{n}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

is equal to zero. We can also draw the functions $f(n) = n$ and $g(n) = n^2$ as in Figure 12.1 together with their ratio $\frac{1}{n}$.

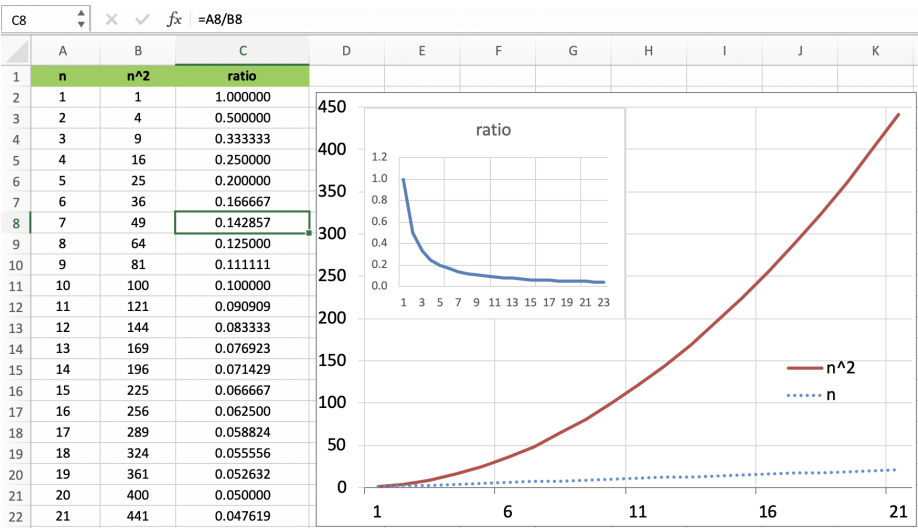


Figure 12.1: Functions $f(n) = n$, $g(n) = n^2$ and their ratio $\frac{n}{n^2}$.

Similarly, we should check if the limit of the ratio of $f(n) = n \cdot \log n$ and $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{n \log n}{n^2} = \lim_{n \rightarrow \infty} \frac{\log n}{n}$$

is equal to zero. For this calculation, we would need the tools of mathematical analysis. Alternatively, we can also draw the functions and their ratio and estimate the limit from the graph, see Figure 12.2.

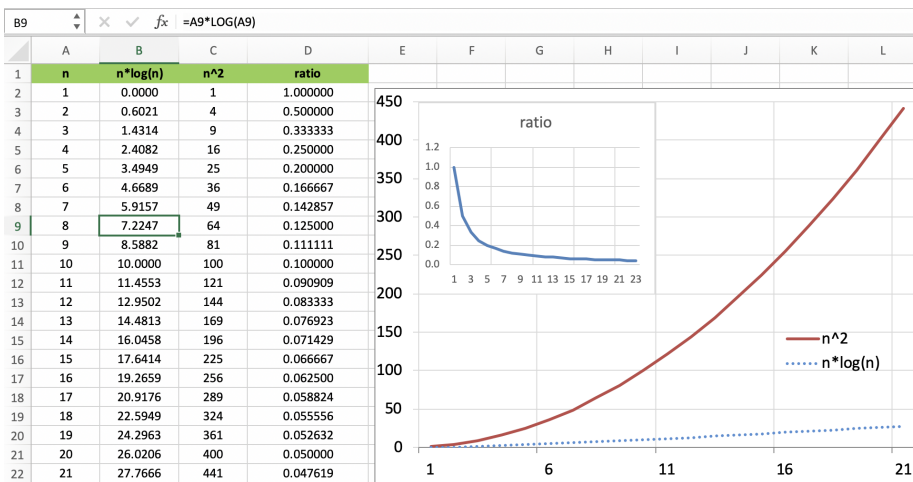


Figure 12.2: Functions $f(n) = n \log n$, $g(n) = n^2$ and their ratio $\frac{\log n}{n}$, which can be used to estimate the limit. Note the similarity to Figure 12.1

Solution:

$$\begin{aligned} n \in \mathfrak{o}(n^2), \quad n^4 \notin \mathfrak{o}(n^2), \quad n^3 + n \notin \mathfrak{o}(n^2), \quad 5n \in \mathfrak{o}(n^2), \\ n + 2 \in \mathfrak{o}(n^2), \quad 7 \in \mathfrak{o}(n^2), \quad n^3 + n + 4 \notin \mathfrak{o}(n^2), \quad \log(n^3) \in \mathfrak{o}(n^2), \\ \log n \in \mathfrak{o}(n^2), \quad n \log n \in \mathfrak{o}(n^2) \end{aligned}$$

5. Which functions belong to $\mathcal{O}(n^3)$ and which do not:
 n , n^4 , $n^3 + n$, $5n$, $n + 2$, 7 , $n^3 + n + 4$, $\log(n^3)$, $\log n$, $n \log n$

Hints: The function f , for which we want to have $f \in \mathcal{O}(g)$, has to be asymptotically smaller or at most as large as g . This is expressed by the big $\mathcal{O}$. Instead of just guessing, we examine the functions using the definition: $f \in \mathcal{O}(g)$ if and only if there exists a $c \in \mathbb{R}$ and $n_0 \in \mathbb{N}$, such

that $\forall n \geq n_0$ we have:

$$|f(n)| \leq c \cdot |g(n)| \quad \text{or} \quad -c \cdot |g(n)| \leq f(n) \leq c \cdot |g(n)|,$$

e.g. for functions $f(n) = 5n$ and $g(n) = n + 2$ we are looking for such $c \in \mathbb{R}$ and $n_0 \in \mathbb{N}$ that $\forall n \geq n_0$ we get the desired inequality.

The inequality is true for example for $c = 6$:

$$|5 \cdot n| \leq c \cdot |n + 2| \quad \text{or} \quad -c \cdot |n + 2| \leq 5 \cdot n \leq c \cdot |n + 2|$$

These functions are illustrated in Figure 12.3.

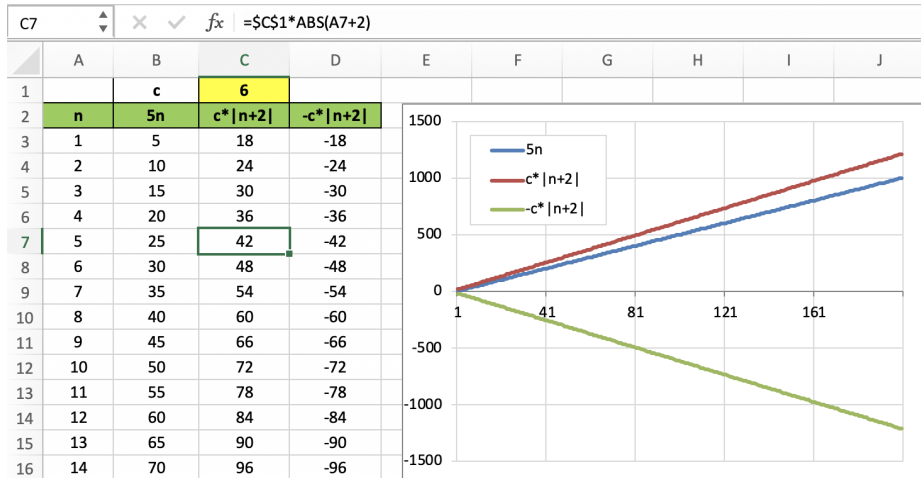


Figure 12.3: Functions $f(n) = 5 \cdot n$, $g(n) = n + 2$ and constant $c = 6$ for the estimate $f(n) \in \mathcal{O}(g(n))$.

Solution:

$$\begin{aligned} n &\in \mathcal{O}(n^3), \quad n^4 \notin \mathcal{O}(n^3), \quad n^3 + n \in \mathcal{O}(n^3), \quad 5n \in \mathcal{O}(n^3), \\ n + 2 &\in \mathcal{O}(n^3), \quad 7 \in \mathcal{O}(n^3), \quad n^3 + n + 4 \in \mathcal{O}(n^3), \quad \log(n^3) \in \mathcal{O}(n^3), \\ \log n &\in \mathcal{O}(n^3), \quad n \log n \in \mathcal{O}(n^3) \end{aligned}$$

10. Which of the functions in the previous problems are $f = \Theta(g)$?

Hint:

We need to find out if the functions can bound each other, i.e. if they are asymptotically the same.

Two functions f and g are in relation Θ , i.e. $f = \Theta(g)$ or $g = \Theta(f)$ if and only if $f \in \mathcal{O}(g)$ and $g \in \mathcal{O}(f)$. Since the functions f and g (with suitable multiplicative constants) are supposed to be bounds for one another, they should not differ too much. An example of such pair are $f(n) = 5n$ and $g(n) = n + 2$. We have already shown that $5n \in \mathcal{O}(n + 2)$. Now we show that also $n + 2 \in \mathcal{O}(5n)$. Here it is sufficient to take $c = 0.6$ and the inequality in the definition is true, see Figure 12.4. This way we showed that $f(n) = 5n$ and $g(n) = n + 2$ are in relation Θ .

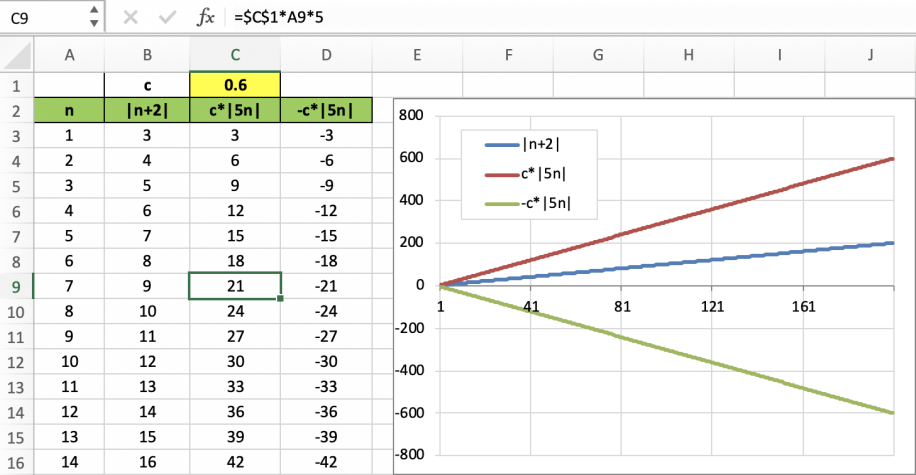


Figure 12.4: Functions $g(n) = n + 2$, $f(n) = 5n$ and constant $c = 0.6$ for the estimate $g \in \mathcal{O}(f)$.

Solution:

Functions $n + 2 \in \Theta(n)$, $5n \in \Theta(n)$. And of course every pair of these three is in relation Θ .

Functions $n^3 + n \in \Theta(n^3)$, $n^3 + n + 4 \in \Theta(n^3)$. And every pair of these three is in relation Θ .

Function $\log(n^3) \in \Theta(\log n)$.

For functions $f(n) = n \log n$, $f(n) = n^4$ and $f(n) = 7$ there are no functions on the list that would be in relation Θ with them.

11. Find out which of the pairs of functions f and g are in relations $f \in \mathfrak{o}(g)$, $g \in \mathfrak{o}(f)$, $f \in \mathcal{O}(g)$, $g \in \mathcal{O}(f)$, $f = \Theta(g)$. In the case of $\mathcal{O}$, find the closest bounding constant c .

Hints: We should use the same approach as in the previous problems. It is also a good idea to train our intuition by guessing first and then verifying the guess by calculations.

Solution: You can find more detailed solutions to these problems in the lecture overview.

- a) For functions $f(n) = 2n^2$, $g(n) = 4n$ we have $g \in \mathfrak{o}(f)$, $g \in \mathcal{O}(f)$ and $f \neq \Theta(g)$, $g \neq \Theta(f)$.
- b) For functions $f(n) = \frac{2n-3}{n+5}$, $g(n) = 54$ we have $g \notin \mathfrak{o}(f)$, and $f \notin \mathfrak{o}(g)$, but $g \in \mathcal{O}(f)$ and $f \in \mathcal{O}(g)$. Therefore we also have $f = \Theta(g)$ and $g = \Theta(f)$.
- c) For functions $f(n) = 1 + (n \sin(n \frac{\pi}{3}))^2$, $g(n) = 4n$ we have $g \notin \mathfrak{o}(f)$ and $f \notin \mathfrak{o}(g)$, but $g \in \mathcal{O}(f)$ and $f \in \mathcal{O}(g)$. Therefore we have both $f = \Theta(g)$ and $g = \Theta(f)$.
- d) For functions $f(n) = 2^n$, $g(n) = 5^n$ we have $g \notin \mathfrak{o}(f)$, but $f \in \mathfrak{o}(g)$. We also have $g \notin \mathcal{O}(f)$ and $f \in \mathcal{O}(g)$. Therefore $f \neq \Theta(g)$ and $g \neq \Theta(f)$.
- e) For functions $f(n) = n^2$, $g(n) = e^n$ we have $g \notin \mathfrak{o}(f)$, but $f \in \mathfrak{o}(g)$. We also have $g \notin \mathcal{O}(f)$ and $f \in \mathcal{O}(g)$. Therefore $f \neq \Theta(g)$ and $g \neq \Theta(f)$.
- f) For functions $f(n) = (\log(n))^k$, $g(n) = n^k$ we have $g \notin \mathfrak{o}(f)$, but $f \in \mathfrak{o}(g)$. We also have $g \notin \mathcal{O}(f)$ and $f \in \mathcal{O}(g)$. Therefore $f \neq \Theta(g)$ and $g \neq \Theta(f)$.

Proofs

12. Prove that $\sqrt{2}$ is irrational, i.e. $\forall a, b \in \mathbb{N} : \frac{a}{b} \neq \sqrt{2}$. What kind of proof can we use? Can we also use it to prove that $\sqrt{3}$ is irrational? How will the proof break down if we attempt to prove that also $\sqrt{4}$ is irrational?

Hints:

- This statement can be proven by contradiction. We can start by assuming that there exists a rational number $\frac{a}{b}$ equal to $\sqrt{2}$, with a and b not having a common divisor. Manipulating the expression

$\sqrt{2} = \frac{a}{b}$ leads to a contradiction: if such two numbers a and b existed, both would have to be divisible by two.

- For $\sqrt{3}$ we get a very similar contradiction with 3 being the common divisor of a and b .
- For $\sqrt{4}$ we get $4|a^2$, but this does not imply $4|a$ as in the previous two cases. We can certainly say that $2|a$ and get that for $a = 2$ and $b = 1$ the fraction $\frac{a}{b}$ is that rational number which is equal to $\sqrt{4}$.

13. Prove the criterion of divisibility by 9, i.e. prove *A natural number is divisible by 9 iff the sum of its digits is divisible by 9*. What kind of proof can we use? How did we use this criterion when drawing the spirolaterals in Section 2.4? Can you modify the proof and use it for proving other divisibility criteria?

Hints:

- A number x divisible by 9 can be written as a multiple of 9, i.e. $x = 9k$.
- A number $x = c_n c_{n-1} c_{n-2} \dots c_2 c_1 c_0$ written in decimal system can be written as $x = c_0 \cdot 1 + c_1 \cdot 10 + c_2 \cdot 100 + \dots + c_n \cdot 10^n$.
- We can divide the number x into a sum of two components $x = y + z$ first of which (y) is divisible by 9. In order for the whole number to be divisible by 9, also the second component (z) has to be divisible by 9.
- The number $10^n - 1$ is divisible by 9. This can be shown by using mathematical induction. Alternatively, we can realise that $10^n - 1 = 99 \dots 99 = 9 \cdot 11 \dots 11$.

Solution: We can set

$$y = c_0 \cdot 0 + c_1 \cdot 9 + c_2 \cdot 99 + \dots + c_n \cdot (10^n - 1)$$

The number y is always divisible by 9. In order for x to be divisible by 9, also

$$z = x - y = c_0 \cdot 1 + c_1 \cdot 1 + c_2 \cdot 1 + \dots + c_n \cdot 1$$

has to be divisible by 9. The number z is the sum of digits of x .

This was a direct proof, but we could have used an indirect proof or proof by contradiction.

When we were drawing the spirolaterals, we needed the sum of digits of the numbers and so we used the remainder after division by 9 (except for the situation when the number was divisible by 9).

We could use this method for example for proving criterion of divisibility by 3.

14. Prove that the following statements define the same set of quadrilaterals, i.e. that the statements are equivalent.
- a) The set (A) of quadrilaterals that have all sides of equal length and all angles 90° .
 - b) The set (B) of quadrilaterals that have all sides of equal length and all angles of the same size.
 - c) The set (C) of quadrilaterals that are both rectangles and diamonds.
 - d) The set (D) of quadrilaterals that have diagonals of equal length. The diagonals are perpendicular to each other and cut each other in half.

Hints:

- Proving equivalence means proving two implications, i.e. show that the statement is true *both ways*. With four sets, we have 6 equivalences to show, which means 12 implications.
- We can save some work by showing that we can get from each of these statements to any other statement, show e.g.

$$a) \Rightarrow b) \Rightarrow c) \Rightarrow d) \Rightarrow a).$$

This is equivalent to proving that all the sets are the same, i.e. $A = B = C = D$, by showing four subset relations

$$A \subset B \subset C \subset D \subset A.$$

- To prove that $A \subset B$ we need to show that every element from the set A belongs also to the set B . For example if there is a quadrilateral that has *all sides of equal length and all angles 90°* , then it also has *sides of equal length and all angles of the same size*. This is true because two right angles are angles of the same size.

15. Prove by mathematical induction $\forall n \in \mathbb{N}$:

a)

$$\sum_{k=1}^n k = \frac{n \cdot (n+1)}{2}$$

b)

$$\sum_{k=1}^n k^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$$

Hints:

The proof by mathematical induction uses the axiom that with every natural number, the set of natural numbers contains its successor, a number greater by 1. Once we show that if a statement is true for some number, then it is true for a number greater by 1, we have shown the statement for all natural numbers (provided that we have shown we can start this *domino effect* somewhere).

1. We prove that the statement is true for $n = 1$.

2. We prove the implication that if the statement is true for some natural number, it is also true for the next natural number.

Solution:

a) We use mathematical induction to prove the formula:

$$\sum_{k=1}^n k = \frac{n \cdot (n+1)}{2}$$

1.

$$\sum_{k=1}^1 k = 1, \quad \frac{1 \cdot (1+1)}{2} = 1$$

2. In the formula for $n+1$:

$$\sum_{k=1}^{n+1} k = \frac{(n+1) \cdot ((n+1)+1)}{2}$$

we start from the left-hand side and use the induction premise

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^n k \right) + (n+1) = \frac{n \cdot (n+1)}{2} + n+1 = \frac{n \cdot (n+1)}{2} + \frac{2(n+1)}{2} =$$

$$= \frac{n \cdot (n+1) + 2(n+1)}{2} = \frac{(n+1)(n+2)}{2} = \frac{(n+1) \cdot ((n+1) + 1)}{2}$$

b) We use mathematical induction to prove the formula:

$$\sum_{k=1}^n k^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$$

1.

$$\sum_{k=1}^1 k^2 = 1, \quad \frac{1 \cdot (1+1) \cdot (2 \cdot 1 + 1)}{6} = 1$$

2. In the formula for $n+1$:

$$\sum_{k=1}^{n+1} k^2 = \frac{(n+1) \cdot (n+2) \cdot (2n+3)}{6}$$

we start from the left-hand side and use the induction premise

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= \frac{n \cdot (n+1) \cdot (2n+1)}{6} + (n+1)^2 = \\ &= \frac{(n+1) \cdot n \cdot (2n+1)}{6} + \frac{6(n+1)^2}{6} = \\ &= \frac{(n+1) \cdot (2n^2 + n + 6n + 6)}{6} = \\ &= \frac{(n+1) \cdot (2n^2 + 7n + 6)}{6} = \frac{(n+1) \cdot (n+2) \cdot (2n+3)}{6} \end{aligned}$$

16. Suppose we have a few lines in a plane. No two of them are parallel, i.e. every pair of lines has an intersection. Moreover, no three lines intersect at a single point. How many intersections are there? Express this number as a function of number of lines and prove it using mathematical induction.

Hints: One line has 0 intersections.

Two lines have 1 intersection.

Three lines that do not intersect at one point have 3 intersections.

Four lines that do not intersect at one point have 6 intersections, see Figure 12.5.

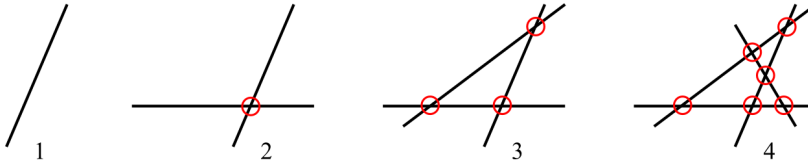


Figure 12.5: Intersections of 1, 2, 3 and 4 lines.

When we add another line, the number of intersections increases. By how much?

We observe that the number of intersections increases by the previous number of lines.

One line has 0 intersections.

Two lines have $0+1$ intersection.

Three lines have $1+2$ intersections.

Four lines have $3+3$ intersections.

Five lines have $6+4$ intersections.

etc.

We thus have a hypothesis that for n lines there are $0+1+2+3+\dots+n-1 = \frac{(0+n-1) \cdot n}{2}$ intersections.

Solution:

n lines have $\frac{(n-1) \cdot n}{2}$ intersections.

Proof by mathematical induction:

1. For $n = 1$ line, the number of intersections is $\frac{(n-1) \cdot n}{2} = 0$.

2. Suppose that for n lines we have $\frac{(n-1) \cdot n}{2}$. When we add one line, it crosses all the n lines already there. Therefore in addition to $\frac{(n-1) \cdot n}{2}$ intersections we have n more. The number of intersections would then be

$$\frac{(n-1) \cdot n}{2} + n = \frac{(n-1) \cdot n + 2n}{2} = \frac{n \cdot (n+1)}{2}.$$

We have shown that for $n+1$ lines, there are $\frac{n \cdot (n+1)}{2}$ intersections and thus if the statement is true for n , it is also true for $n+1$.

17. In the lecture, there is an indirect proof of the statement $2|n^2 \Rightarrow 2|n$. Repeat this proof for divisibility by 3, i.e. prove $3|n^2 \Rightarrow 3|n$. Then, try to prove $4|n^2 \Rightarrow 4|n$.

Hints: The proof for divisibility by 3 works the same way as the proof in the lecture. The proof will not work for divisibility by 4 because 4 is not a prime. We can find a counterexample, e.g. $n = 2$. For this n , the premise of the implication is true, since $4|n^2$, but the conclusion is false, we do not have $4|n$.

12.4 Lab problems

1. Bounding constants

- a) Find a constant c with which you can show that the sequence $f(n) = n + 5$ belongs to $\mathcal{O}(n^2)$, i.e.:

$$|n + 5| \leq c|n^2|, \quad \text{or} \quad -c|n^2| \leq n + 5 \leq c|n^2|.$$

- b) Find a constant c with which you can show that the sequence $f(n) = n^2$ belongs to $\mathcal{O}(n \log n)$, i.e.:

$$|n^2| \leq c|n \log n|, \quad \text{or} \quad -c|n \log n| \leq n^2 \leq c|n \log n|.$$

2. Big $\mathcal{O}$

Use graphs to show that a quadratic sequence is $\mathcal{O}(n^2)$ and a cubic sequence is $\mathcal{O}(n^3)$.

3. Approximation by polynomial expansion

Show how good is the approximation of functions $\sin x$, $\cos x$ and e^x using the first five terms of their expansions into series for $|x| \ll 1$. Which of the calculations is most precise? How does the theoretical estimate of error compare to the calculated error?

4. Computation of new prime numbers

In the proof of the statement that *there are infinitely many primes* we reasoned that *if there was only a finite number of primes, their product incremented by 1 would give a new prime number*. Can we find a set of primes that break this rule, i.e. when we increment their product by 1, we do not get a prime number? If we can find such a set, does it mean that the proof is incorrect?

12.4.1 Hints and solutions

1. Bounding constants

- a) Find a constant c with which you can show that the sequence $f(n) = n + 5$ belongs to $\mathcal{O}(n^2)$, i.e.:

$$|n + 5| \leq c|n^2|, \quad \text{or} \quad -c|n^2| \leq n + 5 \leq c|n^2|.$$

- b) Find a constant c with which you can show that the sequence $f(n) = n^2$ belongs to $\mathcal{O}(n \log n)$, i.e.:

$$|n^2| \leq c|n \log n|, \quad \text{or} \quad -c|n \log n| \leq n^2 \leq c|n \log n|.$$

Hint: We can create an Excel worksheet, where we can input various values of the constant c and try to find a value which shows that the sequence $n + 2$ belongs to $\mathcal{O}(n^2)$ (or that the sequence n^2 belongs to $\mathcal{O}(n \log n)$).

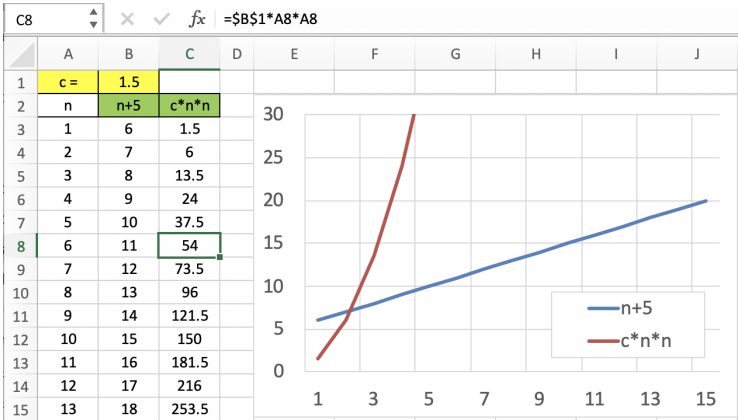


Figure 12.6: Looking for $c|n^2|$ bound for $n + 5$.

Solution:

- a) Note that even if we take a small value of c and the graph of the linear function is initially above the graph of the quadratic function for the first few values of n , eventually they two graphs cross and the quadratic function dominates because it grows faster than a linear function, see Figure 12.6.

- b) No matter how useful the software tool is and not matter how we try, we cannot find such value c for which we would have $|n^2| \leq c|n \log n|$, see Figure 12.7. Therefore $n^2 \notin \mathcal{O}(n \log n)$.



Figure 12.7: Attempts to find $c|n \log n|$ bounds for n^2 for $c = 15$ and $c = 25$.

2. **Big $\mathcal{O}$**

Use graphs to show that a quadratic sequence is $\mathcal{O}(n^2)$ and a cubic sequence is $\mathcal{O}(n^3)$.

Hint: We can start by creating an Excel worksheet with input parameters a, b, c that define the function $f(n) = an^2 + bn + c$. The constant D should also be a variable input. We tabulate and draw the sequences $f(n)$ and Dn^2 for n from 1 to 100. For various values of a, b, c we are looking for the constant D which would satisfy the inequality $|an^2 + bn + c| \leq D|n^2|$. This will manifest by a correct relative position of the graphs.

Similarly, we can show that for any parameters a, b, c, d in a cubic sequence, we can find a constant D , such that all terms of $f(n) = an^3 + bn^2 + cn + d$ are smaller than $D|n^3|$.

Solution: Figure 12.8 shows that a quadratic sequence is $\mathcal{O}(n^2)$.

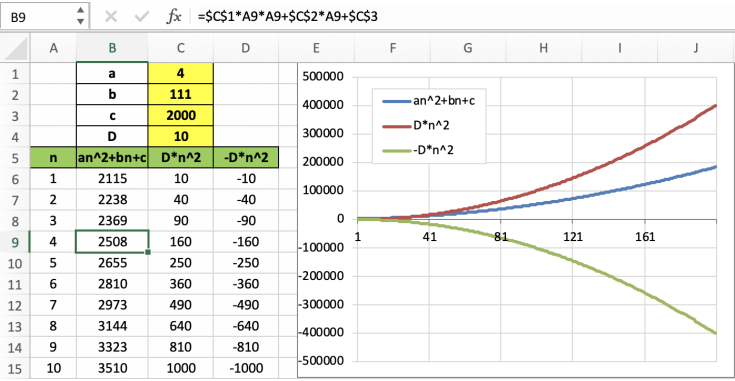


Figure 12.8: Bounding $an^2 + bn + c$ using a sequence Dn^2 .

Figure 12.9 shows that a cubic sequence is $\mathcal{O}(n^3)$.

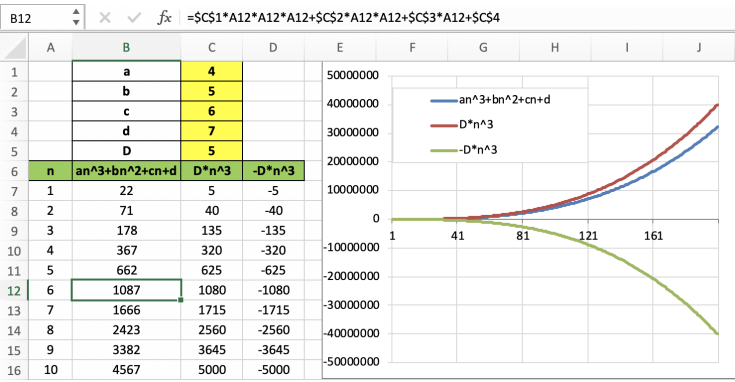


Figure 12.9: Bounding $an^3 + bn^2 + cn + d$ using a sequence Dn^3 .

3. Approximation by polynomial expansion

Show how good is the approximation of functions $\sin x$, $\cos x$ and e^x using the first five terms of their expansions into series for $|x| \ll 1$. Which of the calculations is most precise? How does the theoretical estimate of error compare to the calculated error?

Hint: We have discussed the infinite series for $\sin(x)$, $\cos(x)$, e^x in Chapter 11.

$$\begin{aligned}\sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \\ \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \\ e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots\end{aligned}$$

We start by testing the approximation accuracy for x from the interval $(0, 1)$ and then increasing the values of x . For which functions does the accuracy depend on x ? Does taking more terms of the series always increase the accuracy? On which part of the domain can we approximate the functions using their respective series reasonably well?

Solution: We can approximate the functions well near 0. For the function $\sin(x)$, we have the approximation error $\mathcal{O}(x^{11})$ when using 5 terms close to zero, see Figure 12.10. The error for $\cos(x)$ is $\mathcal{O}(x^{10})$ and the error for e^x , Figure 12.11, is $\mathcal{O}(x^6)$.

4. Computation of new prime numbers

In the proof of the statement that *there are infinitely many primes* we reasoned that *if there was only a finite number of primes, their product incremented by 1 would give a new prime number*. $2 \cdot 3 + 1 = 7$, $2 \cdot 3 \cdot 5 + 1 = 31$, ... Can we find a set of primes that break this rule, i.e. when we increment their product by 1, we do not get a prime number? If we can find such a set, does it mean that the proof is incorrect?

Hint: This method looks promising, but if it was truly working, we could keep building new primes (relatively) easily by multiplying existing primes and adding one. Since this is not happening, it is very likely that there is something wrong with this method. Try applying it to the first few primes (systematically in Excel) and check if all the results are indeed prime. If there was a composite result, could it be a product of new primes?

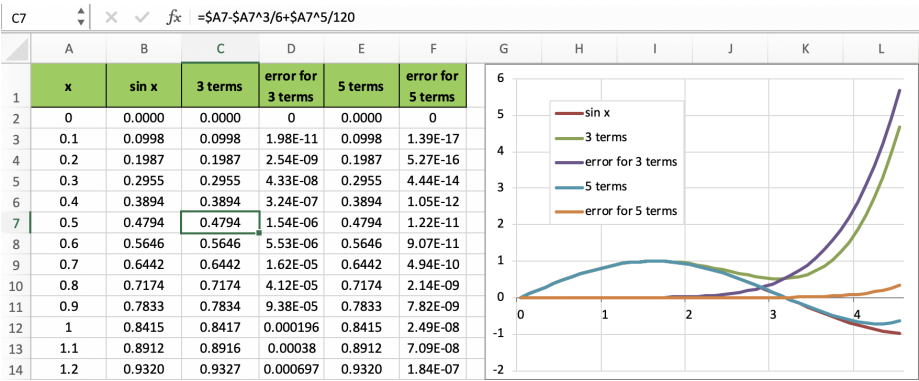


Figure 12.10: Approximation of the function $\sin(x)$ using power series and the error when using 3 and 5 terms.

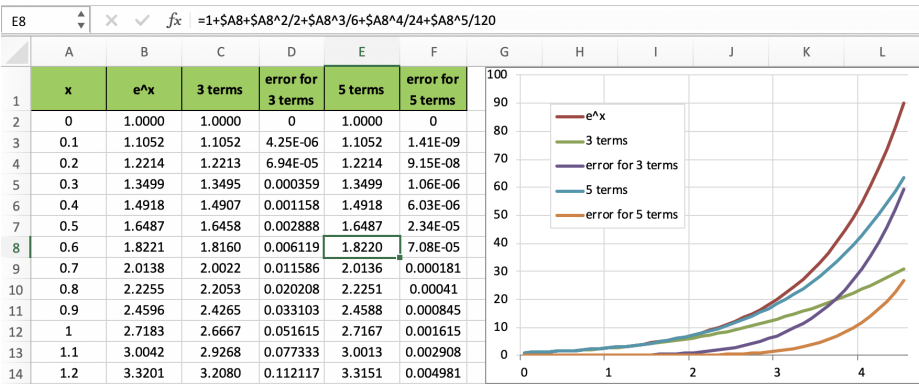


Figure 12.11: Approximation of the function e^x using power series and the error when using 3 and 5 terms.

Solution: A worksheet, such as the one in Figure 12.12, shows that this is not a universal method for creating new primes.

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 509 \cdot 59$$

The proof is still correct, because the implication *if there was only a finite number of primes, their product incremented by 1 would give a new prime*

D7				
	A	B	C	D
1	prime number	product	+1	
2	2	2		
3	3	6	7	
4	5	30	31	
5	7	210	211	
6	11	2310	2311	
7	13	30030	30031	509

Figure 12.12: Multiplication of first six primes incremented by 1 yields 30031, which is divisible by 59.

number was considered under the assumption that the first part *there was only a finite number of primes* is a true statement. It leads to a contradiction, thus it cannot be true. And in such case the implication is true regardless the truth value of its conclusion.

Farewell

Dear reader,

on the previous page, you dealt with the last problem in this book. If you are reading these words, we hope that some of the following statements are true:

- The content of the book was so interesting that you went through it all not only as a passive reader but actively solving the problems and questions that are the core of the book.
- Several times you experienced that the calculation, illustration or simulation on the computer helped you solve a problem or better understand the mathematical terms and principles.
- You also encountered the opposite situations in which mathematical understanding and correct use of mathematical tools simplified or sped up a computer algorithm, defined its main idea, helped avoid an error or understand the cause of an error.
- You are going to transfer the notion of interconnectedness of mathematics and computer science to your next studies and professional life.
- And if necessary, you will gladly return to the relevant chapters of this book, which will serve as a foundation and starting point for your study of more complex problems and other literature.

The final challenge

When you read the book the next time, solve each problem by a different method than before.

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